



DOCTORAL THESIS

Jet physics at the LHC and beyond

*A thesis submitted in fulfilment of the requirements
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Declaration

I, Natasha D’arcy Beaumont Peake, hereby declare that this thesis has not been, and will not be, submitted in whole or in part to another university for the award of any other degree. The work presented in this thesis includes research carried out in collaboration with Andrea Banfi, Melissa van Beekveld, Silvia Ferrario Ravasio, Alexander Karlberg, Matthew Lim, and Daniel Reichelt.

The content of chapter 3 is taken nearly verbatim from the paper [1], currently under peer review:

M. van Beekveld, S. Ferrario Ravasio, A. Karlberg and D. Peake, *A generalised- k_t jet algorithm for Deep Inelastic Scattering*, arXiv:2606.13077.

This algorithm builds on the angular-ordered ($p = 0$) variant introduced by M. van Beekveld and S. Ferrario Ravasio in Ref. [2], while the generalisation to parameter p was developed jointly by all authors. I implemented the initial versions of the algorithm, which were subsequently refined in collaboration with my co-authors, with M. van Beekveld preparing the version released in the plugin. I performed the majority of the computational work underlying the phenomenological results. The phenomenological studies, comparisons, and interpretation of the results were developed collectively. The plotting scripts were produced by S. Ferrario Ravasio using the results of my simulations, and the paper was written jointly by all authors.

Appendix A contains relevant parts of the code taken directly from the DISGenkt plugin.

The content of chapter 5 is based on a project that will be submitted for publication in due course:

D. Peake, A. Banfi, M.A. Lim and D. Reichelt, *NNLL resummation in Z +jet events with the ARES method*.

I performed the majority of the calculations presented in this chapter, with further input from A. Banfi, M.A. Lim and D. Reichelt. I calculated the NNLL corrections to the observable and carried out all validations presented here. Together with A. Banfi, I developed the treatment of initial-state radiation within the ARES formalism.

Brighton,

24th August 2026

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DOCTORAL THESIS

Jet physics at the LHC and beyond

by Natasha D'arcy Beaumont Peake

Abstract

This thesis presents two research projects developing theoretical tools for the description of QCD radiation in high-energy collisions, with applications to jet physics at the LHC and lepton-hadron colliders.

The first project introduces an inclusive generalised- k_t jet algorithm for Deep Inelastic Scattering (DIS), defined in the Breit frame and implemented in `fjcontrib`, a third-party extension of FASTJET. The family of algorithms is governed by the usual parameter p , which controls the transverse-momentum dependence of the algorithm, together with a jet radius parameter R . We investigate phenomenological applications related to the identification of the jet associated with the struck quark, and assess their sensitivity to non-perturbative effects such as hadronisation, alongside comparisons with the recent Centauro algorithm.

The second extends the ARES method, a semi-numerical framework for next-to-next-to-leading logarithmic (NNLL) resummation, to hadronic collisions, considering events with a Z boson recoiling against a high-transverse-momentum jet. We find that the structure of the NNLL resummation is formally identical to that for three-jet events in e^+e^- annihilation, but with different ingredients. In particular, initial-state radiation is incorporated within the ARES framework for the first time, with the associated NNLL effects involving the convolution of splitting functions with parton distribution functions. The results are validated analytically against the known resummation of one-jettiness.

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Chapter 1

Introduction

The Large Hadron Collider (LHC) is, as its name suggests, a particle accelerator designed to collide hadrons at energies far beyond those achieved by previous experiments. Having recently concluded its third run, it has spent over a decade probing physics at the fundamental scale. The Standard Model (SM) is currently our best theory for describing the interactions between elementary particles. It describes the electromagnetic, weak, and strong interactions, with the latter taking centre stage throughout this thesis, titled *Jet physics at the LHC and beyond*. The strong interaction, described by quantum chromodynamics (QCD), plays a central role in almost every measurement made at the LHC. Whether in measurements of the Higgs boson, studies of the top quark, or searches for physics beyond the Standard Model, hard-scattering processes are accompanied by QCD radiation, giving rise to the complex hadronic final states observed in experiments. Reliable theoretical predictions for this radiation are therefore essential for interpreting LHC data and extracting the underlying physics.

Like other quantum field theories, QCD exhibits behaviour that depends on the energy scale at which it is probed. At high energies, the strong coupling becomes small, allowing quarks and gluons to interact only weakly. This property, known as asymptotic freedom, makes perturbation theory applicable and allows systematic predictions to be obtained as an expansion in the strong coupling, α_s . At low energies, in contrast, the strong coupling grows and perturbation theory eventually breaks down as quarks and gluons become confined to hadrons. Non-perturbative methods are then required to describe the physics. Within this thesis, we restrict ourselves to the high-energy regime where perturbative QCD provides an excellent description. Nevertheless, obtaining reliable predictions can still present a number of theoretical challenges.

One such challenge arises from the fact that the quarks and gluons produced in a hard scattering are not observed directly. Instead, they undergo further QCD radiation and ultimately

hadronise, producing collimated sprays of hadrons known as *jets*. A jet is an inherently ambiguous object, and defining one requires a prescription for clustering final-state particles – whether partons in a theoretical calculation or energy deposits in a detector – into jets in a way that can be applied consistently to both. This prescription is known as a *jet algorithm*. Different clustering procedures can lead to different jet configurations, and the choice of algorithm can therefore have a significant impact on the resulting predictions and on the physical information that can be extracted. Designing an algorithm suited to a particular process and observable is consequently a non-trivial task, requiring careful consideration of the underlying kinematics and the properties one wishes the resulting jets to possess. This is particularly relevant for Deep Inelastic Scattering (DIS), where the asymmetric initial state introduces additional features that must be taken into account.

This motivates the first piece of original work presented in this thesis, described in Chapter 3. Here, we introduce a new inclusive generalised- k_t jet algorithm for DIS, formulated in the Breit frame, together with an infrared- and collinear-safe prescription for identifying the jet containing the struck quark. We investigate phenomenological applications of the algorithm, studying the reconstructed jets and assessing their sensitivity to non-perturbative effects such as hadronisation. The algorithm has been implemented as the **DISGenkt** plugin to FASTJET as part of `fjcontrib`, and is presented in Ref. [1].

A second challenge arises in the form of large logarithms. For many observables, logarithmically enhanced contributions arise in regions of phase space where the observable becomes small. These logarithms can become sufficiently large such that successive terms in the fixed-order perturbative expansion are no longer parametrically suppressed, reducing the reliability of fixed-order predictions in these regions. Accurate theoretical predictions therefore require the logarithmically enhanced contributions to be systematically accounted for at all orders in perturbation theory through a procedure known as *resummation*.

As with fixed-order calculations, resummation can be performed at successively higher levels of logarithmic accuracy. Several approaches have been developed to perform such resummations, and in this thesis we focus on the semi-numerical CAESAR and ARES formalisms. The second piece of original work presented in this thesis, described in Chapter 5, extends the ARES framework to hadronic collisions, incorporating initial-state radiation at next-to-next-to-leading logarithmic (NNLL) accuracy. The framework is applied to the process $hh \rightarrow Z + \text{jet}$ and completes the crucial conceptual steps required for its extension to hadronic initial states. The resulting predictions are validated through a combination of direct and indirect checks, all of which have been successful.

The remainder of this thesis is structured as follows. Chapter 2 introduces the theoretical background required for this thesis. We review the basic structure of QCD, perturbation theory, the running of the strong coupling, and the description of high-energy scattering processes, establishing the notation used throughout.

Chapter 3 presents the generalised- k_t jet algorithm for DIS. The relevant DIS kinematics and the development of jet algorithms in this context are introduced before presenting the new algorithm and studying its phenomenological properties.

Chapter 4 provides the background to resummation and outlines the formalism on which Chapter 5 builds. We discuss the origin of logarithmically enhanced contributions, the organisation of logarithmic accuracy, and the CAESAR and ARES approaches to resummation. This provides the necessary foundation for Chapter 5, where the ARES framework is extended to hadronic collisions, including the treatment of initial-state radiation.

Finally, Chapter 6 summarises the main results of this thesis and discusses possible directions for future work and the broader implications of the research presented here.

Chapter 2

Quantum chromodynamics

In this chapter we introduce the QCD background required for the remainder of the thesis. Unless otherwise stated, the material presented here follows Refs. [3–8].

2.1 The QCD Lagrangian

In the 1940s, a large number of unstable, strongly interacting particles, collectively known as *hadrons*, were discovered. In 1964, Gell-Mann [9] and Zweig [10, 11] independently proposed the quark model, in which hadrons are interpreted as composite objects built from fundamental spin- $1/2$ constituents known as *quarks*. The six known quarks are organised into three generations – light, intermediate, and heavy – as shown in Table 2.1.

Quark	Generation	Mass	Electric Charge (e)
u (up)	Light	$2.16 \pm 0.07 \text{ MeV}$	$+2/3$
d (down)	Light	$4.70 \pm 0.07 \text{ MeV}$	$-1/3$
s (strange)	Intermediate	$92.9 \pm 0.7 \text{ MeV}$	$-1/3$
c (charm)	Intermediate	$1.2729 \pm 0.0045 \text{ GeV}$	$+2/3$
b (bottom)	Heavy	$4.186 \pm 0.006 \text{ GeV}$	$-1/3$
t (top)	Heavy	$172.60 \pm 0.27 \text{ GeV}$	$+2/3$

Table 2.1: Properties of the six quarks. Mass values are taken from the Particle Data Group [12], where further details on their definitions and extractions are provided.

Hadrons are broadly classified into two categories: mesons ($q\bar{q}$), which are integer-spin bosons, and baryons (qqq), which are half-integer-spin fermions. Quantum chromodynamics (QCD) is the $SU(3)$ non-Abelian gauge theory governing the strong interaction. It describes how quarks bind together to form hadrons, with the interaction mediated by gauge bosons known as gluons. Throughout this thesis, quarks and gluons are collectively referred to as *partons* and carry a

conserved charge known as colour.

Quarks transform in the fundamental representation of $SU(3)$, with fields denoted by q_a , where $a = 1, 2, 3$ labels the three colour states, conventionally referred to as red, green, and blue. The corresponding antiquark fields are denoted by $\bar{q}^a$ and transform in the conjugate representation, where $\bar{q} = q^\dagger \gamma^0$ is the Dirac adjoint of q . Gluons, denoted by $\mathcal{A}_\mu^A$, transform in the adjoint representation of $SU(3)$, with $A = 1, \dots, 8$ labelling the eight colour degrees of freedom of the gluon field.

The QCD Lagrangian density governing the dynamics of quarks with mass m_f and massless gluons can be written as

$$\mathcal{L}_{\text{QCD}} = \mathcal{L}_{\text{classical}} + \mathcal{L}_{\text{GF}} + \mathcal{L}_{\text{ghost}} . \quad (2.1)$$

The classical contribution contains the kinetic and interaction terms for the quark and gluon fields,

$$\mathcal{L}_{\text{classical}} = -\frac{1}{4} F_{\mu\nu}^A F_A^{\mu\nu} + \sum_{f=1}^{n_f} \bar{q}_a^f (i \not{D} - m_f)_{ab} q_b^f . \quad (2.2)$$

Here n_f denotes the number of quark flavours, with $n_f = 6$ in the Standard Model (see Table 2.1).

The gluon dynamics are encoded in the non-Abelian field-strength tensor

$$F_{\alpha\beta}^A = \partial_\alpha \mathcal{A}_\beta^A - \partial_\beta \mathcal{A}_\alpha^A - g_s f^{ABC} \mathcal{A}_\alpha^B \mathcal{A}_\beta^C , \quad (2.3)$$

where g_s is the strong coupling constant, conventionally expressed through

$$\alpha_s \equiv \frac{g_s^2}{4\pi} . \quad (2.4)$$

The covariant derivative is written using Feynman slash notation, $\not{D} = \gamma_\mu D^\mu$, with

$$(D_\mu)_{ab} = \partial_\mu \delta_{ab} + i g_s (t^C \mathcal{A}_\mu^C)_{ab} . \quad (2.5)$$

The gamma matrices satisfy the anti-commutation relation $\{\gamma^\mu, \gamma^\nu\} = 2g^{\mu\nu}$ and throughout this work we adopt the metric convention $g^{\mu\nu} = \text{diag}(1, -1, -1, -1)$.

The matrices t^C are the generators of the fundamental representation of $SU(3)$, given explicitly by

$$t^C = \frac{1}{2} \lambda^C , \quad (2.6)$$

where λ^C are the eight traceless Hermitian Gell-Mann matrices. These satisfy the commutation relation

$$[t^A, t^B] = i f^{ABC} t^C , \quad (2.7)$$

where f^{ABC} are the structure constants of the $SU(3)$ gauge group. The generators satisfy the Casimir identities

$$\begin{aligned} \text{Tr}(t^A t^B) &= T_R \delta^{AB}, & T_R &= \frac{1}{2}, \\ t_{ab}^A t_{bc}^A &= C_F \delta_{ac}, & C_F &= \frac{4}{3}, \\ f^{ABC} f^{ABD} &= C_A \delta^{CD}, & C_A &= 3, \end{aligned} \quad (2.8)$$

where repeated indices are summed implicitly.

The second term in Eq. (2.1) is the gauge-fixing contribution. For instance, in a covariant gauge, this is

$$\mathcal{L}_{\text{GF}} = -\frac{1}{2\xi} (\partial^\mu \mathcal{A}_\mu^A)^2, \quad (2.9)$$

which is required to remove redundant gauge degrees of freedom associated with the local $SU(3)$ gauge invariance of the QCD Lagrangian. The parameter ξ specifies the choice of gauge: for example, $\xi = 1$ corresponds to the Feynman gauge, while $\xi = 0$ corresponds to the Landau gauge.

The final term in Eq. (2.1) is the ghost contribution,

$$\mathcal{L}_{\text{ghost}} = \partial_\mu \eta^{A\dagger} (D_{AB}^\mu \eta^B), \quad (2.10)$$

where η^A are complex scalar fields known as Faddeev–Popov ghosts. These fields anti-commute and therefore obey Fermi statistics. Ghosts do not correspond to physical states and instead arise as a consequence of the gauge-fixing procedure in non-Abelian gauge theories.

The Feynman rules obtained from the QCD Lagrangian are illustrated in Figs. 2.1 and 2.2. The propagators arise from the kinetic terms, while the interaction terms determine the vertices. Throughout this thesis, a, b, c denote colour indices in the fundamental representation, while A, B, C, D denote colour indices in the adjoint representation and i, j denote Dirac indices. For the calculations presented in this thesis, the rules shown here are sufficient.

$$\begin{aligned} a, i \xrightarrow{p} b, j &= \delta^{ab} \frac{i}{(\not{p} - m + i\epsilon)_{ji}} \\ A, \mu \xrightarrow{p} B, \nu &= \delta^{AB} \frac{i}{p^2 + i\epsilon} \left(-g^{\mu\nu} + (1 - \xi) \frac{p^\mu p^\nu}{p^2} \right) \end{aligned}$$

Figure 2.1: Feynman rules for the quark and gluon propagators.

The figure displays three Feynman diagrams and their corresponding mathematical expressions:

- Quark-gluon vertex:** A wavy line labeled A, μ splits into two straight lines labeled c, j and b, i . The expression is $= -ig_s (t^A)_{cb} (\gamma^\mu)_{ji}$.
- Three-gluon vertex:** A wavy line labeled A, μ with momentum p enters from the left. Two wavy lines labeled B, ν and C, ρ exit to the right with momenta q and r respectively. The expression is $= -g_s f^{ABC} [(p - q)^\rho g^{\mu\nu} + (q - r)^\mu g^{\nu\rho} + (r - p)^\nu g^{\mu\rho}]$.
- Four-gluon vertex:** Four wavy lines labeled A, μ , B, ν , C, ρ , and D, σ meet at a central point. The expression is $= -ig_s^2 [f^{EAC} f^{EBD} (g^{\mu\nu} g^{\rho\sigma} - g^{\mu\sigma} g^{\nu\rho}) + f^{EAD} f^{EBC} (g^{\mu\nu} g^{\rho\sigma} - g^{\mu\rho} g^{\nu\sigma}) + f^{EAB} f^{ECD} (g^{\mu\rho} g^{\nu\sigma} - g^{\mu\sigma} g^{\nu\rho})]$.

Figure 2.2: Feynman rules for the quark-gluon vertex, the three-gluon vertex, and the four-gluon vertex.

2.2 Perturbation theory

With the QCD Lagrangian and its Feynman rules established, we now outline how perturbative calculations are performed in practice. Scattering processes are described by considering all Feynman diagrams contributing to the transition between a given set of incoming and outgoing states. Each diagram is constructed from the propagators and interaction vertices shown in Figs. 2.1 and 2.2, and translated into a mathematical expression according to the corresponding Feynman rules. The sum of all contributing diagrams gives the *scattering amplitude*, or matrix element, $\mathcal{M}$. Cross sections are then obtained from the squared amplitude, integrated over the allowed phase space.

To illustrate the construction of a partonic cross section, consider a hard $2 \rightarrow m$ scattering process involving massless partons,

$$p_a + p_b \rightarrow p_1 + \cdots + p_m.$$

The corresponding partonic cross section is

$$\hat{\sigma} = \frac{1}{2s} \int d\Phi^{(m)}(p_1, \dots, p_m) |\mathcal{M}_m(p_a, p_b; p_1, \dots, p_m)|^2 F_J^{(m)}(p_1, \dots, p_m), \quad (2.11)$$

where the individual components are defined as follows:

- $1/(2s)$ is the Lorentz-invariant flux factor for the two massless incoming partons, with $s = (p_a + p_b)^2$ the centre-of-mass energy squared of the hard process;

- $d\Phi^{(m)}$ denotes the m -parton phase space measure,

$$d\Phi^{(m)} = \left[\prod_{i=1}^m \frac{d^3 p_i}{(2\pi)^3 2E_i} \right] (2\pi)^4 \delta^{(4)} \left(p_a + p_b - \sum_{i=1}^m p_i \right), \quad (2.12)$$

- $|\mathcal{M}_m|^2 = \langle \mathcal{M}_m | \mathcal{M}_m \rangle$ is the squared scattering amplitude for the $2 \rightarrow m$ process. The inner product includes a sum over the colour and spin degrees of freedom of all final-state partons and an average over those of the initial-state partons;
- $F_J^{(m)}$ is the *jet function*, which specifies the observable under consideration. Its precise form depends on the observable and may contain Θ -functions, δ -functions, or numerical factors. The choice $F_J^{(m)} = 1$ reduces Eq. (2.11) to the total partonic cross section.

The structure of the QCD interaction vertices implies that additional gluon emissions or internal loops introduce further powers of the strong coupling α_s . The partonic cross section can therefore be organised as a perturbative expansion in α_s ,

$$\hat{\sigma} = \hat{\sigma}_0 + \frac{\alpha_s}{2\pi} \hat{\sigma}_1 + \left(\frac{\alpha_s}{2\pi} \right)^2 \hat{\sigma}_2 + \dots, \quad (2.13)$$

where $\hat{\sigma}_0$ is the tree-level contribution, referred to as the leading-order (LO) or Born term. The terms $\hat{\sigma}_1, \hat{\sigma}_2, \dots$ represent successive orders in the perturbative expansion, known as next-to-leading order (NLO), next-to-next-to-leading order (NNLO), and so on.

As we will see in Sec. 2.3, α_s becomes small at high energies. This means that successive terms in the perturbative expansion are typically suppressed in this regime. As a result, the expansion can be well approximated by retaining only the first few terms, yielding a *fixed-order* (FO) prediction. This approach is known as perturbation theory, and its systematic application to QCD processes forms the basis of perturbative QCD (pQCD). Throughout this thesis, all calculations are performed within the framework of pQCD.

2.3 The running of the strong coupling α_s

As discussed in Sec. 2.2, the validity of pQCD relies on the strong coupling becoming small at high energies. This behaviour follows from the fact that α_s is not a fixed constant, but instead

runs with the renormalisation scale μ_R – an unphysical scale introduced by the renormalisation procedure, discussed in Sec. 2.4. The scale dependence of the coupling is governed by the renormalisation group equation

$$\mu_R^2 \frac{\partial \alpha_s(\mu_R^2)}{\partial \mu_R^2} = \beta(\alpha_s), \quad (2.14)$$

where the QCD β -function admits a perturbative expansion of the form

$$\beta(\alpha_s) = -(\beta_0 \alpha_s^2 + \beta_1 \alpha_s^3 + \beta_2 \alpha_s^4 + \dots). \quad (2.15)$$

The first two coefficients are

$$\beta_0 = \frac{11C_A - 4T_R n_f}{12\pi}, \quad (2.16a)$$

$$\beta_1 = \frac{17C_A^2 - 10C_A T_R n_f - 6C_F T_R n_f}{24\pi^2}. \quad (2.16b)$$

The coefficients β_0 and β_1 are renormalisation-scheme independent, while higher coefficients depend on the choice of renormalisation scheme, taken throughout to be the $\overline{\text{MS}}$ scheme (discussed in Sec. 2.4.1).

Solving Eq. (2.14) at leading order with reference scale Q gives

$$\alpha_s(\mu_R^2) = \frac{\alpha_s(Q^2)}{1 + \alpha_s(Q^2) \beta_0 \ln\left(\frac{\mu_R^2}{Q^2}\right)}. \quad (2.17)$$

The sign of β_0 and the form of Eq. (2.17) determine the scale dependence of the strong coupling. As $\mu_R \rightarrow \infty$, the coupling decreases, and quarks and gluons behave increasingly like free particles. This behaviour is known as *asymptotic freedom*, a regime in which pQCD provides an excellent approximation. Conversely, as μ_R decreases the running coupling in Eq. (2.17) diverges at Λ_{QCD} , a scale of a few hundred MeV. Close to Λ_{QCD} the coupling grows large, signalling the breakdown of perturbation theory and the emergence of a strongly interacting regime in which quarks and gluons are confined into hadrons. This phenomenon is known as *confinement*.

The running of α_s has been confirmed experimentally across several orders of magnitude in Q , as illustrated in Fig. 2.3, with the five-loop QCD prediction in excellent agreement with data from a wide range of processes. Throughout this thesis we work in the high-energy regime $\mu_R \gg \Lambda_{\text{QCD}}$, thereby exploiting asymptotic freedom.

2.4 Divergences

While perturbation theory is very effective at high energies, individual contributions beyond leading order introduce divergences that must be treated systematically. In QCD, these divergences arise in two distinct forms:

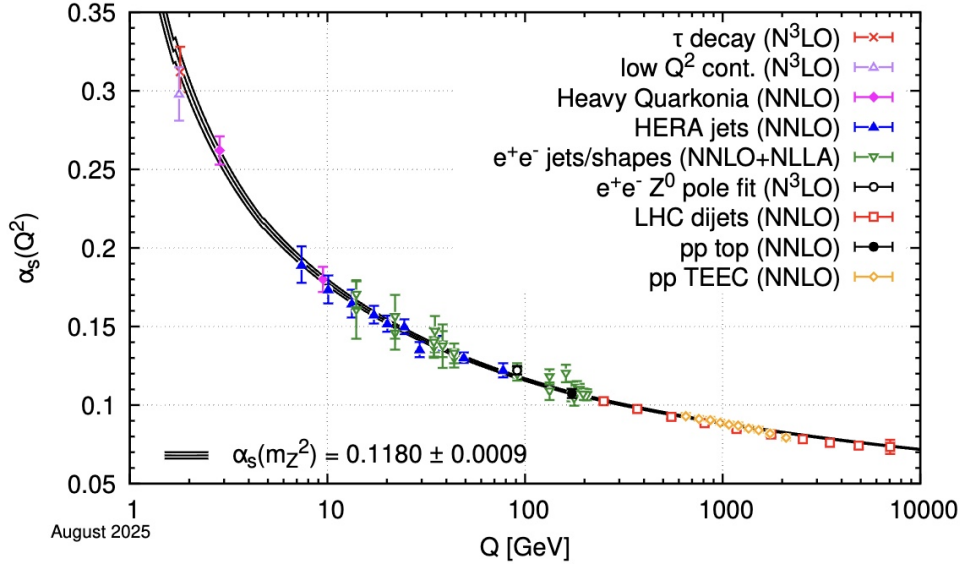


Figure 2.3: Determinations of α_s as a function of the energy scale Q , taken from Ref. [12], compared to the five-loop QCD prediction evaluated at the current Particle Data Group (PDG) average $\alpha_s(m_Z^2) = 0.1180 \pm 0.0009$.

- **Ultraviolet (UV) divergences** arise in loop diagrams when the momentum of the virtual particle becomes large.
- **Infrared (IR) divergences** arise when a gluon emission becomes soft (low energy) or collinear to a hard (high energy) parton.

2.4.1 UV divergences

One Feynman rule not shown in Figs. 2.1 and 2.2 is associated with loop diagrams. Each independent loop momentum introduces an integration of the form

$$\text{loop} \longrightarrow \frac{d^4 q}{(2\pi)^4}, \quad (2.18)$$

where q^μ denotes the momentum of the virtual particle running inside the loop, as illustrated by the fermion bubble diagram in Fig. 2.4.

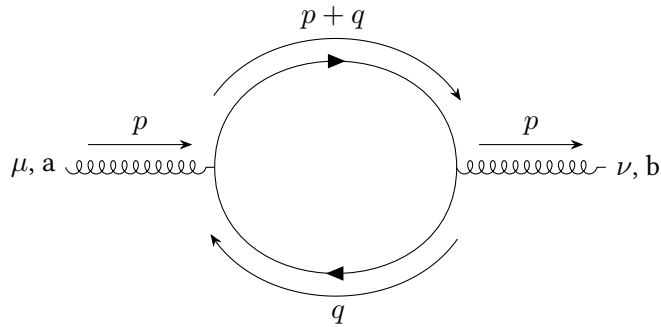


Figure 2.4: Fermion bubble diagram illustrating a one-loop contribution to the gluon self-energy.

Taking the quarks to be massless, as assumed throughout this thesis, the Feynman rules give an integral of the form

$$I = \int d^4q \frac{N^{\mu\nu}}{q^2(p+q)^2}, \quad (2.19)$$

where $N^{\mu\nu}$ denotes a numerator whose explicit form is not required for the present discussion. At large loop momentum, the integral reveals a UV divergence,

$$I \sim \lim_{\Lambda \rightarrow \infty} \int_0^\Lambda \frac{dq}{q} \sim \lim_{\Lambda \rightarrow \infty} \ln \Lambda. \quad (2.20)$$

Such divergences cannot be allowed to remain in the theory, and therefore a systematic method for handling them is required. This is achieved through a procedure known as *regularisation*. Although several regularisation procedures exist, in this thesis we use *dimensional regularisation* [13, 14]. This method preserves both Lorentz and gauge invariance and, as will be discussed in Sec. 2.4.2, can also be applied to infrared divergences. The basic idea is to promote the space-time dimension to $d = 4 - 2\varepsilon$, such that logarithmic divergences appear as poles in ε .

Once these poles have been isolated, the next step is to remove them. This is achieved by a procedure known as *renormalisation* where the parameters of the theory are redefined to absorb the divergent contributions via counterterms. Throughout this thesis we work in the modified minimal subtraction ($\overline{\text{MS}}$) renormalisation scheme [15]. In this scheme, the counterterms subtract not only the $1/\varepsilon$ poles, but also the accompanying constants $\ln 4\pi$ and $\gamma_E \approx 0.5772$ (the Euler–Mascheroni constant), that arise from using dimensional regularisation. The renormalisation procedure introduces the scale μ_R , giving rise to the corresponding scale dependence of the strong coupling α_s , as discussed in Sec. 2.3.

2.4.2 IR divergences

We now turn to infrared (IR) divergences, which arise from the emission of partons in the soft and collinear limits. To illustrate their origin and cancellation, we consider the simplest QCD process, $e^+e^- \rightarrow \text{hadrons}$. At leading order, the hadronic cross section is described by the corresponding partonic process $e^+e^- \rightarrow \gamma^* \rightarrow q\bar{q}$, shown in Fig. 2.5. This approach is justified by quark-hadron duality [16]: in the high-energy regime, the *fully inclusive* hadronic cross section can be approximated sufficiently well by the corresponding partonic cross section computed in perturbation theory ¹.

¹ For differential distributions, one expects a more pronounced difference between parton and hadron level. However, it is found that hadronisation does not significantly reshuffle the momentum of very energetic partons, a feature known as local parton-hadron duality [17, 18].

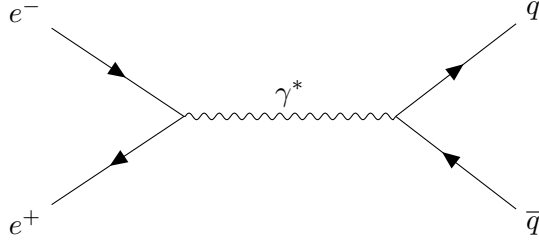


Figure 2.5: Feynman diagram for the leading-order process $e^+e^- \rightarrow \gamma^* \rightarrow q\bar{q}$.

The corresponding leading-order cross section is

$$\sigma_0 = \frac{4\pi\alpha^2}{3s} N_c \sum_f Q_f^2, \quad (2.21)$$

where we have dropped the partonic hat notation for simplicity. Here, Q_f denotes the electric charge of a quark of flavour f , $N_c = 3$ is the number of quark colours, $s = (p_{e^+} + p_{e^-})^2$ is the centre-of-mass energy squared, and α is the QED coupling.

At NLO, the cross section receives both real-emission and virtual corrections. We first consider the real-emission contribution, corresponding to $e^+e^- \rightarrow \gamma^* \rightarrow q\bar{q}g$, with the two possible diagrams shown in Fig. 2.6.

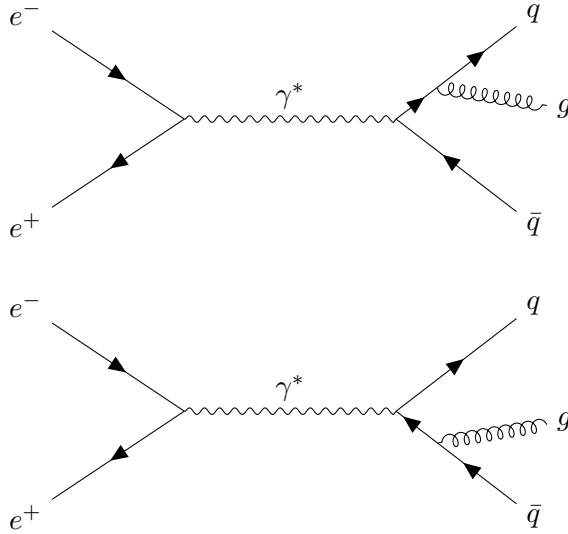


Figure 2.6: Real-emission contributions to $e^+e^- \rightarrow \gamma^* \rightarrow q\bar{q}g$ at NLO.

Writing the three-parton phase space in terms of the energy fractions

$$x_i = \frac{2E_i}{\sqrt{s}}, \quad i = q, \bar{q}, g, \quad (2.22)$$

the corresponding cross section is

$$\sigma^{q\bar{q}g} = \sigma_0 C_F \frac{\alpha_s}{2\pi} \int_0^1 dx_q \int_0^1 dx_{\bar{q}} \frac{x_q^2 + x_{\bar{q}}^2}{(1-x_q)(1-x_{\bar{q}})} \Theta(x_q + x_{\bar{q}} - 1), \quad (2.23)$$

where energy conservation gives $\sum_i x_i = 2$. The Heaviside function enforces the kinematic constraint $x_q + x_{\bar{q}} \geq 1$, which follows directly from energy conservation in the three-particle final state.

The singular structure is apparent from the integrand in Eq. (2.23). In particular, divergences arise in the following limits:

- $x_q \rightarrow 1$ and $x_{\bar{q}} \rightarrow 1$ simultaneously, corresponding to a soft gluon, $E_g \rightarrow 0$;
- $x_q \rightarrow 1$, corresponding to the antiquark and gluon becoming collinear, $\theta_{\bar{q}g} \rightarrow 0$;
- $x_{\bar{q}} \rightarrow 1$, corresponding to the quark and gluon becoming collinear, $\theta_{qg} \rightarrow 0$.

Here, θ_{qg} and $\theta_{\bar{q}g}$ denote the quark–gluon and antiquark–gluon angles, respectively. These soft and collinear singularities are collectively referred to as IR (or IRC) divergences.

To regulate these divergences, we evaluate this integral in $d = 4 - 2\varepsilon$ dimensions using dimensional regularisation (see Sec. 2.4.1).² The regulated expression is

$$\begin{aligned} \sigma^{q\bar{q}g} &= \sigma_0 C_F \frac{\alpha_s}{2\pi} \left(\frac{4\pi\mu_R^2}{s} \right)^\varepsilon \frac{1}{\Gamma(1-\varepsilon)} \int_0^1 dx_q \int_0^1 dx_{\bar{q}} \\ &\times \frac{(x_q^2 + x_{\bar{q}}^2) - \varepsilon(2 - x_q - x_{\bar{q}})^2}{(1 - x_q)^{1+\varepsilon} (1 - x_{\bar{q}})^{1+\varepsilon} (x_q + x_{\bar{q}} - 1)^\varepsilon} \Theta(x_q + x_{\bar{q}} - 1). \end{aligned} \quad (2.25)$$

Performing the integrations gives

$$\sigma^{q\bar{q}g} = \sigma_0 \frac{\alpha_s}{2\pi} \frac{C_F}{\Gamma(1-\varepsilon)} \left(\frac{4\pi\mu_R^2}{s} \right)^\varepsilon \left(\frac{2}{\varepsilon^2} + \frac{3}{\varepsilon} - \pi^2 + \frac{19}{2} + \mathcal{O}(\varepsilon) \right). \quad (2.26)$$

The real-emission contribution is therefore divergent as $\varepsilon \rightarrow 0$. At the same order, however, the cross section also receives a virtual correction to the leading-order process, shown in Fig. 2.7.

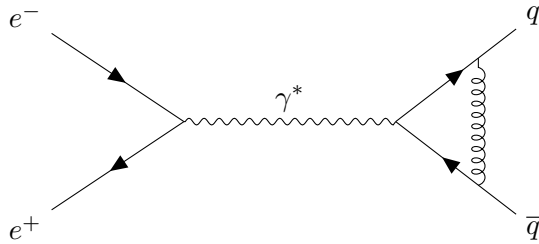


Figure 2.7: Virtual correction to $e^+e^- \rightarrow \gamma^* \rightarrow q\bar{q}$ at NLO.

The virtual contribution is

$$\sigma^{q\bar{q}(g)} = \sigma_0 \frac{\alpha_s}{2\pi} \frac{C_F}{\Gamma(1-\varepsilon)} \left(\frac{4\pi\mu_R^2}{s} \right)^\varepsilon \left(-\frac{2}{\varepsilon^2} - \frac{3}{\varepsilon} + \pi^2 - 8 + \mathcal{O}(\varepsilon) \right). \quad (2.27)$$

² Note that the m -particle phase space in Eq. (2.12) is written in d dimensions as

$$d\Phi^{(m)} = \left[\prod_{i=1}^m \frac{d^d p_i}{(2\pi)^d} (2\pi) \delta(p_i^2) \Theta(p_i^0) \right] (2\pi)^d \delta^{(d)} \left(p_a + p_b - \sum_{j=1}^m p_j \right). \quad (2.24)$$

The poles are equal in magnitude and opposite in sign to those appearing in Eq. (2.26). Consequently, the sum of the real and virtual contributions is finite and is given by

$$\sigma^{q\bar{q}g} + \sigma^{q\bar{q}(g)} = \sigma_0 \frac{\alpha_s}{2\pi} \frac{C_F}{\Gamma(1-\varepsilon)} \left(\frac{4\pi\mu_R^2}{s} \right)^\varepsilon \left(\frac{3}{2} + \mathcal{O}(\varepsilon) \right). \quad (2.28)$$

Expanding in ε and taking the limit $\varepsilon \rightarrow 0$, the total cross section up to NLO accuracy is consequently

$$\sigma = \sigma_0 \left(1 + \frac{\alpha_s}{2\pi} \frac{3}{2} C_F + \mathcal{O}(\alpha_s^2) \right). \quad (2.29)$$

The cancellation of the IR singularities between the real and virtual contributions is a direct consequence of the *KLN theorem* [19, 20], which guarantees the cancellation of IR divergences between real and virtual contributions order by order in perturbation theory provided one sums over all degenerate initial and final states.

2.5 Soft and collinear factorisation of $|\mathcal{M}_m|^2$

In Sec. 2.4.2, we illustrated the cancellation of infrared singularities between real and virtual corrections for $e^+e^- \rightarrow \text{hadrons}$ at NLO. We now consider the general structure of these divergences by examining the behaviour of the squared matrix element $|\mathcal{M}_m|^2$ in the soft and collinear limits, where it factorises into a remarkably simple and universal form [21]. It is this factorisation that underpins the systematic treatment of IR singularities in QCD. The results below describe the emission of a single soft or collinear parton, as required at NLO. The same factorisation holds at higher orders [21–24].

2.5.1 Soft limit

Suppressing the initial-state momenta (p_a, p_b) for brevity, consider the emission of an additional soft parton with momentum q . In the soft limit, the squared tree-level matrix element factorises as

$$|\mathcal{M}_{m+1}^{(0)}(q, p_1, \dots, p_m)|^2 \simeq -4\pi\alpha_s\mu_R^{2\varepsilon} \sum_{i,j=1}^m S_{ij}(q) \left\langle \mathcal{M}_m^{(0)}(p_1, \dots, p_m) | \mathbf{T}_i \cdot \mathbf{T}_j | \mathcal{M}_m^{(0)}(p_1, \dots, p_m) \right\rangle, \quad (2.30)$$

where the superscript (0) denotes the tree-level matrix element. The colour-charge operators $\mathbf{T}_i$ act on the colour indices of parton i and satisfy

$$\begin{aligned} \mathbf{T}_i \cdot \mathbf{T}_j &= \mathbf{T}_j \cdot \mathbf{T}_i, & (i \neq j), \\ \mathbf{T}_i^2 &= C_i, \end{aligned} \quad (2.31)$$

with $C_i = C_F$ for quarks and $C_i = C_A$ for gluons. The eikonal factor $S_{ij}(q)$ contains the kinematic dependence of the soft emission and is given by

$$S_{ij}(q) = \frac{p_i \cdot p_j}{(p_i \cdot q)(p_j \cdot q)}. \quad (2.32)$$

The factorisation shows that the soft emission depends only on the colour charges and directions of the hard legs, not on the details of the hard process.

2.5.2 Collinear limit

We next consider the limit in which two partons become collinear. The momenta of the two partons, i and j , can be parametrised using a Sudakov decomposition,

$$\begin{aligned} p_i^\mu &= z p^\mu + k_\perp^\mu - \frac{k_\perp^2}{z} \frac{n^\mu}{2p \cdot n}, \\ p_j^\mu &= (1-z) p^\mu - k_\perp^\mu - \frac{k_\perp^2}{1-z} \frac{n^\mu}{2p \cdot n}, \end{aligned} \quad (2.33)$$

where p^μ is a light-like vector ($p^2 = 0$) defining the collinear direction, n^μ is an auxiliary light-like vector ($n^2 = 0$), and $k_\perp^\mu$ is the transverse momentum ($k_\perp^2 < 0$, $k_\perp \cdot p = k_\perp \cdot n = 0$). The parameter z is the longitudinal momentum fraction carried by parton i , and the collinear limit corresponds to $k_\perp^\mu \rightarrow 0$.

In this limit, the squared matrix element factorises as

$$\begin{aligned} |\mathcal{M}_{m+1}^{(0)}(p_1, \dots, p_i, p_j, \dots, p_{m+1})|^2 &\simeq \\ &\frac{8\pi\alpha_s\mu_R^{2\varepsilon}}{s_{ij}} \left\langle \mathcal{M}_m^{(0)}(p_1, \dots, p_{ij}, \dots, p_m) \left| \hat{P}_{(ij) \rightarrow i+j}(z, k_\perp; \varepsilon) \right| \mathcal{M}_m^{(0)}(p_1, \dots, p_{ij}, \dots, p_m) \right\rangle, \end{aligned} \quad (2.34)$$

where $p_{ij}^\mu = p_i^\mu + p_j^\mu$ is the momentum of the parent parton (ij) and $s_{ij} \equiv (p_i + p_j)^2$. The function $\hat{P}_{(ij) \rightarrow i+j}(z, k_\perp; \varepsilon)$ is the unregularised tree-level Altarelli–Parisi splitting function for the collinear splitting $ij \rightarrow i + j$. The explicit forms are

$$\begin{aligned} \hat{P}_{g \rightarrow q\bar{q}}^{\mu\nu}(z, k_\perp; \varepsilon) &= \hat{P}_{\bar{q} \rightarrow q\bar{q}}^{\mu\nu}(z, k_\perp; \varepsilon) = T_R \left[-g^{\mu\nu} + 4z(1-z) \frac{k_\perp^\mu k_\perp^\nu}{k_\perp^2} \right], \\ \hat{P}_{q \rightarrow qg}^{ss'}(z, k_\perp; \varepsilon) &= \hat{P}_{\bar{q} \rightarrow \bar{q}g}^{ss'}(z, k_\perp; \varepsilon) = \delta_{ss'} C_F \left[\frac{1+z^2}{1-z} - \varepsilon(1-z) \right], \\ \hat{P}_{q \rightarrow gq}^{ss'}(z, k_\perp; \varepsilon) &= \hat{P}_{\bar{q} \rightarrow g\bar{q}}^{ss'}(z, k_\perp; \varepsilon) = \delta_{ss'} C_F \left[\frac{1+(1-z)^2}{z} - \varepsilon z \right], \\ \hat{P}_{g \rightarrow gg}^{\mu\nu}(z, k_\perp; \varepsilon) &= 2C_A \left[-g^{\mu\nu} \left(\frac{z}{1-z} + \frac{1-z}{z} \right) - 2(1-\varepsilon)z(1-z) \frac{k_\perp^\mu k_\perp^\nu}{k_\perp^2} \right], \end{aligned} \quad (2.35)$$

where the gluon splitting functions $\hat{P}^{\mu\nu}$ carry Lorentz indices μ, ν corresponding to the two polarisation states of the parent gluon, while the quark splitting functions $\hat{P}^{ss'}$ carry spin indices s, s' of the parent fermion. In both cases the indices are contracted with those of the reduced matrix element in the factorisation formula above.

2.6 IRC safety

In Sec. 2.4, we saw through the explicit NLO calculation that the infrared singularities arising from real and virtual contributions cancel, yielding a finite result. This cancellation relies on the fact that the total cross section is a *fully inclusive* observable, for which $F_J^{(m)} = 1$ for any number of final-state particles (see Sec. 2.2).

For more general observables, such a cancellation is not guaranteed. To obtain finite perturbative predictions, the observable must be *infrared and collinear safe* (IRC safe): its value must be insensitive to the emission of additional soft and/or collinear partons. At NLO, these conditions can be expressed in terms of the jet function as

$$\begin{aligned} F_J^{(m+1)}(p_1, \dots, p_i, \dots, p_{m+1}) &\longrightarrow F_J^{(m)}(p_1, \dots, \cancel{p_i}, \dots, p_{m+1}), & E_i \rightarrow 0, \\ F_J^{(m+1)}(p_1, \dots, p_i, p_j, \dots, p_{m+1}) &\longrightarrow F_J^{(m)}(p_1, \dots, \cancel{p_i}, \cancel{p_j}, p_{ij}, \dots, p_{m+1}), & p_i \parallel p_j, \end{aligned} \quad (2.36)$$

with a diagrammatic representation shown in Fig. 2.8.

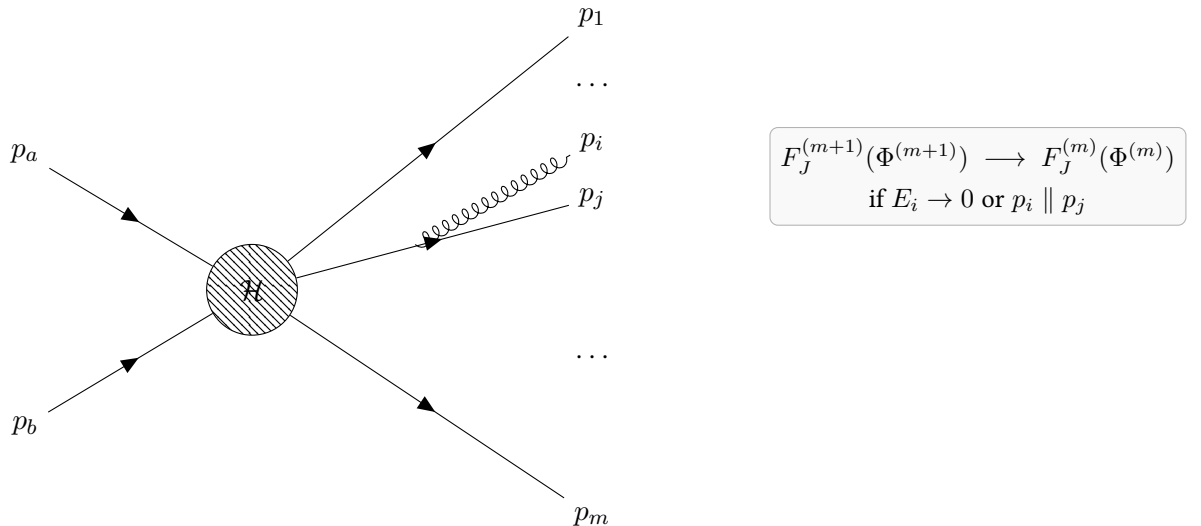


Figure 2.8: Diagrammatic representation of a $2 \rightarrow m$ hard scattering process, $\mathcal{H}$, with an additional soft or collinear emission p_i . IRC safety requires the observable to be unchanged in either limit.

Although Eq. (2.36) is written for a single additional parton, the same condition extends to any number of soft or collinear partons. From this definition it follows immediately that fully

inclusive observables with $F_J^{(m)} = 1$ are IRC safe. A simple example of an IRC-unsafe observable is the number of final-state partons.

Although IRC safety ensures a finite result at every order in perturbation theory, it does not guarantee a well-behaved perturbative expansion. Many observables, while IRC safe, do not display a complete cancellation between real and virtual contributions – only enough to remove divergences. This incomplete cancellation generates large logarithmic terms arising from emissions in the soft and/or collinear regions. Such quantities are referred to as *semi-inclusive* observables [25]. A prominent class of examples are event-shape variables, discussed in Chapter 4. These logarithms can spoil the convergent nature of the perturbative expansion and must be *resummed* – the subject of Chapters 4 and 5.

2.7 The full picture: high-energy collisions

Up to this point, the discussion has largely focused on physics at the parton level. In particular, we have relied on concepts such as quark-hadron duality (cf. Sec. 2.4) to relate partonic cross sections to hadronic observables, and have worked predominantly in the high-energy regime, where asymptotic freedom (cf. Sec. 2.3) allows the use of perturbative QCD.

In realistic collider experiments, however, the situation is significantly more complex. Consider, for example, a hadronic collision at the LHC. The incoming particles are composite hadrons rather than elementary partons, and the final-state particles observed in the detector consist of hadrons produced at energy scales far below that of the hard interaction. A complete description of such a collision therefore necessarily involves physics beyond the perturbative regime.

The aim of this section is to outline the various stages of a high-energy hadronic collision, organised by the relevant energy scale Q . These stages are depicted in Fig. 2.9, which shows a $t\bar{t}h$ event produced in a proton–proton collision as generated by the SHERPA [26] collaboration.

2.7.1 Hard scattering ($Q \gg \Lambda_{\text{QCD}}$)

At the highest energy scale, the dynamics of the collision are governed by the hard scattering, represented by the central red blob in Fig. 2.9. In this process, two partons, denoted i and j , are extracted from the incoming protons P_1 and P_2 and undergo a short-distance interaction. The partonic content of the proton is encoded in parton distribution functions (PDFs), $f_{i/P}(x, Q^2)$, which give the probability of finding a parton of flavour i carrying longitudinal momentum fraction x of the proton P at the energy scale Q^2 .

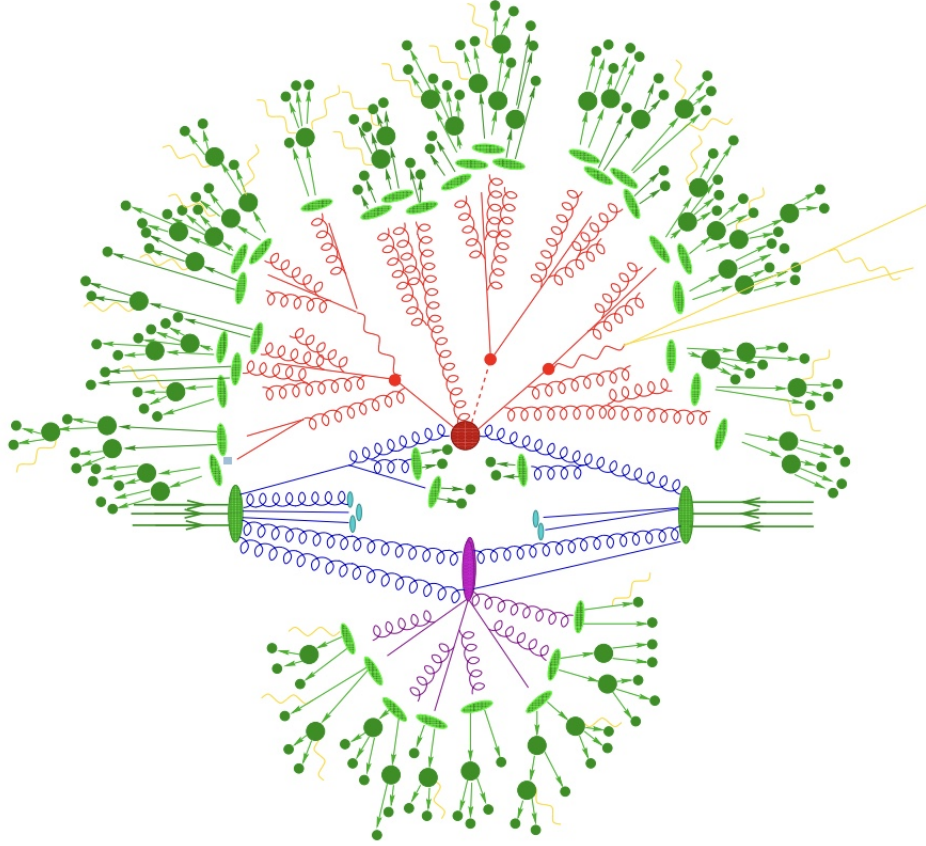


Figure 2.9: Pictorial representation of a $t\bar{t}h$ event produced in a high-energy hadronic collision, taken from Ref. [26].

PDFs are non-perturbative functions and must be extracted from experimental data. However, their scale dependence is governed by perturbative QCD through the Dokshitzer–Gribov–Lipatov–Altarelli–Parisi (DGLAP) evolution equations [27–29],

$$\frac{\partial f_{i/P}(x, Q^2)}{\partial \ln Q^2} = \sum_j \int_x^1 \frac{dz}{z} \frac{\alpha_s(Q^2)}{2\pi} P_{ij}(z, \alpha_s(Q^2)) f_{j/P}\left(\frac{x}{z}, Q^2\right), \quad (2.37)$$

where the Altarelli–Parisi splitting functions $P_{ij}(z)$ admit the perturbative expansion

$$P_{ij}(z, \alpha_s(Q^2)) = P_{ij}^{(0)}(z) + \frac{\alpha_s}{2\pi} P_{ij}^{(1)}(z) + \dots \quad (2.38)$$

The leading-order splitting functions $P_{ij}^{(0)}$ are the regularised counterparts of the unregularised functions $\hat{P}_{(ij) \rightarrow i+j}$ of Eq. (2.35). Here, we consider initial- rather than final-state splittings, for which the two cases coincide at this order. Note that the index ordering follows the opposite convention, $P_{ij} \leftrightarrow \hat{P}_{j \rightarrow i+k}$, with j the parton taken from the proton, i the parton entering the hard scattering, and k the parton radiated into the final state. The $P_{ij}^{(0)}$ are obtained by averaging $\hat{P}_{j \rightarrow i+k}$ over the spin of the parent and setting $\varepsilon = 0$, applying the $(1-z)_+$ prescription to handle the soft $z \rightarrow 1$ divergence, and adding the virtual $\delta(1-z)$ piece from one-loop corrections.

Explicitly,

$$\begin{aligned}
P_{qq}^{(0)}(z) &= C_F \left[\frac{1+z^2}{(1-z)_+} + \frac{3}{2} \delta(1-z) \right], \\
P_{gg}^{(0)}(z) &= T_R [z^2 + (1-z)^2], \\
P_{gq}^{(0)}(z) &= C_F \left[\frac{1+(1-z)^2}{z} \right], \\
P_{qg}^{(0)}(z) &= 2C_A \left[\frac{z}{(1-z)_+} + \frac{1-z}{z} + z(1-z) \right] + 2\pi\beta_0 \delta(1-z).
\end{aligned} \tag{2.39}$$

The $(1-z)_+$ prescription is defined by

$$\int_0^1 dz \frac{f(z)}{(1-z)_+} = \int_0^1 dz \frac{f(z) - f(1)}{1-z}, \tag{2.40}$$

and ensures integrability at $z \rightarrow 1$. The $\delta(1-z)$ terms in $P_{qq}^{(0)}$ and $P_{gg}^{(0)}$ arise from virtual corrections and are present only for the corresponding splittings.

The hard interaction produces the primary particles of the event – in this case a $t\bar{t}h$ system, indicated by the smaller red blobs in Fig. 2.9. Since the energy scale Q is much larger than Λ_{QCD} , predictions at this stage can be reliably obtained using fixed-order perturbation theory (Sec. 2.2). Such calculations are typically provided by matrix-element generators such as MADGRAPH [30] and MCFM [31–33].

2.7.2 QCD evolution ($Q \gtrsim \Lambda_{\text{QCD}}$)

Following the hard scattering, the coloured partons produced in the event undergo further QCD radiation, illustrated by the red quark and gluon propagators in Fig. 2.9. This radiation is dominated by soft and collinear emissions, which for many observables give rise to logarithmically enhanced contributions (see Sec. 2.6). When such logarithms become large, fixed-order perturbation theory is no longer sufficient, and dedicated techniques are required to obtain reliable results in the perturbative regime. The accuracy of these methods is typically characterised in terms of logarithmic accuracy, such as leading logarithmic (LL), next-to-leading logarithmic (NLL), and so on. Common approaches include:

- *Parton showers* model the soft and collinear emissions from the hard scale down to the hadronisation scale. Well-known implementations include SHERPA [26], PYTHIA [34], HERWIG [35, 36], and PANSCALES [37]. While PYTHIA and HERWIG achieve LL accuracy, the SHERPA [38, 39] and PANSCALES [2, 40, 41] showers have demonstrated NLL accuracy for a broad class of processes and observables, with ongoing work aimed at extending to higher logarithmic accuracy.

- *QCD resummation* systematically resums logarithmically enhanced contributions from soft and collinear radiation to all orders in perturbation theory. Semi-numerical resummation frameworks such as CAESAR [42] and ARES [43–46] fall into this category. For e^+e^- , resummation for jet observables has been achieved at NNLL accuracy, while for hadronic collisions the state of the art is generally NLL. The work presented in Chapter 5 focuses on extending the ARES framework to hadronic collisions at NNLL accuracy.
- *Effective field theory (EFT) resummation*, most notably through soft-collinear effective theory (SCET) [47–50], describes soft and collinear degrees of freedom through a systematic observable-dependent factorisation formula. In this framework, the relevant dynamics are encoded in separate hard, collinear, soft, and jet functions, which enables the resummation to be carried out analytically for a wide range of observables. In some cases, this can be performed up to N^3 LL accuracy, as demonstrated in Refs. [51–53].

2.7.3 Hadronisation ($Q \sim \Lambda_{\text{QCD}}$)

As the energy scale of the event approaches Λ_{QCD} , confinement effects dominate and hadronisation occurs. The coloured partons produced by the hard scattering and subsequent QCD evolution (light green blobs in Fig. 2.9) are transformed into colour-singlet hadrons (dark green blobs). These primary hadrons subsequently decay into longer-lived particles (smaller dark green blobs) that are ultimately observed in the detector. Experimentally, these particles appear predominantly as *jets* – collimated sprays of hadrons – whose clustering properties are discussed in Sec. 2.8. Since hadronisation occurs in the non-perturbative regime, pQCD cannot be applied directly; instead, its effects are modelled using phenomenological approaches such as the string [54, 55] and cluster [56] models, as well as analytical power-correction methods [57].

The three stages described above capture the main features of the QCD dynamics in a high-energy hadronic collision, but several additional ingredients contribute to the structure of a realistic event. Electromagnetic radiation, shown as yellow lines in Fig. 2.9, may occur at any stage of the process. Secondary semi-hard interactions between the proton remnants give rise to the *underlying event*, represented by the purple blob in Fig. 2.9. These interactions generate further radiation and hadrons, but are not described by standard QCD factorisation and require dedicated modelling.

Prior to the hard interaction, the incoming partons may radiate, producing *initial-state radiation* (ISR). This radiation evolves down to the hadronisation scale and contributes to the final-state hadrons (dark green blobs). The light blue ovals in Fig. 2.9 represent the beam remnants:

the fragments of the incoming protons that do not participate in the hard scattering.

2.7.4 The QCD factorisation formula

To relate the partonic dynamics of a hard scattering to experimentally measured hadronic cross sections, we employ the QCD factorisation formula,

$$d\sigma(P_1, P_2) = \sum_{i,j} \int_0^1 dx_1 dx_2 f_{i/P_1}(x_1, \mu_F^2) f_{j/P_2}(x_2, \mu_F^2) d\hat{\sigma}_{ij}(\alpha_s(\mu_R^2), \mu_R^2, \mu_F^2) + \mathcal{O}\left(\frac{\Lambda_{\text{QCD}}^2}{Q^2}\right), \quad (2.41)$$

which is illustrated schematically in Fig. 2.10. Here, $f_{i/P_1}(x_1, \mu_F^2)$ and $f_{j/P_2}(x_2, \mu_F^2)$ are the PDFs for partons i and j inside the incoming protons P_1 and P_2 , carrying longitudinal momentum fractions x_1 and x_2 . The sum over i, j runs over all contributing parton flavours, while $\hat{\sigma}_{ij}$ denotes the corresponding partonic cross section.

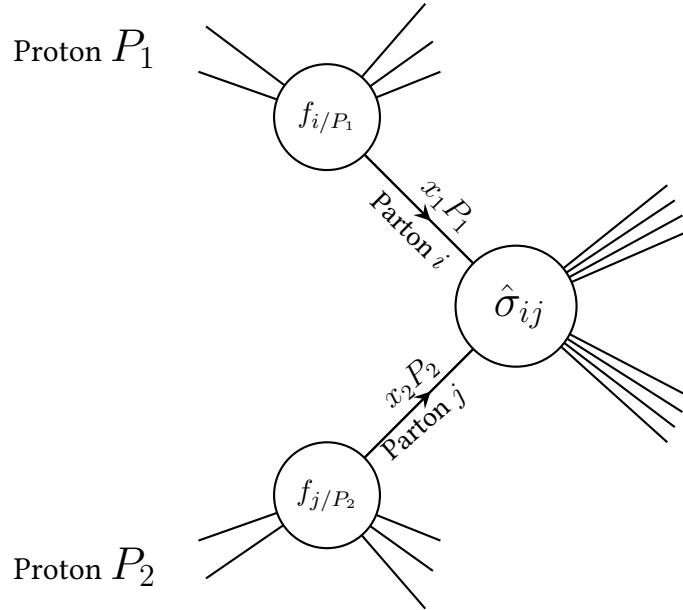


Figure 2.10: Diagrammatic representation of the QCD factorisation formula, Eq. (2.41). The PDFs $f_{i/P_1}(x_1, \mu_F^2)$ and $f_{j/P_2}(x_2, \mu_F^2)$ describe the distributions of partons i and j carrying longitudinal momentum fractions x_1 and x_2 inside the incoming protons P_1 and P_2 . These partons undergo a hard scattering $\hat{\sigma}_{ij}$, with the outgoing lines representing the final-state particles and proton remnants.

The factorisation formula separates the long-distance, non-perturbative physics, encoded in the PDFs, from the short-distance hard interaction, captured by $\hat{\sigma}_{ij}$. This separation is made precise by the factorisation scale μ_F : radiation with transverse momentum below μ_F is absorbed into the PDFs and treated as part of the proton structure, while harder emissions are included explicitly in $\hat{\sigma}_{ij}$. This scale comes about because, beyond leading order, the partonic cross sections contain collinear singularities from radiation off the incoming partons which, unlike their

final-state counterparts, do not cancel against the virtual corrections (cf. Sec. 2.4.2). These singularities are removed by a counterterm absorbed into the definition of the PDFs, giving rise to their dependence on the factorisation scale μ_F :

$$f_{i/P}(x) = f_{i/P}(x, \mu_F^2) + \frac{\alpha_s(\mu_R^2)}{2\pi} \frac{(4\pi)^\varepsilon}{\Gamma(1-\varepsilon)} \left(\frac{\mu_R^2}{\mu_F^2} \right)^\varepsilon \frac{1}{\varepsilon} \sum_j \int_x^1 \frac{dz}{z} P_{ij}^{(0)}(z) f_{j/P}\left(\frac{x}{z}, \mu_F^2\right) + \mathcal{O}(\alpha_s^2). \quad (2.42)$$

The power-suppressed corrections $\mathcal{O}(\Lambda_{\text{QCD}}^2/Q^2)$ in Eq. (2.41) reflect the fact that this description is valid only in the high-energy limit, $Q \gg \Lambda_{\text{QCD}}$, which is satisfied throughout this thesis.

2.8 Jet physics

As discussed in Sec. 2.7.3, jets appear as highly collimated sprays of hadrons resulting from the hadronisation of energetic quarks and gluons produced in high-energy processes. They are among the most frequently observed objects in proton–proton collisions at the LHC, and are inherently complex and ambiguous objects. To be able to describe them in a well-defined and reproducible way, a *jet definition* is required, consisting of three components:

1. a *jet algorithm*: a systematic procedure for clustering final-state particles into jets, i.e. a set of rules that determine which particles belong to the same jet;
2. the algorithm's *parameters*: these specify the proximity required between two particles for them to be recombined into a single jet constituent;
3. a *recombination scheme*: this defines how the four-momenta of two particles are combined when they are merged. The most widely used is the E -scheme, which is a four-momentum sum, $p_i^\mu + p_j^\mu = p_k^\mu$.

2.9 Jet algorithms

In 1990, significant discussion within the high-energy physics community led to the formulation of a set of criteria that a well-defined jet algorithm should satisfy. These requirements were outlined in a document known as the *Snowmass Accord* [58]:

1. simple to implement in an experimental analysis;

2. simple to implement in a theoretical calculation;
3. defined at any order in perturbation theory;
4. yields a finite cross section at any order in perturbation theory;
5. yields a cross section that is relatively insensitive to hadronisation effects.

The final three requirements are ensured by IRC safety (Sec. 2.6), which guarantees that jet definitions at the detector, hadron, and parton levels are essentially equivalent. For IRC-safe observables, the effects of hadronisation are suppressed by inverse powers of the energy scale Q used in the process. Consequently, at higher energies, the correspondence between hadronic and partonic observables becomes increasingly accurate, consistent with quark-hadron duality (Sec. 2.4). Together, these requirements allow us to regard jets as well-defined objects whose properties can be determined either by their constituent hadrons or underlying partons. This enables direct comparison between experimental measurements and theoretical calculations at the parton level.

A wide variety of jet algorithms exist, each with its own advantages and limitations. The choice of algorithm depends on the specific physics question being addressed and the type of information one seeks to extract from an event. In general, jet algorithms can be divided into two broad classes:

- **Cone algorithms** cluster particles whose transverse energy deposits fall within a fixed cone of radius R in (η, ϕ) space. In three dimensions, these regions appear as cones. Different implementations vary in how stable cones are defined and identified, see Refs. [59, 60] for details. These are often described as *top-down* algorithms.
- **Sequential recombination algorithms** iteratively merge the closest pair of particles according to a chosen distance measure, until a stopping criterion is reached. Variations arise from different definitions of the distance measure and stopping condition. These are often referred to as *bottom-up* algorithms.

In this thesis, we focus exclusively on sequential recombination algorithms. For a comprehensive review of jet algorithms across different processes, see Refs. [8, 61].

2.10 Sequential recombination in e^+e^- collisions

To illustrate the basic procedure of sequential recombination algorithms, we first consider their application to e^+e^- collisions. The JADE algorithm, introduced by the JADE Collaboration [62], was the first sequential recombination algorithm to be used experimentally and proceeds as follows:

1. Define a *resolution parameter*, y_{cut} .
2. Compute the *distance measure* for every pair of particles i and j :

$$y_{ij} = \frac{2E_i E_j (1 - \cos \theta_{ij})}{Q^2}, \quad (2.43)$$

where Q is the total centre-of-mass energy, E_i is the energy of particle i , and θ_{ij} is the angle between particles i and j .

3. Identify the pair with the smallest y_{ij} . If $y_{ij} < y_{\text{cut}}$, recombine the two particles into a pseudojet according to the chosen recombination scheme.
4. Repeat the procedure from step 2 until all remaining pairs satisfy $y_{ij} > y_{\text{cut}}$. The surviving objects are defined as jets.

A drawback of the JADE algorithm is its tendency to incorrectly cluster soft partons that are widely separated in angle. This follows from the energy dependence of the distance measure, y_{ij} , in Eq. (2.43), which becomes small whenever either particle is soft, irrespective of their angular separation. Consequently, two soft gluons emitted at widely separated angles can be recombined before either is clustered with the hard parton from which it originated. This can subsequently result in a soft gluon being incorrectly associated with a different hard parton.

An example is shown in Fig. 2.11. The soft gluons k_1 and k_2 are first recombined into the pseudojet $k_1 + k_2$, which then clusters with the hard parton p_1 . As a result, the radiation k_2 is associated with p_1 , despite being emitted from the other hard parton. This behaviour is undesirable, since one would generally expect a jet to consist of a hard parton together with its associated radiation.

The shortcomings of the JADE distance measure motivated modifications designed to improve its treatment of soft radiation. This led to the development of the Durham [63, 64] and Cambridge [65] algorithms, which are discussed in Ch. 3.

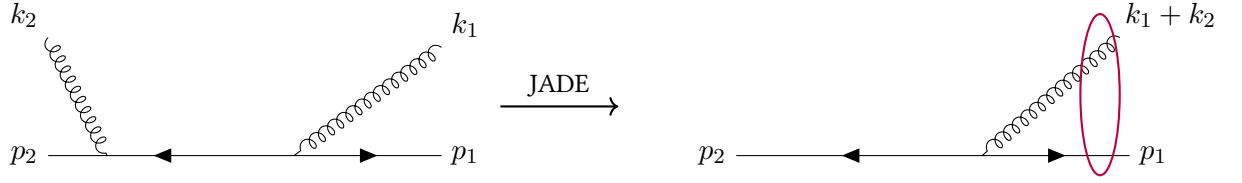


Figure 2.11: The JADE algorithm incorrectly recombining two gluons k_1 and k_2 that are widely separated in angle. The merged pseudojet $k_1 + k_2$ subsequently clusters with the hard parton p_1 , incorrectly associating k_2 with p_1 .

2.11 Jet algorithms beyond e^+e^-

The clustering procedure becomes more involved when moving beyond e^+e^- collisions to processes involving incoming hadrons, such as hadronic collisions and deep inelastic scattering (DIS). In these cases, the algorithm must also account for radiation that becomes collinear to an incoming beam. This is incorporated through an additional *beam distance*, d_{iB} , which determines whether a particle is associated with the beam or recombined with another particle.

The construction of jet algorithms for DIS is particularly interesting because the presence of a single hadronic beam introduces an asymmetry that is absent in both e^+e^- and hadronic collisions. This means that the radiation associated with the proton remnant and the struck quark must be treated carefully within the jet definition. The next chapter discusses this in detail. We first introduce the relevant DIS kinematics and the Breit frame, before reviewing the development of DIS jet algorithms and more recent proposals. We then introduce the inclusive generalised- k_t algorithm for DIS and investigate its properties.

Chapter 3

A generalised- k_t jet algorithm for Deep Inelastic Scattering

3.1 Introduction

Lepton-hadron colliders such as the Hadron Elektron Ringanlage (HERA) and the forthcoming Electron–Ion Collider (EIC) are unique laboratories for the clean investigation of the inner structure of the nucleons and the nuclei that they form. Although the HERA collider has not been operational since 2007, access to previously recorded HERA data has enabled recent re-analyses taking advantage of modern jet algorithms and event-shape observables in Deep Inelastic Scattering (DIS) [66–71]. In the theory community this has spurred a renewed interest in formulating jet algorithms [2, 72, 73], and in performing resummed calculations of event shapes for DIS [74–79], the latter building upon earlier work (see e.g. Refs. [80–85]). Such modern QCD tools will be instrumental for the EIC,¹ currently under construction at Brookhaven National Laboratory, whose operation is scheduled to start in the early 2030s. The EIC will provide new opportunities to study the structure of QCD, including the spin structure of the proton, the dynamics of confinement, and gluon saturation [87, 88]. Since jet production is the dominant process at hadron-lepton colliders, jet reconstruction will play an essential part in many (re)-analyses for both the EIC and HERA.

Most developments in jet algorithms over the past decades have been driven by hadron collider applications, in particular at the LHC, where the anti- k_t algorithm [89] has become the standard choice. Jet algorithms for DIS were developed around the same time as those for hadronic collisions, but unlike for hadron collisions, no single algorithm dominates the land-

¹ And the LHeC [86] should it be approved.

scape. The exclusive (spherically-invariant) k_t algorithm was introduced in Ref. [90] extending the Durham algorithm for e^+e^- collisions [63, 64] to DIS by treating the proton remnant as a particle of infinite momentum. An angular-ordered version of this algorithm was later proposed in Ref. [91]. It modifies the e^+e^- Cambridge algorithm [65] to DIS again by treating the proton remnant as a particle of infinite momentum. A longitudinally invariant (LI) k_t algorithm was introduced in Ref. [92, 93], and was extended to an angular-ordered version (the Aachen algorithm) in Ref. [91]. The latter two algorithms are identical to the hadron-collider versions, except that the clustering takes place in the Breit frame as is the case for all the DIS jet algorithms listed here (cf. Sec. 3.2 for details on the Breit frame). More recently, the Centauro algorithm [72] was proposed, combining features of longitudinally-invariant k_t algorithms with those of spherically-invariant algorithms. Likewise a set of jet definitions suitable for small- x_B DIS were proposed in Ref. [73]. Most of the historical algorithms, including a modified version of the JADE algorithm [62, 94], have been studied extensively at HERA and Fermilab's E665 experiment [95–112], highlighting the importance of jet finding algorithms in DIS. The recent Centauro algorithm has also recently been used by the H1 collaboration to perform the first measurement of groomed event shapes [71] in DIS.

Unlike at the LHC, where one of the main requirements of a good jet algorithm is the insensitivity to pile-up and underlying event, in DIS, where these effects are suppressed, a good jet algorithm is one that preserves a meaningful clustering sequence and allows soft radiation to influence jet boundaries while respecting the natural hemisphere structure of the Breit frame. For this reason it is natural to consider k_t and Cambridge/Aachen (C/A) type algorithms in DIS, in addition to anti- k_t . A C/A-based DIS algorithm designed for the study of Lund variables [79, 113] and parton shower logarithmic accuracy was introduced in Ref. [2], and forms the starting point for an inclusive generalised- k_t algorithm adapted to DIS, which we present and study in more detail in the present chapter. The formulation of the jet algorithm itself follows the structure of the generalised- k_t algorithm used in e^+e^- collisions as implemented and introduced in FASTJET [114], but is formulated in the Breit frame and uses a beam distance that is sensitive to the angular separation between the proton and the particles taking part in the clustering. We also discuss a prescription to select the jet that is most likely to contain the struck quark based on the lightcone fraction that the jet is carrying. This allows for a clean separation between *target* and *current* hemisphere jets. Finally, we study the robustness of the new jet algorithm against hadronisation corrections.

The structure of this chapter is as follows. In Section 3.2 we introduce the kinematics relevant to DIS and define the Breit frame. In Section 3.3 we summarise historical DIS jet algorithms

and the recently proposed Centauro one, while in Section 3.4 we present our new inclusive generalised- k_t jet algorithm. In Section 3.5 we examine the behaviour of the new algorithms and we provide a comparison with the Centauro one. Finally, we present our conclusions in Section 3.6.

3.2 DIS kinematics and the Breit Frame

The DIS process is defined by the partonic scattering process

$$\ell_i(k) + h(P) \rightarrow \ell_f'(k') + X, \quad (3.1)$$

where P^μ is the momentum of the incoming nucleon h , k^μ is the momentum of the lepton beam ℓ_i , k'^μ is the momentum of the outgoing lepton ℓ_f' , and X denotes the hadronic final state. The standard DIS invariants are

$$Q^2 = -q^2, \quad x_B = \frac{Q^2}{2P \cdot q}, \quad y_{\text{DIS}} = \frac{P \cdot q}{P \cdot k}, \quad (3.2)$$

with $q^\mu = k^\mu - k'^\mu$ the momentum of the exchanged virtual photon. At the lowest perturbative order, the hadronic final state consists of a single outgoing parton,

$$\ell_i(k) + p_{\text{in}} \rightarrow \ell_f(k') + p_{\text{out}}, \quad (3.3)$$

where $p_{\text{in}}^\mu = x_B P^\mu$ and $p_{\text{out}}^\mu = q^\mu + p_{\text{in}}^\mu$ are the momenta of the incoming and outgoing partons, respectively. The *Breit frame* plays a central role in many DIS jet algorithms [115].² It is defined by

$$2x_B \vec{P} + \vec{q} = 0, \quad (3.4)$$

such that the virtual photon carries no energy component and is aligned along the z -axis, i.e.

$$q^\mu = (0, 0, 0, -Q), \quad (3.5)$$

and is therefore purely space-like. In this frame, the incoming and outgoing parton momenta take the simple form

$$p_{\text{in}}^\mu = \frac{Q}{2}(1, 0, 0, +1), \quad p_{\text{out}}^\mu = \frac{Q}{2}(1, 0, 0, -1). \quad (3.6)$$

Thus, the proton and virtual photon collide head-on in the Breit frame. Within this frame, the proton remnant appears in the region at positive rapidity (generally referred to as the *target*

² The frame was first introduced in Ref. [116] in the context of relativistic scattering kinematics. The term ‘Breit frame’ was first coined by Ernst, Sachs, and Wali in Refs. [117, 118]. For an explicit implementation we refer the reader to Appendix 7.11 in Ref. [119].

hemisphere), while the struck quark lies in the negative rapidity region (the *current hemisphere*). In practice, the jet clustering in this frame can lead to radiation from the proton remnant being absorbed into the struck-quark jet, making the separation between the two hemispheres an important feature of DIS jet algorithms.

3.3 Existing DIS jet algorithms

Before presenting a generalised- k_t algorithm for DIS in Section 3.4, it is instructive to first discuss existing jet algorithms that have historically been used for DIS, focusing on sequential-recombination jet algorithms (cf. Sec. 2.9). The overview we present here serves to explain how jet algorithms are formulated in general, but will also allow us to highlight similarities and differences between existing jet algorithms and the new generalised k_t jet algorithm for DIS. We formulate the jet algorithms in the Breit frame, but note that those algorithms that are longitudinally boost invariant, such as Centauro and some of the inclusive k_t -algorithms (cf. Secs. 3.3.3 and 3.3.4), can also be formulated in a frame that is connected to the Breit frame through a longitudinal boost.³

3.3.1 Exclusive jet algorithms for e^+e^- collisions

Since some of the existing DIS jet algorithms are adapted from their e^+e^- counterparts, and in fact use e^+e^- clustering explicitly in a re-clustering step, we start by reviewing the e^+e^- variants before turning to DIS.

The Durham jet algorithm

The Durham algorithm was introduced in Ref. [63, 64]. Given a list of objects undergoing the clustering procedure, which are conventionally called pseudojets, one introduces a dimensionless resolution parameter y_{ij} that reads

$$y_{ij} = \frac{\min(E_i^2, E_j^2)}{s} 2(1 - \cos \theta_{ij}), \quad (3.7)$$

with E_i, E_j the energies of the two pseudojets, θ_{ij} the angle between them, and $\sqrt{s}$ the center-of-mass energy of the collision. One also introduces a parameter $y_{\text{cut}} \leq 1$. The algorithm then proceeds as follows.

³ We note that the lab frame in DIS is *not* connected to the Breit frame through only a longitudinal boost, but requires a rotation as well.

1. For every pair of pseudojets compute y_{ij} .
2. The pair with the smallest y_{ij} is clustered together if $y_{ij} < y_{\text{cut}}$, meaning that a pseudojet with momentum $p_{ij} = p_i + p_j$ is inserted in the list of pseudojets, which replaces i and j .⁴
3. Repeat from 1 until all $y_{ij} \geq y_{\text{cut}}$ at which point the remaining pseudo-jets are identified as jets.

After the clustering terminates, each particle in the event is either merged into a jet, or it remains as a single-particle jet, making it an exclusive algorithm.

The Cambridge jet algorithm

The Cambridge algorithm, introduced in Ref. [65], differs from the Durham algorithm by introducing, next to a merging condition, also a distance measure d_{ij} , which reads

$$d_{ij} = 2(1 - \cos \theta_{ij}) . \quad (3.8)$$

This choice was motivated by the aim of producing a clustering sequence that closely follows the coherent pattern of QCD radiation, where a possible merging of two partons that are close in angle is considered before the possible merge of two particles that are more widely separated. The merging of two particles is decided on the Durham distance measure y_{ij} . Secondly, the clustering algorithm itself is modified with a ‘soft freezing’ mechanism, which prevents the softer of two pseudojets to attract any additional pseudojets. The Cambridge algorithm proceeds as follows.

1. Compute d_{ij} for all pairs and select the pair of pseudojets with the smallest d_{ij} .
2. For this pair, if $y_{ij} < y_{\text{cut}}$, delete i and j and introduce a new pseudojet with momentum $p_{ij} = p_i + p_j$. If instead $y_{ij} \geq y_{\text{cut}}$, the softer (i.e. less energetic) pseudojet is removed from the list and is promoted to a jet: this mechanism is called soft freezing.
3. Repeat from 1 until there are no more pseudojets to be clustered.

The last pseudojet that remains in the list of pseudojets is also promoted to a jet.

⁴ This recombination is dubbed the E -scheme but other alternatives exist [62, 63, 94].

3.3.2 Exclusive jet algorithms for DIS collisions

Exclusive k_t -algorithm for DIS

The exclusive k_t -algorithm for DIS was presented in Ref. [90]. It is based on the equivalent e^+e^- variant [63] that we discussed above, but treats also the additional initial-state (and beam remnant) collinear singularities. The algorithm is formulated in the Breit frame. Instead of using the center-of-mass energy $\sqrt{s}$, one introduces a transverse-momentum-like scale E_t satisfying $\Lambda \ll E_t \leq Q$ and defines a resolution variable y_{ij} between two pseudojets as

$$y_{ij} = \frac{\min(E_i^2, E_j^2)}{E_t^2} 2(1 - \cos \theta_{ij}), \quad (3.9)$$

as well as a beam resolution parameter for each pseudojets

$$y_{iB} = \frac{E_i^2}{E_t^2} 2(1 - \cos \theta_{iB}), \quad (3.10)$$

where θ_{ij} is the angular separation between two pseudojets, θ_{iB} is the angle of pseudojet i with respect to the beam, and E_i and E_j are the energies of the corresponding pseudojets. The E_t variable determines how close to the beam direction (in the Breit frame) resulting jets are allowed to be formed, with the transverse momentum of the resulting jets satisfying $p_t > E_t$. The clustering procedure works in two steps. In the first step, a pre-clustering takes place, with the aim of separating particles into a *beam* and a set of *macro-jets*, and operates as follows.

1. For every pseudojets i compute y_{iB} and for every pair i, j compute y_{ij} .
2. Take the smallest value of all y_{ij} and y_{iB} . If the smallest value is a y_{ij} and $y_{ij} < 1$, combine their momenta into a new pseudojets. If $y_{iB} < 1$ is the smallest, include i into the beam jet, and remove it from the list of pseudojets.
3. Repeat the procedure from step 1 for all pseudojets, until all distance measures are $y_{ij}, y_{iB} > 1$.

The resulting objects that remain at this stage are the final-state macro-jets, and does not include the beam jet. In the second step of the algorithm we resolve the structure of the final-state macro-jets. The input for this second step consists of all the constituents of these jets. At this stage, a resolution parameter $y_{\text{cut}} = Q_0^2/E_t^2 < 1$ is introduced, with $Q_0 < E_t$, and the algorithm proceeds by clustering the constituents using the e^+e^- Durham algorithm (Sec. 3.3.1). The objects that remain at the end are the final-state jets. The full two-step procedure produces jets with relative transverse momenta k_t in the range $\sqrt{y_{\text{cut}}}E_t < k_t < E_t$, and with a Breit-frame

transverse momentum $p_t > E_t$. Every particle in the event ends up either in the list of final-state jets or in the beam jet.

Cambridge algorithm for DIS

The Cambridge algorithm for DIS was introduced in Ref. [91]. It is an exclusive algorithm, inspired by the Cambridge algorithm for e^+e^- -collisions [65], and very similar to the exclusive k_t -algorithm presented in Sec. 3.3.2. Like in e^+e^- , one decouples the merging step from the ordering step, now by introducing two distance measures: d_{ij} and d_{iB} , which read

$$d_{ij} = 2(1 - \cos \theta_{ij}), \quad (3.11)$$

and

$$d_{iB} = 2(1 - \cos \theta_{iB}). \quad (3.12)$$

Like for the exclusive k_t -algorithm presented in Sec. 3.3.2, also in this case the algorithm proceeds in two steps. The first step is

1. For every pseudojet i compute d_{iB} and for every pair i, j compute d_{ij} .
2. Take the smallest value of all d_{ij} and d_{iB} . If the smallest value is a d_{ij} , combine their momenta into a new pseudojet if $y_{ij} < 1$ otherwise call the softest (less energetic) a jet (i.e. apply the freezing mechanism). If d_{iB} is the smallest include i as a beam jet, and remove it from the list of pseudojets if $y_{iB} < 1$, otherwise promote it to a jet.
3. Repeat the procedure from step 1 for all pseudojets, until all resolution variables $y_{ij}, y_{iB} > 1$, at which point we promote all the remaining pseudojets to jets and proceed with the second step.

In the second step of the algorithm we undo the clustering of all the jets (excluding the beam jet), and re-cluster the constituents using the e^+e^- Cambridge algorithm using a resolution parameter y_{cut} .

3.3.3 Generalised inclusive k_t -algorithms for DIS

The jet algorithms for DIS discussed so far were all **exclusive**. For an **inclusive** algorithm, when the smallest distance is the one between the beam(s) and a pseudo-jet, the pseudo-jet is promoted to a jet rather than being absorbed into a beam remnant collection. The total number of jets is

therefore an infrared unsafe observable, because it includes jets that can be either very soft or collinear with the beam. Only after introducing a suitable resolution parameter (like a minimum transverse momentum with respect to the beam) and removing jets that fall below this resolution parameter, will it be infrared safe.

The inclusive longitudinally-invariant (LI) k_t -algorithm was first introduced in Ref. [93] (see Ref. [92] for the exclusive version) for hadron-hadron collisions. An angular-ordered variant, often referred to as the Aachen or Cambridge/Aachen algorithm, was since introduced in Ref. [91] and then eventually the family of generalised inclusive k_t -algorithms was introduced in Ref. [89] along with the anti- k_t algorithm. The DIS variant is in this case identical to the hadron-hadron variant, except that the clustering takes place in the Breit frame. We define a set of distance measures d_{ij} and d_{iB} , which are dimensionful for $p \neq 0$, given by

$$d_{ij} = \min(p_{Ti}^{2p}, p_{Tj}^{2p}) \frac{\Delta R_{ij}^2}{R^2}, \quad d_{iB} = p_{Ti}^{2p}, \quad (3.13)$$

where $\Delta R_{ij}^2 = (y_i - y_j)^2 + (\phi_i - \phi_j)^2$, with y_i the rapidity and ϕ_i the azimuthal angle of pseudojet i . The procedure is then given by

1. Compute all the distances d_{ij} and d_{iB} . If a d_{ij} is the smallest, cluster i and j , if d_{iB} is the smallest call i a jet.
2. Repeat until no more pseudojets remain.

We note that the k_t -algorithm corresponds to $p = 1$, C/A to $p = 0$ and anti- k_t to $p = -1$.

Note that one is not restricted to using distances that are natural for pp collisions. Indeed, the inclusive algorithm can be made spherically invariant (SI) instead of LI, which is more appropriate for e^+e^- collisions [114], by replacing the distance measures with

$$d_{ij} = \min(E_i^{2p}, E_j^{2p}) \frac{1 - \cos \theta_{ij}}{f(R)}, \quad d_{iB} = E_i^{2p}, \quad (3.14)$$

with

$$f(R) = \begin{cases} 1 - \cos R & 0 < R < \pi \\ 3 + \cos R & \pi < R < 2\pi. \end{cases} \quad (3.15)$$

In this case, the beam distance d_{iB} is not sensitive to the angular separation between the pseudojet i and the beam, but only on the energy of the pseudo-jet. This feature is not desirable in the context of ep collisions, which is an asymmetrical collision, where we need to distinguish between a collinear and an anti-collinear direction, as there is only one hadronic beam.

3.3.4 Centauro algorithm

The Centauro algorithm [72] is a longitudinally invariant jet algorithm designed to capture jets close to the Born configuration in the Breit frame. Longitudinally invariant k_t -type algorithms such as those presented in Sec. 3.3.3 cluster in the (y, ϕ) plane and are unable to form a jet along the beam axis, where $y \rightarrow -\infty$. Conversely, the spherically invariant algorithms can cluster around the beam axis but do not preserve longitudinal invariance.

The Centauro algorithm is designed to address both of these aspects. It is built around the variable $\bar{\eta}_i$ that behaves differently in the two hemispheres:

- **Current hemisphere:** $\bar{\eta}_i$ decreases as particles become closer in angle, allowing particles in this hemisphere to cluster.
- **Target hemisphere:** $\bar{\eta}_i$ becomes large and diverges. This prevents clustering in the proton beam region, similar to the behaviour of the anti- k_t -algorithm.

Working in the Breit frame, for every momentum p_i^μ , one defines the variable $\bar{\eta}_i$ as

$$\bar{\eta}_i = -2 \frac{Q}{q \cdot n_{\text{out}}} \frac{p_{T,i}}{p_i \cdot n_{\text{in}}}, \quad (3.16)$$

with

$$n_{\text{in}}^\mu = \frac{2}{Q} x_B P^\mu \quad (3.17a)$$

$$n_{\text{out}}^\mu = \frac{2}{Q} (q^\mu + x_B P^\mu) \quad (3.17b)$$

$$p_{T,i}^2 = \frac{2(p_i \cdot n_{\text{in}})(p_i \cdot n_{\text{out}})}{n_{\text{in}} \cdot n_{\text{out}}}, \quad (3.17c)$$

meaning that $p_{T,i}$ is the transverse component of the momentum in the Breit frame (or any other frame that only differs by a longitudinal boost from the Breit frame). From this, one constructs the inter-pseudojet distance measure as ⁵

$$d_{ij} = \frac{(\bar{\eta}_i - \bar{\eta}_j)^2 + 2\bar{\eta}_i\bar{\eta}_j(1 - \cos \Delta\phi_{ij})}{R^2}. \quad (3.18)$$

The distance to the beam is always equal to $d_{iB} = 1$. One then clusters jets according to the procedure given in Sec. 3.3.3.

⁵ Note that we are reporting the definition used in the `fjcontrib` implementation, which is the one we have also used in this work.

3.4 A generalised- k_t algorithm for DIS

3.4.1 The algorithm

We now introduce a generalised- k_t algorithm for DIS, implemented as a FASTJET plugin in `fjcontrib`. Relevant parts of the code are given in appendix A. The algorithm shares the essential structure of its e^+e^- counterpart, introduced at the end of Sec. 3.3.3, with the key modification that the beam distance is now sensitive to the angular separation with respect to the proton beam and the clustering takes place in the Breit frame. The angular-ordered variant of the algorithm ($p = 0$) was already presented in Ref. [2], and used to define Lund Tree Event shapes (cf. Sec. 5.7) with remarkably simple all-orders structure [79]. Here we present the extension for $p \neq 0$, and include two parameters, R and y_{cut} (to be detailed below), that control the behaviour of the algorithm. We first define the distances

$$d_{ij} = \min(E_i^{2p}, E_j^{2p}) \frac{(1 - \cos \theta_{ij})}{1 - \cos R}, \quad (3.19a)$$

$$d_{iB} = E_i^{2p} (1 - \cos \theta_{iB}), \quad (3.19b)$$

where $R < \pi$. The algorithm is then as follows.

1. Compute all d_{ij} and d_{iB} . If the smallest is a d_{iB} , call i a jet, and remove it from the list of pseudojets. If the smallest is a d_{ij} , cluster i and j .
2. Continue until there are no pseudojets left in the list.

The introduction of the distance d_{iB} measuring both the energy and the closeness in angle to the beam direction, allows one to cluster jets in the current hemisphere while promoting remnant hemisphere particles to jets early in the cluster sequence. Note that the C/A variant ($p = 0$) presented in Ref. [2] uses $R = \frac{\pi}{2}$.⁶ For $p = 0$, the energy of a particle does not enter the distance measure, and particles are merged purely based on angular proximity. For $p = 1$, i.e. the k_t variant of the algorithm, soft particles typically lead to smaller d_{ij} and therefore are clustered first. On the contrary, for $p = -1$, i.e. the anti- k_t variant, hard final-state particles will cluster first.

As in the standard inclusive (LI) algorithms for pp collisions, some reconstructed jets may originate entirely from initial-state radiation or from soft large-angle emissions. In pp collisions,

⁶ It is also possible to formulate an exclusive variant of our algorithm by introducing a dimensionless quantity y_{ij} and an associated y_{cut} . The $p = 0$ zero variant will then share some similarities with the algorithm in Sec. 3.3.2 but without the final re-clustering step. We do not explore this variant any further in this thesis.

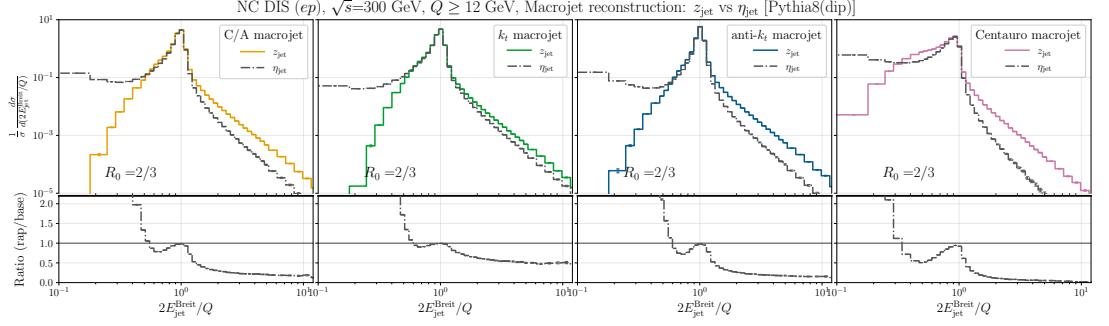


Figure 3.1: Normalised Breit-frame energy distribution of the final-state macrojet for neutral-current DIS, $\sqrt{s}=300$ GeV using, from left to right, the inclusive version of the generalised k_t algorithms with $R_0 = 2/3$ and $p = 0$ (C/A), 1 (k_t), and $p = -1$ (anti- k_t), followed by the Centauro jet algorithm with $R_0 = 2/3$. The final-state macrojet is identified as the one that maximises the light-cone component opposite in direction to the proton (z_{jet} , solid line) or the Breit-frame pseudo-rapidity (η_{jet} , dash-dotted lines). Events are generated using the Pythia 8.3 [34] default shower, with dipole recoil option [120]. See Tab. 3.1 for further settings.

such jets are removed by imposing a transverse-momentum (and a rapidity) cut. This however cannot be applied to DIS, as the jet originating from the fragmentation of the struck quark does not necessarily carry a large transverse momentum (in fact, in the case where no additional radiation is produced, this jet will carry zero transverse momentum). To remove only soft jets or jets collinear to the beam, one can define an ordering variable according to the (scaled) effective transverse momentum

$$y_{iB} = 2 \frac{E_i^2}{Q^2} (1 - \cos \theta_{iB}), \quad (3.20)$$

and discard jets with $y_{iB} < y_{\text{cut}}$. Alternatively, one might use an ordering variable based on the light-cone fraction of the original struck-quark momentum carried by the i^{th} jet, defined by

$$z_i = \frac{2x_B p_i \cdot P}{Q^2} = 2 \frac{E_i}{Q} (1 - \cos \theta_{iB}), \quad (3.21)$$

where P is the proton momentum. One may then discard jets with $z_i < z_{\text{cut}}$, with $0 < z_{\text{cut}} < 1$. Below, we discuss how to use z_i to select what we will call the *macrojet*. Differences between Centauro (Sec. 3.3.4) and the generalised- k_t jet algorithm presented here will be explored in Sec. 3.5.

3.4.2 Identifying the macrojet

This section describes how we identify the *final-state macrojet*, defined as the jet that is most likely to contain the fragments of the struck quark. Our prescription is to select the jet that carries the largest light-cone component z_{jet} along the direction of the original struck quark. To make this precise, we perform a Sudakov decomposition of each jet momentum (see also Sec. 3.2

and Ref. [74])

$$p_{\text{jet}}^\mu = p_{\text{jet}}^- x_B P^\mu + z_{\text{jet}} p_{\text{out}}^\mu + k_\perp^\mu, \quad (3.22)$$

where P^μ is the proton momentum, and as before, $p_{\text{out}}^\mu = q^\mu + x_B P^\mu$ is the original struck-quark momentum. The coefficient p_{jet}^- is given by

$$p_{\text{jet}}^- = \frac{|\vec{k}_\perp|^2 + p_{\text{jet}}^2}{Q^2 z_{\text{jet}}}. \quad (3.23)$$

The vector $k_\perp^\mu$ is a space-like vector orthogonal to P^μ and p_{out}^μ .⁷ Maximising z_{jet} is then equivalent to selecting the jet whose momentum p_{jet}^μ has the largest dot product with P^μ . As discussed in Ref. [2], this prescription is infrared safe. A seemingly natural alternative is to identify the macrojet as the jet with the most negative rapidity. However, this prescription is not infrared safe, as illustrated in Fig. 3.1. There, one can observe that both for the new family of jet algorithms introduced here and for the Centauro jet algorithm, identifying the final-state macrojet by rapidity (dash-dotted lines) leads to a soft tail. This tail is produced by wrongfully identifying soft emissions, generated either from radiation off the DIS hard-scattering process or from the beam remnant, as the final-state macrojet. This means that selecting the jet with the most negative rapidity is not infrared safe because an arbitrarily soft emission at sufficiently negative rapidity can alter the jet assignment discontinuously. We note that for the k_t -algorithm the soft tail is partially suppressed even when selecting the macrojet with the most negative rapidity, but there is no exponential decrease in the tail. Maximising the light-cone component in the direction of the original struck quark (solid curves) exponentially suppresses the tail for all algorithms.

Beyond macrojet identification, the properties of the macrojet itself provide a useful way to characterise the behaviour of the algorithm. In particular, the macrojet offers a direct handle separating the remnant hemisphere from the current hemisphere. We will explore this further in the next section.

We conclude this section by explicitly discussing the infrared and collinear safety of the jet algorithm and the macrojet definition. In the Born limit, the only final-state parton lies in the current hemisphere and is promoted to a single jet with $z_{\text{jet}} = 1$, so that the macrojet prescription reproduces the struck-quark direction. Collinear splittings are clustered by the recombination step before they can affect the resolved jet structure, and the macrojet selection remains unchanged because z_{jet} is additive under such splittings, preserving the total light-cone momentum of the parent jet. Soft emissions either cluster with a nearby hard pseudojet or form

⁷ Note that in Eq. (3.22) we have adopted the notation of Ref. [74] to label the lightcone component along the proton in the ‘−’ direction.

Input Parameter	Value
PDF	NNPDF40MC_nlo_as_01180
$\sqrt{s}$	300 GeV
$Q_{\min}$	12 GeV
Hadronisation effects	On
Beam remnants	On
QED	Off
Tune	Monash [122]

Table 3.1: Input parameters used for the numerical results presented below.

separate soft jets whose light-cone fraction vanishes in the soft limit. They therefore cannot change the largest- z_{jet} assignment except in degenerate configurations of measure zero. The resolved macrojet observables considered below are consequently insensitive to arbitrarily soft emissions and collinear splittings.

3.5 Phenomenological results

In this section we explore a few macrojet distributions obtained with our inclusive jet algorithms. In what follows we will simply label the $p = -1$ variant of the algorithm with anti- k_t , $p = 0$ with C/A, and $p = 1$ with k_t . We compare them to the Centauro algorithm to highlight differences and similarities.

DIS neutral-current events are produced using the `Pythia 8.3` [34] event generator, with the native `Pythia` shower with dipole local recoil [120]. The analysis is performed with `Rivet 3.1.8` [121]. The parameters, unless otherwise stated, used for the results presented in the following sections are summarised in Table 3.1. In Sec. 3.5.1 we investigate the properties of all the reconstructed jets, while in Sec. 3.5.2 and 3.5.3 we focus on the final-state macrojet distributions.

3.5.1 Properties of the reconstructed jets

To better understand general features of the new jet algorithms and how they compare to the Centauro jet algorithm, we show in Fig. 3.2 the (θ, ϕ) distribution of particles from a DIS event obtained after showering (but before hadronisation). The location of the central point in each disc corresponds to the particle’s position in the θ and ϕ plane, while the size of the disc illustrates the particle’s energy, with larger radii corresponding to higher relative energies. Particles in the remnant hemisphere lie on the left-hand side of each figure, while particles in the current

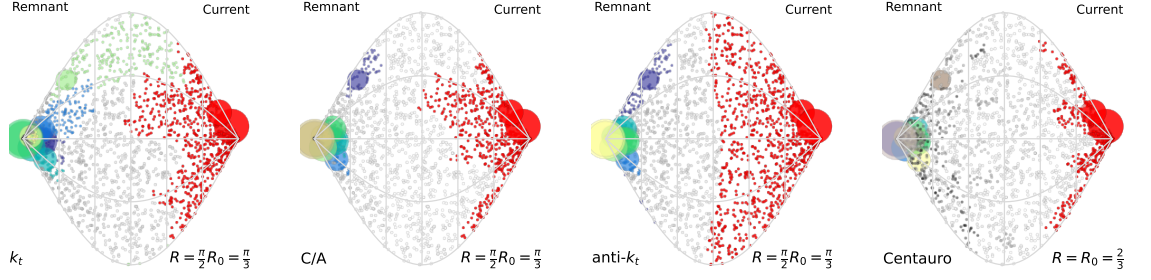


Figure 3.2: From left to right, we show the jet and jet boundaries for the k_t ($p = 1$), C/A ($p = 0$), anti- k_t ($p = -1$) and Centauro jet algorithms for a DIS event with $x_B = 0.2$ and $Q = 114$ GeV. Each particle is represented by a disc whose area is proportional to its energy, while its position corresponds to the direction of its momentum projected onto the unfolded sphere around the hard-scattering vertex. In addition to the physical particles, we have added a sea of 2000 ghosts [123] to better illustrate how the shape of the jet boundaries. The soft particles that are not part of any physical jet are coloured gray. Vertical lines indicate constant θ , and curved lines indicate constant ϕ . The remnant hemisphere lies in the left half of the plot and the current hemisphere in the right half of the plot. All particles clustered into a given jet share the same color, and jets only made out of ghost particles are coloured gray. Particles entering the final-state macro-jet are highlighted in red.

hemisphere lie on the right-hand side. Particles entering the same jet share the same colour. Those forming the final-state macrojet are coloured in red. We have placed the event in a sea of 2000 additional ultra-soft particles, dubbed ghosts in Ref. [123], where the idea of illustrating the jet boundaries was originally introduced. We generate these ghosts with a fixed energy ($E = 10^{-6}$ GeV) and with a uniform distribution in θ and ϕ . We stress here that the exact shape of the jet boundaries depends to some extent on the exact distribution of the soft particles.

Upon inspection of Fig. 3.2, one of the most striking visual features is that of the jet boundary produced by the anti- k_t algorithm. We see that the jet boundary for the final-state macrojet in this case follows the lines of constant θ . This is a direct consequence of the anti- k_t clustering sequence, which always initiates clustering from the hardest particle outward, producing perfectly conical jets in the θ, ϕ plane. This circular symmetry is not shared by the other algorithms: k_t , C/A and Centauro all produce jets with irregular, jagged boundaries.

Following Ref. [72], we adopt

$$R_{\text{generalised-}k_t} = \frac{\pi}{2} R_{\text{centauro}} \equiv \frac{\pi}{2} R_0, \quad (3.24)$$

to compare the generalised- k_t variants with Centauro. In the following phenomenological comparisons we quote radii in terms of R_0 . One other notable feature is that Centauro tends to produce smaller jets with similar-sized jet radius parameter R_0 . This is a consequence of a general property of the Centauro algorithm: it produces comparatively isolated, tight jets. This feature is even more pronounced in the remnant hemisphere, where the size of the jet clusters is much bigger for all variants of the generalised- k_t jet algorithm than for Centauro, which exponentially suppresses clustering of particles in the remnant hemisphere.

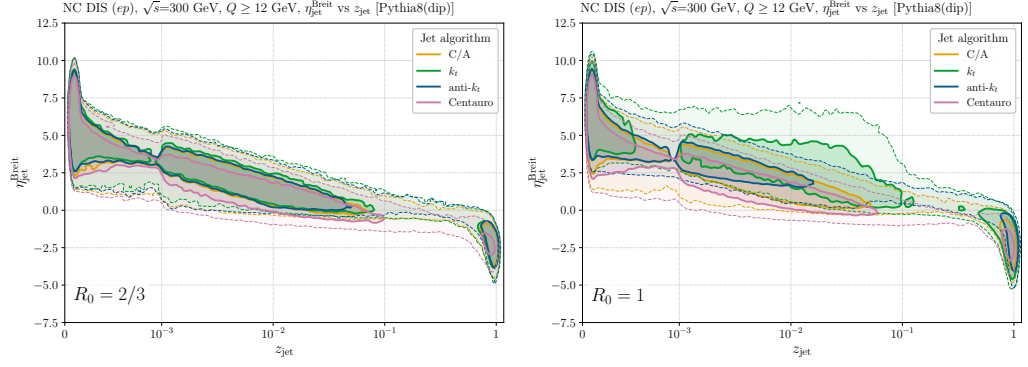


Figure 3.3: The $z - \eta$ -plane for all jets constructed with the Centauro (pink), C/A (orange), k_t (green) and anti- k_t (blue) algorithms for $R_0 = 2/3$ (left) and $R_0 = 1$ (right), using Eq. (3.24). The solid (dashed) curves correspond to 68% (95%) of the jets, in terms of their density in the $z - \eta$ -plane.

Note that for the k_t and C/A variants of the generalised- k_t algorithm, and for Centauro, the exact set of particles clustered into the *macrojet* can change significantly with the choice of R_0 . This occurs because these algorithms cluster particles based on relative softness or angular proximity, so in events with several particles carrying comparable energies, small changes in R_0 can cause different particles to be pulled into or out of the macrojet (as our definition of the macrojet relies on z_{jet}). The anti- k_t variant does not suffer from this, since it always clusters outward from the hardest particle, making the macrojet constituents much more stable against variations in R_0 .

Despite its attractive properties, it is worth pointing out that depending on the choice of R_0 , the anti- k_t variant of our algorithm carries a specific feature. Precisely because it clusters outward from the hardest particle without regard for the hemisphere boundary, it can in principle absorb particles from the remnant hemisphere into the current-hemisphere macrojet. Empirically, values around $R_0 \lesssim 2/3$ are found to significantly reduce contamination of the current-hemisphere macrojet by remnant radiation.

We end this subsection by studying general properties of all jets. Fig. 3.3 shows the $z - \eta$ plane for all final-state jets constructed with the four algorithms considering 10^6 events. Here, z_{jet} , the lightcone-momentum fraction carried by each of the final-state jets, is defined through Eq. (3.22) as

$$z_{\text{jet}} = \frac{P \cdot p_{\text{jet}}}{P \cdot q} = \frac{p_{\text{jet}}^E - p_{\text{jet}}^z}{|Q|}, \quad (3.25)$$

where the last equation holds in the Breit frame. We show contours corresponding to areas that contain 68% (solid) and 95% (dashed) of the total number of jets in the plot, for each of the four considered jet algorithms, and for $R_0 = 2/3$ (left) and $R_0 = 1$ (right).

We observe that the jet algorithms behave relatively similar for $R_0 = 2/3$, where we observe

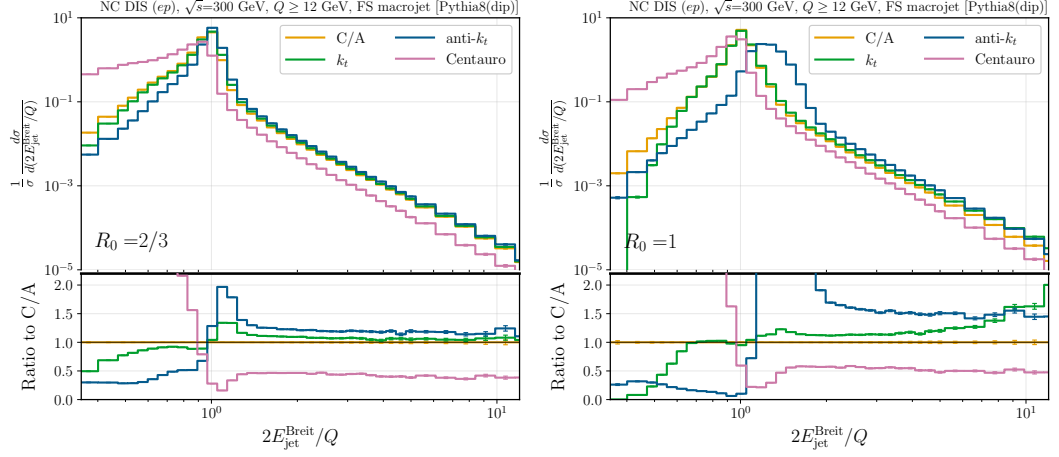


Figure 3.4: Normalised energy distribution of the final-state macrojet in the Breit frame for the Centauro (pink) jet algorithm, as well as the inclusive versions of the C/A ($p = 0$, orange), k_t ($p = 1$, green) and anti- k_t ($p = -1$, blue) algorithms. We show $R_0 = 2/3$ (left) and $R_0 = 1$ (right).

two dense regions: one at small z_{jet} and (large and) positive pseudo-rapidity, and one at large z_{jet} and negative rapidity. The large- z_{jet} region corresponds to the struck quark, whereas the stretched out region in the small- z_{jet} region corresponds to the proton remnant. We do however notice that our algorithms tend to produce harder (larger z_{jet}) jets than Centauro in the current hemisphere. For $R_0 = 1$, there are larger differences between the algorithms. The central region is populated by soft wide-angle radiation that forms stand-alone jets. It is therefore not surprising that the k_t algorithm yields more jets in this region, as it tends to cluster soft particles first. Anti- k_t on the other hand clusters a significant fraction of the partons in the remnant hemisphere with the current hemisphere jets, leading to fewer jets in the remnant hemisphere and harder jets in the current hemisphere. C/A and Centauro seem to be less sensitive to the increased jet radius.

3.5.2 Macrojet observables at hadron level

We now investigate the properties of the final-state macrojet, identified as the jet that maximises the light-cone component of the original struck quark, as explained in Sec. 3.4.2. We begin by examining the Breit-frame energy of the macrojet, E_{jet} , normalised to $Q/2$. At the Born-level (in the absence of beam remnants) this variable is exactly 1. The normalised energy distributions of the macrojet for the Centauro (pink), C/A (orange), k_t (green) and anti- k_t (blue) algorithms, with $R_0 = 2/3$ and $R_0 = 1$, are shown in Fig. 3.4. For both jet radii, the behaviour of the algorithms is very different in the soft region, i.e. for $2E_{\text{jet}}/Q < 1$. Centauro tends to produce a macrojet that is much softer than the other three algorithms. This can be understood by inspecting the distance measure of the Centauro algorithm (cf. Eq. (3.18)), which effectively prevents partons

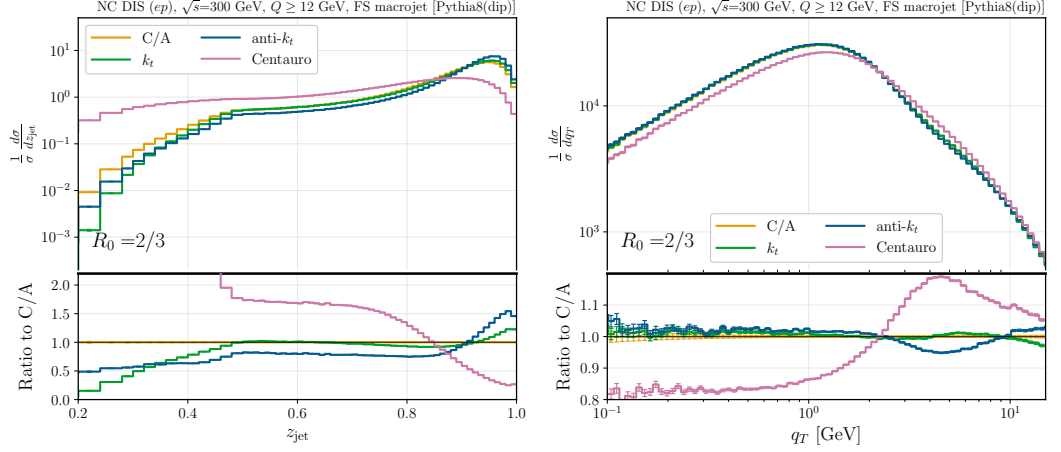


Figure 3.5: Similar to Fig. 3.4, but for z_{jet} (left) and q_T (right, see Eq. (3.26)) with $R_0 = 2/3$.

from clustering in the target hemisphere, and prevents the recoiled struck quark (which may lie at a more central rapidity) from clustering with anything that is not very close in angle. For $R_0 = 2/3$, we observe that the slopes of all four jet algorithms are similar in the hard region, i.e. for $2E_{\text{jet}}/Q > 1$. In the peak region, around $2E_{\text{jet}}/Q \sim 1$, we observe that anti- k_t and k_t shift the peak to slightly larger values compared to C/A, whereas Centauro shifts it to lower values when choosing $R_0 = 2/3$. However, for $R_0 = 1$, we see that the behaviour in the hard region is vastly different for all considered jet algorithms. In particular, all generalised- k_t variants result in a peak that is broader for $R_0 = 1$ with respect to using $R_0 = 2/3$. On the contrary, Centauro shows a peak that is narrower. We stress again that for $R_0 = 1$, the anti- k_t variant will have the tendency to contaminate the final-state macrojet with beam radiation.

To further investigate the behaviour of soft radiation, we examine z_{jet} (cf. Eq. (3.25)). The resulting distributions are shown in the left panel of Fig. 3.5. One can notice the presence of three regions:

- a peak region, around $z_{\text{jet}} \sim 0.95$, characterised by only soft and collinear emissions;
- a “dijet” region, with $0.5 < z_{\text{jet}} \lesssim 0.7$, where the final-state macrojet still carries most of the original momentum, but a non-negligible fraction has been distributed to other jets;
- a “multi-jet” region, with $z_{\text{jet}} < 0.5$, where the struck quark lost the majority of its energy.

We notice that our generalised- k_t algorithms behave quite similarly in the peak and dijet region, which terminates with a shoulder around $z_{\text{jet}} = 0.5$. This similarity is expected, as these regions are dominated by configurations with at most one hard emission, and the two algorithms behave identically at $\mathcal{O}(\alpha_s)$. Below this shoulder the distributions fall off very quickly. The Centauro algorithm produces a quite different distribution, with the peak region being noticeably smeared

towards lower values. The slope of the Centauro distribution agrees with the other three jet algorithms in the dijet region. This is perhaps not surprising, as this is the region that is dominated by two well-separated jets, which means the different treatment of additional soft radiation by the algorithms in their cluster sequences is less impactful. Towards $z_{\text{jet}} \rightarrow 1$, the spectrum differs significantly, which can again be attributed to the behaviour of the $\bar{\eta}$ variable of that algorithm, which prevents some soft radiation carrying away a fraction of the energy from being clustered into the macrojet. Similarly, for $z_{\text{jet}} < 0.5$ the spectrum is dominated by soft jets in the target hemisphere.

The last observable we study is q_T , which is defined as [124]

$$q_T = \frac{|k_T|}{z_{\text{jet}}}, \quad (3.26)$$

where $|k_T|$ is the modulus of the transverse momentum of the jet in the Breit frame (see Eq. (3.22)). This observable is known to have little sensitivity to hadronisation, and can be used to study the transverse momentum distribution of partons inside a proton. We see that all four jet algorithms show a similar slope in the very low- q_T region, but that the three generalised- k_t algorithms result in 20% more events in the region of interest, i.e. $q_T \lesssim Q_{\text{min}}/4 = 3 \text{ GeV}$.

3.5.3 Impact of hadronisation and beam remnants

In this section we study the impact of hadronisation and beam remnants on the final-state macrojet. Given the amount of high quality DIS data that currently exists, with more to come from the EIC, it is worth considering if it is possible to use jets to constrain fundamental QCD parameters, like the value of α_s and parameters of various hadronisation models, as is often done with historical e^+e^- collider data and LHC data. Although it is beyond the scope of this thesis to carry out a fit of such parameters, we here discuss how one might use the jet algorithms to perform them.

In Fig. 3.6 we show the normalised energy distribution and lightcone fraction of the final-state macrojet for the Centauro jet algorithm, as well as the inclusive version of our generalised k_t algorithms, for $R_0 = 2/3$.

For each plot we show results at the parton shower level (PS, dotted), parton shower with beam remnants (PS+remn, dashed), and parton shower with beam remnants and hadronisation (PS+remn+hadr, solid). The latter distributions, which also include unstable hadrons decay, coincide with the ones shown in the previous section. We remind the reader, that unlike in hadron-hadron collisions, there is no underlying event in DIS.

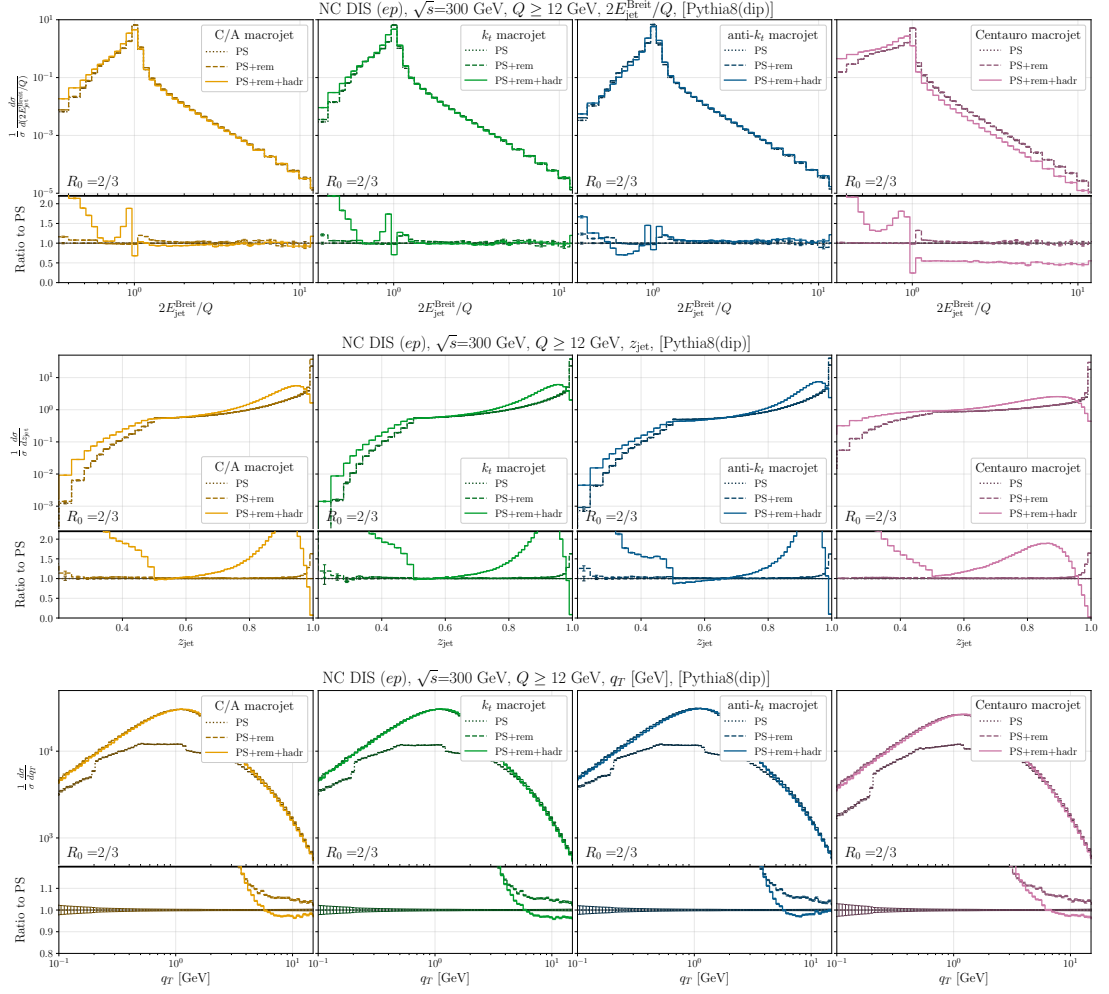


Figure 3.6: Normalised energy (first row), z_{jet} (second row) and q_T (third row) distributions of the final-state macrojet in the Breit frame for the C/A (first column), k_t (second column), anti- k_t (third column), and Centauro (fourth column) jet algorithms, with $R_0 = 2/3$. We show results at the parton shower level (labeled PS, dotted), parton shower with beam remnants (labeled PS+remn, dashed), and parton shower with beam remnants and hadronisation (labeled PS+remn+hadr, solid).

Starting with the normalised energy fraction we see that the beam remnants do not contaminate the jet spectra much when compared to the parton level results, although there is some distortion to the right of the peak. The hadronisation, on the other hand, has a larger effect. To the left of the peak, i.e. the soft region, we observe the expected large distortions for all algorithms. To the right of the peak, the normalisation for the three generalised- k_t algorithms is surprisingly insensitive to hadronisation. Centauro receives very large corrections of $\mathcal{O}(50\%)$ here, but the ratio is rather flat, so we attribute the large shift to the normalisation of the plots.

Turning now to the lightcone component z_{jet} we see again that the beam remnants have very little effect on the plots, except for z_{jet} close to 1. Hadronisation on the other hand has once again a large effect, which induces shape differences to be found across all the spectrum, except for just above $z_{\text{jet}} = 0.5$. We also notice that for all jet algorithms, the peak at $z_{\text{jet}} \approx 1$ present at parton level moves at lower values. This observable can be used to constrain both perturbative and

non-perturbative parameters. The region $0.65 < z_{\text{jet}} < 0.9$ can be used to fit non-perturbative models. On the other hand, the region just above the shoulder, $0.5 < z_{\text{jet}} < 0.65$, is more insensitive to hadronisation effects and could be used to determine α_s by the comparison of data to predictions.

The last observable we show is q_T , cf. Eq. (3.26). Contrary to the previously discussed observables, here we see that the hadronisation has relatively little impact compared to the addition of beam remnants. As said before, the transverse momentum distribution of partons in the proton may be studied using this observable, as can be clearly seen from these results. For the generalised- k_t algorithms we see again that more events enter the region of interest ($q_T \lesssim 3$) compared to the Centauro algorithm. Finally we note that the parton-shower only results are not fully physical, which shows itself through kinks at low q_T , which are associated with various shower cut offs.

3.6 Conclusion

In this chapter we have presented a generalisation of the angular-ordered jet algorithm which was first presented in Ref. [2]. The generalisation follows the idea of introducing an exponent to control how much to weight the energy of each pseudojet in the clustering, providing a tunable p -family familiar from similar lepton- and hadron-collider algorithms. The algorithm is very simple, and gives jets that are naturally separated in the current and target hemispheres. We also compared the angular-ordered, k_t and anti- k_t variants of our new inclusive algorithm to the recently-proposed Centauro jet algorithm. One application that we have in mind is to use the generalised k_t algorithms to re-analyse HERA data, with the aim of providing complementary constraints on α_s from jets, and of constraining hadronisation and intrinsic k_t parameters in event generators. Whereas an α_s extraction is unlikely to be competitive with other extractions on its own, we think there is value in studying non-perturbative physics in DIS data. This is in part due to the fact that DIS allows for studying jets in bins of Q . This could provide a natural handle on the energy dependence of non-perturbative parameters. We leave this to future studies. The algorithm has been implemented as the **DISGenkt** plugin to FASTJET as part of `fjcontrib`.

Chapter 4

Resummation

In this chapter, we discuss *resummation*. As outlined in Sec. 2.7.2, following the initial hard scattering, the coloured partons produced undergo QCD radiation dominated by soft and collinear emissions. For many observables, these emissions give rise to logarithmically enhanced contributions, which become large in the limit where the observable, v , is small. In this regime, fixed-order perturbation theory becomes unreliable and resummation is required to obtain meaningful predictions.

Traditionally, resummation for final-state observables was performed in an observable-dependent manner, requiring separate analytical calculations for each case [125–127]. A first step towards a more general approach was taken in Ref. [128], where a systematic semi-numerical framework was developed to achieve NLL accuracy for a broad class of observables. This framework was subsequently extended and implemented in the computer code CAESAR (*Computer Automated Expert Semi-Analytical Resummer*) [42]. In this approach, the resummation formula is expressed in terms of a set of observable-dependent parameters, which can be determined automatically and numerically for a given observable. The framework has been used to resum a broad class of observables, including event shapes and jet rates. To achieve NNLL resummation, the semi-numerical ARES (*Automated Resummation for Event Shapes*) [43–46] framework was devised for e^+e^- collisions. In this thesis, we work within the ARES framework.

The structure of this chapter is as follows. We begin by introducing event-shape observables and the properties they must satisfy. We then motivate the need for resummation through a fixed-order calculation of a standard event shape, thrust, making explicit the origin and structure of the large logarithms. Finally, we present the NLL resummation formula and the key ingredients required to evaluate it for a given observable. This chapter provides the necessary foundation for the original research presented in Chapter 5.

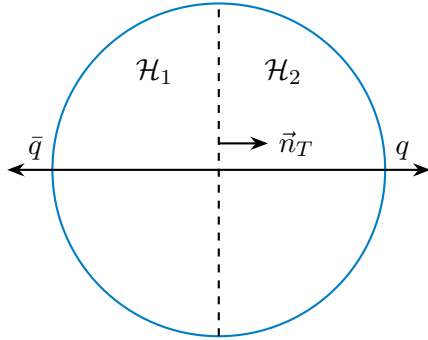
4.1 Event shapes

Event shapes are IRC safe observables constructed from the four-momenta of final-state particles. They characterise the geometry of an event by quantifying the distribution of energy-momentum flow in the hadronic final state. Unlike jet algorithms (see Sec. 2.9), which cluster particles into discrete jets, event shapes provide a continuous description of the event geometry, making them powerful probes of QCD radiation. They have been widely used in precision studies of QCD, most notably in the extraction of the strong coupling constant α_s [51, 52]. Examples include:

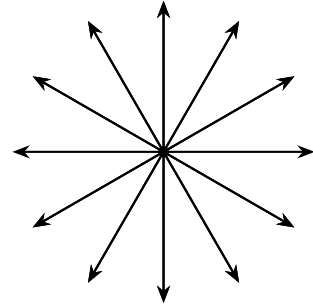
- **Thrust** [129, 130] is one of the most widely studied event-shape observables. It is defined as

$$T \equiv \max_{\vec{n}} \frac{\sum_i |\vec{q}_i \cdot \vec{n}|}{\sum_i |\vec{q}_i|}, \quad \tau \equiv 1 - T, \quad (4.1)$$

where $\vec{q}_i$ denotes the three-momentum of particle i in the final state. The unit vector $\vec{n}$ that maximises this quantity is referred to as the *thrust axis*, $\vec{n}_T$. This axis separates the event into two distinct hemispheres, $\mathcal{H}_1$ and $\mathcal{H}_2$, as shown in Fig. 4.1. In practice, one usually works with the variable τ , as defined in Eq. (4.1). For $e^+e^- \rightarrow$ hadrons, thrust provides a measure of how “pencil-like” an event is: $\tau \rightarrow 0$ corresponds to the Born-level configuration with a perfectly back-to-back $q\bar{q}$ pair, while larger values of τ indicate a more spherically distributed multi-particle final state.



$\tau = 0$: pencil-like configuration



$\tau \rightarrow \frac{1}{2}$: spherically distributed events

Figure 4.1: Diagrammatic representation of thrust. Left: $\tau \rightarrow 0$ corresponds to a pencil-like 2-jet configuration, where the thrust axis $\vec{n}_T$ separates the event into two hemispheres, $\mathcal{H}_1$ and $\mathcal{H}_2$. Right: $\tau \rightarrow 1/2$ corresponds to a spherically distributed, multi-particle final state.

- **Jet broadenings** [131–133] quantify the transverse momentum of final-state hadrons relative to the thrust axis. The event is divided into two hemispheres $\mathcal{H}_1$ and $\mathcal{H}_2$, and for each hemisphere the broadening is defined as

$$B_\ell \equiv \frac{\sum_{i \in \mathcal{H}_\ell} |\vec{q}_i \times \vec{n}_T|}{2 \sum_i |\vec{q}_i|}, \quad \ell = 1, 2. \quad (4.2)$$

The wide-jet broadening, narrow-jet broadening, and total broadening are defined respectively as

$$B_W \equiv \max\{B_1, B_2\}, \quad B_N \equiv \min\{B_1, B_2\}, \quad B_T \equiv B_1 + B_2. \quad (4.3)$$

- **Jet masses** [134, 135] are defined by computing the invariant mass of each hemisphere,

$$\rho_\ell \equiv \frac{\left(\sum_{i \in \mathcal{H}_\ell} q_i\right)^2}{\left(\sum_i E_i\right)^2}, \quad \ell = 1, 2, \quad (4.4)$$

where E_i is the energy of the particle i in the final-state. The heavy-jet mass and light-jet mass are defined respectively as

$$\rho_H \equiv \max\{\rho_1, \rho_2\}, \quad \rho_L \equiv \min\{\rho_1, \rho_2\}. \quad (4.5)$$

While event shapes offer detailed insight into the structure of hadronic final states, their perturbative predictions are complicated by large logarithmic contributions arising from soft and collinear emissions in the limit where the observable approaches zero. To illustrate this explicitly, we now examine how such logarithms emerge in the fixed-order calculation of thrust.

4.2 Motivation: thrust and the need for resummation

To illustrate the need for resummation, we consider a fixed-order calculation of the thrust distribution. At Born level, the process $e^+e^- \rightarrow q\bar{q}$ produces a perfectly back-to-back quark-antiquark pair with $\tau = 0$, giving the distribution

$$\frac{1}{\sigma} \frac{d\sigma}{d\tau} = \delta(\tau), \quad (4.6)$$

where σ is the total hadronic cross section. At the order considered here, we may approximate $\sigma \simeq \sigma_0$, with σ_0 the leading-order cross section defined in Eq. (2.21).

The first non-trivial contribution arises at $\mathcal{O}(\alpha_s)$ from the real-emission process $e^+e^- \rightarrow q\bar{q}g$. For a three-particle final state with energy fractions satisfying $x_q + x_{\bar{q}} + x_g = 2$, the thrust distribution in Eq. (4.1) reduces to

$$\tau = \min[1 - x_q, 1 - x_{\bar{q}}, x_q + x_{\bar{q}} - 1], \quad (4.7)$$

with $\tau_{\max} = 1/3$ for the symmetric configuration $x_q = x_{\bar{q}} = x_g = 2/3$, and $\tau_{\min} = 0$.

Using the real-emission cross section in Eq. (2.23), the differential thrust distribution is

$$\begin{aligned} \frac{1}{\sigma} \frac{d\sigma}{d\tau} &= C_F \frac{\alpha_s}{2\pi} \int_0^1 dx_q \int_0^1 dx_{\bar{q}} \frac{x_q^2 + x_{\bar{q}}^2}{(1-x_q)(1-x_{\bar{q}})} \Theta(x_q + x_{\bar{q}} - 1) \\ &\times \delta(\tau - \min[1 - x_q, 1 - x_{\bar{q}}, x_q + x_{\bar{q}} - 1]) . \end{aligned} \quad (4.8)$$

As discussed in Sec. 2.4.2, this expression contains soft and collinear singularities. Regulating these singularities using dimensional regularisation gives

$$\begin{aligned} \frac{1}{\sigma} \frac{d\sigma}{d\tau} &= C_F \frac{\alpha_s}{2\pi} \left(\frac{4\pi\mu_R^2}{s} \right)^\varepsilon \frac{1}{\Gamma(1-\varepsilon)} \int_0^1 dx_q \int_0^1 dx_{\bar{q}} \frac{(x_q^2 + x_{\bar{q}}^2) - \varepsilon(2 - x_q - x_{\bar{q}})^2}{(1-x_q)^{1+\varepsilon} (1-x_{\bar{q}})^{1+\varepsilon} (x_q + x_{\bar{q}} - 1)^\varepsilon} \\ &\times \Theta(x_q + x_{\bar{q}} - 1) \delta(\tau - \min[1 - x_q, 1 - x_{\bar{q}}, x_q + x_{\bar{q}} - 1]) . \end{aligned} \quad (4.9)$$

The resulting phase-space integrals contain poles in ε , which cancel against the corresponding poles in the virtual contribution. Since the virtual contribution is proportional to $\delta(\tau)$, its finite coefficient can be fixed by requiring that the complete distribution reproduces the total cross section at NLO, given in Eq. (2.29). This gives the complete thrust distribution at $\mathcal{O}(\alpha_s)$,

$$\begin{aligned} \frac{1}{\sigma} \frac{d\sigma}{d\tau} &= \delta(\tau) + C_F \frac{\alpha_s}{2\pi} \left\{ \delta(\tau) \left(\frac{\pi^2}{3} - 1 \right) + \Theta\left(\frac{1}{3} - \tau\right) \right. \\ &\times \left[\left(3(1+\tau)(3\tau-1) + \frac{(4+6\tau(\tau-1)) \ln(1-2\tau)}{1-\tau} \right) \left(\frac{1}{\tau} \right)_+ - \frac{4+6\tau(\tau-1)}{1-\tau} \left(\frac{\ln \tau}{\tau} \right)_+ \right] \Big\} . \end{aligned} \quad (4.10)$$

The plus distributions are defined by

$$\int_0^1 d\tau \left[\frac{\ln^k \tau}{\tau} \right]_+ g(\tau) \equiv \int_0^1 d\tau \frac{\ln^k \tau}{\tau} [g(\tau) - g(0)] , \quad (4.11)$$

so that for $\tau > 0$ they reduce to the ordinary functions $\ln^k \tau / \tau$. In the two-jet limit, $\tau \rightarrow 0$, the fixed-order distribution contains logarithmically enhanced contributions that become increasingly large. Keeping only these terms, Eq. (4.10) gives

$$\left(\frac{1}{\sigma} \frac{d\sigma}{d\tau} \right)_{\log\text{-enh}} = C_F \frac{\alpha_s}{2\pi} \left[-3 \left(\frac{1}{\tau} \right)_+ - 4 \left(\frac{\ln \tau}{\tau} \right)_+ \right] . \quad (4.12)$$

It is useful to consider the cumulative distribution, or cumulant, defined by

$$\Sigma(\tau) = \frac{1}{\sigma} \int_0^\tau d\tau' \frac{d\sigma}{d\tau'} . \quad (4.13)$$

Using

$$\int_0^\tau d\tau' \left(\frac{1}{\tau'} \right)_+ = -L , \quad \int_0^\tau d\tau' \left(\frac{\ln \tau'}{\tau'} \right)_+ = \frac{L^2}{2} , \quad (4.14)$$

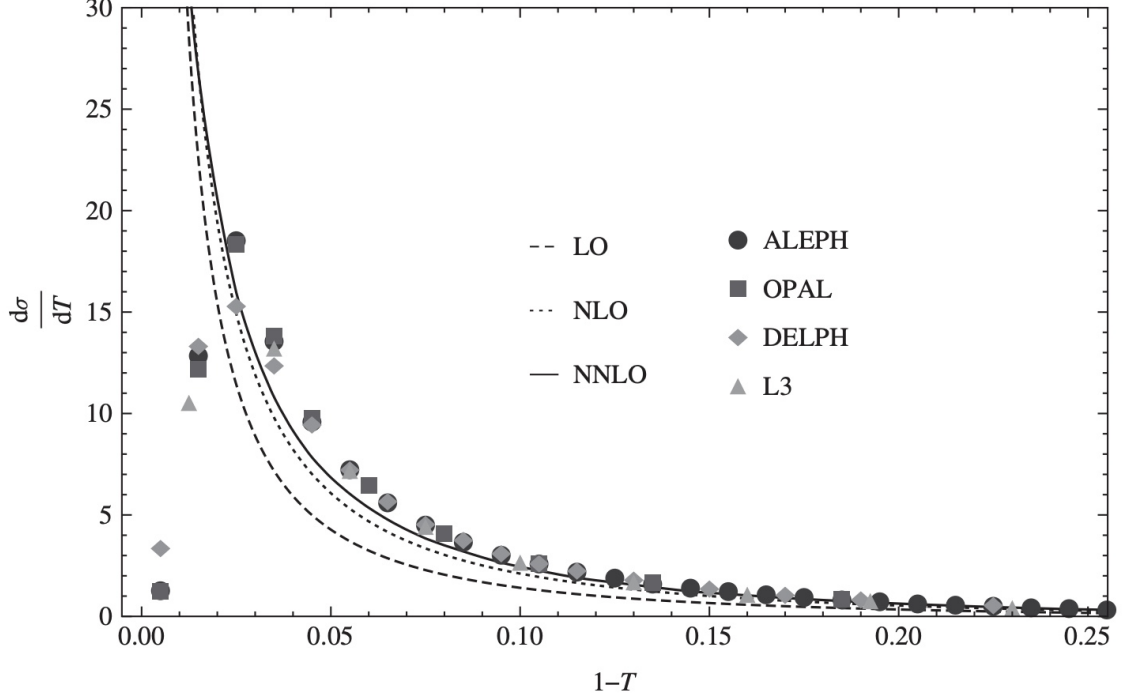


Figure 4.2: Comparison of fixed-order predictions with experimental data from four LEP experiments for the thrust distribution, taken from Ref. [7].

where $L \equiv \ln(1/\tau)$, the logarithmically enhanced contribution to the cumulant is

$$\Sigma_{\log\text{-enh}}(\tau) = C_F \frac{\alpha_s}{2\pi} (-2L^2 + 3L) . \quad (4.15)$$

All remaining terms in Eq. (4.10) remain finite as $\tau \rightarrow 0$ and do not contribute logarithmic enhancements. The logarithmic contributions therefore dominate the behaviour of the fixed-order prediction in the two-jet limit, where $\tau \rightarrow 0$ corresponds to an exactly back-to-back $q\bar{q}$ configuration. At higher orders, the cumulant receives contributions of the form $\alpha_s^n L^m$ with $m \leq 2n$, since each power of α_s can generate up to two logarithms from the soft and collinear divergences. As $\tau \rightarrow 0$, these logarithms become increasingly large and spoil the convergence of the fixed-order expansion.

This behaviour is illustrated in Fig. 4.2, where the fixed-order predictions fail to reproduce the turning point and local maximum seen in the data. Including higher-order corrections, moving from LO to NLO to NNLO, does not resolve this behaviour, since the same logarithmically enhanced contributions appear at every order in perturbation theory. Therefore, fixed-order predictions are not sufficient to describe the two-jet region, and a resummation of these logarithmically enhanced contributions is required.

To restore predictive power in the region $\tau \ll 1$, the perturbative expansion must be reorganised. When $\alpha_s L \sim 1$, successive terms in the perturbative series are no longer suppressed, so an expansion in powers of α_s alone is insufficient. Instead, one must perform a systematic

resummation of the logarithmically enhanced contributions to all orders in α_s . In this region, the cumulant can therefore be reorganised as

$$\Sigma(\tau) \simeq \exp \{ L g_1(\alpha_s L) + g_2(\alpha_s L) + \alpha_s g_3(\alpha_s L) + \cdots \} . \quad (4.16)$$

Each function g_i resums logarithms at a successive level of accuracy, as summarised in Table 4.1.

Function	Logarithms resummed	Accuracy
$g_1(\alpha_s L)$	$\alpha_s^n L^{n+1}$	LL
$g_2(\alpha_s L)$	$\alpha_s^n L^n$	NLL
$g_3(\alpha_s L)$	$\alpha_s^n L^{n-1}$	NNLL

Table 4.1: Logarithms resummed at each order of accuracy in the exponent of Eq. (4.16), where $L = \ln(1/\tau)$. LL, NLL, and NNLL denote leading-logarithmic, next-to-leading-logarithmic, and next-to-next-to-leading-logarithmic accuracy respectively.

It is worth emphasising the distinction between fixed-order and logarithmic accuracy. In a fixed-order calculation, improving the accuracy means computing additional Feynman diagrams at one higher power of α_s . By contrast, each level of logarithmic accuracy already encodes an infinite set of diagrams — those contributing at the corresponding logarithmic order at every power of α_s . Advancing from one logarithmic order to the next therefore requires identifying which physical effects contribute at that order and computing each of their contributions at the necessary accuracy.

Resummation restores predictive power in the two-jet region by systematically accounting for the logarithmically enhanced contributions to all orders in perturbation theory. The resulting Sudakov suppression at small τ , together with the fixed-order behaviour away from the two-jet limit, gives rise to a maximum in the distribution known as the *Sudakov peak*. This suppression is encoded in the Sudakov form factor, which represents the probability of emitting no radiation above the scale τ . Within the ARES formalism, the Sudakov form factor is described by the *radiator*, which will be introduced in Sec. 4.5.

The appearance of large logarithms and the need for resummation are not unique to thrust. The structure of Eq. (4.16) applies broadly to all IRC safe event-shape observables. The semi-numerical framework, implemented in the codes CAESAR and ARES, provides an observable-independent method for performing this resummation systematically. This will be discussed in Sec. 4.5.

4.3 Problem specification

Having motivated the need for resummation using the thrust distribution, we now formulate the general problem. We consider events with n_i incoming partons and an arbitrary number n of final-state particles, whose momenta are denoted $q_1, \dots, q_n$. At Born level the process involves N hard partons in total – including both the n_i incoming legs and the $N - n_i$ outgoing legs – which define the underlying hard configuration. We define an observable $V(q_1, \dots, q_n)$ as a non-negative function of the final-state momenta. We assume the observable to be IRC safe and to vanish smoothly as the event approaches an N -parton configuration. Such observables are commonly referred to as $(N+1)$ -jet observables, where both the outgoing jets and the incoming beam jets are included in the jet counting.

To select events containing N or more well-separated hard jets, we introduce a selection function $\mathcal{H}(q_1, \dots, q_n)$, defined as

$$\mathcal{H}(q_1, \dots, q_n) = \begin{cases} 1, & \text{if the event satisfies the selection criteria,} \\ 0, & \text{otherwise.} \end{cases} \quad (4.17)$$

This can be implemented, for instance, through a jet algorithm that counts the number of jets in the event, or by imposing a cut on a suitable N -jet observable. The total N -jet cross section for selected events is then

$$\sigma_{\mathcal{H}} = \sum_{n=N-n_i}^{\infty} \int d\Phi_n \frac{d\sigma_n}{d\Phi_n} \mathcal{H}(q_1, \dots, q_n), \quad (4.18)$$

where $d\sigma_n/d\Phi_n$ is the differential cross section for producing n final-state particles and $d\Phi_n$ is the corresponding n -particle phase space, defined in Eq. (2.12). The sum begins at $n = N - n_i$, the minimum number of final-state particles required to form the N -parton Born configuration.

We are particularly interested in the cumulative distribution of the observable V for selected events in which V takes a value smaller than v ,

$$\Sigma_{\mathcal{H}}(v) = \frac{1}{\sigma_{\mathcal{H}}} \sum_{n=N-n_i}^{\infty} \int d\Phi_n \frac{d\sigma_n}{d\Phi_n} \Theta(v - V(q_1, \dots, q_n)) \mathcal{H}(q_1, \dots, q_n). \quad (4.19)$$

In the limit $v \ll 1$, where logarithmically enhanced contributions become important, the cumulant $\Sigma_{\mathcal{H}}(v)$ assumes the factorised form [42]

$$\Sigma_{\mathcal{H}}(v) \simeq \frac{1}{\sigma_{\mathcal{H}}} \int d\Phi_N \frac{d\sigma_N}{d\Phi_N} \Sigma(\{p\}, v) \mathcal{H}(p_{n_i+1}, \dots, p_N), \quad (4.20)$$

where $\{p\} \equiv \{p_1, \dots, p_N\}$ denotes the full set of N Born momenta, with $p_1, \dots, p_{n_i}$ the incoming legs and $p_{n_i+1}, \dots, p_N$ the outgoing legs. The term $d\sigma_N/d\Phi_N$ is the leading-order cross

section for the Born configuration, and the integral is performed over the N -particle Born phase space $d\Phi_N$. The factorisation in Eq. (4.20) separates the Born kinematics, encoded in $d\sigma_N/d\Phi_N$ and $\mathcal{H}$, from the effects of soft and collinear radiation, which are captured entirely by $\Sigma(\{p\}, v)$. In the limit $v \ll 1$, the latter contains the logarithmically enhanced contributions that lead to the breakdown of the fixed-order expansion discussed in Sec. 4.2. The remainder of this chapter is devoted to their resummation.

4.4 Observable requirements

Before deriving the resummation formula, it is necessary to specify the class of observables for which the CAESAR and ARES formalisms can be applied. Beyond IRC safety, these observables must satisfy a number of additional requirements. Specifically, for $V(q_1, q_2, \dots)$ these conditions are:

1. **Vanishes smoothly in the soft-collinear limit:** the observable must vanish continuously as a single additional emission of momentum k becomes soft and collinear to a hard leg ℓ . In this limit, the observable can be parametrised as

$$V(\{\tilde{p}\}, k) \simeq V_{\text{sc}}(\{\tilde{p}\}, k) \equiv d_\ell \left(\frac{k_t^{(\ell)}}{Q} \right)^a e^{-b_\ell \eta^{(\ell)}} g_\ell(\phi^{(\ell)}), \quad (4.21)$$

where V_{sc} denotes the observable evaluated in the soft-collinear limit. Here, $\{\tilde{p}\}$ represents the Born momenta after recoil from the emission k , while $k_t^{(\ell)}$, $\eta^{(\ell)}$, and $\phi^{(\ell)}$ denote the transverse momentum, rapidity, and azimuthal angle of k with respect to the emitting leg p_ℓ , respectively. The scale Q is an arbitrary hard scale of the process, while the parameters a , b_ℓ , d_ℓ , and the function $g_\ell(\phi)$ characterise the observable in this limit. These quantities can be determined for each emitting leg and form the basic ingredients of the resummation formula. IRC safety requires $a > 0$ and $b_\ell > -a$.

2. **Recursively IRC (rIRC) safe:** rIRC safety requires that the scaling properties of the observable are insensitive to the addition of an emission that is relatively much softer and/or more collinear than the existing ensemble of soft and collinear emissions. A formal definition is given in Refs. [42, 136]. The key requirements are summarised below.

Consider momenta $\kappa_i(\zeta_i)$ satisfying

$$V(\{\tilde{p}\}, \kappa_i(\zeta_i)) = \zeta_i, \quad (4.22)$$

where in the soft and/or collinear limits $\zeta_i \rightarrow 0$, the azimuthal angle ϕ_i of each emission $\kappa_i(\zeta_i)$ is held fixed. The functions $\kappa_i(\zeta_i)$ may differ for each emission but must satisfy

Eq. (4.22). The collinear splitting of an existing emission is denoted

$$\kappa_i(\zeta) \rightarrow \{\kappa_{i_a}, \kappa_{i_b}\}(\zeta, \mu), \quad (4.23)$$

where μ^2 is the re-scaled invariant mass of the collinear pair,

$$\mu^2 = \frac{(\kappa_{i_a} + \kappa_{i_b})^2}{\kappa_{ti}^2}, \quad \lim_{\mu \rightarrow 0} (\kappa_{i_a} + \kappa_{i_b}) = \kappa_i. \quad (4.24)$$

The conditions for rIRC safety are the following:

a) The limit

$$\lim_{\bar{v} \rightarrow 0} \frac{1}{\bar{v}} V(\{\tilde{p}\}, \kappa_1(\bar{v}\zeta_1), \dots, \kappa_m(\bar{v}\zeta_m)), \quad (4.25)$$

must be finite and non-zero for any choice of non-zero ζ_i . The parameter $\bar{v}$ is introduced to distinguish it from the observable value v we wish to resum. This requirement ensures that the soft-collinear scaling of the observable is independent of the number of emissions.

b) The following two conditions must hold:

i. The addition of a much softer and/or more collinear emission must leave the observable unchanged:

$$\begin{aligned} \lim_{\zeta_{m+1} \rightarrow 0} \lim_{\bar{v} \rightarrow 0} \frac{1}{\bar{v}} V(\{\tilde{p}\}, \kappa_1(\bar{v}\zeta_1), \dots, \kappa_m(\bar{v}\zeta_m), \kappa_{m+1}(\bar{v}\zeta_{m+1})) \\ = \lim_{\bar{v} \rightarrow 0} \frac{1}{\bar{v}} V(\{\tilde{p}\}, \kappa_1(\bar{v}\zeta_1), \dots, \kappa_m(\bar{v}\zeta_m)). \end{aligned} \quad (4.26)$$

ii. Analogously, a collinear splitting of an existing emission must also leave the observable unchanged:

$$\begin{aligned} \lim_{\mu \rightarrow 0} \lim_{\bar{v} \rightarrow 0} \frac{1}{\bar{v}} V(\{\tilde{p}\}, \kappa_1(\bar{v}\zeta_1), \dots, \{\kappa_{i_a}, \kappa_{i_b}\}(\bar{v}\zeta_i, \mu), \dots, \kappa_m(\bar{v}\zeta_m)) \\ = \lim_{\bar{v} \rightarrow 0} \frac{1}{\bar{v}} V(\{\tilde{p}\}, \kappa_1(\bar{v}\zeta_1), \dots, \kappa_m(\bar{v}\zeta_m)). \end{aligned} \quad (4.27)$$

The rIRC safety condition is precisely what allows the leading-logarithmic contributions to exponentiate. Observables that violate rIRC safety do not exponentiate (cf. Eq. (4.16)), a well-known example is the JADE 3-jet resolution parameter in e^+e^- collisions [137].

3. **Continuously global:** an observable is said to be *global* if it is sensitive to emissions in all regions of phase space, i.e. it is non-zero for any additional emission that is not infinitely soft or collinear. For an observable to be *continuously global*, its scaling with respect to the transverse momentum of a soft emission must be uniform across phase space. In terms of

the parametrisation in Eq. (4.21), this requires the parameter a to be the same for all legs. This condition ensures that non-global logarithms [138, 139], which arise when different regions of phase space exhibit different transverse-momentum dependences, do not appear. Non-global logarithms lie beyond the scope of this thesis.

These properties define the class of observables for which the CAESAR and ARES resummation formalisms can be applied. The event-shape observables discussed in Sec. 4.1 satisfy these requirements. With these properties established, we now turn to the derivation of the NLL resummation formula.

4.5 NLL resummation

The cumulant for an observable v in a process with hard legs $\{p\}$ and n real emissions $k_1, \dots, k_n$ is given by

$$\Sigma(\{p\}, v) = \mathcal{V}(\{p\}) \sum_{n=0}^{\infty} S(n) \int \left(\prod_{i=1}^n [dk_i] \right) \mathcal{M}^2(\{\tilde{p}\}, k_1, \dots, k_n) \Theta(v - V(\{\tilde{p}\}, k_1, \dots, k_n)) , \quad (4.28)$$

where:

- $\mathcal{V}(\{p\})$ denotes the virtual corrections to the Born process, normalised to the Born matrix element squared,
- $S(n)$ is the symmetry factor associated with the n -emission configuration,
- $[dk]$ denotes the Lorentz-invariant phase space in $d = 4 - 2\varepsilon$ dimensions,

$$[dk] \equiv \frac{d^d k}{(2\pi)^{d-1}} \delta(k^2) \Theta(k^0) , \quad (4.29)$$

- $\mathcal{M}^2(\{\tilde{p}\}, k_1, \dots, k_n)$ is the squared matrix element for n real emissions off the Born configuration $\{\tilde{p}\}$.

We now introduce the approximations that allow Eq. (4.28) to be evaluated at NLL accuracy.

4.5.1 Approximations at NLL

Colour coherence and the physical coupling. To simplify the matrix element, we use colour coherence: a soft gluon emitted at large angles from a set of collinear partons resolves only

the total colour of the collinear ensemble, not the individual colour charges. This guarantees that emissions widely separated in angle are effectively independent, so that the squared matrix element for n such emissions approximately factorises into a product of independent single-emission contributions.

However, this factorisation is not exact. Correlated gluon clusters, in which a primary emission subsequently branches into further gluons, give rise to corrections to the single-gluon emission probability. At NLL, these corrections can be absorbed into a redefinition of the coupling constant, giving rise to the *physical coupling* α_s^{phys} , defined as the effective strength of soft radiation when all subsequent branchings from an initial emission are included [45]. This is related to the $\overline{\text{MS}}$ coupling α_s by¹

$$\alpha_s^{\text{phys}} = \alpha_s \left(1 + \sum_{n=1}^{\infty} \left(\frac{\alpha_s}{2\pi} \right)^n K^{(n)} \right), \quad (4.30)$$

where the first two coefficients are

$$K^{(1)} = C_A \left(\frac{67}{18} - \frac{\pi^2}{6} \right) - \frac{5}{9} n_f, \quad (4.31a)$$

$$\begin{aligned} K^{(2)} = & C_A^2 \left(\frac{245}{24} - \frac{67}{9} \zeta_2 + \frac{11}{6} \zeta_3 + \frac{11}{5} \zeta_2^2 \right) + C_F n_f \left(-\frac{55}{24} + 2\zeta_3 \right) \\ & + C_A n_f \left(-\frac{209}{108} + \frac{10}{9} \zeta_2 - \frac{7}{3} \zeta_3 \right) - \frac{n_f^2}{27} + \frac{\pi\beta_0}{2} \left(C_A \left(\frac{808}{27} - 28\zeta_3 \right) - \frac{224}{54} n_f \right). \end{aligned} \quad (4.31b)$$

Only the first coefficient, $K^{(1)}$, is required at NLL accuracy.

Factorisation of the matrix element. At NLL accuracy, correlated clusters contribute only through the coupling redefinition above. The squared matrix element for n widely separated soft and collinear emissions factorises into a product of n independent single-emission matrix elements in the soft-collinear limit, $M_{\text{sc}}^2(k_i)$, as defined in Ref. [43]. The observable constraint may likewise be taken in the soft-collinear limit,

$$\frac{V_{\text{sc}}(\{\tilde{p}\}, k_1, \dots, k_n)}{v} \equiv \lim_{v \rightarrow 0} \frac{V(\{\tilde{p}\}, k_1, \dots, k_n)}{v}, \quad (4.32)$$

which is guaranteed to exist for rIRC-safe observables by condition 2a.

At NLL, the virtual corrections exponentiate, $\mathcal{V}(\{p\}) = e^{-\int [dk] M^2(k)}$, ensuring that the inclusive sum over all real and virtual contributions equals unity. Additionally, the n emissions are treated as identical gluons, such that $S(n) = 1/n!$, and:

$$\Sigma(\{p\}, v) = e^{-\int [dk] M^2(k)} \sum_{n=0}^{\infty} \frac{1}{n!} \int \prod_{i=1}^n [dk_i] M_{\text{sc}}^2(k_i) \Theta(v - V_{\text{sc}}(\{\tilde{p}\}, k_1, \dots, k_n)). \quad (4.33)$$

¹ This scheme is an extension of the CMW scheme of Ref. [140].

To evaluate $\Sigma(\{p\}, v)$, we split the exponent in Eq. (4.33) according to whether the observable value for a single emission k exceeds v :

$$\int [dk] M^2(k) = \underbrace{\int [dk] M^2(k) \Theta(V(\{\tilde{p}\}, k) - v)}_{R(v)} + \int [dk] M_{\text{sc}}^2(k) \Theta(v - V_{\text{sc}}(\{\tilde{p}\}, k)) . \quad (4.34)$$

In the second term, the matrix element and observable are replaced by their soft-collinear approximations, M_{sc}^2 and V_{sc} , respectively. At NLL accuracy, only these approximations are required, while the difference between the full and soft-collinear descriptions gives subleading corrections that enter at NNLL accuracy.

The first term defines the *radiator* $R(v)$, and the cumulant factorises as

$$\Sigma(\{p\}, v) = e^{-R(v)} \mathcal{F}(v) , \quad (4.35)$$

where $\mathcal{F}(v)$ collects the real emissions satisfying $V(\{\tilde{p}\}, k) \leq v$, together with the corresponding virtual corrections. We discuss the radiator $R(v)$ and the multiple-emission function $\mathcal{F}(v)$ in turn.

4.5.2 The NLL radiator, R_{NLL}

The radiator is defined as

$$R(v) \equiv \int [dk] M^2(k) \Theta(V(\{\tilde{p}\}, k) - v) , \quad (4.36)$$

which integrates the emission probability over the region of phase space in which the observable exceeds v . Its exponential $e^{-R(v)}$ is the *Sudakov form factor*, giving the probability that no emission with $V(\{\tilde{p}\}, k) > v$ occurs. This factor resums all LL contributions for rIRC safe observables.

At NLL accuracy, the radiator receives contributions from all regions. Accordingly, it can be split into two pieces,

$$R(v) = R_{\text{s}}(v) + R_{\text{hc}}(v) , \quad (4.37)$$

corresponding to the soft and hard-collinear regions, respectively. We treat each piece in turn.

Soft radiator

The soft radiator can be expressed as a sum over colour dipoles $(\ell\bar{\ell})$ formed by pairs of hard legs,

$$R_{\text{s}} = \sum_{\ell} \sum_{\bar{\ell} \neq \ell} C_{\ell\bar{\ell}} \mathcal{R}_{\ell\bar{\ell}}(v) , \quad (4.38)$$

where the dipole colour factor is defined as

$$C_{\ell\bar{\ell}} = -\mathbf{T}_\ell \cdot \mathbf{T}_{\bar{\ell}}. \quad (4.39)$$

For the 2- and 3-jet cases (the latter considered in Ch. 5), the colour structure is sufficiently simple such that the colour factors $C_{\ell\bar{\ell}}$ reduce to scalar quantities rather than matrices. These colour factors satisfy, for any number of hard legs,

$$\sum_{\bar{\ell} \neq \ell} C_{\ell\bar{\ell}} = C_\ell, \quad (4.40)$$

where C_ℓ is the colour factor of leg ℓ (C_F for a quark and C_A for a gluon).

We parametrise the emission kinematics with respect to the $(\ell\bar{\ell})$ dipole in terms of the rapidity

$$\eta^{(\ell\bar{\ell})} \equiv \frac{1}{2} \ln \frac{z^{(\ell)}}{z^{(\bar{\ell})}}, \quad (4.41)$$

where $z^{(\ell)}$ and $z^{(\bar{\ell})}$ are the longitudinal momentum fractions of the emission along legs ℓ and $\bar{\ell}$ respectively, together with the azimuthal angle $\phi^{(\ell\bar{\ell})}$ and the invariant transverse momentum with respect to the dipole,

$$\kappa_{\ell\bar{\ell}}^2 = \frac{(2p_\ell \cdot k)(2k \cdot p_{\bar{\ell}})}{(2p_\ell \cdot p_{\bar{\ell}})}. \quad (4.42)$$

In terms of these variables, the contribution $\mathcal{R}_{\ell\bar{\ell}}(v)$ is obtained by restricting the general radiator to the soft matrix element associated with the $(\ell\bar{\ell})$ dipole,

$$\begin{aligned} \mathcal{R}_{\ell\bar{\ell}}(v) = \int \frac{d\kappa_{\ell\bar{\ell}}^2}{\kappa_{\ell\bar{\ell}}^2} \frac{\alpha_s^{\text{phys}}(\kappa_{\ell\bar{\ell}})}{\pi} \int d\eta^{(\ell\bar{\ell})} \int_0^{2\pi} \frac{d\phi^{(\ell\bar{\ell})}}{2\pi} \Theta \left(\frac{1}{2} \ln \frac{Q_{\ell\bar{\ell}}^2}{\kappa_{\ell\bar{\ell}}^2} - \eta^{(\ell\bar{\ell})} \right) \Theta(Q_{\ell\bar{\ell}} - \kappa_{\ell\bar{\ell}}) \times \\ \times \Theta(\eta^{(\ell\bar{\ell})}) \Theta \left(d_\ell^{(\ell\bar{\ell})} \left(\frac{\kappa_{\ell\bar{\ell}}}{Q_{\ell\bar{\ell}}} \right)^a e^{-b_\ell \eta^{(\ell\bar{\ell})}} g_\ell(\phi^{(\ell\bar{\ell})}) - v \right), \end{aligned} \quad (4.43)$$

where $Q_{\ell\bar{\ell}}^2 = 2(p_\ell \cdot p_{\bar{\ell}})$. The final Heaviside function imposes the observable constraint, using the soft-collinear limit $V_{\text{sc}}^{(\ell\bar{\ell})}$ expressed in terms of the dipole variables. The corresponding dipole coefficient is

$$d_\ell^{(\ell\bar{\ell})} = d_\ell \left(\frac{Q_{\ell\bar{\ell}}}{Q} \right)^a \left(\frac{Q_{\ell\bar{\ell}}}{2E_\ell} \right)^{b_\ell}. \quad (4.44)$$

The function $\Theta(\eta^{(\ell\bar{\ell})})$ is a convention used to split the dipole phase space between its two legs, while $\Theta \left(\frac{1}{2} \ln \frac{Q_{\ell\bar{\ell}}^2}{\kappa_{\ell\bar{\ell}}^2} - \eta^{(\ell\bar{\ell})} \right)$ restricts the rapidity to its kinematically allowed range. The condition $\Theta(Q_{\ell\bar{\ell}} - \kappa_{\ell\bar{\ell}})$ restricts the integration to the soft region by imposing an upper cutoff on $\kappa_{\ell\bar{\ell}}$ at a scale of the same order as the hard process, taken here to be $Q_{\ell\bar{\ell}}$. The $1/\kappa_{\ell\bar{\ell}}^2$ term is the same as the eikonal factor for soft-gluon emission from the dipole, with colour factor $C_{\ell\bar{\ell}}$, as discussed in Sec. 2.5.1.

To simplify the expression for $\mathcal{R}_{\ell\bar{\ell}}(v)$, we separate out the dependence on the Born momenta and on the azimuthal angle by expanding the final Heaviside function in Eq. (4.43) as

$$\Theta\left(d_{\ell}^{(\ell\bar{\ell})}\left(\frac{\kappa_{\ell\bar{\ell}}}{Q_{\ell\bar{\ell}}}\right)^a e^{-b_{\ell}\eta^{(\ell\bar{\ell})}} g_{\ell}(\phi^{(\ell\bar{\ell})}) - v\right) \simeq \Theta\left(\left(\frac{\kappa_{\ell\bar{\ell}}}{Q_{\ell\bar{\ell}}}\right)^a e^{-b_{\ell}\eta^{(\ell\bar{\ell})}} - v\right) + \delta\left(\ln\left[\left(\frac{\kappa_{\ell\bar{\ell}}}{Q_{\ell\bar{\ell}}}\right)^a e^{-b_{\ell}\eta^{(\ell\bar{\ell})}}\right] - \ln v\right) \ln\left(d_{\ell}^{(\ell\bar{\ell})} g_{\ell}(\phi^{(\ell\bar{\ell})})\right) + \dots \quad (4.45)$$

For NLL accuracy, this expansion is truncated at this order, and the omitted terms are of NNLL accuracy and beyond.

Substituting this expansion into Eq. (4.43) gives

$$\mathcal{R}_{\ell\bar{\ell}}(v) \simeq R_{\ell\bar{\ell}}(v, Q_{\ell\bar{\ell}}) + \langle \ln(d_{\ell}^{(\ell\bar{\ell})} g_{\ell}(\phi^{(\ell\bar{\ell})})) \rangle R'_{\ell\bar{\ell}}(v, Q_{\ell\bar{\ell}}) + \dots, \quad (4.46)$$

where the azimuthal average is defined as

$$\langle f \rangle \equiv \int \frac{d\phi}{2\pi} f(\phi), \quad (4.47)$$

and the leading contribution is

$$R_{\ell\bar{\ell}}(v, Q_{\ell\bar{\ell}}) \equiv \int_0^{Q_{\ell\bar{\ell}}^2} \frac{d\kappa_{\ell\bar{\ell}}^2}{\kappa_{\ell\bar{\ell}}^2} \frac{\alpha_s^{\text{phys}}(\kappa_{\ell\bar{\ell}})}{\pi} \int_0^{\infty} d\eta^{(\ell\bar{\ell})} \Theta\left(\ln \frac{Q_{\ell\bar{\ell}}}{\kappa_{\ell\bar{\ell}}} - \eta^{(\ell\bar{\ell})}\right) \Theta\left(\left(\frac{\kappa_{\ell\bar{\ell}}}{Q_{\ell\bar{\ell}}}\right)^a e^{-b_{\ell}\eta^{(\ell\bar{\ell})}} - v\right), \quad (4.48)$$

where $R'_{\ell\bar{\ell}} \equiv -v dR_{\ell\bar{\ell}}/dv$.

Hard-collinear radiator

The hard-collinear radiator arises from virtual corrections collinear to each hard leg, integrated from the characteristic hard-collinear scale $Qv^{1/(a+b_{\ell})}$ up to the hard scale Q . It can be written as a sum over hard legs,

$$R_{\text{hc}}(v) = \sum_{\ell} R_{\text{hc},\ell}(v), \quad (4.49)$$

where

$$R_{\text{hc},\ell}(v) = - \int_{Q^2 v^{2/(a+b_{\ell})}}^{Q^2} \frac{dk^2}{k^2} \gamma_{\ell}^{(0)} \frac{\alpha_s^{\text{phys}}(k)}{2\pi}, \quad (4.50)$$

and $\gamma_{\ell}^{(0)}$ is the coefficient of the $\delta(1-z)$ term in the relevant splitting function, as defined in Eq. (2.39):

$$\gamma_{\ell}^{(0)} = \begin{cases} \frac{3}{2} C_F, & \text{if } \ell \text{ is a quark,} \\ 2\pi\beta_0, & \text{if } \ell \text{ is a gluon.} \end{cases} \quad (4.51)$$

To evaluate Eqs. (4.48) and (4.50), we solve the renormalisation group equation in Eq. (2.14) using the ansatz

$$\alpha_s(k) = \sum_{n=1}^{\infty} \left(\frac{\alpha_s(Q)}{1+t} \right)^n f_n(t), \quad t \equiv \alpha_s(Q) \beta_0 \ln \frac{k^2}{Q^2}, \quad (4.52)$$

with

$$f_1(t) = 1, \quad (4.53a)$$

$$f_2(t) = -\frac{\beta_1}{\beta_0} \ln(1+t), \quad (4.53b)$$

$$f_3(t) = -\frac{\beta_2}{\beta_0} t + \left(\frac{\beta_1}{\beta_0} \right)^2 [t + (\ln(1+t) - 1)] \ln(1+t). \quad (4.53c)$$

At NLL accuracy only f_1 and f_2 are required, while f_3 enters at NNLL. The coefficients β_0 and β_1 are given in Eq. (2.16) and the parameter λ is defined as

$$\lambda \equiv \alpha_s(Q) \beta_0 L, \quad (4.54)$$

where $L = \ln(1/v)$. At NLL, one finds

$$R_{\ell\bar{\ell}}(v, Q_{\ell\bar{\ell}}) = -\frac{\lambda}{\alpha_s(Q) \beta_0} g_1^{(\ell)}(\lambda) - g_2^{(\ell)}(\lambda) - \mathfrak{g}_2^{(\ell)}(\lambda, Q_{\ell\bar{\ell}}), \quad (4.55a)$$

$$R'_{\ell\bar{\ell}}(v, Q_{\ell\bar{\ell}}) = \alpha_s(Q) \beta_0 \frac{d}{d\lambda} \left(-\frac{\lambda}{\alpha_s(Q) \beta_0} g_1^{(\ell)}(\lambda) \right), \quad (4.55b)$$

$$R_{\text{hc},\ell}(v) = -h_2^{(\ell)}(\lambda). \quad (4.55c)$$

Assembling the contributions in Eq. (4.37), the full NLL radiator is

$$\begin{aligned} R_{\text{NLL}}(v) = & -\sum_{\ell} h_2^{(\ell)}(\lambda) + \sum_{\ell} C_{\ell} \left(-\frac{\lambda}{\alpha_s(Q) \beta_0} g_1^{(\ell)}(\lambda) - g_2^{(\ell)}(\lambda) \right) \\ & - \sum_{\ell} \sum_{\bar{\ell} \neq \ell} C_{\ell\bar{\ell}} \left(\mathfrak{g}_2^{(\ell)}(\lambda, Q_{\ell\bar{\ell}}) - \alpha_s(Q) \beta_0 \frac{d}{d\lambda} \left(-\frac{\lambda}{\alpha_s(Q) \beta_0} g_1^{(\ell)}(\lambda) \right) \langle \ln(d_{\ell}^{(\ell\bar{\ell})} g_{\ell}) \rangle \right), \end{aligned} \quad (4.56)$$

where the functions $g_1^{(\ell)}$, $g_2^{(\ell)}$, $\mathfrak{g}_2^{(\ell)}$, and $h_2^{(\ell)}$ are given by

$$g_1^{(\ell)}(\lambda) = \frac{(a + b_{\ell} - 2\lambda) \ln\left(1 - \frac{2\lambda}{a+b_{\ell}}\right) - (a - 2\lambda) \ln\left(1 - \frac{2\lambda}{a}\right)}{2\pi b_{\ell} \beta_0 \lambda}, \quad (4.57a)$$

$$\begin{aligned} g_2^{(\ell)}(\lambda) = & \frac{K^{(1)}}{4\pi^2 b_{\ell} \beta_0^2} \left[a \ln\left(1 - \frac{2\lambda}{a}\right) - (a + b_{\ell}) \ln\left(1 - \frac{2\lambda}{a+b_{\ell}}\right) \right] \\ & + \frac{\beta_1(a + b_{\ell})}{4\pi b_{\ell} \beta_0^3} \ln^2\left(1 - \frac{2\lambda}{a+b_{\ell}}\right) + \frac{\beta_1(a + b_{\ell})}{2\pi b_{\ell} \beta_0^3} \ln\left(1 - \frac{2\lambda}{a+b_{\ell}}\right) \\ & - \frac{\beta_1 a}{4\pi b_{\ell} \beta_0^3} \ln\left(1 - \frac{2\lambda}{a}\right) \left[\ln\left(1 - \frac{2\lambda}{a}\right) + 2 \right], \end{aligned} \quad (4.57b)$$

$$\mathfrak{g}_2^{(\ell)}(\lambda, Q_{\ell\bar{\ell}}) = -\lambda^2 \frac{dg_1^{(\ell)}(\lambda)}{d\lambda} \ln\left(\frac{Q_{\ell\bar{\ell}}^2}{Q^2}\right), \quad (4.57c)$$

$$h_2^{(\ell)}(\lambda) = -\frac{\gamma_{\ell}^{(0)}}{2\pi \beta_0} \ln\left(1 - \frac{2\lambda}{a+b_{\ell}}\right). \quad (4.57d)$$

The derivative of the NLL radiator decomposes as a sum over legs, $R'_{\text{NLL}}(v) = \sum_{\ell} R'_{\text{NLL},\ell}(v)$, where

$$R'_{\text{NLL},\ell}(v) = C_{\ell} \alpha_s(Q) \beta_0 \frac{d}{d\lambda} \left(-\frac{\lambda}{\alpha_s(Q) \beta_0} g_1^{(\ell)}(\lambda) \right) = \frac{C_{\ell}}{b_{\ell} \pi \beta_0} \left[\ln \left(1 - \frac{2\lambda}{a + b_{\ell}} \right) - \ln \left(1 - \frac{2\lambda}{a} \right) \right]. \quad (4.58)$$

4.5.3 The NLL $\mathcal{F}$ -function, $\mathcal{F}_{\text{NLL}}$

Having established the radiator, we now turn to the multiple-emission function $\mathcal{F}(v)$. Combining Eqs. (4.34) and (4.35), it is given by

$$\mathcal{F}(v) = e^{-\int [dk] M_{\text{sc}}^2(k) \Theta(v - V_{\text{sc}}(\{\tilde{p}\}, k))} \sum_{n=0}^{\infty} \frac{1}{n!} \int \prod_{i=1}^n [dk_i] M_{\text{sc}}^2(k_i) \Theta(v - V_{\text{sc}}(\{\tilde{p}\}, k_1, \dots, k_n)). \quad (4.59)$$

To evaluate this expression, it is convenient to express the soft-collinear matrix element in terms of the radiator derivative. Defining

$$R'(v) \equiv -v \frac{d}{dv} R(v), \quad (4.60)$$

and introducing, for each emission, the rescaled variable

$$\zeta = \frac{V_{\text{sc}}(\{\tilde{p}\}, k)}{v}, \quad (4.61)$$

we have

$$\int [dk] M^2(k) = \int_0^{\infty} \frac{d\zeta}{\zeta} R'(\zeta v), \quad (4.62)$$

and, in particular,

$$\int [dk] M^2(k) \Theta(v - V(\{\tilde{p}\}, k)) = \int_0^1 \frac{d\zeta}{\zeta} R'(\zeta v). \quad (4.63)$$

Using the fact that $R'(v)$ decomposes as a sum over hard legs, $R'(v) \equiv \sum_{\ell} R'_{\ell}(v)$, and changing variables, we obtain

$$[dk] M_{\text{sc}}^2(k) = \sum_{\ell} \frac{d\zeta}{\zeta} \frac{d\phi^{(\ell)}}{2\pi} R'_{\ell}(\zeta v). \quad (4.64)$$

The terms neglected in taking this limit contribute only at NNLL and beyond. At NLL, each emission k_i is therefore parametrised by $\{\ell_i, \phi_i^{(\ell_i)}, \zeta_i\}$, with ζ_i defined as in Eq. (4.61).

Substituting Eq. (4.64) into Eq. (4.59) gives

$$\mathcal{F}(v) = e^{-\int_{\epsilon}^1 \frac{d\zeta}{\zeta} R'(\zeta v)} \sum_{n=0}^{\infty} \frac{1}{n!} \int_{\epsilon}^{\infty} \prod_{i=1}^n \left(\sum_{\ell_i} R'_{\ell_i}(\zeta_i v) \frac{d\zeta_i}{\zeta_i} \frac{d\phi_i^{(\ell_i)}}{2\pi} \right) \Theta(v - V_{\text{sc}}(\{\tilde{p}\}, k_1, \dots, k_n)),$$

(4.65)

where the cutoff $\epsilon \ll 1$ ensures that emissions with $\zeta_i < \epsilon$ cancel between the real sum and the virtual exponent, up to power corrections in ϵ that vanish because of rIRC safety. Such emissions contribute negligibly to the observable on their own, and are referred to as *unresolved*. Emissions with $\zeta_i > \epsilon$ are referred to as *resolved*, and their contribution to the observable must be accounted for.

Approximating $R'(\zeta v) \simeq R'(v)$, thereby neglecting subleading effects due to corrections of relative order α_s , the exponent evaluates to $\epsilon^{R'}$, giving

$$\mathcal{F}(v) \simeq \epsilon^{R'} \sum_{n=0}^{\infty} \frac{1}{n!} \int_{\epsilon}^{\infty} \prod_{i=1}^n \left(\frac{d\zeta_i}{\zeta_i} \sum_{\ell_i} R'_{\ell_i}(v) \frac{d\phi_i^{(\ell_i)}}{2\pi} \right) \Theta(v - V_{\text{sc}}(\{\tilde{p}\}, k_1, \dots, k_n)) . \quad (4.66)$$

This can be written compactly as

$$\mathcal{F}_{\text{NLL}}(R'_{\text{NLL}}(v)) = \int d\mathcal{Z}[\{R'_{\text{NLL},\ell_i}, \{k_i\}\}] \Theta\left(1 - \frac{V_{\text{sc}}(\{\tilde{p}\}, \{k_i\})}{v}\right) , \quad (4.67)$$

where the soft-collinear measure $d\mathcal{Z}[\{R'_{\text{NLL},\ell_i}, \{k_i\}\}]$ is defined by

$$\begin{aligned} & \int d\mathcal{Z}[\{R'_{\text{NLL},\ell_i}, \{k_i\}\}] G(\{k_i\}) \\ & \equiv \epsilon^{R'_{\text{NLL}}} \sum_{n=0}^{\infty} \frac{1}{n!} \int \prod_{i=1}^n \left(\sum_{\ell_i} R'_{\text{NLL},\ell_i} \int_{\epsilon}^{\infty} \frac{d\zeta_i}{\zeta_i} \int_0^{2\pi} \frac{d\phi_i^{(\ell_i)}}{2\pi} \right) G(k_1, \dots, k_n) , \end{aligned} \quad (4.68)$$

for any function G of the emissions.

4.5.4 The NLL cumulant

Combining the radiator and multiple-emission function derived above, the NLL cumulant is

$$\Sigma(\{p\}, v) = e^{-R_{\text{NLL}}(v)} \mathcal{F}_{\text{NLL}}(R'_{\text{NLL}}(v)) , \quad R'_{\text{NLL}} \simeq -v \frac{dR_{\text{NLL}}}{dv} , \quad (4.69)$$

where only the terms required for NLL accuracy are retained in the derivative.

This result applies to any continuously global, rIRC-safe observable. Its NLL resummation requires only the parameters a , b_{ℓ} , d_{ℓ} and $g_{\ell}(\phi)$, together with the soft-collinear limit V_{sc} , all of which are determined by the definition of the observable. The existence of this limit follows from rIRC safety and it can be evaluated using suitable ensembles of soft and collinear emissions.

This procedure is implemented in the CAESAR computer code [42]. The code generates emissions using a simplified parton-shower event generator and passes them to a subroutine that evaluates V_{sc} . It can also be used to test the rIRC safety of an observable directly from its

definition in terms of final-state momenta. Its successor, ARES [43–46], is a first step towards extending the framework to NNLL accuracy (cf. Ch. 5). The two approaches also differ in how the multiple-emission function is evaluated. CAESAR determines $\mathcal{F}_{\text{NLL}}$ entirely numerically, whereas ARES determines V_{sc} analytically and evaluates the remaining integrations using Monte Carlo methods. Figure 4.3 compares the results from both codes with the analytical result for thrust derived below, showing excellent agreement.

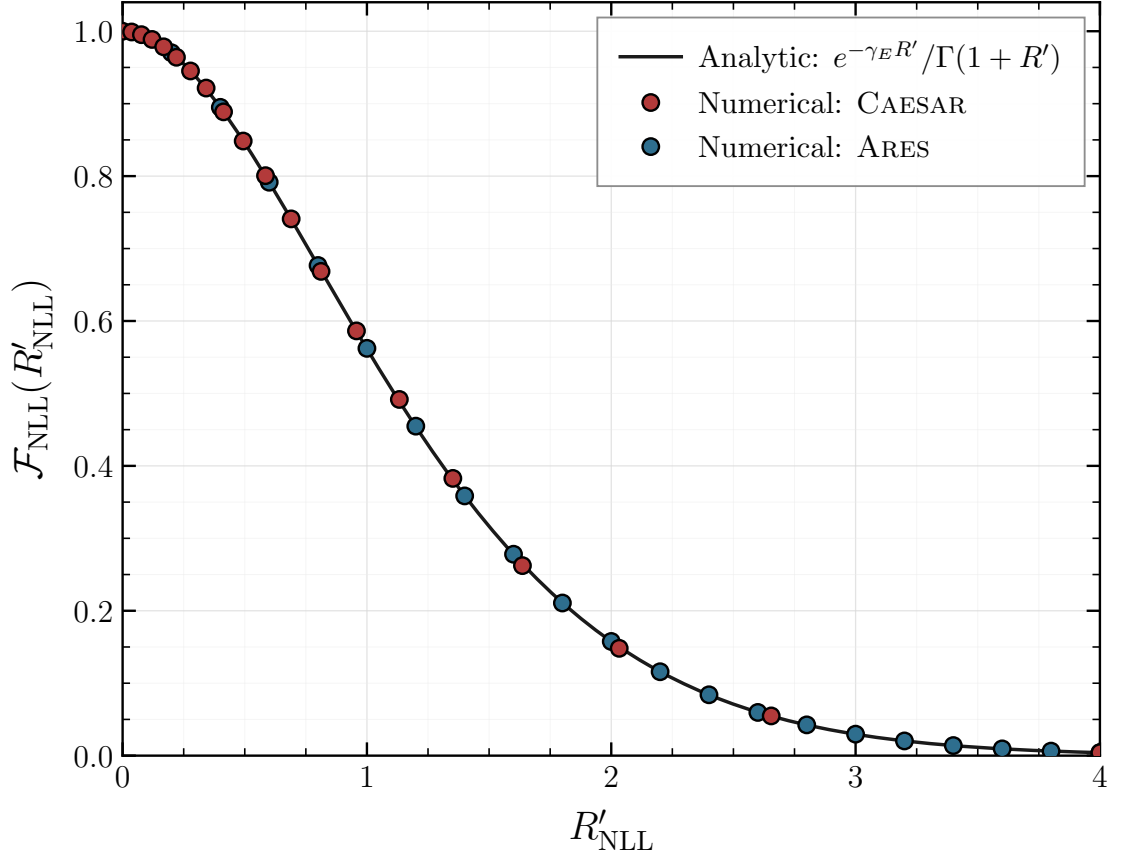


Figure 4.3: Comparison of the analytical result for $\mathcal{F}_{\text{NLL}}$ with the numerical result obtained from the CAESAR and ARES codes for thrust.

4.5.5 Additive observables

A particularly important class of observables are those that are *additive*, for which the soft-collinear limit can be written as a sum of independent contributions from individual emissions,

$$V_{\text{sc}}(\{\tilde{p}\}, k_1, \dots, k_n) = \sum_{i=1}^n V_{\text{sc}}(\{\tilde{p}\}, k_i). \quad (4.70)$$

This property allows the multiple-emission function to be evaluated analytically. Remarkably, the resulting expression is the same for all additive observables, with the observable dependence

entering only through R'_{NLL} . Examples include thrust and one-jettiness, the latter being the observable studied in Ch. 5.

Substituting Eq. (4.70) into Eq. (4.67), and recalling the rescaled variable ζ_i of Eq. (4.61), we obtain

$$\mathcal{F}_{\text{NLL}}(R'_{\text{NLL}}) = \lim_{\epsilon \rightarrow 0} \epsilon^{R'_{\text{NLL}}} \sum_{n=0}^{\infty} \frac{(R'_{\text{NLL}})^n}{n!} \prod_{i=1}^n \left(\int_{\epsilon}^{\infty} \frac{d\zeta_i}{\zeta_i} \int_0^{2\pi} \frac{d\phi_i}{2\pi} \right) \Theta \left(1 - \sum_{i=1}^n \zeta_i \right), \quad (4.71)$$

where $R'_{\text{NLL}} \equiv \sum_{\ell_i} R'_{\text{NLL}, \ell_i}$ and $\phi_i^{(\ell_i)} \rightarrow \phi_i$ for brevity.

The Heaviside function can be expressed using the Mellin transform

$$\Theta \left(1 - \sum_{i=1}^n \zeta_i \right) = \int_{c-i\infty}^{c+i\infty} \frac{d\nu}{2\pi i \nu} e^{\nu} \prod_{i=1}^n e^{-\nu \zeta_i}, \quad c > 0, \quad (4.72)$$

so that Eq. (4.71) becomes,

$$\mathcal{F}_{\text{NLL}}(R'_{\text{NLL}}) = \lim_{\epsilon \rightarrow 0} \epsilon^{R'_{\text{NLL}}} \int_{c-i\infty}^{c+i\infty} \frac{d\nu}{2\pi i \nu} e^{\nu} \sum_{n=0}^{\infty} \frac{(R'_{\text{NLL}})^n}{n!} \prod_{i=1}^n \left(\int_{\epsilon}^{\infty} \frac{d\zeta_i}{\zeta_i} \int_0^{2\pi} \frac{d\phi_i}{2\pi} e^{-\nu \zeta_i} \right). \quad (4.73)$$

The azimuthal integrals are trivial, while the sum over emissions exponentiates, giving

$$\mathcal{F}_{\text{NLL}}(R'_{\text{NLL}}) = \int_{c-i\infty}^{c+i\infty} \frac{d\nu}{2\pi i \nu} e^{\nu} \lim_{\epsilon \rightarrow 0} \epsilon^{R'_{\text{NLL}}} \exp \left[R'_{\text{NLL}} \int_{\epsilon}^{\infty} \frac{d\zeta}{\zeta} e^{-\nu \zeta} \right]. \quad (4.74)$$

The $\epsilon \rightarrow 0$ limit is finite only after combining the prefactor $\epsilon^{R'_{\text{NLL}}}$ with the exponent. Writing

$$\epsilon^{R'_{\text{NLL}}} = \exp \left[-R'_{\text{NLL}} \int_{\epsilon}^1 \frac{d\zeta}{\zeta} \right], \quad (4.75)$$

and combining the two exponents leaves the integrand $e^{-\nu \zeta} - \Theta(1 - \zeta)$, which vanishes as $\zeta \rightarrow 0$. The limit may now be taken, giving

$$\mathcal{F}_{\text{NLL}}(R'_{\text{NLL}}) = \int_{c-i\infty}^{c+i\infty} \frac{d\nu}{2\pi i \nu} e^{\nu} \exp \left[R'_{\text{NLL}} \int_0^{\infty} \frac{d\zeta}{\zeta} \left(e^{-\nu \zeta} - \Theta(1 - \zeta) \right) \right]. \quad (4.76)$$

The remaining ζ -integral evaluates to

$$\int_0^{\infty} \frac{d\zeta}{\zeta} \left(e^{-\nu \zeta} - \Theta(1 - \zeta) \right) = -\gamma_E - \ln \nu, \quad (4.77)$$

so that

$$\mathcal{F}_{\text{NLL}}(R'_{\text{NLL}}) = e^{-\gamma_E R'_{\text{NLL}}} \int_{c-i\infty}^{c+i\infty} \frac{d\nu}{2\pi i \nu} \frac{e^{\nu}}{\nu^{R'_{\text{NLL}}}}. \quad (4.78)$$

Applying the standard inverse Mellin formula

$$\int_{c-i\infty}^{c+i\infty} \frac{d\nu}{2\pi i \nu} \frac{e^{\nu}}{\nu^z} = \frac{1}{\Gamma(1+z)}, \quad (4.79)$$

we arrive at the final result

$$\mathcal{F}_{\text{NLL}}(R'_{\text{NLL}}) = \frac{e^{-\gamma_E R'_{\text{NLL}}}}{\Gamma(1 + R'_{\text{NLL}})}, \quad (4.80)$$

in agreement with the numerical results shown in Fig. 4.3.

Having established the NLL resummation framework, we now turn to its extension to NNLL accuracy through the ARES method. The method was originally developed for e^+e^- annihilation, first for jet rates and jet observables in dijet processes [43–45], and subsequently for event shapes in three-jet production [46]. In the following chapter, we extend the ARES framework to hadronic collisions for the first time.

Chapter 5

NNLL resummation in Z +jet events with the ARES method

5.1 Introduction

To date, no general framework has been established for performing next-to-next-to-leading logarithmic (NNLL) resummation of jet observables in hadronic collisions. Considerable effort is being devoted to extending parton showers to this accuracy, but the state of the art remains at NLL, as demonstrated by the SHERPA [38, 39] and PANSCALES [2, 40, 41] collaborations. Within Soft-Collinear Effective Theory (SCET), resummation has been achieved up to N³LL accuracy for a number of observables [51–53]. However, these calculations rely on observable-specific factorisation theorems, which limits their generality. Recent progress has been made towards a more general treatment [79], but a general framework for NNLL resummation of jet observables in hadronic collisions is still not available.

In this chapter, we extend the ARES formalism for NNLL resummation to hadronic collisions, considering the process $hh \rightarrow Z + \text{jet}$, where the Z boson recoils against a jet with high transverse momentum. We find that the structure of the NNLL resummation is the same as that for three-jet events in e^+e^- annihilation [46], with the differences contained in the corresponding ingredients of the master formula. The main new ingredient is the treatment of initial-state radiation. At NNLL accuracy, this gives rise to contributions involving convolutions of splitting functions with parton distribution functions, which have no analogue in the e^+e^- case. These contributions therefore provide the main extension required to apply ARES to hadronic collisions.

As a first application of the formalism, we consider the event-shape observable one-jettiness.

This observable has been extensively studied and resummed to N³LL accuracy [53], providing a benchmark against which our results can be validated. Furthermore, one-jettiness is an additive observable (see Sec. 4.5.5), making it particularly straightforward to calculate within ARES. The aim of this application is therefore not to provide new phenomenological predictions, but rather to demonstrate the implementation of the extension to hadronic collisions and validate the resulting resummation against known results [141]. The structure of the initial-state radiation contributions is validated independently by comparing the corresponding NNLL terms with the results obtained for Lund tree shapes in Ref. [79].

Unlike the NLL discussion presented in the previous chapter, we do not derive the NNLL master formula from first principles. Instead, we focus on the extension to hadronic collisions, in particular on the treatment of initial-state radiation. The general derivation of the NNLL master formula can be found in Refs. [43, 45, 46], to which we refer the reader for a detailed discussion.

The remainder of this chapter is organised as follows. We first introduce the ARES NNLL master formula for hadronic collisions and derive the corresponding initial-state radiation contributions. We then apply the formalism to one-jettiness and validate the resulting expressions against existing calculations in the literature.

5.2 General setup

We consider the production of a Z boson and a hard jet in hadron collisions,

$$h_1(P_1) + h_2(P_2) \rightarrow Z(q) + \text{jet}(p_J) + \dots, \quad (5.1)$$

where h_1 and h_2 denote the incoming hadrons with momenta P_1 and P_2 , and the dots represent accompanying hadrons. At Born level, the process proceeds through two incoming partons with momenta $p_1 = x_1 P_1$ and $p_2 = x_2 P_2$, producing a Z boson with momentum q and a single hard parton with momentum p_3 . Beyond Born level, additional QCD radiation causes the hard parton to recoil, modifying its momentum to $\tilde{p}_3$. The recoiling parton is subsequently clustered together with the accompanying radiation to form the observed jet with momentum p_J . In the Born limit, the two coincide, so that $p_J = p_3$.

We consider event shapes satisfying the conditions set out in Sec. 4.4, denoted generally as functions $V(q_1, \dots, q_n)$ of the hadron momenta $q_1, \dots, q_n$. As an example, we consider one-jettiness, an observable that has been widely studied in SCET. It corresponds to the $N = 1$ case of the more general N -jettiness observable [142], which quantifies how well the final state can

be described by N narrow jets. It is defined as

$$\mathcal{T}_1 \equiv \sum_i \min \left\{ \frac{2\bar{p}_1 \cdot q_i}{Q_1}, \frac{2\bar{p}_2 \cdot q_i}{Q_2}, \frac{2p_J \cdot q_i}{Q_3} \right\}, \quad (5.2)$$

where q_i is the four-momentum of a *hadron*, and Q_1, Q_2, Q_3 are three momentum scales. The two “incoming” momenta, $\bar{p}_1$ and $\bar{p}_2$, are defined as

$$\bar{p}_1^\mu = \frac{1}{2} n_2 \cdot (p_J + q) n_1^\mu, \quad \bar{p}_2^\mu = \frac{1}{2} n_1 \cdot (p_J + q) n_2^\mu, \quad (5.3)$$

where

$$n_1 = (1, 0, 0, 1), \quad n_2 = (1, 0, 0, -1). \quad (5.4)$$

When $\mathcal{T}_1 \rightarrow 0$, we are in the *one-jet region*, in which the outgoing jet p_J recoils against the Z boson. In this limit, $\bar{p}_1$ and $\bar{p}_2$ coincide with the Born momenta p_1 and p_2 .

We also introduce the normalised observable $\tau_1 \equiv \mathcal{T}_1/M$, where M denotes a generic hard scale. Choosing

$$Q_1 = \frac{2\bar{p}_1 \cdot p_J}{p_{T,J}}, \quad Q_2 = \frac{2\bar{p}_2 \cdot p_J}{p_{T,J}}, \quad Q_3 = 2p_{T,J}, \quad (5.5)$$

where $p_{T,J}$ is the transverse momentum of the outgoing jet with respect to the beam, provides a definition of $\mathcal{T}_1$ that is manifestly invariant under boosts along the beam direction,

$$\mathcal{T}_1 \equiv \sum_i k_{t,i} \min \{ e^{(\eta_i - \eta_J)}, e^{-(\eta_i - \eta_J)}, \cosh(\eta_i - \eta_J) - \cos(\phi_i - \phi_J) \}, \quad (5.6)$$

where $k_{t,i}$ and η_i denote the transverse momentum and rapidity of hadron q_i with respect to the beam, while η_J is the rapidity of the jet.

To reduce the impact of experimental systematics associated with the measurements of particle momenta, we can set $M = H_T$, with H_T the scalar sum of all hadron transverse momenta. This leads to the dimensionless observable

$$\tau_1 \equiv \sum_i \frac{k_{t,i}}{H_T} \min \{ e^{(\eta_i - \eta_J)}, e^{-(\eta_i - \eta_J)}, \cosh(\eta_i - \eta_J) - \cos(\phi_i - \phi_J) \}. \quad (5.7)$$

As discussed in Sec. 4.3, we consider the cumulant of an event shape, imposing the constraint $V(q_1, q_2, \dots, q_n) < v$. For one-jettiness, the one-jet region corresponds to the limit $v \rightarrow 0$. In this limit, we fix the Born kinematics by considering the cross section differential in η_J , and in the transverse momentum q_T and rapidity y_Z of the Z boson. The differential cross section then takes the factorised form of Eq. (4.20), specialised to the present process,

$$\frac{d\sigma(v)}{dq_T dy_Z d\eta_J} \simeq \sum_{i,j} \int_0^1 dx_1 f_i(x_1, \mu_F) \int_0^1 dx_2 f_j(x_2, \mu_F) \frac{d\hat{\sigma}_{ij}}{dq_T dy_Z d\eta_J} \Sigma_{ij}(\{p_1, p_2, p_3\}, v). \quad (5.8)$$

Here $d\hat{\sigma}_{ij}$ is the partonic cross section for the production of a Z boson and an extra jet via the partonic subprocess ij , where i and j each represent a quark, an antiquark of any light flavour (u, d, c, s, b , considered massless), or a gluon.

The cumulative distribution $\Sigma_{ij}(\{p_1, p_2, p_3\}, v)$ contains logarithms of v that become large in the one-jet region. The largest of these contributions are double logarithms of the form $(\alpha_s L^2)^n$, with $L \equiv \ln(1/v)$. Provided $V(\{\tilde{p}\}, \{k_i\})$ is rIRC safe, Σ_{ij} can be reorganised in the region $\alpha_s L \sim 1$ as

$$\Sigma_{ij}(\{p_1, p_2, p_3\}, v) = e^{Lg_1(\alpha_s L)} [G_2(\alpha_s L) + \alpha_s G_3(\alpha_s L) + \dots] . \quad (5.9)$$

This organisation differs from the form in Eq. (4.16), where the logarithmic contributions are fully exponentiated. In ARES, the leading logarithms are contained in an exponentiated radiator, while the subleading contributions are accounted for in the prefactor (cf. Eq. (4.69)). As illustrated in Table 4.1, the function $g_1(\alpha_s L)$ resums all leading logarithmic (LL) terms, $\alpha_s^n L^{n+1}$, in $\ln \Sigma_{ij}$. For rIRC safe observables, g_1 exponentiates. The function $G_2(\alpha_s L)$ resums next-to-leading logarithmic (NLL) contributions ($\alpha_s^n L^n$) to $\ln \Sigma_{ij}$, and so on. Here we work to NNLL accuracy, calculating g_1 , G_2 and G_3 .

5.3 General form of NNLL resummation

The main differences between event shape resummation in hadronic collisions and e^+e^- annihilation are

- the presence of initial-state radiation;
- the fact that there is no single natural hard scale.

Nevertheless, as will be made clear in Sec. 5.4, the structure of the NNLL resummation for $\Sigma_{ij}(\{p_1, p_2, p_3\}, v)$ is the same as for e^+e^- annihilation, although several ingredients appearing in the resummation formula are modified to account for initial-state radiation. In this section, we present the general structure of the NNLL resummation, and discuss the physical meaning of each ingredient. For each partonic channel ij , we have

$$\begin{aligned} \Sigma_{ij}(\{p_1, p_2, p_3\}, v) &= e^{-R_{\text{NNLL}}(v)} \frac{f_i(x_1, v^{\frac{1}{a+b_1}} Q)}{f_i(x_1, \mu_F)} \frac{f_j(x_2, v^{\frac{1}{a+b_2}} Q)}{f_j(x_2, \mu_F)} \times \\ &\times \left(\mathcal{F}_{\text{NLL}}^{(ij)}(\lambda) \left(1 + \frac{\alpha_s(Q)}{2\pi} H_{ij}^{(1)}(Q, \mu_R) + \sum_{\ell} \frac{\alpha_s(Q v^{\frac{1}{a+b_\ell}})}{2\pi} C_{\text{hc}, \ell}^{(1)} \right) + \frac{\alpha_s(Q)}{\pi} \delta \mathcal{F}_{\text{NNLL}}^{(ij)}(\lambda) \right) , \end{aligned} \quad (5.10)$$

where μ_R, μ_F are the renormalisation and factorisation scales in the $\overline{\text{MS}}$ scheme, and λ is defined in Eq. (4.54).

The parameters a, b_ℓ are determined by the behaviour of observable V in the presence of a single soft emission collinear to leg ℓ , as given in Eq. (4.21). For hadronic collisions, the resummation scale Q is arbitrary. The physical interpretation of the ingredients in Eq. (5.10) can be recovered from Refs. [42, 45, 46]. We briefly summarise them below, highlighting the differences with respect to the e^+e^- case:

- The NNLL radiator $R_{\text{NNLL}}(v)$ includes all virtual corrections of *soft and/or collinear origin*. These are divergent, but their divergences cancel with those of real radiation. In order to cancel the singularities of virtual corrections analytically, the radiator also includes a subset of real corrections for suitably defined unresolved emissions. The procedure to define those emissions is not unique (cf. Sec. 4.5). A proposal adequate for three emitting legs is given in Ref. [46], leading to a master formula for $R_{\text{NNLL}}(v)$ whose explicit expression depends only on the single-emission parameters a, b_ℓ, d_ℓ , and $g_\ell(\phi)$. For completeness, we illustrate how the NNLL radiator extends from the NLL result in Sec. 5.3.2.
- Parton distribution functions (PDFs) enter through the ratios $f_i(x_\ell, v^{1/(a+b_\ell)}Q)/f_i(x_\ell, \mu_F)$, where $f_i(x_\ell, \mu_F)$ is the PDF for the incoming parton p_ℓ entering the hard scattering, carrying momentum fraction x_ℓ . Between the factorisation scale μ_F and the veto scale $Qv^{1/(a+b_\ell)}$, the incoming parton can emit *hard-collinear* gluons that are too collinear to be resolved by the observable. These unresolved emissions are absent from $\mathcal{F}_{\text{NLL}}$ and instead they modify the effective PDF seen by the hard process. The ratio $f_i(x_\ell, v^{1/(a+b_\ell)}Q)/f_i(x_\ell, \mu_F)$ resums precisely this effect via DGLAP evolution from μ_F down to the veto scale. This contribution is absent in e^+e^- annihilation, where there are no incoming partons, and is a genuinely new ingredient in hadronic collisions.
- Resolved *soft and collinear* emissions widely separated in angle, together with the appropriate virtual corrections, build up the NLL function $\mathcal{F}_{\text{NLL}}(\lambda)$. Its derivation is given in Sec. 4.5, and its explicit form in Eq. (4.67). This contribution is unchanged from the e^+e^- case.
- Finite virtual corrections of relative order α_s are encoded in the hard coefficient $H_{ij}^{(1)}$. For a generic process, $H_{ij}^{(1)}$ depends on the hard scale Q , the relevant kinematic invariants and the renormalisation scale μ_R . This contribution is process dependent. Its general form is derived in Sec. 5.3.1.

- A single unresolved *hard-collinear* splitting of leg ℓ , together with the corresponding virtual correction and, for initial-state radiation, the collinear counterterm associated with PDF renormalisation (cf. Sec. 2.7.4), gives the coefficient $C_{\text{hc},\ell}^{(1)}$. It is the finite remainder after the cancellation of the collinear $1/\varepsilon$ singularities. For final-state radiation, the singularities cancel completely between real and virtual contributions. For initial-state radiation, this is not the case, and the remaining singularity can be reabsorbed into a redefinition of the PDF. The final-state coefficient is identical to that of the e^+e^- case given in Ref. [46], while the initial-state coefficient is a new result of this work, derived in Sec. 5.4.
- All remaining resolved emissions and their associated virtual corrections are collected in the NNLL function $\delta\mathcal{F}_{\text{NNLL}}(\lambda)$,

$$\delta\mathcal{F}_{\text{NNLL}} = \delta\mathcal{F}_{\text{sc}} + \delta\mathcal{F}_{\text{hc}} + \delta\mathcal{F}_{\text{rec}} + \Delta\mathcal{F}_{\text{rec}} + \delta\mathcal{F}_{\text{wa}} + \Delta\mathcal{F}_{\text{wa}} + \delta\mathcal{F}_{\text{correl}}. \quad (5.11)$$

These terms describe configurations containing one special *soft and/or collinear* emission together with an arbitrary ensemble of widely separated soft-collinear emissions and the corresponding virtual corrections. Each term reflects the region of phase space spanned by the special emission and can be expressed as an integral over the soft-collinear measure $d\mathcal{Z}[\{R'_{\text{NNLL},\ell_i}, \{k_i\}\}]$ with the appropriate weight function $G(\{k_i\})$ (see Eq. (4.68)).

The contributions of soft origin, $\delta\mathcal{F}_{\text{sc}}$, $\delta\mathcal{F}_{\text{wa}}$, $\Delta\mathcal{F}_{\text{wa}}$ and $\delta\mathcal{F}_{\text{correl}}$, are identical to those derived in Ref. [46] and receive no additional contributions from initial-state radiation. This follows directly from the soft matrix-element factorisation described in Sec. 2.5.1, according to which each soft emission depends only on the colour charge and direction of the emitting leg, and is independent of its momentum fraction, flavour or whether it is incoming or outgoing. Consequently, the soft corrections are universal between ISR and FSR.

The collinear contributions, by contrast, distinguish initial- and final-state radiation. An emission collinear to an incoming parton carries a finite fraction of its longitudinal momentum, leading to collinear singularities that are absorbed into the PDFs and regulated through the PDF counterterm. The resulting factorisation therefore involves splitting functions convoluted with PDFs and depends explicitly on the flavour and momentum fraction of the incoming parton. This modifies the hard-collinear contributions $\delta\mathcal{F}_{\text{hc}}$, $\delta\mathcal{F}_{\text{rec}}$ and $\Delta\mathcal{F}_{\text{rec}}$. While these terms retain the same formal structure as in the e^+e^- case, their ingredients are modified for ISR through the appearance of PDF convolutions.

For the one-jettiness observable considered in this chapter, the spin-correlation contribution to $\Delta\mathcal{F}_{\text{rec}}$ vanishes for both initial- and final-state radiation. Therefore, the ISR

extension considered here only requires the derivation of the contributions to $\delta\mathcal{F}_{\text{hc}}$ and $\delta\mathcal{F}_{\text{rec}}$. For generic observables, spin-correlation effects are present, and so for completeness the corresponding initial-state contribution is derived in Appendix B. The remaining ISR corrections relevant to the present calculation are derived in Sec. 5.4.

In the following two subsections we discuss $H_{ij}^{(1)}(Q, \mu_R)$ and $R_{\text{NNLL}}(v)$ in more detail. The radiator can in principle be taken directly from Ref. [46], since the Born-level process $hh \rightarrow Z + \text{jet}$ has the same three-leg colour structure as $e^+e^- \rightarrow q\bar{q}g$. We nevertheless derive it here to show how the NLL radiator is extended to NNLL and to establish the conventions used throughout this chapter. The hard coefficient $H_{ij}^{(1)}$, by contrast, is process dependent and cannot be taken from Ref. [46]. Its general form is derived in the following subsection.

5.3.1 The hard coefficient $H_{ij}^{(1)}(Q, \mu_R)$

The hard coefficient $H_{ij}^{(1)}(Q, \mu_R)$ encodes the finite contribution from one-loop virtual corrections to the hard scattering amplitude. Its derivation allows us to set the conventions used throughout this chapter and introduce the web function formalism that also underlies the derivation of the NNLL radiator (cf. Sec. 5.3.2). We follow the notation of Ref. [143], slightly adapted to our needs.

The starting point for the computation of $H_{ij}^{(1)}$ is the following representation of UV-renormalised virtual corrections [45, 46]:

$$\begin{aligned} \mathcal{V}(Q; \varepsilon) = & H_{ij}(\alpha_s(\mu_R), Q) \exp \left\{ \int^{Q^2} \frac{dk^2}{k^2} \sum_{\ell=1}^3 \gamma_\ell(\alpha_s(k; \varepsilon)) \right\} \times \\ & \times \exp \left\{ - \int \frac{d^d k}{(2\pi)^d} \sum_{\ell=1}^3 \sum_{\bar{\ell} \neq \ell} w_{\ell\bar{\ell}}(m^2, m^2 + \kappa_{\ell\bar{\ell}}^2; \varepsilon) \Theta \left(\frac{1}{2} \ln \frac{Q_{\ell\bar{\ell}}^2}{m^2 + \kappa_{\ell\bar{\ell}}^2} - |\eta^{(\ell\bar{\ell})}| \right) \Theta(Q_{\ell\bar{\ell}} - \kappa_{\ell\bar{\ell}}) \right\}, \end{aligned} \quad (5.12)$$

where $Q_{\ell\bar{\ell}}^2 = 2p_\ell \cdot p_{\bar{\ell}}$ and dimensional regularisation is used to regulate soft and collinear divergences, with $d = 4 - 2\varepsilon$.

The virtual corrections are separated into hard-collinear and soft contributions. The hard-collinear singularities are described by the γ_ℓ exponential, where γ_ℓ is the collinear anomalous dimension associated with external leg ℓ . The second exponential contains the web function $w_{\ell\bar{\ell}}$, which describes soft virtual corrections arising from the exchange of an arbitrary number of soft partons between the dipole $(\ell\bar{\ell})$. The hard function $H_{ij}(\alpha_s, Q)$ contains the finite contribution remaining after these IR-divergent terms have been extracted.

The key ingredients in this representation are:

- the “hard” function $H_{ij}(\alpha_s, Q)$ for incoming channel ij , which has the following expansion in powers of $\alpha_s(Q)$:

$$H_{ij}(\alpha_s, Q) = 1 + \frac{\alpha_s}{2\pi} H_{ij}^{(1)}(Q) + \left(\frac{\alpha_s}{2\pi}\right)^2 H_{ij}^{(2)}(Q) + \dots \quad (5.13)$$

- the d -dimensional strong coupling $\alpha_s(k^2; \varepsilon)$, defined as the solution of the renormalisation-group equation

$$k^2 \frac{d\alpha_s}{dk^2} = -\varepsilon \alpha_s + \beta(\alpha_s), \quad (5.14)$$

where $\beta(\alpha_s)$ is the QCD beta function, and renormalisation is performed in the $\overline{\text{MS}}$ scheme.

The collinear anomalous dimension $\gamma_\ell(\alpha_s)$ has the expansion

$$\gamma_\ell(\alpha_s) = \gamma_\ell^{(0)} \frac{\alpha_s}{2\pi} + \gamma_\ell^{(1)} \left(\frac{\alpha_s}{2\pi}\right)^2 + \dots \quad (5.15)$$

where $\gamma_\ell^{(0)}$ is given in Eq. (4.51), and

$$\gamma_\ell^{(1)} = \begin{cases} \frac{C_F}{2} \left(C_F \left(\frac{3}{4} - \pi^2 + 12\zeta_3 \right) + C_A \left(\frac{17}{12} + \frac{11\pi^2}{9} - 6\zeta_3 \right) - n_f \left(\frac{1}{6} + \frac{2\pi^2}{9} \right) \right), & \text{if } \ell \text{ is a quark,} \\ C_A^2 \left(\frac{8}{3} + 3\zeta_3 \right) - \frac{n_f}{2} C_F - \frac{2}{3} n_f C_A, & \text{if } \ell \text{ is a gluon;} \end{cases} \quad (5.16)$$

- the “web” function $w_{\ell\bar{\ell}}(m^2, m^2 + \kappa_{\ell\bar{\ell}}^2; \varepsilon)$ which consistently resums the contribution of multiple soft virtual corrections from the dipole $(\ell\bar{\ell})$. Physically, the web function is the weight associated with the exchange of a single gluon of mass m , inclusive with respect to all its splittings. Since Feynman rules are invariant under rescaling of the emitters’ four-momenta, the web function can only depend on variables that are invariant under such rescalings. These are m^2 and $\kappa_{\ell\bar{\ell}}^2$ (defined in Eq. (4.42)). The web function admits an expansion in powers of the strong coupling,

$$w_{\ell\bar{\ell}} = \alpha_s w_{\ell\bar{\ell}}^{(1)} + \alpha_s^2 w_{\ell\bar{\ell}}^{(2)} + \dots, \quad (5.17)$$

where

$$w_{\ell\bar{\ell}}^{(1)}(m^2, m^2 + \kappa^2; \varepsilon) = (4\pi)^2 \frac{C_{\ell\bar{\ell}}}{\kappa_{\ell\bar{\ell}}^2} \mu_R^{2\varepsilon} \frac{e^{\varepsilon\gamma_E}}{(4\pi)^\varepsilon} \delta(m^2). \quad (5.18)$$

The dipole colour factor $C_{\ell\bar{\ell}}$ is given in Eq. (4.39), such that for the present case, $C_{\ell\bar{\ell}} = C_A/2$ if either leg is a gluon and $C_{\ell\bar{\ell}} = C_F - C_A/2$ otherwise.

- the d -dimensional phase space, given by

$$\frac{d^d k}{(2\pi)^d} = \frac{d\eta^{(\ell\bar{\ell})}}{2\pi} dm^2 \frac{d\kappa_{\ell\bar{\ell}}^2}{(4\pi)^2} \left(\frac{\pi}{\kappa_{\ell\bar{\ell}}^2} \right)^\varepsilon \frac{\Gamma(1-\varepsilon)}{\Gamma(1-2\varepsilon)} \frac{d\phi}{\pi} (\sin^2 \phi)^{-\varepsilon} \Theta(\sin \phi). \quad (5.19)$$

We extract $H_{ij}^{(1)}$ by expanding both sides of Eq. (5.12) to order α_s and comparing. Evaluating the web function integral gives

$$\begin{aligned} -\alpha_s(\mu_R) \int \frac{d^d k}{(2\pi)^d} w_{\ell\bar{\ell}}^{(1)}(m^2, m^2 + \kappa_{\ell\bar{\ell}}^2; \varepsilon) \Theta \left(\frac{1}{2} \ln \frac{Q_{\ell\bar{\ell}}^2}{m^2 + \kappa_{\ell\bar{\ell}}^2} - |\eta^{(\ell\bar{\ell})}| \right) \Theta(Q_{\ell\bar{\ell}} - \kappa_{\ell\bar{\ell}}) \\ = -\frac{\alpha_s(\mu_R)}{2\pi} \left(\frac{\mu_R^2}{Q_{\ell\bar{\ell}}^2} \right)^\varepsilon \frac{e^{\varepsilon\gamma_E}}{\Gamma(1-\varepsilon)} \frac{C_{\ell\bar{\ell}}}{\varepsilon^2}, \end{aligned} \quad (5.20)$$

and the anomalous dimension integral gives

$$\int_0^{Q^2} \frac{dk^2}{k^2} \gamma_\ell^{(0)} \frac{\alpha_s(k^2; \varepsilon)}{2\pi} = \frac{\alpha_s(\mu_R)}{2\pi} \gamma_\ell^{(0)} \int_0^{Q^2} \frac{dk^2}{k^2} \left(\frac{\mu_R^2}{k^2} \right)^\varepsilon = -\frac{1}{\varepsilon} \left(\frac{\mu_R^2}{Q^2} \right)^\varepsilon \gamma_\ell^{(0)} \frac{\alpha_s(\mu_R)}{2\pi}. \quad (5.21)$$

Using the fact that $\mathcal{V}(Q; \varepsilon)$ has the universal structure [143]¹

$$\mathcal{V}(Q; \varepsilon) = 1 + \frac{\alpha_s(\mu_R)}{2\pi} \left[-\frac{e^{\varepsilon\gamma_E}}{\Gamma(1-\varepsilon)} \sum_\ell \left(\frac{1}{\varepsilon^2} + \frac{1}{\varepsilon} \frac{\gamma_\ell^{(0)}}{C_\ell} \right) \sum_{\bar{\ell} \neq \ell} C_{\ell\bar{\ell}} \left(\frac{\mu_R^2}{Q_{\ell\bar{\ell}}^2} \right)^\varepsilon + \mathcal{H}_1 \right], \quad (5.22)$$

and comparing the above result with the right-hand side of Eq. (5.12),

$$\mathcal{V}(Q; \varepsilon) = 1 + \frac{\alpha_s(\mu_R)}{2\pi} \left\{ -\frac{e^{\varepsilon\gamma_E}}{\Gamma(1-\varepsilon)} \sum_\ell \left[\sum_{\bar{\ell} \neq \ell} \left(\frac{\mu_R^2}{Q_{\ell\bar{\ell}}^2} \right)^\varepsilon \frac{C_{\ell\bar{\ell}}}{\varepsilon^2} + \left(\frac{\mu_R^2}{Q^2} \right)^\varepsilon \frac{\gamma_\ell^{(0)}}{\varepsilon} \right] + H_1 \right\}, \quad (5.23)$$

we find

$$H_{ij}^{(1)}(Q, \mu_R) = \mathcal{H}_1 + \sum_\ell \frac{\gamma_\ell^{(0)}}{C_\ell} \sum_{\bar{\ell} \neq \ell} C_{\ell\bar{\ell}} \ln \left(\frac{Q_{\ell\bar{\ell}}^2}{Q^2} \right), \quad (5.24)$$

where the finite constant piece, $\mathcal{H}_1$, is process dependent and determined by the one-loop hard scattering amplitude.

5.3.2 The NNLL radiator

The NNLL radiator extends the NLL result derived in Sec. 4.5.2, whose notation and kinematic variables we retain throughout. It is obtained from the singular part of the virtual corrections in Eq. (5.12), by cutting off hard-collinear contributions down to the scale $Qv^{1/(a+b_\ell)}$ and including suitably defined unresolved soft emissions. Restricted to this region, the web function integral

¹ Note that the general formula of [143] includes also phase factors (Coulomb phases) that arise when the two legs ℓ and $\bar{\ell}$ are both incoming or outgoing. In the case of three emitters, these phase factors give rise to a finite contribution, included in $\mathcal{H}_1$.

of Eq. (5.20) and the anomalous dimension integral of Eq. (5.21) give the soft and hard-collinear pieces respectively, so that, as at NLL, the radiator decomposes as

$$R_{\text{NNLL}}(v) = R_s(v) + R_{\text{hc}}(v), \quad (5.25)$$

where terms beyond NNLL accuracy are neglected.

Soft radiator

The soft radiator is given by a sum over colour dipoles,

$$R_s(v) = \sum_{\ell} \sum_{\bar{\ell} \neq \ell} C_{\ell\bar{\ell}} \mathcal{R}_{\ell\bar{\ell}}(v), \quad (5.26)$$

where the contribution from each dipole is obtained by restricting the web-function integral to unresolved soft emissions,

$$\begin{aligned} \mathcal{R}_{\ell\bar{\ell}}(v) = \int \frac{d^4k}{(2\pi)^4} w_{\ell\bar{\ell}}(m^2, m^2 + \kappa_{\ell\bar{\ell}}^2; 0) \Theta \left(\frac{1}{2} \ln \frac{Q_{\ell\bar{\ell}}^2}{m^2 + \kappa_{\ell\bar{\ell}}^2} - \eta^{(\ell\bar{\ell})} \right) \Theta(Q_{\ell\bar{\ell}} - \kappa_{\ell\bar{\ell}}) \times \\ \times \Theta(\eta^{(\ell\bar{\ell})}) \Theta \left(d_{\ell}^{(\ell\bar{\ell})} \left(\frac{\kappa_{\ell\bar{\ell}}}{Q_{\ell\bar{\ell}}} \right)^a e^{-b_{\ell}\eta^{(\ell\bar{\ell})}} g_{\ell}(\phi) - v \right). \end{aligned} \quad (5.27)$$

This differs from the NLL soft radiator of Eq. (4.43) in two respects. Firstly, the physical rapidity boundary, $\eta^{(\ell\bar{\ell})} < \ln \left(Q_{\ell\bar{\ell}} / \sqrt{m^2 + \kappa_{\ell\bar{\ell}}^2} \right)$, replaces the massless approximation used at NLL, and generates the “mass correction” $\delta R_{\ell\bar{\ell}}$. Secondly, NNLL accuracy requires the azimuthal expansion shown in Eq. (4.46) to be carried to the second derivative $R''_{\ell\bar{\ell}}$, rather than stopping at $R'_{\ell\bar{\ell}}$ as at NLL. Apart from the rapidity boundary, all the remaining constraints are a matter of convention. Different choices modify the definition of the NLL function $\mathcal{F}_{\text{NLL}}$ and the NNLL correction $\delta\mathcal{F}_{\text{NNLL}}$, while leaving the final resummed prediction unchanged.

To make the NNLL contributions explicit, we reorganise $\mathcal{R}_{\ell\bar{\ell}}(v)$ as

$$\mathcal{R}_{\ell\bar{\ell}}(v) = \int \frac{d\phi}{2\pi} R_{\ell\bar{\ell}} \left(\frac{v}{d_{\ell\bar{\ell}} g_{\ell}(\phi)}, Q_{\ell\bar{\ell}} \right) + \delta R_{\ell\bar{\ell}}(v), \quad (5.28)$$

where $R_{\ell\bar{\ell}}(v, Q_{\ell\bar{\ell}})$ is the “massless” radiator, identical to the NLL radiator derived previously in Eq. (4.48). Expanding the first term to NNLL accuracy gives

$$\begin{aligned} \int \frac{d\phi}{2\pi} R_{\ell\bar{\ell}} \left(\frac{v}{d_{\ell\bar{\ell}} g_{\ell}(\phi)}, Q_{\ell\bar{\ell}} \right) \simeq R_{\ell\bar{\ell}}(v, Q_{\ell\bar{\ell}}) \\ + \langle \ln(d_{\ell\bar{\ell}} g_{\ell}) \rangle R'_{\ell\bar{\ell}}(v, Q_{\ell\bar{\ell}}) + \frac{1}{2} \langle \ln^2(d_{\ell\bar{\ell}} g_{\ell}) \rangle R''_{\ell\bar{\ell}}(v, Q_{\ell\bar{\ell}}), \end{aligned} \quad (5.29)$$

where the azimuthal average is given in Eq. (4.47), and the derivatives are defined as $R'_{\ell\bar{\ell}} \equiv -v dR_{\ell\bar{\ell}}/dv$, and $R''_{\ell\bar{\ell}} \equiv -v dR'_{\ell\bar{\ell}}/dv$.

The remaining term in Eq. (5.28) is the mass correction,

$$\begin{aligned} \delta R_{\ell\bar{\ell}}(v) = & - \int \frac{d^4 k}{(2\pi)^4} w_{\ell\bar{\ell}}(m^2, m^2 + \kappa_{\ell\bar{\ell}}^2; 0) \Theta(Q_{\ell\bar{\ell}} - \kappa_{\ell\bar{\ell}}) \Theta(\kappa_{\ell\bar{\ell}} - v^{\frac{1}{a+b_\ell}} Q_{\ell\bar{\ell}}) \times \\ & \times \Theta \left(\eta^{(\ell\bar{\ell})} - \frac{1}{2} \ln \frac{Q_{\ell\bar{\ell}}^2}{m^2 + \kappa_{\ell\bar{\ell}}^2} \right) \Theta \left(\ln \frac{Q_{\ell\bar{\ell}}}{\kappa_{\ell\bar{\ell}}} - \eta^{(\ell\bar{\ell})} \right), \end{aligned} \quad (5.30)$$

which accounts for the correct rapidity bound.

Hard-collinear radiator

The hard-collinear radiator resums virtual corrections associated with radiation collinear to each emitting leg. As in the NLL case (see Eq. (4.50)), the integration extends from the observable-dependent hard-collinear scale $Qv^{1/(a+b_\ell)}$ up to the hard scale Q ,

$$R_{\text{hc}}(v) = - \sum_{\ell} \int_{Q^2 v^{\frac{2}{a+b_\ell}}}^{Q^2} \frac{dk^2}{k^2} \gamma_{\ell}(\alpha_s(k)), \quad (5.31)$$

where $\gamma_{\ell}(\alpha_s)$ is given in Eq. (5.15). Retaining only $\gamma_{\ell}^{(0)}$ recovers the NLL result of Eq. (4.50), while the $\gamma_{\ell}^{(1)}$ term provides the additional NNLL contribution.

Following the same procedure as at NLL, we employ the ansatz of Eq. (4.52) to obtain

$$R_{\ell\bar{\ell}}(v) = - \frac{\lambda}{\alpha_s(Q)\beta_0} g_1^{(\ell)}(\lambda) - g_2^{(\ell)}(\lambda) - \mathfrak{g}_2^{(\ell)}(\lambda, Q_{\ell\bar{\ell}}) - \frac{\alpha_s(Q)}{\pi} \left(g_3^{(\ell)}(\lambda) + \mathfrak{g}_3^{(\ell)}(\lambda, Q_{\ell\bar{\ell}}) \right), \quad (5.32a)$$

$$\delta R_{\ell\bar{\ell}}(v) = - \frac{\alpha_s(Q)}{\pi} \delta g_3^{(\ell)}(\lambda), \quad (5.32b)$$

$$R_{\text{hc},\ell}(v) = - h_2^{(\ell)}(\lambda) - \frac{\alpha_s(Q)}{\pi} h_3^{(\ell)}(\lambda), \quad (5.32c)$$

where $g_1^{(\ell)}$, $g_2^{(\ell)}$, $\mathfrak{g}_2^{(\ell)}$, and $h_2^{(\ell)}$ are given in Eq. (4.57). The remaining functions are:

$$g_3^{(\ell)}(\lambda) = \left[K^{(1)} \frac{\beta_1 \left(a^2(a+b_\ell-2\lambda) \ln \left(1 - \frac{2\lambda}{a} \right) - (a+b_\ell)^2(a-2\lambda) \ln \left(1 - \frac{2\lambda}{a+b_\ell} \right) + 6b_\ell\lambda^2 \right)}{4\pi b_\ell \beta_0^3 (a-2\lambda)(a+b_\ell-2\lambda)} \right. \\ + \frac{\left(\beta_1^2(a+b_\ell)^2(a-2\lambda) \ln^2 \left(1 - \frac{2\lambda}{a+b_\ell} \right) - 4b_\ell\lambda^2 (\beta_0\beta_2 + \beta_1^2) \right)}{4b_\ell\beta_0^4(a-2\lambda)(a+b_\ell-2\lambda)} \\ - \frac{a \ln \left(1 - \frac{2\lambda}{a} \right) (2\beta_0\beta_2(a-2\lambda) + a\beta_1^2 \ln \left(1 - \frac{2\lambda}{a} \right) + 4\beta_1^2\lambda)}{4b_\ell\beta_0^4(a-2\lambda)} \\ + \frac{(a+b_\ell) \ln \left(1 - \frac{2\lambda}{a+b_\ell} \right) (\beta_0\beta_2(a+b_\ell-2\lambda) + 2\beta_1^2\lambda)}{2b_\ell\beta_0^4(a+b_\ell-2\lambda)} \\ \left. - K^{(2)} \frac{2\lambda^2}{8\pi^2(a-2\lambda)(a+b_\ell-2\lambda)\beta_0^2} \right], \quad (5.33a)$$

$$\mathfrak{g}_3^{(\ell)}(\lambda, Q_{\ell\bar{\ell}}) = \pi\beta_0\lambda^2 \left(\frac{dg_1^{(\ell)}(\lambda)}{d\lambda} + \frac{\lambda}{2} \frac{d^2g_1^{(\ell)}(\lambda)}{d\lambda^2} \right) \ln^2 \left(\frac{Q_{\ell\bar{\ell}}^2}{Q^2} \right) \\ - \pi\beta_0\lambda \left(\frac{dg_2^{(\ell)}(\lambda)}{d\lambda} + \frac{\beta_1}{\beta_0^2} \lambda \frac{dg_1^{(\ell)}(\lambda)}{d\lambda} \right) \ln \left(\frac{Q_{\ell\bar{\ell}}^2}{Q^2} \right), \quad (5.33b)$$

$$\delta g_3^{(\ell)}(\lambda) = -\zeta_2 \frac{\lambda}{(a+b_\ell-2\lambda)}, \quad (5.33c)$$

$$h_3^{(\ell)}(\lambda) = -\gamma_\ell^{(0)} \frac{\beta_1 \left((a+b_\ell) \ln \left(1 - \frac{2\lambda}{a+b_\ell} \right) + 2\lambda \right)}{2\beta_0^2(a+b_\ell-2\lambda)} + \gamma_\ell^{(1)} \frac{\lambda}{2\pi\beta_0(a+b_\ell-2\lambda)}. \quad (5.33d)$$

The derivatives of the NNLL radiator decompose as sums over the emitting legs, $R'_{\text{NNLL}}(\lambda) = \sum_\ell R'_{\text{NNLL},\ell}(\lambda)$ and $R''_{\text{NNLL}}(\lambda) = \sum_\ell R''_{\text{NNLL},\ell}(\lambda)$, where

$$R'_{\text{NNLL},\ell}(\lambda) = C_\ell \frac{\alpha_s(Q)}{b_\ell \pi^2 \beta_0^2 (a-2\lambda)(a+b_\ell-2\lambda)} \left[b_\ell \beta_0 \lambda K^{(1)} - 2\pi b_\ell \beta_1 \lambda \right. \\ \left. - \pi a(a+b_\ell-2\lambda) \beta_1 \ln \left(1 - \frac{2\lambda}{a} \right) + \pi(a+b_\ell)(a-2\lambda) \beta_1 \ln \left(1 - \frac{2\lambda}{a+b_\ell} \right) \right], \quad (5.34a)$$

$$R''_{\text{NNLL},\ell}(\lambda) \equiv \alpha_s(Q) \beta_0 \frac{dR'_{\text{NNLL},\ell}}{d\lambda} = 2C_\ell \frac{\alpha_s(Q)}{\pi} \frac{1}{(a-2\lambda)(a+b_\ell-2\lambda)}. \quad (5.34b)$$

With the hard function $H_{ij}^{(1)}$ and the NNLL radiator R_{NNLL} determined, we now turn to the remaining NNLL corrections. The next section addresses the main focus of this work, namely the derivation of initial-state radiation contributions within the ARES formalism.

5.4 NNLL resummation: initial-state radiation

No general treatment of initial-state radiation at NNLL is currently available. Results exist observable by observable, for one-jettiness in deep-inelastic scattering [84, 144], for N -jettiness in hadronic collisions [141, 142], and for selected observables in colour-singlet production [79].

We restrict ourselves to event-shape variables and, as is customary in the ARES approach, we identify the relevant emissions contributing at NNLL: a single hard and collinear emission k , accompanied by an arbitrary number of soft and collinear emissions $k_1, \dots, k_n$. The emission k carries a fraction $(1 - z)/z$ of the momentum p_ℓ entering the hard scattering,

$$k = \frac{1 - z}{z} p_\ell + k_t^{(\ell)} + \mathcal{O}\left(\left(k_t^{(\ell)}\right)^2\right). \quad (5.35)$$

The extension to initial-state radiation rests on the same two principles as the final-state case:

- we consider one collinear emission k and the corresponding virtual corrections, accompanied by an arbitrary number of soft and collinear emissions $\{k_i\}$;
- we subtract the double counting occurring when k is also soft, since such contributions are already included at NLL in $\mathcal{F}_{\text{NLL}}$.

The procedure involves contributions that are not separately finite in four dimensions, and so we use dimensional regularisation to make each contribution finite in $4 - 2\epsilon$ dimensions.

Throughout, we assume that soft emissions factorise coherently from collinear radiation, such that a soft gluon resolves only the total colour charge of harder emissions at smaller angles. The coherence-violating corrections identified recently [145, 146] first enter at $\mathcal{O}(\alpha_s^4)$ and require a dedicated treatment beyond the scope of this thesis.²

Four possible collinear splittings can produce the parton entering the hard scattering, as shown in Fig. 5.1. The upper row contains the two splittings in which a quark enters the hard scattering, $q \rightarrow qg$ and $g \rightarrow q\bar{q}$, while the lower row contains the two splittings in which a gluon enters, $q \rightarrow qg$ and $g \rightarrow gg$. We derive the NNLL contributions explicitly for the $q \rightarrow qg$ splitting shown in Fig. 5.1(a), and obtain the remaining channels by generalising the result.

The two splittings in which a gluon enters the hard scattering, shown in Figs. 5.1(c) and 5.1(d), also generate spin correlations, which contribute to $\Delta\mathcal{F}_{\text{rec}}$. Since one-jettiness has no azimuthal

² These corrections enter Σ_{ij} as a separate additive term, so the structure of the NNLL cumulant derived below is the same even with these effects present.

dependence, these contributions vanish for the observable considered here (see Sec. 5.5.2). The general treatment of these corrections is deferred to Appendix B.

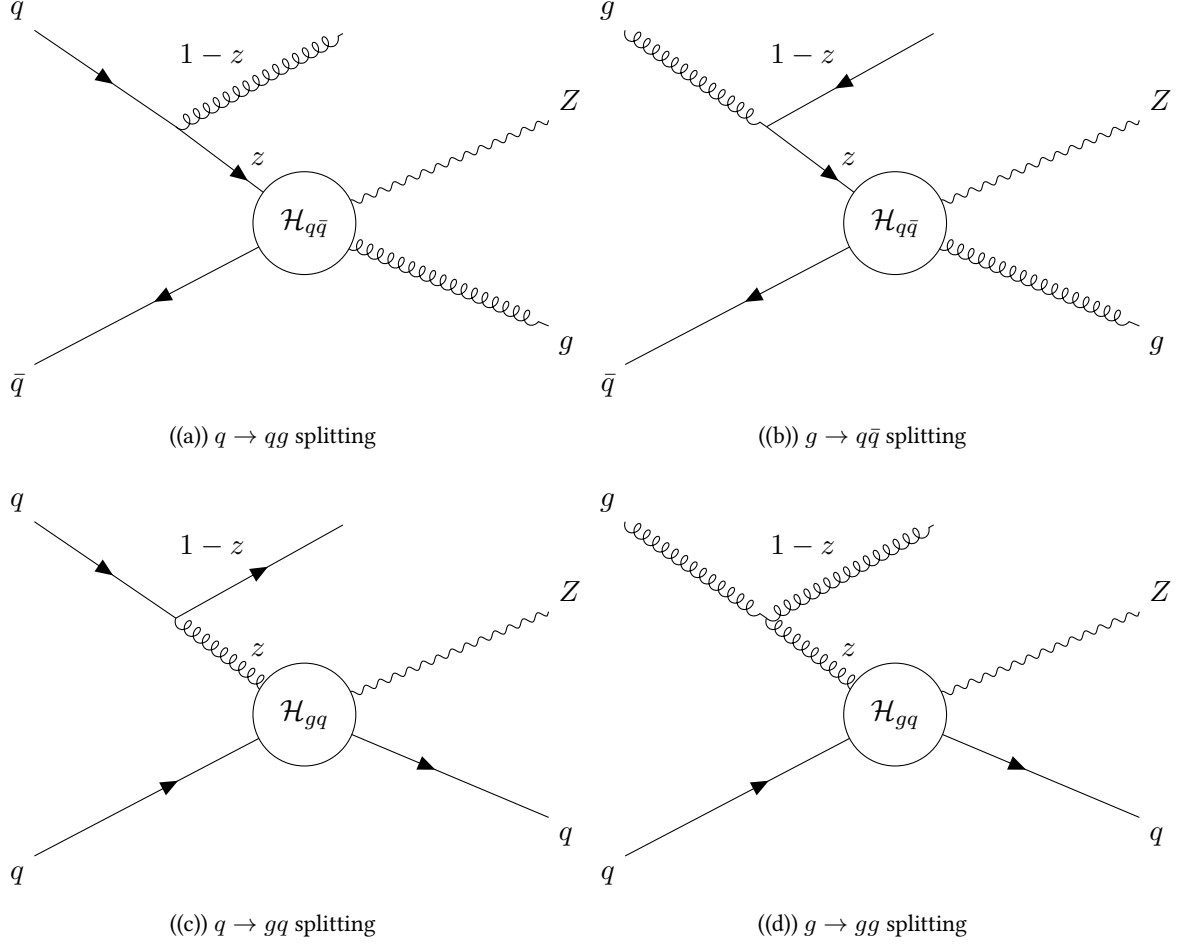


Figure 5.1: Initial-state collinear splittings contributing to Z +jet production. The upper row shows the splittings in which a quark enters the hard scattering, while the lower row shows those in which a gluon enters. The hard scattering is denoted by $\mathcal{H}_{ij}$, where the subscripts indicate the incoming partons. In each diagram, the daughter parton entering the hard scattering carries a fraction z of the parent momentum, with the parent carrying momentum fraction x_ℓ/z and the emitted parton carrying the remaining fraction $1 - z$.

Consider an incoming quark carrying a fraction x_ℓ/z of the proton momentum. It emits a collinear gluon taking a fraction $1 - z$ of that momentum and enters the hard scattering with a fraction x_ℓ . The contribution of this emission to $\Sigma_{ij}(\{\tilde{p}\}, v)$ defines a function $\mathcal{F}_{qq}$, which we reorganise into the separate NNLL contributions. In the $\overline{\text{MS}}$ scheme, it is

$$\begin{aligned}
 \mathcal{F}_{qq} = & \int d\mathcal{Z}[\{R'_{\text{NLL}, \ell_i}, \{k_i\}\}] \int_0^{2\pi} \frac{d\phi}{2\pi} \left\{ \int_0^{Q^2} \frac{dk_t^2}{k_t^2} \left(\frac{\mu_R^2}{k_t^2} \right)^\epsilon \frac{\alpha_s(k_t)}{2\pi} \right. \\
 & \times \left[\int_{x_\ell}^1 \frac{dz}{z} \hat{P}_{qq}(z; \epsilon) \Theta \left(1 - \frac{V_{\text{hc}}(\{\tilde{p}\}, k, \{k_i\})}{v} \right) \frac{f_q(x_\ell/z, \mu_F)}{f_q(x_\ell, \mu_F)} - \int_0^1 dz \frac{2C_\ell}{1-z} \Theta \left(1 - \frac{V_{\text{sc}}(\{\tilde{p}\}, k, \{k_i\})}{v} \right) \right] \\
 & + \left[-\gamma_\ell^{(0)} \int_0^{Q^2 v^{2/(a+b_\ell)}} \frac{dk^2}{k^2} \left(\frac{\mu_R^2}{k^2} \right)^\epsilon \frac{\alpha_s(k)}{2\pi} + \frac{1}{\epsilon} \left(\frac{\mu_R^2}{\mu_F^2} \right)^\epsilon \frac{\alpha_s(\mu_R)}{2\pi} \int_{x_\ell}^1 \frac{dz}{z} P_{qq}(z) \frac{f_q(x_\ell/z, \mu_F)}{f_q(x_\ell, \mu_F)} \right] \\
 & \times \Theta \left(1 - \frac{V_{\text{sc}}(\{\tilde{p}\}, \{k_i\})}{v} \right) \Bigg\} ,
 \end{aligned}$$

(5.36)

where we have introduced the regularised Altarelli-Parisi (AP) splitting function

$$\hat{P}_{ik}(z; \varepsilon) = \hat{P}_{ik}(z; 0) + \varepsilon P_{ik}^{(\varepsilon)}(z), \quad k = q, g. \quad (5.37)$$

In the present case,

$$\hat{P}_{qq}(z; 0) = C_F \frac{1+z^2}{1-z}, \quad P_{qq}^{(\varepsilon)}(z) = -C_F(1-z), \quad (5.38)$$

and $P_{qq}(z) = (\hat{P}_{qq}(z; 0))_+$ is the full AP splitting function.

We now comment on the physical meaning of the ingredients within Eq. (5.36). The second line represents the splitting of a quark emitting a collinear gluon k in $4 - 2\varepsilon$ dimensions, minus the contribution when the gluon is soft, already accounted for in $\mathcal{F}_{\text{NLL}}$. The third line contains virtual corrections together with the collinear counterterm needed to renormalise the parton distribution function. The parameter b_ℓ corresponds to the leg identified by the quark entering the hard process.

The expression in Eq. (5.36) contains single logarithms of the form $\alpha_s \ln[Qv^{1/(a+b_\ell)}/\mu_R]$ and $\alpha_s \ln[Qv^{1/(a+b_\ell)}/\mu_F]$. These can be resummed into the PDF ratio in Eq. (5.10) by fixing $\mu_R = \mu_F = Qv^{1/(a+b_\ell)}$. At NNLL, we may set $\alpha_s(k) = \alpha_s(k_t) = \alpha_s(Qv^{1/(a+b_\ell)})$ throughout, leaving a finite expression up to subleading terms of $\mathcal{O}(\alpha_s^2)$.

To evaluate $\mathcal{F}_{qq}$ in four dimensions we add and subtract the counterterm

$$\begin{aligned} \text{CT} = & \int dZ[\{R'_{\text{NLL}, \ell_i}, \{k_i\}\}] \frac{\alpha_s(Qv^{1/(a+b_\ell)})}{2\pi} \int \frac{dk_t^2}{k_t^2} \left(\frac{\mu_R^2}{k_t^2} \right)^\varepsilon \int_0^{2\pi} \frac{d\phi}{2\pi} \\ & \times \left[\int_{x_\ell}^1 \frac{dz}{z} \hat{P}_{qq}(z; \varepsilon) \frac{f_q(x_\ell/z, Qv^{\frac{1}{a+b_\ell}})}{f_q(x_\ell, Qv^{\frac{1}{a+b_\ell}})} - \int_0^1 dz \frac{2C_\ell}{1-z} \right] \Theta\left(1 - \frac{V_{\text{sc}}(\{\tilde{p}\}, k)}{v}\right) \Theta\left(1 - \frac{V_{\text{sc}}(\{\tilde{p}\}, \{k_i\})}{v}\right). \end{aligned} \quad (5.39)$$

Adding and subtracting CT recasts $\mathcal{F}_{qq}$ in terms of the ingredients of Eq. (5.10),

$$\mathcal{F}_{qq} = \frac{\alpha_s(Q)}{\pi} (\delta\mathcal{F}_{\text{rec}, \text{qq}} + \delta\mathcal{F}_{\text{hc}, \text{qq}}) + \frac{\alpha_s(Qv^{1/(a+b_\ell)})}{2\pi} C_{\text{hc}, \text{qq}}^{(1)} \mathcal{F}_{\text{NLL}}. \quad (5.40)$$

Here $C_{\text{hc}, \text{qq}}^{(1)}$ plays the role of an integrated counterterm in a fixed-order calculation, while $\delta\mathcal{F}_{\text{rec}, \text{qq}} + \delta\mathcal{F}_{\text{hc}, \text{qq}}$ is the real-emission contribution after CT has been subtracted, leaving a finite integrand in four dimensions.

We now proceed to the detailed derivation of Eq. (5.40). Adding CT to the last two lines of Eq. (5.36) gives

$$\begin{aligned}
C_{\text{hc,qq}}^{(1)} \mathcal{F}_{\text{NLL}}(\lambda) &= \int d\mathcal{Z}[\{R'_{\text{NLL},\ell_i}, \{k_i\}\}] \int_0^{2\pi} \frac{d\phi}{2\pi} \Theta\left(1 - \frac{V_{\text{sc}}(\{\tilde{p}\}, \{k_i\})}{v}\right) \times \\
&\times \left\{ -\gamma_\ell^{(0)} \int_0^{Q^2 v^{2/(a+b_\ell)}} \frac{dk_t^2}{k_t^2} \left(\frac{Q^2 v^{2/(a+b_\ell)}}{k_t^2} \right)^\varepsilon + \frac{1}{\varepsilon} \int_{x_\ell}^1 \frac{dz}{z} P_{qq}(z) \frac{f_q(x_\ell/z, Qv^{1/(a+b_\ell)})}{f_q(x_\ell, Qv^{1/(a+b_\ell)})} \right. \\
&+ \left. \int_0^{Q^2} \frac{dk_t^2}{k_t^2} \left(\frac{Q^2 v^{2/(a+b_\ell)}}{k_t^2} \right)^\varepsilon \left[\int_{x_\ell}^1 \frac{dz}{z} \hat{P}_{qq}(z; \varepsilon) \frac{f_q(x_\ell/z, Qv^{\frac{1}{a+b_\ell}})}{f_q(x_\ell, Qv^{\frac{1}{a+b_\ell}})} - \int_0^1 dz \frac{2C_\ell}{1-z} \right] \Theta\left(1 - \frac{V_{\text{sc}}(\{\tilde{p}\}, k)}{v}\right) \right\}
\end{aligned} \tag{5.41}$$

We identify the first term on the right-hand side as $\mathcal{F}_{\text{NLL}}(\lambda)$, by comparison with Eq. (4.67). Parameterising $V_{\text{sc}}(\{\tilde{p}\}, k)$ in terms of the splitting variables,

$$V_{\text{sc}}(\{\tilde{p}\}, k) = d_\ell g_\ell(\phi) \left(\frac{Q}{2E_\ell} \right)^{b_\ell} \left(\frac{k_t}{Q} \right)^{a+b_\ell} \frac{1}{(1-z)^{b_\ell}}, \tag{5.42}$$

and using Eq. (5.38), we find

$$\begin{aligned}
C_{\text{hc,qq}}^{(1)} &= \int_{x_\ell}^1 \frac{dz}{z} \frac{f_q(x_\ell/z, Qv^{\frac{1}{a+b_\ell}})}{f_q(x_\ell, Qv^{\frac{1}{a+b_\ell}})} \times \\
&\times C_F \left[\frac{2}{a+b_\ell} \left(b_\ell \left(\frac{\ln(1-z)}{1-z} \right)_+ - \frac{1}{(1-z)_+} \left(\langle \ln d_\ell g_\ell \rangle - b_\ell \ln \frac{2E_\ell}{Q} \right) \right) (1+z^2) + (1-z) \right].
\end{aligned} \tag{5.43}$$

Using the regularised AP function in Eq. (5.38), the above expression can be written in the more compact form

$$\begin{aligned}
C_{\text{hc,qq}}^{(1)} &= \int_{x_\ell}^1 \frac{dz}{z} \frac{f_q(x_\ell/z, Qv^{\frac{1}{a+b_\ell}})}{f_q(x_\ell, Qv^{\frac{1}{a+b_\ell}})} \times \\
&\times \left[\frac{2}{a+b_\ell} \left(b_\ell \left(\frac{\ln(1-z)}{1-z} \right)_+ - \frac{1}{(1-z)_+} \left(\langle \ln d_\ell g_\ell \rangle - b_\ell \ln \frac{2E_\ell}{Q} \right) \right) (1-z) \hat{P}_{qq}(z; 0) - P_{qq}^{(\varepsilon)}(z) \right].
\end{aligned} \tag{5.44}$$

The remaining contribution to Eq. (5.40) follows in the same way, by subtracting CT from

the first two lines of Eq. (5.36),

$$\begin{aligned}
\frac{\alpha_s(Q)}{\pi} (\delta\mathcal{F}_{\text{rec,qq}} + \delta\mathcal{F}_{\text{hc,qq}}) &= \int d\mathcal{Z}[\{R'_{\text{NLL},\ell_i}, \{k_i\}\}] \frac{\alpha_s(Qv^{\frac{1}{a+b_\ell}})}{2\pi} \int_0^{Q^2} \frac{dk_t^2}{k_t^2} \left(\frac{Q^2 v^{\frac{2}{a+b_\ell}}}{k_t^2} \right)^\varepsilon \int_0^{2\pi} \frac{d\phi}{2\pi} \\
&\times \left\{ \left[\int_{x_\ell}^1 \frac{dz}{z} \hat{P}_{qq}(z; \varepsilon) \Theta \left(1 - \frac{V_{\text{hc}}(\{\tilde{p}\}, k, \{k_i\})}{v} \right) \frac{f_q(x_\ell/z, Qv^{\frac{1}{a+b_\ell}})}{f_q(x_\ell, Qv^{\frac{1}{a+b_\ell}})} + \right. \right. \\
&- \int_0^1 dz \frac{2C_\ell}{1-z} \Theta \left(1 - \frac{V_{\text{sc}}(\{\tilde{p}\}, k, \{k_i\})}{v} \right) \Big] + \\
&- \left[\int_{x_\ell}^1 \frac{dz}{z} \hat{P}_{qq}(z; \varepsilon) \frac{f_q(x_\ell/z, Qv^{\frac{1}{a+b_\ell}})}{f_q(x_\ell, Qv^{\frac{1}{a+b_\ell}})} - \int_0^1 dz \frac{2C_\ell}{1-z} \right] \times \\
&\times \Theta \left(1 - \frac{V_{\text{sc}}(\{\tilde{p}\}, k)}{v} \right) \Theta \left(1 - \frac{V_{\text{sc}}(\{\tilde{p}\}, \{k_i\})}{v} \right) \Big\}. \tag{5.45}
\end{aligned}$$

We now introduce the rescaled variable $\zeta \equiv V_{\text{sc}}(\{\tilde{p}\}, k)/v$ and change variables using Eq. (5.42), which gives

$$\frac{dk_t^2}{k_t^2} \simeq \frac{2}{(a+b_\ell)} \frac{d\zeta}{\zeta}. \tag{5.46}$$

With this substitution, and in the limit $\varepsilon \rightarrow 0$, Eq. (5.45) becomes

$$\begin{aligned}
\frac{\alpha_s(Q)}{\pi} (\delta\mathcal{F}_{\text{rec,qq}} + \delta\mathcal{F}_{\text{hc,qq}}) &= \int d\mathcal{Z}[\{R'_{\text{NLL},\ell_i}, \{k_i\}\}] \frac{\alpha_s(Qv^{1/(a+b_\ell)})}{\pi(a+b_\ell)} \left\{ \int_0^\infty \frac{d\zeta}{\zeta} \int_0^{2\pi} \frac{d\phi}{2\pi} \times \right. \\
&\times \left(\int_{x_\ell}^1 \frac{dz}{z} \hat{P}_{qq}(z; 0) \frac{f_q(x_\ell/z, Qv^{1/(a+b_\ell)})}{f_q(x_\ell, Qv^{1/(a+b_\ell)})} \left[\Theta \left(1 - \frac{V_{\text{hc}}(\{\tilde{p}\}, k, \{k_i\})}{v} \right) + \right. \right. \\
&- \Theta(1-\zeta) \Theta \left(1 - \frac{V_{\text{sc}}(\{\tilde{p}\}, \{k_i\})}{v} \right) \Big] + \\
&- \int_0^1 dz \frac{2C_\ell}{1-z} \left[\Theta \left(1 - \frac{V_{\text{sc}}(\{\tilde{p}\}, k, \{k_i\})}{v} \right) - \Theta(1-\zeta) \Theta \left(1 - \frac{V_{\text{sc}}(\{\tilde{p}\}, \{k_i\})}{v} \right) \right] \Big\}. \tag{5.47}
\end{aligned}$$

We observe that the second and third line give two contributions that are separately finite in the collinear limit $\zeta \rightarrow 0$. However, in order to take the limit $z \rightarrow 1$ and obtain a finite result, one needs to compute the expression in Eq. (5.47) as a whole. A further simplification can be achieved by adding and subtracting the following term

$$\begin{aligned}
&\int d\mathcal{Z}[\{R'_{\text{NLL},\ell_i}, \{k_i\}\}] \frac{\alpha_s(Qv^{1/(a+b_\ell)})}{\pi(a+b_\ell)} \int_0^\infty \frac{d\zeta}{\zeta} \int_0^{2\pi} \frac{d\phi}{2\pi} \int_{x_\ell}^1 \frac{dz}{z} \hat{P}_{qq}(z; 0) \frac{f_q(x_\ell/z, Qv^{1/(a+b_\ell)})}{f_q(x_\ell, Qv^{1/(a+b_\ell)})} \\
&\times \Theta \left(1 - \frac{V_{\text{sc}}(\{\tilde{p}\}, k, \{k_i\})}{v} \right). \tag{5.48}
\end{aligned}$$

This separates Eq. (5.47) into $\delta\mathcal{F}_{\text{rec},\text{qq}}$ and $\delta\mathcal{F}_{\text{hc},\text{qq}}$:

$$\delta\mathcal{F}_{\text{rec},\text{qq}} = \frac{\alpha_s(Qv^{1/(a+b_\ell)})}{\alpha_s(Q)(a+b_\ell)} \int_0^\infty \frac{d\zeta}{\zeta} \int_0^{2\pi} \frac{d\phi}{2\pi} \int d\mathcal{Z}[\{R'_{\text{NLL},\ell_i}, \{k_i\}\}] \int_{x_\ell}^1 \frac{dz}{z} \hat{P}_{qq}(z; 0) \frac{f_q(x_\ell/z, Qv^{1/(a+b_\ell)})}{f_q(x_\ell, Qv^{1/(a+b_\ell)})} \times \left[\Theta\left(1 - \frac{V_{\text{hc}}(\{\tilde{p}\}, k, \{k_i\})}{v}\right) - \Theta\left(1 - \frac{V_{\text{sc}}(\{\tilde{p}\}, k, \{k_i\})}{v}\right) \right], \quad (5.49)$$

$$\delta\mathcal{F}_{\text{hc},\text{qq}} = \frac{\alpha_s(Qv^{1/(a+b_\ell)})}{\alpha_s(Q)(a+b_\ell)} \left(\int_{x_\ell}^1 \frac{dz}{z} \frac{f_q(x_\ell/z, Qv^{1/(a+b_\ell)})}{f_q(x_\ell, Qv^{1/(a+b_\ell)})} \left(P_{qq}(z) - \gamma_\ell^{(0)} \delta(1-z) \right) \right) \int_0^\infty \frac{d\zeta}{\zeta} \int_0^{2\pi} \frac{d\phi}{2\pi} \times \int d\mathcal{Z}[\{R'_{\text{NLL},\ell_i}, \{k_i\}\}] \left[\Theta\left(1 - \frac{V_{\text{sc}}(\{\tilde{p}\}, k, \{k_i\})}{v}\right) - \Theta(1-\zeta) \Theta\left(1 - \frac{V_{\text{sc}}(\{\tilde{p}\}, \{k_i\})}{v}\right) \right]. \quad (5.50)$$

The $q \rightarrow qg$ channel is one instance of a general structure. Surprisingly, each of the three objects in Eq. (5.40) takes the same form for initial- and final-state legs, the two cases differing only in their ingredients.

The hard-collinear coefficient for leg ℓ takes the general form

$$C_{\text{hc},\ell}^{(1)} = \sum_k \int_{x_\ell}^1 \frac{dz}{z} C_{ik}^{(1)}(z) \frac{f_k(x_\ell/z, Qv^{\frac{1}{a+b_\ell}})}{f_i(x_\ell, Qv^{\frac{1}{a+b_\ell}})}, \quad (5.51)$$

where i is the flavour of leg ℓ , and Eq. (5.44) is its $i = k = q$ term. The coefficients $C_{ik}^{(1)}(z)$ follow from the regularised splitting functions of Eq. (5.37),

$$C_{ik}^{(1)}(z) = -P_{ik}^{(\varepsilon)}(z) + \frac{2}{a+b_\ell} \left(b_\ell \left(\frac{\ln(1-z)}{1-z} \right)_+ - \frac{1}{(1-z)_+} \left(\langle \ln d_\ell g_\ell \rangle - b_\ell \ln \frac{2E_\ell}{Q} \right) \right) (1-z) \hat{P}_{ik}(z; 0). \quad (5.52)$$

The explicit expressions of the splitting functions are

$$\hat{P}_{qq}(z; \varepsilon) = C_F \frac{1+z^2}{1-z} - \varepsilon C_F (1-z), \quad (5.53a)$$

$$\hat{P}_{qg}(z; \varepsilon) = T_R [z^2 + (1-z)^2] - \varepsilon T_R [2z(1-z)], \quad (5.53b)$$

$$\hat{P}_{gq}(z; \varepsilon) = C_F \frac{1+(1-z)^2}{z} - \varepsilon C_F z, \quad (5.53c)$$

$$\hat{P}_{gg}(z; \varepsilon) = 2C_A \left[\frac{z}{1-z} + \frac{1-z}{z} + z(1-z) \right]. \quad (5.53d)$$

For the final-state leg ($\ell = 3$), $C_{\text{hc},3}^{(1)}$ depends only on the flavour of the outgoing parton and can be obtained by the replacement

$$\frac{1}{z} \frac{f_k(x_\ell/z, Qv^{\frac{1}{a+b_\ell}})}{f_i(x_\ell, Qv^{\frac{1}{a+b_\ell}})} \Theta(z - x_\ell) \rightarrow 1 \quad (5.54)$$

and integrating over z . This reproduces the result of Ref. [46], such that for a quark or antiquark,

$$C_{\text{hc},3}^{(1)} = C_F \left(\frac{7}{2} \frac{b_3}{a+b_3} + \frac{3}{a+b_3} \left(\langle \ln d_3 g_3(\phi) \rangle - b_3 \ln \frac{2E_3}{Q} \right) + \frac{1}{2} \right), \quad (5.55)$$

while for a gluon

$$C_{\text{hc},3}^{(1)} = \left(\frac{67}{18} C_A - \frac{13}{9} T_R n_f \right) \frac{b_3}{a+b_3} + \frac{1}{a+b_3} \left(\frac{11}{3} C_A - \frac{4}{3} T_R n_f \right) \left(\langle \ln d_3 g_3(\phi) \rangle - b_3 \ln \frac{2E_3}{Q} \right) + \frac{1}{3} T_R n_f. \quad (5.56)$$

The hard-collinear correction, $\delta\mathcal{F}_{\text{hc}}$ takes the general form

$$\delta\mathcal{F}_{\text{hc}}(\lambda) = \sum_{\ell} \frac{\alpha_s(Q v^{1/(a+b_{\ell})})}{\alpha_s(Q)(a+b_{\ell})} B_{\ell} \int_0^{\infty} \frac{d\zeta}{\zeta} \int_0^{2\pi} \frac{d\phi}{2\pi} \int d\mathcal{Z}[\{R'_{\text{NLL},\ell_i}, \{k_i\}\}] \times \\ \times \left[\Theta \left(1 - \frac{V_{\text{sc}}(\{\tilde{p}\}, k, \{k_i\})}{v} \right) - \Theta(1-\zeta) \Theta \left(1 - \frac{V_{\text{sc}}(\{\tilde{p}\}, k, \{k_i\})}{v} \right) \right], \quad (5.57)$$

with, for an initial-state leg,

$$B_{\ell} = \sum_k \int_{x_{\ell}}^1 \frac{dz}{z} \left(P_{ik}(z) - \gamma_{\ell}^{(0)} \delta_{ik} \delta(1-z) \right) \frac{f_k(x_{\ell}/z, Q v^{1/(a+b_{\ell})})}{f_i(x_{\ell}, Q v^{1/(a+b_{\ell})})}, \quad (5.58)$$

where $P_{ik}(z)$ are the full Altarelli-Parisi splitting functions,

$$P_{qq}(z) = C_F \left[\frac{1+z^2}{1-z} \right]_+, \quad (5.59a)$$

$$P_{qg}(z) = T_R [z^2 + (1-z)^2], \quad (5.59b)$$

$$P_{gq}(z) = C_F \left[\frac{1+(1-z)^2}{z} \right], \quad (5.59c)$$

$$P_{gg}(z) = 2C_A \left[\frac{z}{(1-z)_+} + \frac{1-z}{z} + z(1-z) \right] + 2\pi\beta_0\delta(1-z). \quad (5.59d)$$

For final-state radiation, the same substitution can be used as in Eq. (5.54), such that $B_{\ell} = -\gamma_{\ell}^{(0)}$, with $\gamma_{\ell}^{(0)}$ defined in Eq. (4.51).

The recoil correction, $\delta\mathcal{F}_{\text{rec}}$ takes the general form,

$$\delta\mathcal{F}_{\text{rec}}(\lambda) = \sum_{\ell} \frac{\alpha_s(Q v^{1/(a+b_{\ell})})}{\alpha_s(Q)(a+b_{\ell})} \int_0^1 dz P_{\ell}(z) \int_0^{\infty} \frac{d\zeta}{\zeta} \int_0^{2\pi} \frac{d\phi}{2\pi} \int d\mathcal{Z}[\{R'_{\text{NLL},\ell_i}, \{k_i\}\}] \times \\ \times \left[\Theta \left(1 - \frac{V_{\text{hc}}(\{\tilde{p}\}, k, \{k_i\})}{v} \right) - \Theta \left(1 - \frac{V_{\text{sc}}(\{\tilde{p}\}, k, \{k_i\})}{v} \right) \right], \quad (5.60)$$

with, for an initial-state leg,

$$P_{\ell}(z) = \sum_k \hat{P}_{ik}(z; 0) \frac{f_k(x_{\ell}/z, Q v^{1/(a+b_{\ell})})}{z f_i(x_{\ell}, Q v^{1/(a+b_{\ell})})} \Theta(z - x_{\ell}), \quad (5.61)$$

where $\hat{P}_{ik}(z; 0)$ are defined in Eq. (5.37). For a final-state leg,

$$P_{\ell}(z) = \begin{cases} C_F \frac{1+z^2}{1-z}, & \text{if } \ell \text{ is a quark,} \\ C_A \left(\frac{2z}{1-z} + z(1-z) \right) + T_R [z^2 + (1-z)^2], & \text{if } \ell \text{ is a gluon.} \end{cases} \quad (5.62)$$

5.5 NNLL resummation: one-jettiness

We now apply the general NNLL formalism to one-jettiness τ_1 , introduced in Sec. 5.2. One-jettiness is an rIRC-safe, continuously global observable that vanishes at Born level, and satisfies the scaling conditions for an arbitrary number of soft and/or collinear emissions discussed in Sec. 4.4. It therefore belongs to the class of observables to which the ARES formalism applies.

We begin by considering the behaviour of the observable in the presence of a single soft emission collinear to leg ℓ . By definition, $k \cdot p_\ell / E_\ell = k_t^{(\ell)} e^{-\eta^{(\ell)}}$, where $k_t^{(\ell)}$ is the relative transverse momentum with respect to the emitting leg ℓ and $\eta^{(\ell)}$ is the corresponding rapidity. Using $\tau_1 = \mathcal{T}_1/M$ with $\mathcal{T}_1$ defined in Eq. (5.2), the soft-collinear limit takes the same form for each leg:

$$\tau_1(\{\tilde{p}\}, k) \simeq \tau_1^{\text{sc}}(\{\tilde{p}\}, k) = \frac{2E_\ell}{Q_\ell} \frac{k_t^{(\ell)}}{M} e^{-\eta^{(\ell)}}. \quad (5.63)$$

From Eq. (5.63), we immediately identify the single-emission parameters,

$$a = 1, \quad b_\ell = 1, \quad d_\ell = \frac{Q}{M} \frac{2E_\ell}{Q_\ell}, \quad g_\ell(\phi) = 1, \quad \ell = 1, 2, 3. \quad (5.64)$$

These parameters determine the ingredients entering the NNLL cumulant. In particular, Eqs. (5.24), (5.25), and (5.51) determine the hard coefficient $H_{ij}^{(1)}$, the NNLL radiator R_{NNLL} , and the hard-collinear coefficient $C_{\text{hc},\ell}^{(1)}$, respectively. Since one-jettiness is additive, the NLL multiple-emission function $\mathcal{F}_{\text{NLL}}$ is given by the additive expression in Eq. (4.80). The remaining NNLL terms are contained in $\delta\mathcal{F}_{\text{NNLL}}$, given in Eq. (5.11). For an additive observable, these simplify considerably, and the corresponding expressions are collected in Appendix C.

In the following subsections we evaluate each contribution separately, organising the discussion according to the kinematic region from which the correction originates.

5.5.1 Soft and collinear contributions

The NNLL corrections associated with the soft-collinear region are $\delta\mathcal{F}_{\text{sc}}$ and $\delta\mathcal{F}_{\text{correl}}$. Both are expressed in terms of the soft-collinear limit $V_{\text{sc}}(\{\tilde{p}\}, \{k_i\})/v$, defined in Eq. (4.32), which describes an ensemble of soft and collinear emissions that are widely separated in angle.

The correction $\delta\mathcal{F}_{\text{sc}}$ corresponds to the addition of a single soft-collinear emission k , parametrised by $\zeta = V_{\text{sc}}(\{\tilde{p}\}, k)/v$, the emitting leg ℓ , and its azimuthal angle $\phi^{(\ell)}$. It accounts for running coupling corrections to real emissions in the physical scheme, as well as the correct rapidity boundary for a single soft-collinear emission. The general expression is

$$\delta\mathcal{F}_{\text{sc}}(\lambda) = \frac{\pi}{\alpha_s(Q)} \int_0^\infty \frac{d\zeta}{\zeta} \int_0^{2\pi} \frac{d\phi}{2\pi} \sum_{\ell=1}^3 \left[\Delta R'_{\ell,\text{NNLL}} + R''_{\ell,\text{NNLL}} \left(\ln \frac{d_\ell g_\ell(\phi)}{\zeta} - b_\ell \ln \frac{2E_\ell}{Q} \right) \right] \times \\ \times \int d\mathcal{Z}[\{R'_{\text{NLL},\ell_i}, \{k_i\}\}] \left[\Theta \left(1 - \frac{V_{\text{sc}}(\{\tilde{p}\}, k, \{k_i\})}{v} \right) - \Theta \left(1 - \frac{V_{\text{sc}}(\{\tilde{p}\}, \{k_i\})}{v} \right) \Theta(1 - \zeta) \right]. \quad (5.65)$$

Substituting the one-jettiness parameters in Eq. (5.64) into the additive expression for $\delta\mathcal{F}_{\text{sc}}$ given in Appendix C, we obtain

$$\delta\mathcal{F}_{\text{sc}}(\lambda) = -\mathcal{F}_{\text{NLL}}(\lambda) \frac{\pi}{\alpha_s(Q)} \sum_{\ell=1}^3 \left[\left(\Delta R'_{\ell,\text{NNLL}} + R''_{\ell,\text{NNLL}} \ln \frac{Q^2}{MQ_\ell} \right) \left(\psi^{(0)}(1 + R'_{\text{NLL}}) + \gamma_E \right) \times \right. \\ \left. + \frac{R''_{\ell,\text{NNLL}}}{2} \left(\left(\psi^{(0)}(1 + R'_{\text{NLL}}) + \gamma_E \right)^2 - \psi^{(1)}(1 + R'_{\text{NLL}}) + \frac{\pi^2}{6} \right) \right], \quad (5.66)$$

where R'_{NLL} and $R''_{\ell,\text{NNLL}}$ are given in Eqs. (4.58) and (5.34b), respectively, while

$$\Delta R'_{\ell,\text{NNLL}}(v) \equiv C_\ell \left(-\alpha_s(Q) \beta_0 \frac{dg_2^{(\ell)}(\lambda)}{d\lambda} \right), \quad (5.67)$$

with $g_2^{(\ell)}$ given in Eq. (4.57).

The correlated correction $\delta\mathcal{F}_{\text{correl}}$ accounts for the contribution of correlated double emissions. In this case, the additional soft-collinear emission undergoes a splitting into two partons, k_a and k_b , whose kinematics are described by the variables introduced in Eq. (5.68). This correction therefore accounts for correlated splittings into either a $q\bar{q}$ pair or a pair of gluons. The general expression is

$$\delta\mathcal{F}_{\text{correl}}(\lambda) = \sum_{\ell=1}^3 \frac{\lambda R''_{\ell,\text{NNLL}}}{2a\beta_0\alpha_s(Q)} \int_0^\infty \frac{d\zeta}{\zeta} \int_0^{2\pi} \frac{d\phi^{(\ell)}}{2\pi} \int_0^\infty \frac{d\mu^2}{\mu^2(1+\mu^2)} \int_0^1 dz \int_0^{2\pi} \frac{d\phi}{2\pi} \frac{1}{2!} \mathcal{A}^2(z, \mu, \phi) \times \\ \int d\mathcal{Z}[\{R'_{\text{NLL},\ell_i}, \{k_i\}\}] \left[\Theta \left(1 - \frac{V_{\text{sc}}(\{\tilde{p}\}, k_a, k_b, \{k_i\})}{v} \right) - \lim_{\mu^2 \rightarrow 0} \Theta \left(1 - \frac{V_{\text{sc}}(\{\tilde{p}\}, k_a + k_b, \{k_i\})}{v} \right) \right], \quad (5.68)$$

where $k = k_a + k_b$, $\mu^2 = m^2/k_t^2$ is the rescaled invariant mass of the splitting, and $\mathcal{A}^2$ is the double-emission function defined in Ref. [46].

Since one-jettiness has $a = b_\ell = 1$ and $g_\ell(\phi) = 1$, the soft-collinear parameters entering the correlated correction are identical to those in the angularity result of Ref. [45]. We can therefore use that result, evaluated at $x = 0$, to obtain

$$\delta\mathcal{F}_{\text{correl}}(\lambda) = \frac{\pi}{2\alpha_s(Q)} \lambda R'' \zeta_2 \mathcal{F}_{\text{NLL}}(\lambda). \quad (5.69)$$

5.5.2 Hard and collinear contributions

The NNLL corrections associated with the hard-collinear region are $\delta\mathcal{F}_{\text{hc}}$, $\delta\mathcal{F}_{\text{rec}}$, and $\Delta\mathcal{F}_{\text{rec}}$. These arise from the emission of a hard-collinear parton and its effect on the soft-collinear ensemble.

The hard-collinear correction, $\delta\mathcal{F}_{\text{hc}}$, accounts for the emission of a hard-collinear parton at the level of the squared matrix element and depends on the same soft-collinear limit as $\delta\mathcal{F}_{\text{sc}}$, given in Eq. (5.65). The general expression is

$$\delta\mathcal{F}_{\text{hc}} = \sum_{\ell} \frac{\alpha_s(Q v^{1/(a+b_{\ell})})}{\alpha_s(Q) (a+b_{\ell})} B_{\ell} \int_0^{\infty} \frac{d\zeta}{\zeta} \int_0^{2\pi} \frac{d\phi}{2\pi} \int d\mathcal{Z}[\{R'_{\text{NLL},\ell_i}, \{k_i\}\}] \times \left[\Theta\left(1 - \frac{V_{\text{sc}}(\{\tilde{p}\}, k, \{k_i\})}{v}\right) - \Theta(1-\zeta) \Theta\left(1 - \frac{V_{\text{sc}}(\{\tilde{p}\}, \{k_i\})}{v}\right) \right]. \quad (5.70)$$

Substituting the one-jettiness parameters in Eq. (5.64) into the additive expression for $\delta\mathcal{F}_{\text{hc}}$ given in Appendix C, we obtain

$$\delta\mathcal{F}_{\text{hc}}(\lambda) = \frac{\alpha_s(\sqrt{\tau_1} Q)}{2\alpha_s(Q)} \frac{d\mathcal{F}_{\text{NLL}}}{dR'_{\text{NLL}}} \sum_{\ell} B_{\ell}. \quad (5.71)$$

For the final-state leg, $\ell = 3$, this reduces to the $e^+e^- \rightarrow 3\text{-jet}$ result of Ref. [46]:

$$\delta\mathcal{F}_{\text{hc}}|_{\ell=3} = -\frac{\alpha_s(\sqrt{\tau_1} Q)}{2\alpha_s(Q)} \frac{d\mathcal{F}_{\text{NLL}}}{dR'_{\text{NLL}}} \gamma_3^{(0)}, \quad (5.72)$$

while for the initial-state contributions, using Eq. (5.58), the z integration cannot be performed analytically due to the presence of PDFs:

$$\delta\mathcal{F}_{\text{hc}}|_{\ell=1,2} = \frac{\alpha_s(\sqrt{\tau_1} Q)}{2\alpha_s(Q)} \frac{d\mathcal{F}_{\text{NLL}}}{dR'_{\text{NLL}}} \sum_k \int_{x_{\ell}}^1 \frac{dz}{z} \left(P_{ik}(z) - \gamma_{\ell}^{(0)} \delta_{ik} \right) \frac{f_k(x_{\ell}/z, Q v^{1/(a+b_{\ell})})}{f_i(x_{\ell}, Q v^{1/(a+b_{\ell})})}, \quad (5.73)$$

where $P_{ik}(z)$ are given in Eq. (5.59).

The recoil corrections additionally require the hard-collinear limit of the observable,

$$\frac{V_{\text{hc}}(\{\tilde{p}\}, k_{\text{hc}}, k_1, \dots, k_n)}{v} \equiv \lim_{v \rightarrow 0} \frac{V(\{\tilde{p}\}, k_{\text{hc}}, k_1, \dots, k_n)}{v}, \quad (5.74)$$

where k_{hc} denotes a single hard-collinear emission, while $k_1, \dots, k_n$ represent an ensemble of independently emitted soft and collinear emissions.

The recoil correction, $\delta\mathcal{F}_{\text{rec}}$, describes the kinematic recoil induced by a hard-collinear emission against the soft-collinear ensemble. The general expression is

$$\delta\mathcal{F}_{\text{rec}}(\lambda) = \sum_{\ell} \frac{\alpha_s(Q v^{1/(a+b_{\ell})})}{\alpha_s(Q) (a+b_{\ell})} \int_0^1 dz P_{\ell}(z) \int_0^{\infty} \frac{d\zeta}{\zeta} \int_0^{2\pi} \frac{d\phi}{2\pi} \int d\mathcal{Z}[\{R'_{\text{NLL},\ell_i}, \{k_i\}\}] \times \left[\Theta\left(1 - \frac{V_{\text{hc}}(\{\tilde{p}\}, k_{\text{hc}}, \{k_i\})}{v}\right) - \Theta\left(1 - \frac{V_{\text{sc}}(\{\tilde{p}\}, k, \{k_i\})}{v}\right) \right]. \quad (5.75)$$

Using the additive result for $\delta\mathcal{F}_{\text{rec}}$ in Appendix C, the final-state contribution agrees with the corresponding e^+e^- three-jet result [46]. For a final-state quark,

$$\delta\mathcal{F}_{\text{rec}}|_{\ell=3} = \mathcal{F}_{\text{NLL}} \frac{\alpha_s(Q\sqrt{\tau_1})}{2\alpha_s(Q)} C_F \left(\frac{5}{4} - \frac{\pi^2}{3} \right), \quad (5.76)$$

while for a final-state gluon,

$$\delta\mathcal{F}_{\text{rec}}|_{\ell=3} = \mathcal{F}_{\text{NLL}} \frac{\alpha_s(Q\sqrt{\tau_1})}{2\alpha_s(Q)} \left[C_A \left(\frac{67}{36} - \frac{\pi^2}{3} \right) - T_R n_f \frac{13}{18} \right]. \quad (5.77)$$

For the initial-state contributions, the z integrations cannot be performed analytically. Using Eq. (5.61), we obtain

$$\delta\mathcal{F}_{\text{rec}}|_{\ell=1,2} = \mathcal{F}_{\text{NLL}} \frac{\alpha_s(Q\sqrt{\tau_1})}{2\alpha_s(Q)} \int_0^1 dz \sum_k \hat{P}_{ik}(z;0) \frac{f_k(x_\ell/z, Qv^{1/(a+b_\ell)})}{zf_i(x_\ell, Qv^{1/(a+b_\ell)})} \Theta(z-x_\ell) \ln z, \quad (5.78)$$

where $\hat{P}_{ik}(z;0)$ is defined in Eq. (5.37)

The final contribution, $\Delta\mathcal{F}_{\text{rec}}$, accounts for spin correlations between a hard-collinear gluon and the event plane. The general expression is

$$\begin{aligned} \Delta\mathcal{F}_{\text{rec}}(\lambda) = & \sum_{\ell_g} \frac{\alpha_s(Qv^{1/(a+b_{\ell_g})})}{\alpha_s(Q)(a+b_{\ell_g})} \int_0^\infty \frac{d\zeta}{\zeta} \int_0^\pi \frac{d\phi^{(\ell_g)}}{\pi} \left[2 \cos^2 \phi^{(\ell_g)} - 1 \right] \int_0^1 dz^{(\ell_g)} \Delta P_g(z^{(\ell_g)}) \\ & \times \int d\mathcal{Z}[\{R'_{\text{NLL},\ell_i}, \{k_i\}\}] \left[\Theta \left(1 - \frac{V_{\text{hc}}(\{\tilde{p}\}, k, \{k_i\})}{v} \right) - \Theta(1-\zeta) \Theta \left(1 - \frac{V_{\text{sc}}(\{\tilde{p}\}, \{k_i\})}{v} \right) \right], \end{aligned} \quad (5.79)$$

where the sum runs over gluon legs, ℓ_g .

Substituting the one-jettiness parameters into the corresponding additive expression in Appendix C, this contribution vanishes for all legs since $g(\phi) = 1$:

$$\Delta\mathcal{F}_{\text{rec}}(\lambda) = 0. \quad (5.80)$$

5.5.3 Soft wide angle contributions

The final class of NNLL corrections originates from soft radiation emitted at wide angles. The two contributions in this region are $\delta\mathcal{F}_{\text{wa}}$ and $\Delta\mathcal{F}_{\text{wa}}$.

The wide-angle correction, $\delta\mathcal{F}_{\text{wa}}$, accounts for the difference between the full observable and its soft-collinear parametrisation for a single soft wide-angle emission accompanied by multiple soft-collinear emissions.

This correction is naturally expressed in terms of dipoles $(\ell\bar{\ell})$. Accordingly, the momentum of a single emission k can be decomposed along any pair of light-like momenta $(p_\ell, p_{\bar{\ell}})$ forming

the $(\ell\bar{\ell})$ dipole,

$$k^{(\ell\bar{\ell})} = z^{(\ell)} p_\ell + z^{(\bar{\ell})} p_{\bar{\ell}} + \kappa_{\ell\bar{\ell}} \cos \phi^{(\ell\bar{\ell})} n_{\text{in}}^{(\ell\bar{\ell})} + \kappa_{\ell\bar{\ell}} \sin \phi^{(\ell\bar{\ell})} n_{\text{out}}^{(\ell\bar{\ell})}, \quad (5.81)$$

where

$$n_{\text{in}}^{(\ell\bar{\ell})} = \left(\cot \frac{\theta_{\ell\bar{\ell}}}{2}, \frac{\vec{n}_\ell + \vec{n}_{\bar{\ell}}}{\sin \theta_{\ell\bar{\ell}}} \right), \quad n_{\text{out}}^{(\ell\bar{\ell})} = \left(0, \frac{n_\ell \times n_{\bar{\ell}}}{\sin \theta_{\ell\bar{\ell}}} \right), \quad (5.82)$$

with $\vec{n}_{\ell/\bar{\ell}} = \vec{p}_{\ell/\bar{\ell}}/E_{\ell/\bar{\ell}}$. The remaining variables entering this decomposition are defined in Sec. 4.5.2.

This correction requires the soft wide-angle limit of the observable,

$$\frac{V_{\text{wa}}(\{\tilde{p}\}, k^{(\ell\bar{\ell})}, k_1, \dots, k_n)}{v} \equiv \lim_{v \rightarrow 0} \frac{V(\{\tilde{p}\}, k^{(\ell\bar{\ell})}, k_1, \dots, k_n)}{v}, \quad (5.83)$$

where the emission $k^{(\ell\bar{\ell})}$ denotes a soft emission at wide angle with respect to the Born legs.

The general expression is

$$\begin{aligned} \delta\mathcal{F}_{\text{wa}}(\lambda) &= \frac{\alpha_s(v^{1/a}Q)}{a\alpha_s(Q)} 2 \sum_{(\ell\bar{\ell})} C_{\ell\bar{\ell}} \int_0^\infty \frac{d\zeta}{\zeta} \int_{-\infty}^\infty d\eta^{(\ell\bar{\ell})} \int_0^{2\pi} \frac{d\phi^{(\ell\bar{\ell})}}{2\pi} \times \\ &\times \int d\mathcal{Z}[\{R'_{\text{NLL},\ell_i}, \{k_i\}\}] \left[\Theta \left(1 - \frac{V_{\text{wa}}(\{\tilde{p}\}, k^{(\ell\bar{\ell})}, \{k_i\})}{v} \right) - \Theta \left(1 - \frac{V_{\text{sc}}(\{\tilde{p}\}, k^{(\ell\bar{\ell})}, \{k_i\})}{v} \right) \right], \end{aligned} \quad (5.84)$$

where the sum runs over all the dipoles $(\ell\bar{\ell})$ that build up the Born event.

Using the additive result for $\delta\mathcal{F}_{\text{wa}}$ in Appendix C, we obtain

$$\delta\mathcal{F}_{\text{wa}}(\lambda) = \mathcal{F}_{\text{NLL}}(\lambda) \frac{\alpha_s(\tau_1 Q)}{\alpha_s(Q)} \sum_{(\ell\bar{\ell})} C_{\ell\bar{\ell}} \left[\ln^2 \frac{Q_{\ell\bar{\ell}}}{Q_\ell} - 2(I_{\ell\bar{\ell},\hat{\ell}} + I_{\bar{\ell}\ell,\hat{\ell}}) \right], \quad (5.85)$$

where $I_{\ell\bar{\ell},\hat{\ell}}$ are the integrals defined in Eq. (51) of Ref. [53] (see also Ref. [147]), and $\hat{\ell}$ denotes the leg which is neither ℓ nor $\bar{\ell}$.

The second contribution, $\Delta\mathcal{F}_{\text{wa}}$, is associated with resolved soft wide-angle radiation and takes the same soft-collinear limit as $\delta\mathcal{F}_{\text{sc}}$, given in Eq. (5.65). The general expression is

$$\begin{aligned} \Delta\mathcal{F}_{\text{wa}} &= 2 \sum_{(\ell\bar{\ell})} C_{\ell\bar{\ell}} \frac{\alpha_s(v^{1/a}Q)}{a\alpha_s(Q)} \ln \frac{Q_{\ell\bar{\ell}}}{Q} \int_0^\infty \frac{d\zeta}{\zeta} \int_0^{2\pi} \frac{d\phi}{2\pi} \int d\mathcal{Z}[\{R'_{\text{NLL},\ell_i}, \{k_i\}\}] \times \\ &\times \left[\Theta \left(1 - \frac{V_{\text{sc}}(\{\tilde{p}\}, k, \{k_i\})}{v} \right) - \Theta \left(1 - \frac{V_{\text{sc}}(\{\tilde{p}\}, \{k_i\})}{v} \right) (1 - \zeta) \right]. \end{aligned} \quad (5.86)$$

Using the additive expression for Substituting the one-jettiness parameters in Eq. (5.64) into the additive expression for $\Delta\mathcal{F}_{\text{wa}}$ given in Appendix C, we obtain

$$\Delta\mathcal{F}_{\text{wa}}(\lambda) = \frac{\alpha_s(\tau_1 Q)}{\alpha_s(Q)} \frac{d\mathcal{F}_{\text{NLL}}}{dR'_{\text{NLL}}} 2 \sum_{(\ell\bar{\ell})} C_{\ell\bar{\ell}} \ln \left(\frac{Q_{\ell\bar{\ell}}}{Q} \right). \quad (5.87)$$

This completes the evaluation of all NNLL corrections entering the resummation of one-jettiness in Z +jet production. In the following sections, these results are validated against two independent calculations: the N -jettiness subtraction framework of Ref. [141] in Sec. 5.6, and the Lund-tree shape resummation of Ref. [79] in Sec. 5.7.

5.6 Comparison with N -jettiness

Having derived the NNLL cumulant for hadronic collisions, we now validate our resummation formula by comparing its logarithmic expansion with the corresponding results obtained in the SCET-based N -jettiness subtraction framework of Ref. [141].

Expanding the master formula, Σ_{ij} , in Eq. (5.10), order by order in α_s and L , we write [42]

$$\Sigma_{ij}(v) = \sum_{n=1}^{\infty} \sum_{m=0}^{n+1} h_{nm} \bar{\alpha}_s^n L^m, \quad (5.88)$$

where $\bar{\alpha}_s \equiv \alpha_s(Q)/(2\pi)$. The coefficients h_{nm} are given explicitly for one-jettiness in Appendix D.

These expansion coefficients are compared with the subtraction coefficients, $\mathcal{C}_n^{(m)}$, appearing in the logarithmic expansion of the SCET-based N -jettiness subtraction formula. Agreement between the two calculations provides an independent check of the NNLL resummation derived in this chapter, and in particular of the treatment of initial-state radiation.

The subtraction coefficients, $\mathcal{C}_n^{(m)}(\{p\}, M, \mu_R)$, are dimensionless functions of the Born momenta, the invariant mass of the Z boson, and the renormalisation scale. They are constructed from the hard, beam, jet, and soft function coefficients, collectively referred to as the subtraction ingredients.

To facilitate the comparison, we adapt the notation of Ref. [141] to the conventions used throughout this thesis. The incoming partons on legs $\ell = 1$ and $\ell = 2$ are denoted by κ_1 and κ_2 , with flavours i and j , respectively, while the outgoing parton on leg $\ell = 3$ is labelled by κ_3 . The beam-function coefficients $B_{\kappa_1,n}^{(m)}(x_1, \mu_R, \lambda_F, \lambda_1)$ and $B_{\kappa_2,n}^{(m)}(x_2, \mu_R, \lambda_F, \lambda_2)$ describe collinear radiation from the incoming partons with momentum fractions x_1 and x_2 . The jet-function coefficient $J_{\kappa_3,n}^{(m)}(\lambda_3)$ describes collinear radiation from the outgoing parton, while the soft-function coefficient $\widehat{S}_{\kappa,n}^{(m)}(\{\hat{p}_\ell\}, \lambda_s)$, with $\kappa \equiv \{\kappa_1, \kappa_2, \kappa_3\}$, describes soft radiation between the coloured legs and depends on the normalised momenta $\hat{p}_\ell = p_\ell/Q_\ell$.

The dimensionless variables entering the subtraction ingredients are

$$\lambda_1 = \frac{Q_1 M}{\mu_R}, \quad \lambda_2 = \frac{Q_2 M}{\mu_R}, \quad \lambda_3 = \frac{Q_3 M}{\mu_R^2}, \quad \lambda_s = \frac{M}{\mu_R}, \quad \lambda_F = \frac{\mu_R^2}{\mu_F^2}, \quad (5.89)$$

while their explicit expressions are collected in Appendix D.2.

For later convenience, we also adopt the notation of Ref. [141] and expand each subtraction ingredient in logarithms of its corresponding λ variable. For example, the jet function is written as

$$J_{\kappa_3,n}^{(m)}(\lambda_3) = J_{\kappa_3,n}^{(m)} + \sum_{r=1}^{2m-1-n} \frac{(n+r)!}{n! r!} J_{\kappa_3,n+r}^{(m)} \ln^r \lambda_3, \quad (5.90a)$$

$$J_{\kappa_3,-1}^{(m)}(\lambda_3) = J_{\kappa_3,-1}^{(m)} + \sum_{n=0}^{2m-1} J_{\kappa_3,n}^{(m)} \frac{\ln^{n+1} \lambda_3}{n+1}, \quad (5.90b)$$

with analogous expansions holding for the beam and soft functions.

To relate the two sets of coefficients, we note that the ARES coefficients are obtained from the integrated cumulant, whereas the subtraction coefficients are defined for the differential distribution. Accounting for this difference and for the different normalisations of the strong coupling, 2π versus 4π , gives

$$h_{mn} = \frac{C_{n-1}^{(m)}}{n} \frac{(-1)^n}{2m}. \quad (5.91)$$

For $C_{-1}^{(m)}$, the denominator is taken to be $n = 1$.

For brevity, we suppress the arguments of the subtraction coefficients throughout the remainder of this section. We also partially suppress the arguments of the beam functions $B_{\kappa_1,n}^{(m)}(x_1, \lambda_1)$ and $B_{\kappa_2,n}^{(m)}(x_2, \lambda_2)$, as well as the soft functions $\widehat{S}_{\kappa,n}^{(m)}(\lambda_s)$.

5.6.1 Leading-logarithmic coefficients

At LL accuracy, the first subtraction coefficient is

$$C_1^{(1)} = \frac{B_{\kappa_1,1}^{(1)}(x_1, \lambda_1)}{f_i(x_1, \mu_F)} + \frac{B_{\kappa_2,1}^{(1)}(x_2, \lambda_2)}{f_j(x_2, \mu_F)} + J_{\kappa_3,1}^{(1)}(\lambda_3) + \widehat{S}_{\kappa,1}^{(1)}(\lambda_s). \quad (5.92)$$

Using the explicit expressions for the individual ingredients in Eq. (D.4), this reduces to

$$C_1^{(1)} = -4 \sum_{\ell} C_{\ell}. \quad (5.93)$$

Applying the relation between the two sets of coefficients in Eq. (5.91), we obtain

$$h_{12} = - \sum_{\ell} C_{\ell}, \quad (5.94)$$

in agreement with the corresponding ARES coefficient in Eq. (D.1a).

The second subtraction coefficient contributing at LL accuracy is

$$\begin{aligned} \mathcal{C}_3^{(2)} = & \frac{B_{\kappa_1,3}^{(2)}(x_1, \lambda_1)}{f_i(x_1, \mu_F)} + \frac{B_{\kappa_2,3}^{(2)}(x_2, \lambda_2)}{f_j(x_2, \mu_F)} + J_{\kappa_3,3}^{(2)}(\lambda_3) + \widehat{S}_{\kappa,3}^{(2)}(\lambda_s) \\ & + \left(\frac{B_{\kappa_1,1}^{(1)}(x_1, \lambda_1)}{f_i(x_1, \mu_F)} + \frac{B_{\kappa_2,1}^{(1)}(x_2, \lambda_2)}{f_j(x_2, \mu_F)} \right) \left(J_{\kappa_1,1}^{(1)}(\lambda_3) + \widehat{S}_{\kappa,1}^{(1)}(\lambda_s) \right) + \frac{B_{\kappa_1,1}^{(1)}(x_1, \lambda_1)}{f_i(x_1, \mu_F)} \frac{B_{\kappa_2,1}^{(1)}(x_2, \lambda_2)}{f_j(x_2, \mu_F)}. \end{aligned} \quad (5.95)$$

Substituting the explicit expressions from Eqs. (D.4) and (D.5), this reduces to

$$\mathcal{C}_3^{(2)} = 8 \left(\sum_{\ell} C_{\ell} \right)^2. \quad (5.96)$$

Using Eq. (5.91), we then obtain

$$h_{24} = \frac{1}{2} \left(\sum_{\ell} C_{\ell} \right)^2, \quad (5.97)$$

again in agreement with the corresponding ARES expansion coefficient given in Eq. (D.1b).

5.6.2 Next-to-leading-logarithmic coefficients

The first subtraction coefficient contributing at NLL accuracy is

$$\mathcal{C}_0^{(1)} = \frac{B_{\kappa_1,0}^{(1)}(x_1, \lambda_1)}{f_i(x_1, \mu_F)} + \frac{B_{\kappa_2,0}^{(1)}(x_2, \lambda_2)}{f_j(x_2, \mu_F)} + J_{\kappa_3,0}^{(1)}(\lambda_3) + \widehat{S}_{\kappa,0}^{(1)}(\lambda_s). \quad (5.98)$$

Using the explicit expressions for the subtraction ingredients given in Eqs. (D.4) and (D.6), this becomes

$$\begin{aligned} \mathcal{C}_0^{(1)} = & -2 \sum_{\ell} \gamma_{\ell}^{(0)} + \sum_k \int \frac{dz}{z} 2P_{ik}(z) \frac{f_k(x_1/z, \mu_F)}{f_i(x_1, \mu_F)} + \sum_k \int \frac{dz}{z} 2P_{jk}(z) \frac{f_k(x_2/z, \mu_F)}{f_j(x_2, \mu_F)} \\ & + 4C_1 \ln \frac{Q_1 M}{\mu_R^2} + 4C_2 \ln \frac{Q_2 M}{\mu_R^2} + 4C_3 \ln \frac{Q_3 M}{\mu_R^2} \\ & - 4 \left[(C_1 - C_2 - C_3) \ln \frac{Q_{23}^2}{Q_2 Q_3} + (C_2 - C_1 - C_3) \ln \frac{Q_{13}^2}{Q_1 Q_3} + (C_3 - C_1 - C_2) \ln \frac{Q_{12}^2}{Q_1 Q_2} \right] \\ & - 8 \ln \frac{M}{\mu_R} \sum_{\ell} C_{\ell}, \end{aligned} \quad (5.99)$$

which can be simplified to

$$\begin{aligned} \mathcal{C}_0^{(1)} = & -2 \sum_{\ell} \gamma_{\ell}^{(0)} + \sum_k \int \frac{dz}{z} 2P_{ik}(z) \frac{f_k(x_1/z, \mu_F)}{f_i(x_1, \mu_F)} + \sum_k \int \frac{dz}{z} 2P_{jk}(z) \frac{f_k(x_2/z, \mu_F)}{f_j(x_2, \mu_F)} \\ & + 4C_1 \ln \frac{Q_{12}^2 Q_{13}^2}{Q_{23}^2 M Q_1} + 4C_2 \ln \frac{Q_{12}^2 Q_{23}^2}{Q_{13}^2 M Q_2} + 4C_3 \ln \frac{Q_{13}^2 Q_{23}^2}{Q_{12}^2 M Q_3}. \end{aligned}$$

(5.100)

Using Eq. (5.91), we obtain the corresponding ARES expansion coefficient

$$h_{11} = \sum_{\ell} \gamma_{\ell}^{(0)} - \sum_k \int \frac{dz}{z} P_{ik}(z) \frac{f_k(x_1/z, \mu_F)}{f_i(x_1, \mu_F)} - \sum_k \int \frac{dz}{z} P_{jk}(z) \frac{f_k(x_2/z, \mu_F)}{f_j(x_2, \mu_F)} - 2C_1 \ln \frac{Q_{12}^2 Q_{13}^2}{Q_{23}^2 M Q_1} - 2C_2 \ln \frac{Q_{12}^2 Q_{23}^2}{Q_{13}^2 M Q_2} - 2C_3 \ln \frac{Q_{13}^2 Q_{23}^2}{Q_{12}^2 M Q_3}, \quad (5.101)$$

which agrees with the ARES result in Eq. (D.2a).

The remaining subtraction coefficient entering at NLL accuracy is

$$\begin{aligned} C_2^{(2)} = & \frac{B_{\kappa_1,2}^{(2)}(x_1, \lambda_1)}{f_i(x_1, \mu_F)} + \frac{B_{\kappa_2,2}^{(2)}(x_2, \lambda_2)}{f_j(x_2, \mu_F)} + J_{\kappa_3,2}^{(2)}(\lambda_3) + \widehat{S}_{\kappa,2}^{(2)}(\lambda_s) \\ & + \frac{3}{2} \left[\frac{B_{\kappa_1,0}^{(1)}(x_1, \lambda_1)}{f_i(x_1, \mu_F)} \left(\frac{B_{\kappa_2,1}^{(1)}(x_2, \lambda_2)}{f_j(x_2, \mu_F)} + J_{\kappa_3,1}^{(1)}(\lambda_3) + \widehat{S}_{\kappa,1}^{(1)}(\lambda_s) \right) \right. \\ & + \frac{B_{\kappa_2,0}^{(1)}(x_2, \lambda_2)}{f_j(x_2, \mu_F)} \left(\frac{B_{\kappa_1,1}^{(1)}(x_1, \lambda_1)}{f_i(x_1, \mu_F)} + J_{\kappa_3,1}^{(1)}(\lambda_3) + \widehat{S}_{\kappa,1}^{(1)}(\lambda_s) \right) \\ & + J_{\kappa_3,0}^{(1)}(\lambda_3) \left(\frac{B_{\kappa_1,1}^{(1)}(x_1, \lambda_1)}{f_i(x_1, \mu_F)} + \frac{B_{\kappa_2,1}^{(1)}(x_2, \lambda_2)}{f_j(x_2, \mu_F)} + \widehat{S}_{\kappa,1}^{(1)}(\lambda_s) \right) \\ & \left. + \widehat{S}_{\kappa,0}^{(1)}(\lambda_s) \left(\frac{B_{\kappa_1,1}^{(1)}(x_1, \lambda_1)}{f_i(x_1, \mu_F)} + \frac{B_{\kappa_2,1}^{(1)}(x_2, \lambda_2)}{f_j(x_2, \mu_F)} + J_{\kappa_3,1}^{(1)}(\lambda_3) \right) \right]. \quad (5.102) \end{aligned}$$

Substituting the explicit expressions for the ingredients from Eqs. (D.6), (D.4), and (D.7), the expression reduces to

$$C_2^{(2)} = 24\pi\beta_0 \sum_{\ell} C_{\ell} + 12 \left(-\frac{C_0^{(1)}}{2} \right) \sum_{\ell} C_{\ell}, \quad (5.103)$$

where $C_0^{(1)}$ is given in Eq. (5.100).

Applying Eq. (5.91), this gives

$$h_{23} = -2\pi\beta_0 \sum_{\ell} C_{\ell} - h_{11} \sum_{\ell} C_{\ell}, \quad (5.104)$$

which agrees with the corresponding ARES expansion coefficient given in Eq. (D.2b).

5.6.3 Next-to-next-to-leading-logarithmic coefficients

The first subtraction coefficient entering at NNLL accuracy is $C_{-1}^{(1)}$, defined as

$$C_{-1}^{(1)} = H_{\text{SCET}}^{(1)} + \frac{B_{\kappa_1,-1}^{(1)}(x_1, \lambda_1)}{f_i(x_1, \mu_F)} + \frac{B_{\kappa_2,-1}^{(1)}(x_2, \lambda_2)}{f_j(x_2, \mu_F)} + J_{\kappa_3,-1}^{(1)}(\lambda_c) + \widehat{S}_{\kappa,-1}^{(1)}(\lambda_s), \quad (5.105)$$

where the individual subtraction ingredients are given in Eqs. (D.8), (D.6), and (D.4). The hard function is given by

$$H_{\text{SCET}}^{(1)} \equiv \frac{\left(\vec{C}^{\dagger(1)}\vec{C}^{(0)} + \vec{C}^{\dagger(0)}\vec{C}^{(1)}\right)}{|\vec{C}^{(0)}|^2}, \quad (5.106)$$

where $\vec{C}^{(0)}$ and $\vec{C}^{(1)}$ are the tree-level and one-loop hard Wilson coefficients, obtained by matching the QCD amplitude onto the corresponding SCET operator basis.

Both $H_{\text{SCET}}^{(1)}$ and the ARES hard coefficient $H_{ij}^{(1)}$ require the corresponding one-loop amplitude. Their relation, however, follows from the universal infrared structure of one-loop amplitudes, which depends only on the colour charges and directions of the external legs. Since the partonic channels of $Z + \text{jet}$ and $H + \text{jet}$ production have the same coloured-parton configurations, we may use the one-loop amplitude for $hh \rightarrow H + \text{jet}$ given explicitly in Ref. [148]. The extraction is given for the $q\bar{q}$ channel in Appendix E, with the remaining channels following analogously.

Using Eq. (E.2) together with the subtraction ingredients listed in Eq. (D.8), we obtain for the $q\bar{q}$ channel

$$\begin{aligned} \mathcal{C}_{-1}^{(1)} = & H_{\text{SCET}}^{(1)} + \sum_k \int \frac{dz}{z} \frac{f_k(x_1/z, \mu_F)}{f_i(x_1, \mu_F)} \left(2I_{qk}^{(1)}(z) + 2P_{qk}(z) \ln \frac{Q_1 M}{\mu_F^2} \right) \\ & + \sum_k \int \frac{dz}{z} \frac{f_k(x_2/z, \mu_F)}{f_j(x_2, \mu_F)} \left(2I_{\bar{q}k}^{(1)}(z) + 2P_{\bar{q}k}(z) \ln \frac{Q_2 M}{\mu_F^2} \right) \\ & - 3C_F \ln \frac{Q_1 M}{\mu_R^2} + 2C_F \ln^2 \frac{Q_1 M}{\mu_R^2} - 3C_F \ln \frac{Q_2 M}{\mu_R^2} + 2C_F \ln^2 \frac{Q_2 M}{\mu_R^2} \\ & + C_A \left(\frac{4}{3} - \pi^2 \right) + \frac{20}{3} \pi \beta_0 - 4\pi \beta_0 \ln \frac{Q_3 M}{\mu_R^2} + 2C_A \ln^2 \frac{Q_3 M}{\mu_R^2} \\ & + (C_A - 2C_F) \left[\ln^2 \frac{Q_{12}^2}{Q_1 Q_2} - \frac{\pi^2}{6} + 2(I_{12,3} + I_{21,3}) \right] \\ & - C_A \left[\ln^2 \frac{Q_{13}^2}{Q_1 Q_3} - \frac{\pi^2}{6} + 2(I_{13,2} + I_{31,2}) \right] \\ & - C_A \left[\ln^2 \frac{Q_{23}^2}{Q_2 Q_3} - \frac{\pi^2}{6} + 2(I_{23,1} + I_{32,1}) \right] \\ & - 4 \left[(C_A - 2C_F) \ln \frac{Q_{12}^2}{Q_1 Q_2} - C_A \ln \frac{Q_{13}^2}{Q_1 Q_3} - C_A \ln \frac{Q_{23}^2}{Q_2 Q_3} \right] \ln \frac{M}{\mu_R} \\ & - 4(2C_F + C_A) \ln^2 \frac{M}{\mu_R}. \end{aligned} \quad (5.107)$$

The corresponding ARES expansion coefficient h_{10} , given in Eq. (D.3a), is

$$\begin{aligned}
h_{10} = & H_{q\bar{q}}^{(1)} + \sum_k \int \frac{dz}{z} \frac{f_k(x_1/z, \mu_F)}{f_i(x_1, \mu_F)} \left(I_{qk}^{(1)}(z) + P_{qk}(z) \ln \frac{Q_1 M}{\mu_F^2} \right) \\
& + \sum_k \int \frac{dz}{z} \frac{f_k(x_2/z, \mu_F)}{f_j(x_2, \mu_F)} \left(I_{\bar{q}k}^{(1)}(z) + P_{\bar{q}k}(z) \ln \frac{Q_2 M}{\mu_F^2} \right) \\
& - \frac{3}{2} C_F \ln \frac{Q_1 M}{Q^2} - \frac{3}{2} C_F \ln \frac{Q_2 M}{Q^2} - 2\pi\beta_0 \ln \frac{M Q_3}{Q^2} \\
& - \frac{2C_F - C_A}{2} \left(\ln^2 \frac{Q_{12}^2}{Q_1 M} + \ln^2 \frac{Q_{12}^2}{Q_2 M} \right) - \frac{C_A}{2} \left(\ln^2 \frac{Q_{13}^2}{Q_1 M} + \ln^2 \frac{Q_{13}^2}{Q_3 M} \right) \quad (5.108) \\
& - \frac{C_A}{2} \left(\ln^2 \frac{Q_{23}^2}{Q_2 M} + \ln^2 \frac{Q_{23}^2}{Q_3 M} \right) + \frac{2C_F - C_A}{2} \left[\ln^2 \frac{Q_2}{Q_1} - 2(I_{12,3} + I_{21,3}) \right] \\
& + \frac{C_A}{2} \left[\ln^2 \frac{Q_3}{Q_1} - 2(I_{13,2} + I_{31,2}) \right] + \frac{C_A}{2} \left[\ln^2 \frac{Q_3}{Q_2} - 2(I_{23,1} + I_{32,1}) \right] \\
& + \frac{67C_A - 20T_R n_f}{18} - \frac{\pi^2}{3} (C_F + C_A),
\end{aligned}$$

where we have used the relation

$$I_{ik}^{(1)}(z) = -P_{ik}^{(\varepsilon)}(z) + \hat{P}_{ik}(z; 0) \left[\left(\frac{\ln(1-z)}{1-z} \right)_+ (1-z) - \ln z \right] + \frac{\pi^2}{6} \delta_{ik} C_i \delta(1-z), \quad (5.109)$$

to express the SCET matching functions $I_{qk}^{(1)}(z)$ and $I_{\bar{q}k}^{(1)}(z)$ in terms of the corresponding ARES ingredients.

The remaining comparison between these coefficients involves a lengthy rearrangement of the logarithmic and finite terms and is therefore not shown explicitly here. Instead, the symbolic implementation of this comparison is provided in the MATHEMATICA notebook of Ref. [149], where it is verified that $\mathcal{C}_{-1}^{(1)} = 2h_{10}$, as required by Eq. (5.91).

Finally, we consider the remaining subtraction coefficients contributing at NNLL accuracy, $\mathcal{C}_1^{(2)}$ and $\mathcal{C}_0^{(2)}$, which correspond to the ARES expansion coefficients h_{22} and h_{21} , respectively³.

³ Note that h_{22} formally enters at NLL' accuracy.

Their explicit expressions are the following:⁴

$$\begin{aligned}
C_1^{(2)} = & \frac{B_{\kappa_1,1}^{(2)}(x_1, \lambda_1)}{f_i(x_1, \mu_F)} + \frac{B_{\kappa_2,1}^{(2)}(x_2, \lambda_2)}{f_j(x_2, \mu_F)} + J_{\kappa_3,1}^{(2)}(\lambda_3) + \widehat{S}_{\kappa,1}^{(2)}(\lambda_s) \\
& + H_{\text{SCET}}^{(1)} \left(\frac{B_{\kappa_1,1}^{(1)}(x_1, \lambda_1)}{f_i(x_1, \mu_F)} + \frac{B_{\kappa_2,1}^{(1)}(x_2, \lambda_2)}{f_j(x_2, \mu_F)} + J_{\kappa_3,1}^{(1)}(\lambda_3) + \widehat{S}_{\kappa,1}^{(1)}(\lambda_s) \right) \\
& + 2 \left[\left(\frac{B_{\kappa_1,0}^{(1)}(x_1, \lambda_1)}{f_i(x_1, \mu_F)} + \frac{B_{\kappa_2,0}^{(1)}(x_2, \lambda_2)}{f_j(x_2, \mu_F)} \right) \left(J_{\kappa_3,0}^{(1)}(\lambda_3) + \widehat{S}_{\kappa,0}^{(1)}(\lambda_s) \right) \right. \\
& \quad \left. + \frac{B_{\kappa_1,0}^{(1)}(x_1, \lambda_1)}{f_i(x_1, \mu_F)} \frac{B_{\kappa_2,0}^{(1)}(x_2, \lambda_2)}{f_j(x_2, \mu_F)} + J_{\kappa_3,0}^{(1)}(\lambda_3) \widehat{S}_{\kappa,0}^{(1)}(\lambda_s) \right] \\
& - \frac{\pi^2}{3} \left[\left(\frac{B_{\kappa_1,1}^{(1)}(x_1, \lambda_1)}{f_i(x_1, \mu_F)} + \frac{B_{\kappa_2,1}^{(1)}(x_2, \lambda_2)}{f_j(x_2, \mu_F)} \right) \left(J_{\kappa_3,1}^{(1)}(\lambda_3) + \widehat{S}_{\kappa,1}^{(1)}(\lambda_s) \right) \right. \\
& \quad \left. + \frac{B_{\kappa_1,1}^{(1)}(x_1, \lambda_1)}{f_i(x_1, \mu_F)} \frac{B_{\kappa_2,1}^{(1)}(x_2, \lambda_2)}{f_j(x_2, \mu_F)} + J_{\kappa_3,1}^{(1)}(\lambda_3) \widehat{S}_{\kappa,1}^{(1)}(\lambda_s) \right] \\
& + \frac{B_{\kappa_1,-1}^{(1)}(x_1, \lambda_1)}{f_i(x_1, \mu_F)} \left(\frac{B_{\kappa_2,1}^{(1)}(x_2, \lambda_2)}{f_j(x_2, \mu_F)} + J_{\kappa_3,1}^{(1)}(\lambda_3) + \widehat{S}_{\kappa,1}^{(1)}(\lambda_s) \right) \\
& + \frac{B_{\kappa_2,-1}^{(1)}(x_2, \lambda_2)}{f_j(x_2, \mu_F)} \left(\frac{B_{\kappa_1,1}^{(1)}(x_1, \lambda_1)}{f_i(x_1, \mu_F)} + J_{\kappa_3,1}^{(1)}(\lambda_3) + \widehat{S}_{\kappa,1}^{(1)}(\lambda_s) \right) \\
& + J_{\kappa_3,-1}^{(1)}(\lambda_3) \left(\frac{B_{\kappa_1,1}^{(1)}(x_1, \lambda_1)}{f_i(x_1, \mu_F)} + \frac{B_{\kappa_2,1}^{(1)}(x_2, \lambda_2)}{f_j(x_2, \mu_F)} + \widehat{S}_{\kappa,1}^{(1)}(\lambda_s) \right) \\
& + \widehat{S}_{\kappa,-1}^{(1)}(\lambda_s) \left(\frac{B_{\kappa_1,1}^{(1)}(x_1, \lambda_1)}{f_i(x_1, \mu_F)} + \frac{B_{\kappa_2,1}^{(1)}(x_2, \lambda_2)}{f_j(x_2, \mu_F)} + J_{\kappa_3,1}^{(1)}(\lambda_3) \right),
\end{aligned} \tag{5.110}$$

⁴ The coefficients as printed in Ref. [141] omit the final four lines of both Eqs. (5.110) and (5.111), which appears to be a misprint since these terms are required for the coefficients to be complete.

and

$$\begin{aligned}
C_0^{(2)} = & \frac{B_{\kappa_1,0}^{(2)}(x_1, \lambda_1)}{f_i(x_1, \mu_F)} + \frac{B_{\kappa_2,0}^{(2)}(x_2, \lambda_2)}{f_j(x_2, \mu_F)} + J_{\kappa_3,0}^{(2)}(\lambda_3) + \widehat{S}_{\kappa,0}^{(2)}(\lambda_s) \\
& + H_{\text{SCET}}^{(1)} \left(\frac{B_{\kappa_1,0}^{(1)}(x_1, \lambda_1)}{f_i(x_1, \mu_F)} + \frac{B_{\kappa_2,0}^{(1)}(x_2, \lambda_2)}{f_j(x_2, \mu_F)} + J_{\kappa_3,0}^{(1)}(\lambda_3) + \widehat{S}_{\kappa,0}^{(1)}(\lambda_s) \right) \\
& - \frac{\pi^2}{6} \left[\frac{B_{\kappa_1,0}^{(1)}(x_1, \lambda_1)}{f_i(x_1, \mu_F)} \left(\frac{B_{\kappa_2,1}^{(1)}(x_2, \lambda_2)}{f_j(x_2, \mu_F)} + J_{\kappa_3,1}^{(1)}(\lambda_3) + \widehat{S}_{\kappa,1}^{(1)}(\lambda_s) \right) \right. \\
& + \frac{B_{\kappa_2,0}^{(1)}(x_2, \lambda_2)}{f_j(x_2, \mu_F)} \left(\frac{B_{\kappa_1,1}^{(1)}(x_1, \lambda_1)}{f_i(x_1, \mu_F)} + J_{\kappa_3,1}^{(1)}(\lambda_3) + \widehat{S}_{\kappa,1}^{(1)}(\lambda_s) \right) \\
& + J_{\kappa_3,0}^{(1)}(\lambda_3) \left(\frac{B_{\kappa_1,1}^{(1)}(x_1, \lambda_1)}{f_i(x_1, \mu_F)} + \frac{B_{\kappa_2,1}^{(1)}(x_2, \lambda_2)}{f_j(x_2, \mu_F)} + \widehat{S}_{\kappa,1}^{(1)}(\lambda_s) \right) \\
& \left. + \widehat{S}_{\kappa,0}^{(1)}(\lambda_s) \left(\frac{B_{\kappa_1,1}^{(1)}(x_1, \lambda_1)}{f_i(x_1, \mu_F)} + \frac{B_{\kappa_2,1}^{(1)}(x_2, \lambda_2)}{f_j(x_2, \mu_F)} + J_{\kappa_3,1}^{(1)}(\lambda_3) \right) \right] \\
& + 2\zeta_3 \left[\left(\frac{B_{\kappa_1,1}^{(1)}(x_1, \lambda_1)}{f_i(x_1, \mu_F)} + \frac{B_{\kappa_2,1}^{(1)}(x_2, \lambda_2)}{f_j(x_2, \mu_F)} \right) \left(J_{\kappa_3,1}^{(1)}(\lambda_3) + \widehat{S}_{\kappa,1}^{(1)}(\lambda_s) \right) \right. \\
& \left. + \frac{B_{\kappa_1,1}^{(1)}(x_1, \lambda_1)}{f_i(x_1, \mu_F)} \frac{B_{\kappa_2,1}^{(1)}(x_2, \lambda_2)}{f_j(x_2, \mu_F)} + J_{\kappa_3,1}^{(1)}(\lambda_3) \widehat{S}_{\kappa,1}^{(1)}(\lambda_s) \right] \\
& + \frac{B_{\kappa_1,-1}^{(1)}(x_1, \lambda_1)}{f_i(x_1, \mu_F)} \left(\frac{B_{\kappa_2,0}^{(1)}(x_2, \lambda_2)}{f_j(x_2, \mu_F)} + J_{\kappa_3,0}^{(1)}(\lambda_3) + \widehat{S}_{\kappa,0}^{(1)}(\lambda_s) \right) \\
& + \frac{B_{\kappa_2,-1}^{(1)}(x_2, \lambda_2)}{f_j(x_2, \mu_F)} \left(\frac{B_{\kappa_1,0}^{(1)}(x_1, \lambda_1)}{f_i(x_1, \mu_F)} + J_{\kappa_3,0}^{(1)}(\lambda_3) + \widehat{S}_{\kappa,0}^{(1)}(\lambda_s) \right) \\
& + J_{\kappa_3,-1}^{(1)}(\lambda_3) \left(\frac{B_{\kappa_1,0}^{(1)}(x_1, \lambda_1)}{f_i(x_1, \mu_F)} + \frac{B_{\kappa_2,0}^{(1)}(x_2, \lambda_2)}{f_j(x_2, \mu_F)} + \widehat{S}_{\kappa,0}^{(1)}(\lambda_s) \right) \\
& + \widehat{S}_{\kappa,-1}^{(1)}(\lambda_s) \left(\frac{B_{\kappa_1,0}^{(1)}(x_1, \lambda_1)}{f_i(x_1, \mu_F)} + \frac{B_{\kappa_2,0}^{(1)}(x_2, \lambda_2)}{f_j(x_2, \mu_F)} + J_{\kappa_3,0}^{(1)}(\lambda_3) \right). \tag{5.111}
\end{aligned}$$

These expressions are considerably more involved than those encountered at LL and NLL accuracy, so we perform the comparison with the ARES expansion coefficients symbolically using MATHEMATICA. Details of the implementation are provided in Appendix F, while the complete notebook is available in Ref. [149].

In the next section, we provide an alternative analytic validation of the treatment of initial-state radiation.

5.7 Validation: Lund-tree observables

In this section we validate the ISR hard-collinear corrections derived in Sec. 5.4 by comparing the ARES master formula in Eq. (5.10) with the NNLL resummation of Ref. [79] for a class of observables known as Lund-tree Shapes (LTS). This provides a particularly clean test of the ISR treatment, since the hadron–hadron results of Ref. [79] are restricted to colour-singlet production without any final-state jets ($N = 0$). In this configuration, the only coloured legs are the two incoming partons, and so the comparison directly tests the treatment of initial-state radiation.

Restricting the ARES master formula in Eq. (5.10) to the same $N = 0$ configuration leaves only the two incoming legs $\ell = 1, 2$, with the only Born channel being $q\bar{q} \rightarrow Z$. The NNLL cumulant becomes

$$\begin{aligned} \Sigma_{q\bar{q}}(\{p\}, v) &= e^{-R_{\text{NNLL}}(v)} \frac{f_q(x_1, v^{\frac{1}{a+b_1}} Q)}{f_q(x_1, \mu_F)} \frac{f_{\bar{q}}(x_2, v^{\frac{1}{a+b_2}} Q)}{f_{\bar{q}}(x_2, \mu_F)} \times \\ &\times \left(\mathcal{F}_{\text{NLL}}^{(q\bar{q})}(\lambda) \left(1 + \frac{\alpha_s(Q)}{2\pi} H_{q\bar{q}}^{(1)}(Q, \mu_R) + \sum_{\ell} \frac{\alpha_s(Q v^{\frac{1}{a+b_\ell}})}{2\pi} C_{\text{hc}, \ell}^{(1)} \right) + \frac{\alpha_s(Q)}{\pi} \delta \mathcal{F}_{\text{NNLL}}^{(q\bar{q})}(\lambda) \right), \end{aligned} \quad (5.112)$$

where

$$\delta \mathcal{F}_{\text{NNLL}}^{(q\bar{q})} = \delta \mathcal{F}_{\text{sc}} + \delta \mathcal{F}_{\text{hc}} + \delta \mathcal{F}_{\text{wa}} + \delta \mathcal{F}_{\text{correl}} + \delta \mathcal{F}_{\text{rec}}. \quad (5.113)$$

The terms $\Delta \mathcal{F}_{\text{wa}}$ and $\Delta \mathcal{F}_{\text{rec}}$ are absent from Eq. (5.113), since these corrections were introduced for three-jet configurations in Ref. [46] and require the presence of a final-state coloured leg.

We restrict ourselves to the additive LTS observable $S_b^{(0)}$, for which the NNLL resummation contains a comparable ISR term.⁵ This allows a term-by-term comparison of the ISR contributions. The observable is

$$S_b^{(0)} = \sum_{j \in \mathcal{I}_0} v_b^j, \quad v_b^j = \frac{p_{t,i}}{M} e^{-b|y_i|}, \quad (5.114)$$

where $(p_{t,i}, y_i)$ are the transverse momentum and rapidity of jet i with respect to the beam axis, M is the invariant mass of the colour singlet (taken as the hard scale), and $\mathcal{I}_0 \equiv \{j_1, j_2, \dots, j_m\}$ is the list of jets from the C/A algorithm of radius R (cf. Sec. 3.3.3 for more details on this algorithm).

Setting $Q = M$ fixes the single-emission parameters to

$$a = 1, \quad b_\ell = b, \quad d_\ell = 1, \quad g_\ell(\phi) = 1, \quad \ell = 1, 2. \quad (5.115)$$

⁵ The other observable they define, $M_b^{(0)} = \max_{j \in \mathcal{I}_0} v_b^j$ is not an event shape as it is not collinear safe and so will not offer a reasonable comparison.

Using the ARES notation and setting $Q = M$, the NNLL cumulant of Ref. [79] can be written as

$$\begin{aligned} \Sigma^{\text{LTS}}(\{p\}, v) = & e^{-R_{\text{NNLL}}(v)} \frac{f_q(x_1, \mu_F v^{\frac{1}{1+b}}) f_{\bar{q}}(x_2, \mu_F v^{\frac{1}{1+b}})}{f_q(x_1, \mu_F) f_{\bar{q}}(x_2, \mu_F)} \left\{ \mathcal{F}_{\text{NLL}}(v) \left[1 + \frac{\alpha_s(\mu_R)}{2\pi} H_{q\bar{q}}^{(1)}(Q, \mu_R) + \right. \right. \\ & + \frac{\alpha_s(\mu_R)}{2\pi} \frac{1+b}{1+b-2\lambda} \sum_k \left(\int_{x_1}^1 \frac{dz}{z} C_{qk}^{(1),\text{LTS}}(z) \frac{f_k(x_1/z, \mu_F v^{\frac{1}{1+b}})}{f_q(x_1, \mu_F v^{\frac{1}{1+b}})} \right. \\ & \left. \left. + \int_{x_2}^1 \frac{dz}{z} C_{\bar{q}k}^{(1),\text{LTS}}(z) \frac{f_k(x_2/z, \mu_F v^{\frac{1}{1+b}})}{f_{\bar{q}}(x_2, \mu_F v^{\frac{1}{1+b}})} \right) \right] + \frac{\alpha_s(\mu_R)}{\pi} \delta \mathcal{F}_{\text{NNLL}}^{\text{LTS}}(v) \left. \right\} + \delta \Sigma_{\text{DGLAP}}(v), \end{aligned} \quad (5.116)$$

where

$$\delta \mathcal{F}_{\text{NNLL}}^{\text{LTS}}(v) = \delta \mathcal{F}_{\text{sc}}^{\text{LTS}}(v) + \delta \mathcal{F}_{\text{wa}}^{\text{LTS}}(v) + \delta \mathcal{F}_{\text{hc}}^{\text{LTS}}(v) + \delta \mathcal{F}_{\text{correl}}^{\text{LTS}}(v) + \delta \mathcal{F}_{\text{clust}}^{\text{LTS}}(v), \quad (5.117)$$

and

$$\begin{aligned} \delta \Sigma_{\text{DGLAP}}(v) = & e^{-R_{\text{NNLL}}(v)} \frac{\alpha_s(\mu_R)}{\pi} \frac{1}{1+b-2\lambda} \frac{d\mathcal{F}_{\text{NLL}}}{dR'_{\text{NLL}}} \\ & \times \sum_k \left(\int_{x_1}^1 \frac{dz}{z} P_{qk}(z) \frac{f_k(x_1/z, \mu_F v^{1/(1+b)})}{f_q(x_1, \mu_F v^{1/(1+b)})} + \int_{x_2}^1 \frac{dz}{z} P_{\bar{q}k}(z) \frac{f_k(x_2/z, \mu_F v^{1/(1+b)})}{f_{\bar{q}}(x_2, \mu_F v^{1/(1+b)})} \right), \end{aligned} \quad (5.118)$$

where $P_{qk}(z)$ and $P_{\bar{q}k}(z)$ are the full Altarelli–Parisi splitting functions defined in Eq. (5.59).

The two cumulants have a similar general structure, differing only in how they organise the NNLL corrections. The relation between the different contributions in the two formulations can be summarised as follows.

The radiator R_{NNLL} , hard coefficient $H_{q\bar{q}}^{(1)}$, and NLL function $\mathcal{F}_{\text{NLL}}$ are identical in the two formulations. Furthermore, the NNLL corrections of soft origin, $\delta \mathcal{F}_{\text{sc}}$, $\delta \mathcal{F}_{\text{wa}}$, and $\delta \mathcal{F}_{\text{correl}}$, are identical. As discussed in Sec. 2.5.1, these contributions depend only on the soft-emission kinematics and colour structure, and are therefore unaffected by whether the emitting leg is in the initial or final state. They are universal between ISR and FSR and receive no additional ISR-dependent contributions. These corrections have already been validated in previous ARES resummations and are treated identically in Ref. [79], so they do not provide an independent test of the ISR formalism developed here.

The additional correction $\delta \mathcal{F}_{\text{clust}}^{\text{LTS}}$ in Eq. (5.117) originates from the dependence of the observable $S_b^{(0)}$ on the jet-clustering procedure. This is a genuine feature of jet-rate observables, and the same type of correction appears in ARES jet-rate resummations [44]. Since this term is also of soft origin it is again not part of the present validation.

The non-trivial comparison therefore concerns the hard-collinear contributions associated with initial-state radiation. In ARES these are organised into $C_{\text{hc},\ell}^{(1)}$, $\delta\mathcal{F}_{\text{hc}}$, and $\delta\mathcal{F}_{\text{rec}}$, whereas Ref. [79] introduces $C_{qk}^{(1),\text{LTS}}$, $\delta\mathcal{F}_{\text{hc}}^{\text{LTS}}$, and $\delta\Sigma_{\text{DGLAP}}$. Although the two formulations organise these terms differently, we show below that they contain the same ISR hard-collinear corrections.

The coefficient function of Ref. [79] is⁶

$$\sum_j \int_{x_1}^1 \frac{dz}{z} C_{qk}^{(1),\text{LTS}}(z) \frac{f_k(x_1/z, \mu_F v^{\frac{1}{1+b}})}{f_q(x_1, \mu_F v^{\frac{1}{1+b}})} = \sum_k \int_{x_1}^1 \frac{dz}{z} \frac{f_k(x_1/z, \mu_F v^{\frac{1}{1+b}})}{f_q(x_1, \mu_F v^{\frac{1}{1+b}})} \left[-P_{qk}^{(\varepsilon)}(z) + \frac{2b}{1+b} \left(\left(\frac{\ln(1-z)}{1-z} \right)_+ - \frac{\ln z}{1-z} \right) (1-z) \hat{P}_{qk}(z; 0) + \ln \frac{M^2}{\mu_F^2} P_{qk}(z) \right], \quad (5.119)$$

and analogously for $C_{\bar{q}k}^{(1),\text{LTS}}$. The first two terms correspond to the hard-collinear coefficient $C_{\text{hc},\ell}^{(1)}$ and recoil correction $\delta\mathcal{F}_{\text{rec}}$ in ARES. Specialising Eq. (5.51) to the parameters in Eq. (5.115), with $Q = M$ and $E_\ell = M/2$, gives

$$\begin{aligned} \frac{\alpha_s(M v^{\frac{1}{1+b}})}{2\pi} \sum_{\ell=1}^2 C_{\text{hc},\ell}^{(1)} &= \\ &= \sum_k \left[\int_{x_1}^1 \frac{dz}{z} \frac{f_k(x_1/z, M v^{\frac{1}{1+b}})}{f_q(x_1, M v^{\frac{1}{1+b}})} \left(-P_{qk}^{(\varepsilon)}(z) + \frac{2b}{1+b} \left(\frac{\ln(1-z)}{1-z} \right)_+ (1-z) \hat{P}_{qk}(z; 0) \right) + \left\{ \frac{x_1 \leftrightarrow x_2}{q \leftrightarrow \bar{q}} \right\} \right]. \end{aligned} \quad (5.120)$$

Using Eq. (5.60) for additive observables with the parameters of Eq. (5.115),

$$\frac{\alpha_s(M)}{\pi} \delta\mathcal{F}_{\text{rec}}(\lambda) = \frac{\alpha_s(M v^{\frac{1}{1+b}})}{2\pi} \sum_k \left[\int_{x_1}^1 \frac{dz}{z} \frac{f_k(x_1/z, \mu_F v^{\frac{1}{1+b}})}{f_q(x_1, \mu_F v^{\frac{1}{1+b}})} \left(-\frac{2b}{1+b} \hat{P}_{qk}(z; 0) \ln z \right) + \left\{ \frac{x_1 \leftrightarrow x_2}{q \leftrightarrow \bar{q}} \right\} \right]. \quad (5.121)$$

Combining Eqs. (5.120) and (5.121) reproduces the first two terms of Eq. (5.119) exactly. The remaining term, $\ln(M^2/\mu_F^2) P_{qk}(z)$, accounts for the different PDF scales used in the two formulations: ARES evaluates the PDFs at $M v^{1/(1+b)}$, whereas Ref. [79] uses $\mu_F v^{1/(1+b)}$. It is precisely the DGLAP evolution between these scales. The prefactor $(1+b)/(1+b-2\lambda)$ in Eq. (5.116) similarly results from expressing $\alpha_s(M v^{1/(1+b)})$ in terms of $\lambda = \alpha_s(\mu_R) \beta_0 \ln(1/v)$ at one loop.

The remaining hard-collinear correction in the LTS formulation is

$$\delta\mathcal{F}_{\text{hc}}^{\text{LTS}}(v) = -\frac{d\mathcal{F}_{\text{NLL}}}{dR_{\text{NLL}}} \frac{1}{1+b-2\lambda} \sum_{\ell} \gamma_{\ell}^{(0)}. \quad (5.122)$$

Combining this with $\delta\Sigma_{\text{DGLAP}}$ in Eq. (5.118), we find that at NNLL this gives rise to:

⁶ In Ref. [79] the final term is written as $\ln(M^2/\mu_F^2) \hat{P}_{ij}(z; 0)$, which appears to be a misprint since it introduces an uncancelled collinear singularity – this has been confirmed with the authors of the paper.

$$\begin{aligned} & \frac{\alpha_s(\mu_R)}{\pi} \delta\mathcal{F}_{\text{hc}}^{\text{LTS}}(v) + \delta\Sigma_{\text{DGLAP}} \simeq \\ & \frac{\alpha_s(\mu_R)}{\pi(1+b)} \frac{1+b}{1+b-2\lambda} \frac{d\mathcal{F}_{\text{NLL}}}{dR_{\text{NLL}}} \left[\int_{x_1}^1 \frac{dz}{z} \left(P_{qk}(z) - \gamma_1^{(0)} \delta_{qk} \right) \frac{f_k(x_1/z, \mu_F v^{1/(1+b)})}{f_q(x_1, \mu_F v^{1/(1+b)})} + \left\{ \begin{smallmatrix} x_1 \leftrightarrow x_2 \\ q \leftrightarrow \bar{q} \end{smallmatrix} \right\} \right]. \end{aligned} \quad (5.123)$$

This is equivalent to the ARES contribution $\delta\mathcal{F}_{\text{hc}}$. This can be seen explicitly by evaluating Eq. (5.57) for the additive observable,

$$\frac{\alpha_s(Q)}{\pi} \delta\mathcal{F}_{\text{hc}} = \frac{\alpha_s(Q v^{1/(1+b)})}{\pi(1+b)} \frac{d\mathcal{F}_{\text{NLL}}}{dR'_{\text{NLL}}} \sum_k \left[\int_{x_1}^1 \frac{dz}{z} \left(P_{qk}(z) - \gamma_1^{(0)} \delta_{qk} \right) \frac{f_k(x_1/z, Q v^{1/(a+b_\ell)})}{f_q(x_1, Q v^{1/(a+b_\ell)})} + \left\{ \begin{smallmatrix} x_1 \leftrightarrow x_2 \\ q \leftrightarrow \bar{q} \end{smallmatrix} \right\} \right]. \quad (5.124)$$

This is exactly the combination of the DGLAP term with $\delta\mathcal{F}_{\text{hc}}^{\text{LTS}}$. Consequently, the additional DGLAP term introduced in Ref. [79] does not represent a new physical effect, but rather a different organisation of the hard-collinear corrections. Once all terms are combined, the resummed contributions due to ISR in the two formalisms are the same.

5.8 Conclusion

In this chapter we extended the ARES formalism to NNLL resummation in hadronic collisions, applying it to $hh \rightarrow Z + \text{jet}$. The central result is that the NNLL master formula retains the same overall structure as the $e^+e^- \rightarrow 3\text{-jet}$ result of Ref. [46], while the individual ingredients do not. The principal new ingredient is the treatment of initial-state radiation, where a collinear splitting on an incoming leg convolutes with the parton distribution functions, generating contributions with no counterpart in e^+e^- annihilation.

The resulting formalism was validated using the additive observable one-jettiness, for which independent resummed results are available and therefore provide a stringent test of the resummation framework. The validation was performed at two levels. First, the LL, NLL, and NNLL expansion coefficients were compared directly with the N -jettiness SCET subtraction framework of Ref. [141], finding exact agreement order by order. Second, the full resummed cumulant, restricted to colour-singlet production was compared with the Lund-Tree Shape resummation of Ref. [79]. Although the two approaches organise their hard-collinear contributions differently, we demonstrated that they contain identical ISR corrections once all terms are combined. Together, these checks confirm the ISR hard-collinear formalism developed in this chapter.

A natural next step would be to obtain the one-loop corrections for $Z + \text{jet}$ production, available for example through MCFM [31–33] or via analytic continuation of the $e^+e^- \rightarrow q\bar{q}g$

amplitude. This would provide the process-specific hard coefficient required for phenomenological predictions, replacing the $H + \text{jet}$ coefficient employed here, and complete the ingredients needed for matching the NNLL resummation to fixed-order calculations. Further interesting observables include the additive observable one minus transverse thrust and the non-additive thrust minor, the latter an out-of-plane observable, both sensitive to the spin-correlation effects derived in appendix B. The formalism could also be extended to N³LL accuracy and to processes involving a larger number of hard coloured legs. The latter presents a significant challenge, since for $\ell \geq 4$ the wide-angle corrections become matrices in colour space, substantially increasing the complexity of the calculation. An extension to deep inelastic scattering should be comparatively straightforward, as this involves only a single incoming coloured leg. Applications to Higgs- or W -boson production would require a dedicated calculation of the corresponding process-dependent hard functions.

The broader motivation for this work is precision phenomenology at the LHC. A general NNLL resummation for hadronic collisions is particularly valuable given the wide availability of high-order fixed-order results. It supplies the resummed ingredient needed for reliable predictions across the full kinematic range, and offers a benchmark for validating parton-shower algorithms aiming at NNLL accuracy. The next steps towards phenomenological applications are to match the resummation to fixed-order calculations, include hadronisation corrections, and compare with experimental data to extract the value of the strong coupling.

Chapter 6

Conclusion

High-energy collisions are accompanied by QCD radiation, giving rise to the complex hadronic final states observed in experiments. Providing reliable theoretical predictions therefore requires an accurate description of this radiation, which presents a number of theoretical challenges. This thesis has worked towards addressing two of them — the clustering of QCD radiation into jets and the resummation of the large logarithms that arise from it.

In Chapter 3 we introduced a new family of inclusive generalised- k_t jet algorithms for Deep Inelastic Scattering, formulated in the Breit frame and presented in Ref. [1]. We described the clustering procedure and introduced an ordering variable for the reconstructed jets — the light-cone fraction z_i of Eq. (3.21) — which provides an infrared-safe prescription for identifying the final-state macrojet. We also reviewed the existing jet algorithms used in DIS, outlining their development and the ideas that led to our algorithm.

To illustrate the features of the algorithm, we performed a phenomenological study comparing the angular-ordered, k_t and anti- k_t variants with the more recently developed Centauro algorithm [72]. The comparison showed differences in the treatment of soft radiation and in the amount of struck-quark radiation retained within the final-state macrojet. We also investigated the effects of hadronisation, which vary considerably across the distributions, and identified those best suited to probing non-perturbative physics in DIS. The algorithms have been implemented as the DISGenkt plugin to FASTJET, distributed as part of `fjcontrib`.

Further work could investigate the non-perturbative effects identified in this study, for example through a re-analysis of existing HERA data. The development of the Electron–Ion Collider also provides an opportunity to apply these algorithms to future DIS measurements. A natural extension of this work is the resummation of jet rates. These observables contain large logarithms of the resolution parameter, which make fixed-order predictions unreliable in pre-

cisely the region where the clustering properties of the algorithm matter most. As discussed in Sec. 5.7, this could be achieved within the ARES framework by including the additional correction $\delta\mathcal{F}_{\text{clust}}$ and a single initial-state radiation leg.

In Chapter 5, we extended the ARES framework to hadronic collisions. We established the ingredients required for NNLL resummation in the presence of initial-state radiation and showed that the ARES master formula retains the same overall structure as in e^+e^- collisions, with the modifications associated with initial-state radiation contained in the corresponding ingredients of the formalism. We derived the treatment of initial-state radiation in detail and applied it to one-jettiness. The resulting expressions were validated against the N -jettiness SCET subtraction coefficients of Ref. [141] and, for colour-singlet production, against the Lund tree-shape resummation of Ref. [79].

The formalism established here provides the basis for NNLL resummation of a broad class of observables in hadronic collisions. The next step towards phenomenological predictions is the calculation of the process-dependent hard coefficient for $hh \rightarrow Z + \text{jet}$, which would complete the ingredients required for matching the resummed result to fixed-order calculations. Beyond $Z + \text{jet}$ production, the same formalism can be applied to other observables and processes. In particular, progress has already been made towards its application to thrust minor and one minus transverse thrust. The formalism could also be extended to higher logarithmic accuracy and to processes with more hard coloured legs, although the latter introduces additional complications through the colour structure of the wide-angle corrections.

The two areas of work in this thesis address different aspects of the same underlying problem, namely how QCD radiation is organised and how its effects can be calculated reliably. The jet-algorithm work develops new tools for reconstructing jets in DIS, while the resummation work extends the reach of NNLL predictions to hadronic collisions. Together, these results contribute to the development of reliable theoretical predictions for QCD at current and future collider experiments.

Appendix A

DISGenkt FastJet contrib

The generalised- k_t jet algorithm for Deep Inelastic Scattering presented in Chapter 3 has been implemented as the **DISGenkt** plugin to FASTJET as part of `fjcontrib`. Relevant parts of the implementation are outlined below.

The pairwise distance d_{ij} and beam distance d_{iB} , defined in Eq. (3.19), are evaluated by the `DISBriefJet` class.

```

1 double DISBriefJet::distance(const DISBriefJet * jet) const {
2     double dij = 1 - _nx*jet->_nx
3                 - _ny*jet->_ny
4                 - _nz*jet->_nz;
5     dij = dij * fmin(_E2p, jet->_E2p) / _dij_norm;
6     return dij;
7
8 double DISBriefJet::beam_distance() const {
9     return (1 - _beam_sign * _nz)*_E2p;
10 }

```

Listing A.1: Implementation of the pairwise distance and beam distance.

The clustering sequence is performed using the FASTJET nearest-neighbour heuristic (NNH) class. If the minimum is d_{ij} ($j \geq 0$), jets i and j are merged, whereas if it is d_{iB} ($j < 0$), jet i is declared final and removed from the clustering.

```

1 void DISGenktPlugin::run_clustering(
2     fastjet::ClusterSequence & cs) const {
3     int njets = cs.jets().size();
4     DISJetInfo info(p(), beam_sign(), R());
5     NNH<DISBriefJet, DISJetInfo> nnh(cs.jets(), &info);
6
7     while (njets > 0) {
8         int i, j, k;
9         double dij = nnh.dij_min(i, j);
10        if (j >= 0) {
11            cs.plugin_record_ij_recombination(i, j, dij, k);
12            nnh.merge_jets(i, j, cs.jets()[k], k);
13        } else {
14            cs.plugin_record_iB_recombination(i, dij);
15            nnh.remove_jet(i);

```

```

16     }
17     njets--;
18 }
19 }

```

Listing A.2: Main clustering loop.

The macrojet is then identified as the jet with the largest DIS longitudinal projection, $E - \text{beam_sign} \cdot p_z$.

```

1  int DISGenktPlugin::find_idx_macrojet(
2      const std::vector<PseudoJet>& jets) const {
3      double best = 0.0;
4      int idx = -1;
5      for (size_t i = 0; i < jets.size(); ++i) {
6          double proj = jets[i].E() - beam_sign() * jets[i].pz();
7          if (proj > best) {
8              best = proj;
9              idx = static_cast<int>(i);
10         }
11     }
12     return idx;
13 }

```

Listing A.3: Identification of the macrojet.

A function is also provided to sort jets by decreasing DIS longitudinal projection.

```

1  std::vector<PseudoJet> DISGenktPlugin::sorted_by_zjet(
2      const std::vector<PseudoJet> & jets) const {
3      std::vector<double> zproj(jets.size());
4      for (size_t i = 0; i < jets.size(); i++) {
5          zproj[i] = -(jets[i].E() - beam_sign() * jets[i].pz());
6      }
7      return objects_sorted_by_values(jets, zproj);
8  }

```

Listing A.4: Sorting jets by DIS longitudinal projection.

This is similar to the `sorted_by_pt(...)` function in FASTJET which sorts the jets by decreasing p_t .

Appendix B

ISR spin-correlation corrections

This appendix derives the additional NNLL correction that arises when a gluon enters the hard scattering. In addition to the ISR hard-collinear corrections $C_{\text{hc},\ell}^{(1)}$, $\delta\mathcal{F}_{\text{hc}}$, and $\delta\mathcal{F}_{\text{rec}}$ derived in Sec. 5.4, an incoming gluon gives rise to a further contribution due to spin correlations. This contribution, $\Delta\mathcal{F}_{\text{rec}}$, vanishes for one-jettiness but is required for the general ISR formalism and for observables with non-trivial azimuthal dependence.

To account for this, we introduce the spin-density matrix for the hard process in $4 - 2\varepsilon$ dimensions. Neglecting terms that are antisymmetric in μ, ν , the general form is

$$\rho^{\mu\nu} = \frac{d^{\mu\nu}}{2(1-\varepsilon)} + \mathcal{T}_\ell(\{p\}; \varepsilon) \left(2n_{\text{in}}^\mu n_{\text{in}}^\nu - \frac{d^{\mu\nu}}{1-\varepsilon} \right). \quad (\text{B.1})$$

Here $d^{\mu\nu}$ is chosen to be the sum over physical polarisations transverse to both p_1 and p_2 ,

$$d^{\mu\nu} = -g^{\mu\nu} + \frac{p_1^\mu p_2^\nu + p_2^\mu p_1^\nu}{p_1 \cdot p_2}, \quad (\text{B.2})$$

and n_{in} is a space-like unit vector that lies in the plane identified by the beam and the Z boson, satisfying $n_{\text{in}}^2 = -1$ and $n_{\text{in}} \cdot p_1 = n_{\text{in}} \cdot p_2 = 0$. The tensors above satisfy

$$d^{\mu\nu} d_{\mu\nu} = -g^{\mu\nu} d_{\mu\nu} = 2(1-\varepsilon), \quad (\text{B.3a})$$

$$d_{\mu\nu} \left(2n_{\text{in}}^\mu n_{\text{in}}^\nu - \frac{d^{\mu\nu}}{1-\varepsilon} \right) = -g^{\mu\nu} \left(2n_{\text{in}}^\mu n_{\text{in}}^\nu - \frac{d^{\mu\nu}}{1-\varepsilon} \right) = 0, \quad (\text{B.3b})$$

$$\left(2n_{\text{in}}^\mu n_{\text{in}}^\nu - \frac{d^{\mu\nu}}{1-\varepsilon} \right) \left(2n_{\text{in},\mu} n_{\text{in},\nu} - \frac{d_{\mu\nu}}{1-\varepsilon} \right) = 2 \frac{1-2\varepsilon}{1-\varepsilon}. \quad (\text{B.3c})$$

This gives

$$\mathcal{T}_\ell(\{\tilde{p}\}; \varepsilon) = \frac{1-\varepsilon}{2(1-2\varepsilon)} \left(2n_{\text{in}}^\mu n_{\text{in}}^\nu - \frac{d^{\mu\nu}}{1-\varepsilon} \right) \rho_{\mu\nu}. \quad (\text{B.4})$$

The collinear splitting probability is obtained by contracting $\rho^{\mu\nu}$ with the unaveraged splitting functions

$$\left[\hat{P}_{ga}^{(1)}(z, \vec{k}_t; \varepsilon) \right]^{\mu\nu} = -g^{\mu\nu} \hat{P}_{ga}(z; \varepsilon) + \left(\frac{g^{\mu\nu}}{1 - \varepsilon} + 2 \frac{k_\perp^\mu k_\perp^\nu}{k_t^2} \right) 2\hat{G}_{ga}(z; \varepsilon), \quad a = g, q, \quad (\text{B.5})$$

with $k_\perp \cdot k_\perp = -k_t^2$, and

$$d_{\mu\nu} \left(\frac{g^{\mu\nu}}{1 - \varepsilon} + 2 \frac{k_\perp^\mu k_\perp^\nu}{k_t^2} \right) = 0. \quad (\text{B.6})$$

The azimuthally averaged splitting functions $\hat{P}_{ga}(z; \varepsilon)$ with $a = q, g$ are those present in Eqs. (5.53c) and (5.53d) respectively. The spin-correlated functions are

$$\hat{G}_{gg}(z) = -(1 - \varepsilon) C_A \frac{1 - z}{z}, \quad \hat{G}_{gq}(z) = C_F \frac{1 - z}{z}. \quad (\text{B.7})$$

Using $k_\perp \cdot n_{\text{in}} = -k_t \cos \phi$, the contraction gives

$$\rho^{\mu\nu} \left[\hat{P}_{ga}(z, \vec{k}_t; \varepsilon) \right]_{\mu\nu} = \hat{P}_{ga}(z; \varepsilon) + \mathcal{T}_\ell(\{\tilde{p}\}; \varepsilon) \hat{G}_{ga}(z; \varepsilon) [2(1 - \varepsilon) \cos^2 \phi - 1], \quad a = g, q. \quad (\text{B.8})$$

The above equation gives the probability for a splitting $a \rightarrow ga$. The first term can be treated as in Sec. 5.4, giving the ISR hard-collinear coefficient $C_{\text{hc},\ell}^{(1)}$ and the corrections $\delta\mathcal{F}_{\text{hc}}$ and $\delta\mathcal{F}_{\text{rec}}$. The second term contains the spin-correlation dependence. Following the derivation of Ref. [46], this gives the additional NNLL correction

$$\begin{aligned} \Delta\mathcal{F}_{\text{rec}}(\lambda) = & \sum_{\ell_g} \frac{\alpha_s(Q v^{1/(a+b_{\ell_g})})}{\alpha_s(Q)(a+b_{\ell_g})} \int_0^\infty \frac{d\zeta}{\zeta} \int_0^\pi \frac{d\phi}{\pi} [2 \cos^2 \phi^{(\ell_g)} - 1] \int_0^1 dz \Delta P_g(z^{(\ell_g)}) \\ & \times \int d\mathcal{Z}[\{R'_{\text{NLL},\ell_i}, \{k_i\}\}] \left[\Theta \left(1 - \frac{V_{\text{hc}}(\{\tilde{p}\}, k, \{k_i\})}{v} \right) - \Theta(1 - \zeta) \Theta \left(1 - \frac{V_{\text{sc}}(\{\tilde{p}\}, \{k_i\})}{v} \right) \right], \end{aligned} \quad (\text{B.9})$$

where the sum runs over the gluon legs ℓ_g . For initial-state radiation,

$$\Delta P_g(z) = 4 \left(\frac{1 - z}{z} \right) \frac{\Theta(z - x_\ell)}{z f_g(x_\ell, Q v^{1/(a+b_\ell)})} \left[C_A f_g \left(\frac{x_\ell}{z}, Q v^{1/(a+b_\ell)} \right) + C_F f_q \left(\frac{x_\ell}{z}, Q v^{1/(a+b_\ell)} \right) \right], \quad (\text{B.10})$$

while for final-state radiation,

$$\Delta P_g(z) = 4z(1 - z) \left[\frac{C_A}{2} - T_R n_f \right]. \quad (\text{B.11})$$

Appendix C

Additive observables

For additive observables, the terms entering $\delta\mathcal{F}_{\text{NNLL}}$, given in Eq. (5.11), take the general form derived in [46]. They are written in terms of $\lambda \equiv \alpha_s(Q)\beta_0 \ln(1/v)$ and the NLL function $\mathcal{F}_{\text{NLL}}(\lambda)$, whose additive form is given in Eq. (4.80).

The soft-collinear correction is

$$\begin{aligned} \delta\mathcal{F}_{\text{sc}}(\lambda) = & -\mathcal{F}_{\text{NLL}}(\lambda) \frac{\pi}{\alpha_s(Q)} \sum_{\ell=1}^3 \left[\left(\Delta R'_{\ell,\text{NNLL}} + R''_{\ell,\text{NNLL}} \left(\langle \ln d_\ell g_\ell(\phi) \rangle - b_\ell \ln \frac{2E_\ell}{Q} \right) \right) \right. \\ & \times \left(\psi^{(0)} (1 + R'_{\text{NLL}}) + \gamma_E \right) + \frac{R''_{\ell,\text{NNLL}}}{2} \left(\left(\psi^{(0)} (1 + R'_{\text{NLL}}) + \gamma_E \right)^2 - \psi^{(1)} (1 + R'_{\text{NLL}}) + \frac{\pi^2}{6} \right) \Big] , \end{aligned} \quad (\text{C.1})$$

where R'_{NLL} , $R''_{\ell,\text{NNLL}}$, and $\Delta R'_{\ell,\text{NNLL}}$ are defined in Eqs. (4.58), (5.34b), and (5.67), respectively.

The correlated correction is

$$\begin{aligned} \delta\mathcal{F}_{\text{correl}}(\lambda) = & -\mathcal{F}_{\text{NLL}}(\lambda) \sum_{\ell=1}^3 \frac{\lambda R''_{\ell,\text{NNLL}}}{2a\beta_0\alpha_s(Q)} \\ & \times \int_0^{2\pi} \frac{d\phi^{(\ell)}}{2\pi} \int_0^\infty \frac{d\mu^2}{\mu^2(1+\mu^2)} \int_0^1 dz \int_0^{2\pi} \frac{d\phi}{2\pi} \frac{1}{2!} \mathcal{A}^2(z, \mu, \phi) \ln f_{\text{correl}}^{(\ell)}(z, \mu, \phi, \phi^{(\ell)}) , \end{aligned} \quad (\text{C.2})$$

where

$$\frac{V_{\text{sc}}(\{\tilde{p}\}, k_a, k_b, \{k_i\})}{v} = \zeta f_{\text{correl}}(z, \mu, \phi, \phi^{(\ell)}) + \sum_{i=1}^n \zeta_i . \quad (\text{C.3})$$

The hard-collinear contribution is

$$\delta\mathcal{F}_{\text{hc}} = -\mathcal{F}_{\text{NLL}}(\lambda) \left(\psi^{(0)} (1 + R'_{\text{NLL}}) + \gamma_E \right) \sum_{\ell} \frac{\alpha_s(Q v^{1/(a+b_\ell)})}{\alpha_s(Q) (a + b_\ell)} B_\ell , \quad (\text{C.4})$$

where $B_\ell = \gamma_\ell^{(0)}$ for final-state legs, while for initial-state legs it is defined in Eq. (5.58).

The recoil correction $\delta\mathcal{F}_{\text{rec}}$ is

$$\delta\mathcal{F}_{\text{rec}} = \mathcal{F}_{\text{NLL}}(\lambda) \sum_{\ell} \frac{\alpha_s(Q v^{1/(a+b_{\ell})})}{\alpha_s(Q)(a+b_{\ell})} \int_0^{2\pi} \frac{d\phi^{(\ell)}}{2\pi} \int_0^1 dz^{(\ell)} P_{\ell}(z) \ln \frac{f_{\text{sc}}^{(\ell)}(z^{(\ell)}, \phi^{(\ell)})}{f_{\text{hc}}^{(\ell)}(z^{(\ell)}, \phi^{(\ell)})} \quad (\text{C.5})$$

where $P_{\ell}(z)$ is given in Eqs. (5.61) and (5.62) for initial- and final-state radiation, respectively.

The observable-dependent functions are defined through

$$\frac{V_{\text{hc}}(\{\tilde{p}\}, k, \{k_i\})}{v} = \zeta f_{\text{hc}}^{(\ell)}(z^{(\ell)}, \phi^{(\ell)}) + \sum_i \zeta_i, \quad (\text{C.6a})$$

$$\frac{V_{\text{sc}}(\{\tilde{p}\}, k, \{k_i\})}{v} = \zeta f_{\text{sc}}^{(\ell)}(z^{(\ell)}, \phi^{(\ell)}) + \sum_i \zeta_i. \quad (\text{C.6b})$$

The recoil correction $\Delta\mathcal{F}_{\text{rec}}$ is

$$\begin{aligned} \Delta\mathcal{F}_{\text{rec}}(\lambda) = & -\mathcal{F}_{\text{NLL}}(\lambda) \sum_{\ell_g} \frac{\alpha_s(Q v^{1/(a+b_{\ell_g})})}{\alpha_s(Q)(a+b_{\ell_g})} \\ & \times \int_0^{\pi} \frac{d\phi^{(\ell_g)}}{\pi} \left(2 \cos^2 \phi^{(\ell_g)} - 1\right) \int_0^1 dz^{(\ell_g)} \Delta P_g(z^{(\ell_g)}) \ln f_{\text{hc}}^{(\ell_g)}(z^{(\ell_g)}, \phi^{(\ell_g)}), \end{aligned} \quad (\text{C.7})$$

where the sum runs over the gluon legs ℓ_g . The functions $\Delta P_g(z)$ are given by Eq. (B.10) for initial-state radiation and Eq. (B.11) for final-state radiation.

The wide-angle correction $\delta\mathcal{F}_{\text{wa}}(\lambda)$ is

$$\delta\mathcal{F}_{\text{wa}}(\lambda) = \mathcal{F}_{\text{NLL}}(\lambda) \sum_{(\ell\bar{\ell})} \frac{2C_{\ell\bar{\ell}}}{a} \frac{\alpha_s(v^{1/a}Q)}{\alpha_s(Q)} \int_{-\infty}^{\infty} d\eta^{(\ell\bar{\ell})} \int_0^{2\pi} \frac{d\phi^{(\ell\bar{\ell})}}{2\pi} \ln \frac{f_{\text{sc}}^{(\ell\bar{\ell})}(\eta^{(\ell\bar{\ell})}, \phi^{(\ell\bar{\ell})})}{f_{\text{wa}}^{(\ell\bar{\ell})}(\eta^{(\ell\bar{\ell})}, \phi^{(\ell\bar{\ell})})}, \quad (\text{C.8})$$

where

$$\frac{V_{\text{wa}}(\{\tilde{p}\}, k^{(\ell\bar{\ell})}, \{k_i\})}{v} = \zeta f_{\text{wa}}^{(\ell\bar{\ell})}(\eta^{(\ell\bar{\ell})}, \phi^{(\ell\bar{\ell})}) + \sum_i \zeta_i. \quad (\text{C.9})$$

Finally, the wide-angle correction $\Delta\mathcal{F}_{\text{wa}}(\lambda)$ is

$$\Delta\mathcal{F}_{\text{wa}}(\lambda) = -\mathcal{F}_{\text{NLL}}(\lambda) \left(\psi^{(0)} (1 + R'_{\text{NLL}}) + \gamma_E \right) \frac{\alpha_s(v^{1/a}Q)}{a\alpha_s(Q)} 2 \sum_{(\ell\bar{\ell})} C_{\ell\bar{\ell}} \ln \left(\frac{Q_{\ell\bar{\ell}}}{Q} \right). \quad (\text{C.10})$$

Appendix D

One-jettiness resummation coefficients

In this appendix we collect the ingredients used to validate the one-jettiness resummation presented in Sec. 5.6. We first give the expansion coefficients of the ARES resummation formula, followed by the corresponding N -jettiness SCET ingredients used to construct the subtraction coefficients extracted from Ref. [141].

D.1 ARES expansion coefficients

The ARES expansion coefficients entering Eq. (5.88) for one-jettiness are given below.

The LL coefficients are:

$$h_{12} = - \sum_{\ell} C_{\ell}, \quad (\text{D.1a})$$

$$h_{24} = \frac{1}{2} \left(\sum_{\ell} C_{\ell} \right)^2. \quad (\text{D.1b})$$

The NLL coefficients are:

$$\begin{aligned} h_{11} = & \sum_{\ell} \left(\gamma_{\ell}^{(0)} - 2 \sum_{\ell} \sum_{\bar{\ell} \neq \ell} C_{\ell \bar{\ell}} \ln d_{\ell}^{(\ell \bar{\ell})} \right) \\ & - \sum_k \int \frac{dz}{z} P_{ik}(z) \frac{f_k(x_1/z, \mu_F)}{f_i(x_1, \mu_F)} - \sum_k \int \frac{dz}{z} P_{jk}(z) \frac{f_k(x_2/z, \mu_F)}{f_j(x_2, \mu_F)}, \end{aligned} \quad (\text{D.2a})$$

$$h_{23} = -2\pi\beta_0 \sum_{\ell} C_{\ell} - h_{11} \sum_{\ell} C_{\ell}. \quad (\text{D.2b})$$

The NNLL coefficients are:

$$\begin{aligned}
h_{10} = & H_{ij}^{(1)} - \sum_{\ell} \sum_{\bar{\ell} \neq \ell} C_{\ell\bar{\ell}} \ln^2 d_{\ell}^{(\ell\bar{\ell})} + \frac{\alpha_s(Q)}{\alpha_s(\tau_1 Q)} \frac{\delta \mathcal{F}_{\text{wa}}}{\mathcal{F}_{\text{NLL}}} + \sum_{\ell} C_{\text{hc},\ell}^{(1)} + \frac{\alpha_s(Q)}{\alpha_s(Q\sqrt{\tau_1})} \frac{\delta \mathcal{F}_{\text{rec}}}{\mathcal{F}_{\text{NLL}}} \\
& + \sum_k \int \frac{dz}{z} \frac{f_k(x_1/z, \mu_F)}{f_i(x_1, \mu_F)} P_{ik}(z) \ln \frac{Q^2}{\mu_F^2} + \sum_k \int \frac{dz}{z} \frac{f_k(x_2/z, \mu_F)}{f_j(x_2, \mu_F)} P_{jk}(z) \ln \frac{Q^2}{\mu_F^2},
\end{aligned} \tag{D.3a}$$

$$\begin{aligned}
h_{22} = & h_{10} \cdot h_{12} + \frac{1}{2} h_{11}^2 + 4\pi \ln \frac{\mu_R}{Q} h_{12} - \frac{\pi^2}{3} \left(\sum_{\ell} C_{\ell} \right)^2 - K^{(1)} \sum_{\ell} C_{\ell} \\
& + 2\pi\beta_0 \left[-4 \sum_{\ell \neq \bar{\ell}} C_{\ell\bar{\ell}} \ln \frac{Q_{\ell\bar{\ell}}^2}{Q^2} + \sum_{\ell} \left(\frac{\gamma_{\ell}^{(0)}}{2} - 3C_{\ell} \ln \frac{Q^2}{MQ_{\ell}} \right) \right. \\
& \left. - \frac{1}{2} \sum_k \int \frac{dz}{z} P_{ik}(z) \frac{f_k(x_1/z, \mu_F)}{f_i(x_1, \mu_F)} - \frac{1}{2} \sum_k \int \frac{dz}{z} P_{jk}(z) \frac{f_k(x_2/z, \mu_F)}{f_j(x_2, \mu_F)} \right],
\end{aligned} \tag{D.3b}$$

$$\begin{aligned}
h_{21} = & h_{11} \cdot h_{10} + 4\pi\beta_0 \ln \frac{\mu_R}{Q} h_{11} - \sum_{\ell} \gamma_{\ell}^{(1)} + 2 \left(\sum_{\ell} C_{\ell} \right)^2 \psi^{(2)}(1) + 2\pi\beta_0 \frac{\pi^2}{6} \sum_{\ell} C_{\ell} \\
& + \sum_{\ell} \sum_{\bar{\ell} \neq \ell} C_{\ell\bar{\ell}} \left(-2K^{(1)} \ln d_{\ell}^{(\ell\bar{\ell})} - 6\pi\beta_0 \ln^2 d_{\ell}^{(\ell\bar{\ell})} - \frac{\pi^3\beta_0}{3} + 2\pi\beta_0 \ln d_{\ell}^{(\ell\bar{\ell})} \ln \frac{Q_{\ell\bar{\ell}}^2}{Q^2} \right) \\
& + 2\pi\beta_0 \left(\frac{\alpha_s(Q)}{\alpha_s(\tau_1 Q)} \frac{\delta \mathcal{F}_{\text{wa}}}{\mathcal{F}_{\text{NLL}}} + \sum_{\ell} C_{\text{hc},\ell}^{(1)} + \frac{\alpha_s(Q)}{\alpha_s(Q\sqrt{\tau_1})} \frac{\delta \mathcal{F}_{\text{rec}}}{\mathcal{F}_{\text{NLL}}} \right) \\
& - \frac{\pi^2}{3} \sum_{\ell} C_{\ell} \left[\sum_{\ell} \left(B_{\ell} + C_{\ell} \ln \frac{Q^2 M}{Q_{\ell}} + 8C_{\ell\bar{\ell}} \ln \frac{Q_{\ell\bar{\ell}}}{Q} \right) \right] \\
& - \sum_k \int \frac{dz}{z} P_{ik}^{(1)}(z) \frac{f_k(x_1/z, \mu_F)}{f_i(x_1, \mu_F)} - \sum_k \int \frac{dz}{z} P_{jk}^{(1)}(z) \frac{f_k(x_2/z, \mu_F)}{f_j(x_2, \mu_F)}.
\end{aligned} \tag{D.3c}$$

D.2 SCET subtraction ingredients

The set of partons is labelled $\kappa \equiv \{\kappa_1, \kappa_2, \kappa_3\}$, where κ_1 and κ_2 denote the incoming partons on legs $\ell = 1, 2$, with flavours i and j , respectively, while κ_3 denotes the outgoing parton on leg $\ell = 3$.

The ingredients for the LL coefficients are:

$$B_{\kappa_1,1}^{(1)}(x_1, \lambda_1) = \sum_k \int \frac{dz}{z} 4C_1 \delta_{ik} \delta(1-z) f_k \left(\frac{x_1}{z}, \mu_F \right), \quad (\text{D.4a})$$

$$B_{\kappa_2,1}^{(1)}(x_2, \lambda_2) = \sum_k \int \frac{dz}{z} 4C_2 \delta_{jk} \delta(1-z) f_k \left(\frac{x_2}{z}, \mu_F \right), \quad (\text{D.4b})$$

$$J_{\kappa_3,1}^{(1)}(\lambda_3) = 4C_3, \quad (\text{D.4c})$$

$$\widehat{S}_{\kappa,1}^{(1)}(\lambda_s) = -8 \sum_{\ell} C_{\ell}. \quad (\text{D.4d})$$

$$B_{\kappa_1,3}^{(2)}(x_1, \lambda_1) = \sum_k \int \frac{dz}{z} 8C_1^2 \delta_{ik} \delta(1-z) f_k \left(\frac{x_1}{z}, \mu_F \right) \quad (\text{D.5a})$$

$$B_{\kappa_2,3}^{(2)}(x_2, \lambda_2) = \sum_k \int \frac{dz}{z} 8C_2^2 \delta_{jk} \delta(1-z) f_k \left(\frac{x_2}{z}, \mu_F \right) \quad (\text{D.5b})$$

$$J_{\kappa_3,3}^{(2)}(\lambda_3) = 8C_3^2, \quad (\text{D.5c})$$

$$\widehat{S}_{\kappa,3}^{(2)}(\lambda_s) = 32 \left(\sum_{\ell} C_{\ell} \right)^2. \quad (\text{D.5d})$$

The ingredients first entering the NLL coefficients are:

$$B_{\kappa_1,0}^{(1)}(x_1, \lambda_1) = \sum_k \int \frac{dz}{z} \left(2P_{ik}(z) - 2\gamma_1^{(0)} \delta_{ik} \delta(1-z) \right) f_k \left(\frac{x_1}{z}, \mu_F \right), \quad (\text{D.6a})$$

$$B_{\kappa_2,0}^{(1)}(x_2, \lambda_2) = \sum_k \int \frac{dz}{z} \left(2P_{jk}(z) - 2\gamma_2^{(0)} \delta_{jk} \delta(1-z) \right) f_k \left(\frac{x_2}{z}, \mu_F \right), \quad (\text{D.6b})$$

$$J_{\kappa_3,0}^{(1)}(\lambda_3) = -2\gamma_3^{(0)}, \quad (\text{D.6c})$$

$$\widehat{S}_{\kappa,0}^{(1)}(\lambda_s) = 4 \sum_{\ell} \sum_{\bar{\ell} \neq \ell} C_{\ell\bar{\ell}} \ln \frac{Q_{\ell\bar{\ell}}^2}{Q_{\ell} Q_{\bar{\ell}}} \quad (\text{D.6d})$$

where $P_{ik}(z)$, $P_{jk}(z)$ are the Altarelli-Parisi splitting functions found in Eq. (5.59) and $\gamma_{\ell}^{(0)}$ are given in Eq. (4.51).

$$B_{\kappa_1,2}^{(2)}(x_1, \lambda_1) = \sum_k \int \frac{dz}{z} 4C_1 \left[3P_{ik}(z) - \left(3\gamma_1^{(0)} + 2\pi\beta_0 \right) \delta_{ik} \delta(1-z) \right] f_k \left(\frac{x_1}{z}, \mu_F \right), \quad (\text{D.7a})$$

$$B_{\kappa_2,2}^{(2)}(x_2, \lambda_2) = \sum_k \int \frac{dz}{z} 4C_2 \left[3P_{jk}(z) - \left(3\gamma_2^{(0)} + 2\pi\beta_0 \right) \delta_{jk} \delta(1-z) \right] f_k \left(\frac{x_2}{z}, \mu_F \right), \quad (\text{D.7b})$$

$$J_{\kappa_3,2}^{(2)}(\lambda_3) = -2C_3 \left(6\gamma_3^{(0)} + 4\pi\beta_0 \right), \quad (\text{D.7c})$$

$$\widehat{S}_{\kappa,2}^{(2)}(\lambda_s) = 4 \sum_{\ell} C_{\ell} \left(8\pi\beta_0 - 12 \sum_{\ell} \sum_{\bar{\ell} \neq \ell} C_{\ell\bar{\ell}} \ln \frac{Q_{\ell\bar{\ell}}^2}{Q_{\ell} Q_{\bar{\ell}}} \right). \quad (\text{D.7d})$$

The ingredients first entering the NNLL coefficients are:

$$B_{\kappa_1,-1}^{(1)}(x_1, \lambda_1) = \sum_k \int \frac{dz}{z} \left(2I_{ik}^{(1)}(z) + 2 \ln \frac{\mu_R^2}{\mu_F^2} P_{ik}(z) \right) f_k \left(\frac{x_1}{z}, \mu_F \right), \quad (\text{D.8a})$$

$$B_{\kappa_2,-1}^{(1)}(x_2, \lambda_2) = \sum_k \int \frac{dz}{z} \left(2I_{jk}^{(1)}(z) + 2 \ln \frac{\mu_R^2}{\mu_F^2} P_{jk}(z) \right) f_k \left(\frac{x_2}{z}, \mu_F \right), \quad (\text{D.8b})$$

$$J_{\kappa_3,-1}^{(1)}(\lambda_3) = \begin{cases} C_F(7 - \pi^2), & \text{if } \ell = 3 \text{ is a quark,} \\ C_A \left(\frac{4}{3} - \pi^2 \right) + \frac{20}{3} \pi\beta_0, & \text{if } \ell = 3 \text{ is a gluon,} \end{cases} \quad (\text{D.8c})$$

$$\widehat{S}_{\kappa,-1}^{(1)}(\lambda_s) = - \sum_{\ell} \sum_{\bar{\ell} \neq \ell} C_{\ell\bar{\ell}} \left(\ln^2 \frac{Q_{\ell\bar{\ell}}^2}{Q_{\ell} Q_{\bar{\ell}}} - \frac{\pi^2}{6} + 4 \sum_{\hat{\ell} \neq \ell, \bar{\ell}} I_{\ell\bar{\ell},\hat{\ell}} \right). \quad (\text{D.8d})$$

where the matching functions, $I_{ak}^{(1)}(z)$ and $I_{bk}^{(1)}(z)$, are defined in Refs. [150, 151].

$$\begin{aligned}
B_{\kappa_1,1}^{(2)}(x_1, \lambda_1) = \sum_k \int \frac{dz}{z} \left\{ 8 \left[C_1 K^{(1)} - \frac{\pi^2}{3} C_1^2 + \gamma_1^{(0)} \left(\frac{\gamma_1^{(0)}}{2} + \pi \beta_0 \right) \right] \delta_{ik} \delta(1-z) \right. \\
+ 8C_1 \left(I_{ik}^{(1)}(z) + \ln \frac{\mu_R^2}{\mu_F^2} P_{ik}(z) \right) + 4 \sum_m P_{im}(z) \otimes_z P_{mk}(z) \\
\left. - 2 \left(4\gamma_1^{(0)} + 4\pi\beta_0 \right) P_{ik}(z) \right\} f_k \left(\frac{x_1}{z}, \mu_F \right), \quad (\text{D.9a})
\end{aligned}$$

$$\begin{aligned}
B_{\kappa_2,1}^{(2)}(x_2, \lambda_2) = \sum_k \int \frac{dz}{z} \left\{ 8 \left[C_2 K^{(1)} - \frac{\pi^2}{3} C_2^2 + \gamma_2^{(0)} \left(\frac{\gamma_2^{(0)}}{2} + \pi \beta_0 \right) \right] \delta_{jk} \delta(1-z) \right. \\
+ 8C_2 \left(I_{jk}^{(1)}(z) + \ln \frac{\mu_R^2}{\mu_F^2} P_{jk}(z) \right) + 4 \sum_m P_{jm}(z) \otimes_z P_{mk}(z) \\
\left. - 2 \left(4\gamma_2^{(0)} + 4\pi\beta_0 \right) P_{jk}(z) \right\} f_k \left(\frac{x_2}{z}, \mu_F \right), \quad (\text{D.9b})
\end{aligned}$$

$$J_{\kappa_3,1}^{(2)}(\lambda_3) = 8C_3 K^{(1)} - \frac{8\pi^2}{3} C_3^2 + 2\gamma_3^{(0)} \left(2\gamma_3^{(0)} + 4\pi\beta_0 \right) + 4C_3 J_{\kappa_3,-1}^{(1)}, \quad (\text{D.9c})$$

$$\begin{aligned}
\widehat{S}_{\kappa,1}^{(2)}(\lambda_s) = 16 \left(\sum_{\ell} \sum_{\bar{\ell} \neq \ell} C_{\ell\bar{\ell}} \ln \frac{Q_{\ell\bar{\ell}}^2}{Q_{\ell} Q_{\bar{\ell}}} \right)^2 - 32\pi\beta_0 \sum_{\ell} \sum_{\bar{\ell} \neq \ell} C_{\ell\bar{\ell}} \ln \frac{Q_{\ell\bar{\ell}}^2}{Q_{\ell} Q_{\bar{\ell}}} \\
- \frac{32\pi^2}{3} \left(\sum_{\ell} C_{\ell} \right)^2 - \left(8\widehat{S}_{\kappa,-1}^{(1)} + 16K^{(1)} \right) \sum_{\ell} C_{\ell}. \quad (\text{D.9d})
\end{aligned}$$

The convolution $\otimes_z$ is defined by $P_{ij}(z) \otimes_z P_{jk}(z) = \int_z^1 \frac{dw}{w} P_{ij}(w) P_{jk}\left(\frac{z}{w}\right)$, where the index j is not summed over, here or in $I_{ij}^{(1)} \otimes_z P_{jk}$ below.

$$\begin{aligned}
B_{\kappa_1,0}^{(2)}(x_1, \lambda_1) = & \sum_k \int \frac{dz}{z} \left\{ \left(16C_1^2 \zeta_3 + \frac{4\pi^2}{3} C_1 \gamma_1^{(0)} - \frac{\gamma_{1,s}^{(1)}}{2} \right) \delta_{ik} \delta(1-z) \right. \\
& - \frac{4\pi^2}{3} C_1 P_{ik}(z) - \left(4\gamma_1^{(0)} + 8\pi\beta_0 \right) I_{ik}^{(1)}(z) + 4 \sum_m I_{im}^{(1)}(z) \otimes_z P_{mk}(z) + 4P_{ik}^{(1)}(z) \\
& \left. + \ln \frac{\mu_R^2}{\mu_F^2} \left[-4\gamma_1^{(0)} P_{ik}(z) + 4 \sum_m P_{im}(z) \otimes_z P_{mk}(z) + 4P_{ik}^{(1)}(z) \right] \right\} f_k \left(\frac{x_1}{z}, \mu_F \right),
\end{aligned} \tag{D.10a}$$

$$\begin{aligned}
B_{\kappa_2,0}^{(2)}(x_2, \lambda_2) = & \sum_k \int \frac{dz}{z} \left\{ \left(16C_2^2 \zeta_3 + \frac{4\pi^2}{3} C_2 \gamma_2^{(0)} - \frac{\gamma_{2,s}^{(1)}}{2} \right) \delta_{jk} \delta(1-z) \right. \\
& - \frac{4\pi^2}{3} C_2 P_{jk}(z) - \left(4\gamma_2^{(0)} + 8\pi\beta_0 \right) I_{jk}^{(1)}(z) + 4 \sum_m I_{jm}^{(1)}(z) \otimes_z P_{mk}(z) + 4P_{jk}^{(1)}(z) \\
& \left. + \ln \frac{\mu_R^2}{\mu_F^2} \left[-4\gamma_2^{(0)} P_{jk}(z) + 4 \sum_m P_{jm}(z) \otimes_z P_{mk}(z) + 4P_{jk}^{(1)}(z) \right] \right\} f_k \left(\frac{x_2}{z}, \mu_F \right),
\end{aligned} \tag{D.10b}$$

$$J_{\kappa_3,0}^{(2)}(\lambda_3) = 16C_3^2 \zeta_3 + \frac{4\pi^2}{3} C_3 \gamma_3^{(0)} - \left(4\gamma_3^{(0)} + 4\pi\beta_0 \right) J_{\kappa_3,-1}^{(1)} - \frac{\gamma_{3,s}^{(1)}}{2}, \tag{D.10c}$$

$$\begin{aligned}
\widehat{S}_{\kappa,0}^{(2)}(\lambda_s) = & 64\zeta_3 \left(\sum_{\ell} C_{\ell} \right)^2 - \gamma_S^{(1)} \sum_{\ell} C_{\ell} - 8\pi\beta_0 \widehat{S}_{\kappa,-1}^{(1)} \\
& + 4 \sum_{\ell} \sum_{\bar{\ell} \neq \ell} C_{\ell\bar{\ell}} \ln \frac{Q_{\ell\bar{\ell}}^2}{Q_{\ell} Q_{\bar{\ell}}} \left(\frac{4\pi^2}{3} \sum_{\ell} C_{\ell} + \widehat{S}_{\kappa,-1}^{(1)} + K^{(1)} \right).
\end{aligned} \tag{D.10d}$$

The noncusp soft anomalous dimension is

$$\gamma_S^{(1)} = C_A \left(-\frac{64}{9} + 28\zeta_3 \right) + 4\pi\beta_0 \left(-\frac{56}{9} + \frac{\pi^2}{3} \right), \tag{D.11}$$

while the corresponding anomalous dimensions for the beam and jet functions, denoted by $\gamma_{\ell,s}^{(1)}$, are

$$\gamma_{\ell,s}^{(1)} = C_F \left[C_A \left(\frac{146}{9} - 80\zeta_3 \right) + C_F (3 - 4\pi^2 + 48\zeta_3) + 4\pi\beta_0 \left(\frac{121}{9} + \frac{2\pi^2}{3} \right) \right] \tag{D.12}$$

for a quark, and

$$\gamma_{\ell,s}^{(1)} = C_A \left[C_A \left(\frac{182}{9} - 32\zeta_3 \right) + 4\pi\beta_0 \left(\frac{94}{9} - \frac{2\pi^2}{3} \right) \right] \tag{D.13}$$

for a gluon. The subscript s indicates that these anomalous dimensions are defined in the N -jettiness SCET convention, which differs from the convention used in the ARES formalism, where the corresponding quantities are given in Eq. (5.16).

Appendix E

One-loop hard function

Following the argument of Sec. 5.6.3, we extract the hard coefficient $H_{ij}^{(1)}$ of Eq. (5.22) and the one-loop hard function $H_{\text{SCET}}^{(1)}$ of Eq. (5.106) for the $q\bar{q} \rightarrow Hg$ channel, using the known one-loop virtual corrections to $hh \rightarrow H + \text{jet}$ [148]. The relation between the two hard functions, like the infrared poles themselves, is fixed by the colour charges and directions of the external partons. Since the partonic channels of $H + \text{jet}$ and $Z + \text{jet}$ production share the same colour configurations, the Higgs amplitude determines both hard functions here. The remaining partonic channels follow analogously.

Adopting the convention given in Sec. 5.3.1, the one-loop virtual corrections for $q\bar{q} \rightarrow Hg$ are

$$\begin{aligned}
 \mathcal{V}(Q; \varepsilon) = 1 + \frac{\alpha_s(\mu_R)}{2\pi} \Bigg[& -\frac{C_A + 2C_F}{\varepsilon^2} + \frac{1}{\varepsilon} \left(-\frac{11}{6}C_A - 3C_F + \frac{n_f}{3} + (C_A - 2C_F) \ln \frac{\mu_R^2}{Q_{12}^2} \right. \\
 & - C_A \ln \frac{\mu_R^2}{Q_{13}^2} - C_A \ln \frac{\mu_R^2}{Q_{23}^2} \Big) - n_f \left(\frac{5}{9} + \frac{1}{3} \ln \frac{\mu_R^2}{Q_{12}^2} \right) + \frac{\pi^2}{12} (18C_F - 7C_A) \\
 & + C_A \left(\frac{38}{9} + \text{Li}_2 \left(\frac{Q_{12}^2 - M^2}{Q_{12}^2} \right) - \frac{11}{6} \ln \frac{Q_{12}^2}{\mu_R^2} + \frac{1}{4} \ln^2 \frac{\mu_R^2}{Q_{12}^2} - \frac{1}{4} \ln^2 \frac{\mu_R^2}{Q_{13}^2} \right. \\
 & - \frac{1}{4} \ln^2 \frac{\mu_R^2}{Q_{23}^2} + \frac{1}{2} \ln \frac{Q_{12}^2}{Q_{13}^2} \ln \frac{Q_{12}^2}{Q_{23}^2} \Big) + C_F \left(-4 + \text{Li}_2 \left(\frac{-Q_{13}^2}{M^2} \right) + \text{Li}_2 \left(\frac{-Q_{23}^2}{M^2} \right) \right. \\
 & - \frac{1}{2} \ln^2 \frac{Q_{13}^2 + M^2}{Q_{13}^2} - \frac{1}{2} \ln^2 \frac{Q_{23}^2 + M^2}{Q_{23}^2} + \frac{1}{2} \ln^2 \frac{M^2 + Q_{13}^2}{M^2} + \frac{1}{2} \ln^2 \frac{M^2 + Q_{23}^2}{M^2} \\
 & \left. \left. - \frac{1}{2} \ln^2 \frac{Q_{12}^2}{\mu_R^2} + \frac{3}{2} \ln \frac{Q_{12}^2}{\mu_R^2} + \frac{1}{2} \ln^2 \frac{Q_{13}^2}{Q_{23}^2} \right) \right].
 \end{aligned} \tag{E.1}$$

The SCET hard function is obtained from the finite terms of Eq. (E.1). By the definition in

Eq. (5.106), it is twice the one-loop correction,

$$\begin{aligned}
H_{\text{SCET}}^{(1)} = & \pi^2 \left(3C_F - \frac{7}{6}C_A \right) - n_f \left(\frac{10}{9} + \frac{2}{3} \ln \frac{\mu_R^2}{Q_{12}^2} \right) + C_A \left(\frac{76}{9} + 2 \text{Li}_2 \left(\frac{Q_{12}^2 - M^2}{Q_{12}^2} \right) \right. \\
& - \frac{11}{3} \ln \frac{Q_{12}^2}{\mu_R^2} + \frac{1}{2} \ln^2 \frac{\mu_R^2}{Q_{12}^2} - \frac{1}{2} \ln^2 \frac{\mu_R^2}{Q_{13}^2} - \frac{1}{2} \ln^2 \frac{\mu_R^2}{Q_{23}^2} + \ln \frac{Q_{12}^2}{Q_{13}^2} \ln \frac{Q_{12}^2}{Q_{23}^2} \Big) \\
& + C_F \left(-8 + 2 \text{Li}_2 \left(\frac{-Q_{13}^2}{M^2} \right) + 2 \text{Li}_2 \left(\frac{-Q_{23}^2}{M^2} \right) - \ln^2 \frac{Q_{13}^2 + M^2}{Q_{13}^2} - \ln^2 \frac{Q_{23}^2 + M^2}{Q_{23}^2} \right. \\
& \left. + \ln^2 \frac{M^2 + Q_{13}^2}{M^2} + \ln^2 \frac{M^2 + Q_{23}^2}{M^2} - \ln^2 \frac{Q_{12}^2}{\mu_R^2} + 3 \ln \frac{Q_{12}^2}{\mu_R^2} + \ln^2 \frac{Q_{13}^2}{Q_{23}^2} \right). \tag{E.2}
\end{aligned}$$

While $H_{\text{SCET}}^{(1)}$ retains all finite terms of the virtual correction, the ARES coefficient does not. Instead, starting from the finite terms of Eq. (E.1), we subtract the finite remainder of the pole structure in Eq. (5.22). Using Eq. (5.24), the ARES hard coefficient for the $q\bar{q}$ channel is

$$H_{q\bar{q}}^{(1)} = \mathcal{H}_1 + 3 \left(C_F - \frac{C_A}{2} \right) \ln \frac{Q_{12}^2}{Q^2} + \left(\frac{3C_A}{4} + \pi\beta_0 \right) \ln \frac{Q_{13}^2}{Q^2} + \left(\frac{3C_A}{4} + \pi\beta_0 \right) \ln \frac{Q_{23}^2}{Q^2}, \tag{E.3}$$

where

$$\begin{aligned}
\mathcal{H}_1 = & \frac{2\pi^2}{3} (2C_F - C_A) - n_f \left(\frac{5}{9} + \frac{1}{3} \ln \frac{\mu_R^2}{Q_{12}^2} \right) + \pi\beta_0 \left(\ln \frac{\mu_R^2}{Q_{13}^2} + \ln \frac{\mu_R^2}{Q_{23}^2} \right) \\
& + C_A \left(\frac{38}{9} + \text{Li}_2 \left(\frac{Q_{12}^2 - M^2}{Q_{12}^2} \right) + \frac{1}{4} \ln^2 \frac{Q_{13}^2 Q_{23}^2}{\mu_R^2 Q_{12}^2} + \frac{1}{3} \ln \frac{\mu_R^2}{Q_{12}^2} + \frac{3}{4} \left(\ln \frac{\mu_R^2}{Q_{13}^2} + \ln \frac{\mu_R^2}{Q_{23}^2} \right) \right) \\
& + C_F \left(-4 + \text{Li}_2 \left(-\frac{Q_{13}^2}{M^2} \right) + \text{Li}_2 \left(-\frac{Q_{23}^2}{M^2} \right) + \frac{3}{2} \ln \frac{\mu_R^2}{Q_{12}^2} + \frac{1}{2} \ln^2 \frac{Q_{12}^2}{\mu_R^2} + \frac{1}{2} \ln^2 \frac{Q_{13}^2}{Q_{23}^2} \right. \\
& \left. - \frac{1}{2} \ln^2 \frac{Q_{13}^2 + M^2}{Q_{13}^2} - \frac{1}{2} \ln^2 \frac{Q_{23}^2 + M^2}{Q_{23}^2} + \frac{1}{2} \ln^2 \frac{M^2 + Q_{13}^2}{M^2} + \frac{1}{2} \ln^2 \frac{M^2 + Q_{23}^2}{M^2} \right). \tag{E.4}
\end{aligned}$$

The two hard functions therefore differ only by terms fixed by the colour charges and directions of the external legs, which are unchanged for $q\bar{q} \rightarrow Zg$. The relation obtained here can then be used in Sec. 5.6.3 to compare the ARES and SCET expansion coefficients.

Appendix F

Symbolic validation of h_{22} and h_{21}

This appendix presents the symbolic validation of the ARES expansion coefficients h_{22} and h_{21} against the corresponding SCET subtraction coefficients $\mathcal{C}_1^{(2)}$ and $\mathcal{C}_0^{(2)}$. The comparison is performed using MATHEMATICA, and the complete notebook is available in Ref. [149].

The notation used in the MATHEMATICA implementation is summarised in Table F.1. Uppercase symbols denote the full expansion of a subtraction ingredient, following Eq. (5.90). The lowercase symbols are the corresponding coefficients defined in Appendix D.

Notebook	LaTeX	Notebook	LaTeX
J[m, n]	$J_{\kappa_{3,n}}^{(m)}$	Ck[1]	C_ℓ
B1[m, n], B2[m, n]	$B_{\kappa_{1,2,n}}^{(m)}/f_{i,j}$	Ctot	$\sum_\ell C_\ell$
S[m, n]	$\widehat{S}_{\kappa,n}^{(m)}$	g[1, m]	$\gamma_\ell^{(m)}$ (SCET)
Q[1], Q[1, lbar]	$Q_\ell, Q_{\ell\bar{\ell}}$	gA[1, m]	$\gamma_\ell^{(m)}$ (ARES)
Qresum	Q	gs1	$\gamma_S^{(1)}$
H1SCET	$H_{\text{SCET}}^{(1)}$	P[1, m], P[2, m]	$P_{ik}^{(m)}(z), P_{jk}^{(m)}(z)$

Table F.1: Correspondence between the symbols used in the MATHEMATICA notebook and the notation of this thesis.

The subtraction coefficients $\mathcal{C}_1^{(2)}$ and $\mathcal{C}_0^{(2)}$ are implemented directly from Eqs. (5.110) and (5.111), using the shorthand `tot` shown below:

```
tot[m_, n_] := J[m, n] + B1[m, n] + B2[m, n] + S[m, n];

C21 = (tot[2, 1] + H1SCET tot[1, 1]
+ 2 (B1[1, 0] J[1, 0] + B1[1, 0] B2[1, 0] + ... )
- \[Pi]^2/3 (B1[1, 1] J[1, 1] + B1[1, 1] B2[1, 1] + ... )
+ B1[1, -1] J[1, 1] + B2[1, -1] J[1, 1] + ... );

C20 = (tot[2, 0] + H1SCET tot[1, 0]
- \[Pi]^2/6 (B1[1, 0] J[1, 1] + B2[1, 0] J[1, 1] + ... )
+ 2 Zeta[3] (B1[1, 1] J[1, 1] + B1[1, 1] B2[1, 1] + ... )
+ B1[1, -1] J[1, 0] + B2[1, -1] J[1, 0] + ... );
```

Listing F.1: Implementation of $\mathcal{C}_1^{(2)}$ and $\mathcal{C}_0^{(2)}$.

The expansion coefficients are implemented according to Eqs. (D.3b) and (D.3c), with the cross-terms $h_{10} \cdot h_{12}$ and $h_{10} \cdot h_{11}$ omitted for simplicity. The coefficients h_{10} , h_{11} , and h_{12} entering these terms have already been validated in Sec. 5.6:

```
h22 = -\[Pi]^2/3 Ctot^2 - K1 Ctot
      + 2 \[Pi] \[Beta]0 (-2 (Ck[1] + Ck[2] - Ck[3])
      Log[Q[1, 2]^2/Qresum^2]
      + g[1, 0]/2 - 3 Ck[1] Log[Qresum^2/(M Q[1])])
      - 1/2 P[1, 0] + ... )
      + 4 \[Pi] Ctot \[Beta]0 Log[Qresum/\[Mu]R]
      + ... ;

h21 = -K1 ((Ck[1] + Ck[2] - Ck[3])
      (Log[Q[1, 2]^2/(M Q[1])])
      + Log[Q[1, 2]^2/(M Q[2])]) + ... )
      + 2 \[Pi] \[Beta]0 ( ... )
      + 2 Ctot^2 PolyGamma[2, 1]
      - gA[1, 1] - gA[2, 1] - gA[3, 1]
      - (P[1, 1] + P[2, 1]);
```

Listing F.2: Implementation of h_{22} and h_{21} with the cross-terms omitted.

The corresponding cross-terms are then removed from the SCET coefficients:

```
C11 = tot[1, 1];
C10 = tot[1, 0];
C1m1 = tot[1, -1] + H1SCET;

C21Rem = C21 - C11 C1m1;
C20Rem = C20 - C10 C1m1;
```

Listing F.3: SCET coefficients with the cross-terms omitted.

Using Eq. (5.91), the h_{22} comparison requires $C_1^{(2)} = 8h_{22}$:

```
C21Rem - 8 h22 // FullSimplify
(* Out: 0 *)
```

Listing F.4: Validation of h_{22} .

For h_{21} , Eq. (5.91) requires $C_0^{(2)} = -4h_{21}$. The corresponding result is

```
g1Res = C20Rem + 4 h21 // FullSimplify
(* Out: (1/2)(-g[1, 1] - 8 gA[1, 1] - g[2, 1] - 8 gA[2, 1]
      - g[3, 1] - 8 gA[3, 1] - 2 Ctot gs1) *)
```

Listing F.5: Validation of h_{21} , leaving a residual in $\gamma_\ell^{(1)}$.

The result differs only through the conventions used for $\gamma_\ell^{(1)}$ in the SCET and ARES expressions, as shown in Eqs. (5.16), (D.12), and (D.13). Substituting the explicit values for the $q\bar{q} \rightarrow Zg$ channel gives

```
g1Res /. {g[1, 1] -> ..., gA[1, 1] -> ..., gs1 -> ...,
      Ck[1] -> CF, Ck[2] -> CF, Ck[3] -> CA} // FullSimplify
(* Out: 0 *)
```

Listing F.6: Evaluation of the residual for the $q\bar{q} \rightarrow Zg$ channel.

The remaining partonic channels follow analogously.

Bibliography

- [1] Melissa van Beekveld, Silvia Ferrario Ravasio, Alexander Karlberg, and Darcy Peake. A generalised- k_t jet algorithm for Deep Inelastic Scattering. 6 2026.
- [2] Melissa van Beekveld and Silvia Ferrario Ravasio. Next-to-leading-logarithmic PanScales showers for Deep Inelastic Scattering and Vector Boson Fusion. *JHEP*, 02:001, 2024.
- [3] R. Keith Ellis, W. James Stirling, and B. R. Webber. *QCD and collider physics*, volume 8. Cambridge University Press, 2 2011.
- [4] John Campbell, Joey Huston, and Frank Krauss. *The Black Book of Quantum Chromodynamics : a Primer for the LHC Era*. Oxford University Press, 2018.
- [5] Michael E. Peskin and Daniel V. Schroeder. *An Introduction to quantum field theory*. Addison-Wesley, Reading, USA, 1995.
- [6] Gionata Luisoni and Simone Marzani. QCD resummation for hadronic final states. *J. Phys. G*, 42(10):103101, 2015.
- [7] Matthew D. Schwartz. *Quantum Field Theory and the Standard Model*. Cambridge University Press, 3 2014.
- [8] Andrea Banfi. Hadronic Jets: An Introduction. 2 2016.
- [9] Murray Gell-Mann. A Schematic Model of Baryons and Mesons. *Phys. Lett.*, 8:214–215, 1964.
- [10] G. Zweig. An $SU(3)$ model for strong interaction symmetry and its breaking. Version 1. 1 1964.
- [11] G. Zweig. *An $SU(3)$ model for strong interaction symmetry and its breaking. Version 2*, pages 22–101. 2 1964.
- [12] F. Takahashi et al. Review of Particle Physics. *Int. J. Mod. Phys. A*, 41:2630011, 2026.

-
- [13] Gerard 't Hooft and M. J. G. Veltman. Regularization and Renormalization of Gauge Fields. *Nucl. Phys. B*, 44:189–213, 1972.
 - [14] C. G. Bollini and J. J. Giambiagi. Dimensional Renormalization: The Number of Dimensions as a Regularizing Parameter. *Nuovo Cim. B*, 12:20–26, 1972.
 - [15] William A. Bardeen, A. J. Buras, D. W. Duke, and T. Muta. Deep Inelastic Scattering Beyond the Leading Order in Asymptotically Free Gauge Theories. *Phys. Rev. D*, 18:3998, 1978.
 - [16] E. C. Poggio, Helen R. Quinn, and Steven Weinberg. Smearing the Quark Model. *Phys. Rev. D*, 13:1958, 1976.
 - [17] Mikhail A. Shifman. Quark hadron duality. In *8th International Symposium on Heavy Flavor Physics*, volume 3, pages 1447–1494, Singapore, 7 2000. World Scientific.
 - [18] Yakov I. Azimov, Yuri L. Dokshitzer, Valery A. Khoze, and S. I. Troyan. Similarity of Parton and Hadron Spectra in QCD Jets. *Z. Phys. C*, 27:65–72, 1985.
 - [19] T. Kinoshita. Mass singularities of Feynman amplitudes. *J. Math. Phys.*, 3:650–677, 1962.
 - [20] T. D. Lee and M. Nauenberg. Degenerate Systems and Mass Singularities. *Phys. Rev.*, 133:B1549–B1562, 1964.
 - [21] Stefano Catani and Massimiliano Grazzini. Infrared factorization of tree level QCD amplitudes at the next-to-next-to-leading order and beyond. *Nucl. Phys. B*, 570:287–325, 2000.
 - [22] Dimitri Colferai, Stefano Catani, and Alessandro Torriani. Triple (and quadruple) soft-gluon radiation in QCD hard scattering. *PoS, LL2022:064*, 2022.
 - [23] Claude Duhr and Thomas Gehrmann. The two-loop soft current in dimensional regularization. *Phys. Lett. B*, 727:452–455, 2013.
 - [24] Franz Herzog, Yao Ma, Bernhard Mistlberger, and Adi Suresh. Single-soft emissions for amplitudes with two colored particles at three loops. *JHEP*, 12:023, 2023.
 - [25] A. Bassetto, M. Ciafaloni, and G. Marchesini. Jet Structure and Infrared Sensitive Quantities in Perturbative QCD. *Phys. Rept.*, 100:201–272, 1983.
 - [26] T. Gleisberg, Stefan. Hoeche, F. Krauss, M. Schonherr, S. Schumann, F. Siegert, and J. Winter. Event generation with SHERPA 1.1. *JHEP*, 02:007, 2009.
 - [27] V. N. Gribov and L. N. Lipatov. Deep inelastic $e p$ scattering in perturbation theory. *Sov. J. Nucl. Phys.*, 15:438–450, 1972.

-
- [28] Yuri L. Dokshitzer. Calculation of the Structure Functions for Deep Inelastic Scattering and $e^+ e^-$ Annihilation by Perturbation Theory in Quantum Chromodynamics. *Sov. Phys. JETP*, 46:641–653, 1977.
- [29] Guido Altarelli and G. Parisi. Asymptotic Freedom in Parton Language. *Nucl. Phys. B*, 126:298–318, 1977.
- [30] J. Alwall, R. Frederix, S. Frixione, V. Hirschi, F. Maltoni, O. Mattelaer, H. S. Shao, T. Stelzer, P. Torrielli, and M. Zaro. The automated computation of tree-level and next-to-leading order differential cross sections, and their matching to parton shower simulations. *JHEP*, 07:079, 2014.
- [31] John M. Campbell and R. Keith Ellis. An Update on vector boson pair production at hadron colliders. *Phys. Rev. D*, 60:113006, 1999.
- [32] John M. Campbell, R. Keith Ellis, and Ciaran Williams. Vector Boson Pair Production at the LHC. *JHEP*, 07:018, 2011.
- [33] John Campbell and Tobias Neumann. Precision Phenomenology with MCFM. *JHEP*, 12:034, 2019.
- [34] Christian Bierlich et al. A comprehensive guide to the physics and usage of PYTHIA 8.3. *SciPost Phys. Codeb.*, 2022:8, 2022.
- [35] M. Bahr et al. Herwig++ Physics and Manual. *Eur. Phys. J. C*, 58:639–707, 2008.
- [36] Johannes Bellm et al. Herwig 7.0/Herwig++ 3.0 release note. *Eur. Phys. J. C*, 76(4):196, 2016.
- [37] Melissa van Beekveld et al. Introduction to the PanScales framework, version 0.1. *SciPost Phys. Codeb.*, 2024:31, 2024.
- [38] Florian Herren, Stefan Höche, Frank Krauss, Daniel Reichelt, and Marek Schoenherr. A new approach to color-coherent parton evolution. *JHEP*, 10:091, 2023.
- [39] Enrico Bothmann et al. Event generation with Sherpa 3. *JHEP*, 12:156, 2024.
- [40] Melissa van Beekveld, Silvia Ferrario Ravasio, Gavin P. Salam, Alba Soto-Ontoso, Gregory Soyez, and Rob Verheyen. PanScales parton showers for hadron collisions: formulation and fixed-order studies. *JHEP*, 11:019, 2022.

-
- [41] Melissa van Beekveld, Silvia Ferrario Ravasio, Keith Hamilton, Gavin P. Salam, Alba Soto-Ontoso, Gregory Soyez, and Rob Verheyen. PanScales showers for hadron collisions: all-order validation. *JHEP*, 11:020, 2022.
- [42] Andrea Banfi, Gavin P. Salam, and Giulia Zanderighi. Principles of general final-state resummation and automated implementation. *JHEP*, 03:073, 2005.
- [43] Andrea Banfi, Heather McAslan, Pier Francesco Monni, and Giulia Zanderighi. A general method for the resummation of event-shape distributions in e^+e^- annihilation. *JHEP*, 05:102, 2015.
- [44] Andrea Banfi, Heather McAslan, Pier Francesco Monni, and Giulia Zanderighi. The two-jet rate in e^+e^- at next-to-next-to-leading-logarithmic order. *Phys. Rev. Lett.*, 117(17):172001, 2016.
- [45] Andrea Banfi, Basem Kamal El-Menoufi, and Pier Francesco Monni. The Sudakov radiator for jet observables and the soft physical coupling. *JHEP*, 01:083, 2019.
- [46] Luke Arpino, Andrea Banfi, and Basem Kamal El-Menoufi. Near-to-planar three-jet events at NNLL accuracy. *JHEP*, 07:171, 2020.
- [47] Christian W. Bauer, Sean Fleming, and Michael E. Luke. Summing Sudakov logarithms in $B \rightarrow X_s \gamma$ in effective field theory. *Phys. Rev. D*, 63:014006, 2000.
- [48] Christian W. Bauer, Sean Fleming, Dan Pirjol, and Iain W. Stewart. An Effective field theory for collinear and soft gluons: Heavy to light decays. *Phys. Rev. D*, 63:114020, 2001.
- [49] Christian W. Bauer and Iain W. Stewart. Invariant operators in collinear effective theory. *Phys. Lett. B*, 516:134–142, 2001.
- [50] Christian W. Bauer, Dan Pirjol, and Iain W. Stewart. Soft collinear factorization in effective field theory. *Phys. Rev. D*, 65:054022, 2002.
- [51] Thomas Becher and Matthew D. Schwartz. A precise determination of α_s from LEP thrust data using effective field theory. *JHEP*, 07:034, 2008.
- [52] Riccardo Abbate, Michael Fickinger, Andre H. Hoang, Vicent Mateu, and Iain W. Stewart. Thrust at N^3LL with Power Corrections and a Precision Global Fit for $\alpha_s(m_Z)$. *Phys. Rev. D*, 83:074021, 2011.
- [53] Simone Alioli, Guido Bell, Georgios Billis, Alessandro Broggio, Bahman Dehnadi, Matthew A. Lim, Giulia Marinelli, Riccardo Nagar, Davide Napoletano, and Rudi Rahn. N₃LL

- resummation of one-jettiness for Z-boson plus jet production at hadron colliders. *Phys. Rev. D*, 109(9):094009, 2024.
- [54] Torbjorn Sjostrand. The Lund Monte Carlo for Jet Fragmentation. *Comput. Phys. Commun.*, 27:243, 1982.
- [55] Bo Andersson, G. Gustafson, and B. Soderberg. A General Model for Jet Fragmentation. *Z. Phys. C*, 20:317, 1983.
- [56] B. R. Webber. A QCD Model for Jet Fragmentation Including Soft Gluon Interference. *Nucl. Phys. B*, 238:492–528, 1984.
- [57] Yuri L. Dokshitzer, G. Marchesini, and B. R. Webber. Dispersive approach to power behaved contributions in QCD hard processes. *Nucl. Phys. B*, 469:93–142, 1996.
- [58] John E. Huth et al. Toward a Standardization of Jet Definitions. In *1990 DPF Summer Study on High-energy Physics: Research Directions for the Decade (Snowmass 90)*, pages 0134–136, 12 1990.
- [59] Gavin P. Salam and Gregory Soyez. A Practical Seedless Infrared-Safe Cone jet algorithm. *JHEP*, 05:086, 2007.
- [60] Nikolaos Kidonakis, Gianluca Oderda, and George F. Sterman. Threshold resummation for dijet cross-sections. *Nucl. Phys. B*, 525:299–332, 1998.
- [61] Gavin P. Salam. Towards Jetography. *Eur. Phys. J. C*, 67:637–686, 2010.
- [62] S. Bethke et al. Experimental Investigation of the Energy Dependence of the Strong Coupling Strength. *Phys. Lett. B*, 213:235–241, 1988.
- [63] S. Catani, Yuri L. Dokshitzer, M. Olsson, G. Turnock, and B. R. Webber. New clustering algorithm for multi - jet cross-sections in $e^+ e^-$ annihilation. *Phys. Lett. B*, 269:432–438, 1991.
- [64] N. Brown and W. James Stirling. Finding jets and summing soft gluons: A New algorithm. *Z. Phys. C*, 53:629–636, 1992.
- [65] Yuri L. Dokshitzer, G. D. Leder, S. Moretti, and B. R. Webber. Better jet clustering algorithms. *JHEP*, 08:001, 1997.
- [66] I. Abt et al. Measurement of jet production in deep inelastic scattering and NNLO determination of the strong coupling at ZEUS. *Eur. Phys. J. C*, 83(11):1082, 2023.

-
- [67] V. Andreev et al. Unbinned deep learning jet substructure measurement in high Q_{2ep} collisions at HERA. *Phys. Lett. B*, 844:138101, 2023.
 - [68] I. Abt et al. The azimuthal correlation between the leading jet and the scattered lepton in deep inelastic scattering at HERA. *Eur. Phys. J. C*, 84(12):1334, 2024.
 - [69] V. Andreev et al. Observation and differential cross section measurement of neutral current DIS events with an empty hemisphere in the Breit frame. *Eur. Phys. J. C*, 84(7):720, 2024.
 - [70] V. Andreev et al. Measurement of the 1-jettiness event shape observable in deep-inelastic electron-proton scattering at HERA. *Eur. Phys. J. C*, 84(8):785, 2024.
 - [71] V. Andreev et al. Measurement of groomed event shape observables in deep-inelastic electron-proton scattering at HERA. *Eur. Phys. J. C*, 84(7):718, 2024.
 - [72] Miguel Arratia, Yiannis Makris, Duff Neill, Felix Ringer, and Nobuo Sato. Asymmetric jet clustering in deep-inelastic scattering. *Phys. Rev. D*, 104(3):034005, 2021.
 - [73] Paul Caucal, Edmond Iancu, A. H. Mueller, and Feng Yuan. Jet Definition and Transverse-Momentum-Dependent Factorization in Semi-Inclusive Deep-Inelastic Scattering. *Phys. Rev. Lett.*, 134(6):061903, 2025.
 - [74] Yiannis Makris. Revisiting the role of grooming in DIS. *Phys. Rev. D*, 103(5):054005, 2021.
 - [75] Max Knobbe, Daniel Reichelt, and Steffen Schumann. (N)NLO+NLL’ accurate predictions for plain and groomed 1-jettiness in neutral current DIS. *JHEP*, 09:194, 2023.
 - [76] Shen Fang, Mei-Sen Gao, Hai Tao Li, and Ding Yu Shao. $N^3\text{LL} + \mathcal{O}(\alpha_s^2)$ predictions of lepton-jet azimuthal angular distribution in deep-inelastic scattering. *JHEP*, 01:029, 2025.
 - [77] Haotian Cao, Zhong-Bo Kang, Xiaohui Liu, and Sonny Mantry. One-jettiness DIS event shape at $N^3\text{LL} + \mathcal{O}(\alpha_s^2)$. *Phys. Rev. D*, 110(1):014045, 2024.
 - [78] June-Haak Ee, Daekyoung Kang, Christopher Lee, and Iain W. Stewart. Precision DIS thrust predictions for HERA and EIC. *JHEP*, 07:240, 2025.
 - [79] Melissa van Beekveld, Luca Buonocore, Silvia Ferrario Ravasio, Pier Francesco Monni, Alba Soto-Ontoso, and Gregory Soyez. A new suite of Lund-tree observables to resolve jets. 11 2025.
 - [80] Vito Antonelli, Mrinal Dasgupta, and Gavin P. Salam. Resummation of thrust distributions in DIS. *JHEP*, 02:001, 2000.

-
- [81] M. Dasgupta and G. P. Salam. Resummation of the jet broadening in DIS. *Eur. Phys. J. C*, 24:213–236, 2002.
- [82] Mrinal Dasgupta and Gavin P. Salam. Event shapes in e^+e^- annihilation and deep inelastic scattering. *J. Phys. G*, 30:R143, 2004.
- [83] Daekyoung Kang, Christopher Lee, and Iain W. Stewart. Using 1-Jettiness to Measure 2 Jets in DIS 3 Ways. *Phys. Rev. D*, 88:054004, 2013.
- [84] Zhong-Bo Kang, Xiaohui Liu, and Sonny Mantry. 1-jettiness DIS event shape: NNLL+NLO results. *Phys. Rev. D*, 90(1):014041, 2014.
- [85] Zexuan Chu, Yunlu Wang, June-Haak Ee, Jinhui Chen, and Daekyoung Kang. 1-jettiness with jet axis at $O(\alpha_s)$ in deep inelastic scattering. *JHEP*, 06:111, 2022.
- [86] P. Agostini et al. The Large Hadron–Electron Collider at the HL-LHC. *J. Phys. G*, 48(11):110501, 2021.
- [87] A. Accardi et al. Electron Ion Collider: The Next QCD Frontier: Understanding the glue that binds us all. *Eur. Phys. J. A*, 52(9):268, 2016.
- [88] R. Abdul Khalek et al. Science Requirements and Detector Concepts for the Electron-Ion Collider: EIC Yellow Report. *Nucl. Phys. A*, 1026:122447, 2022.
- [89] Matteo Cacciari, Gavin P. Salam, and Gregory Soyez. The anti- k_t jet clustering algorithm. *JHEP*, 04:063, 2008.
- [90] S. Catani, Yuri L. Dokshitzer, and B. R. Webber. The K^- perpendicular clustering algorithm for jets in deep inelastic scattering and hadron collisions. *Phys. Lett. B*, 285:291–299, 1992.
- [91] M. Wobisch and T. Wengler. Hadronization corrections to jet cross-sections in deep inelastic scattering. In *Workshop on Monte Carlo Generators for HERA Physics (Plenary Starting Meeting)*, pages 270–279, 4 1998.
- [92] Stephen D. Ellis and Davison E. Soper. Successive combination jet algorithm for hadron collisions. *Phys. Rev. D*, 48:3160–3166, 1993.
- [93] S. Catani, Yuri L. Dokshitzer, M. H. Seymour, and B. R. Webber. Longitudinally invariant K_t clustering algorithms for hadron hadron collisions. *Nucl. Phys. B*, 406:187–224, 1993.
- [94] W. Bartel et al. Experimental Studies on Multi-Jet Production in e^+e^- Annihilation at PETRA Energies. *Z. Phys. C*, 33:23, 1986.

-
- [95] M. R. Adams et al. First measurements of Jet Production Rates in Deep Inelastic Lepton Proton Scattering. *Phys. Rev. Lett.*, 69:1026–1029, 1992.
- [96] M. R. Adams et al. Q^2 dependence of the average squared transverse energy of jets in deep inelastic muon - nucleon scattering with comparison to QCD. *Phys. Rev. Lett.*, 72:466–469, 1994.
- [97] T. Ahmed et al. Determination of the strong coupling constant from jet rates in deep inelastic scattering. *Phys. Lett. B*, 346:415–425, 1995.
- [98] M. Derrick et al. Observation of jet production in deep inelastic scattering with a large rapidity gap at HERA. *Phys. Lett. B*, 332:228–243, 1994.
- [99] M. Derrick et al. Measurement of α_s from jet rates in deep inelastic scattering at HERA. *Phys. Lett. B*, 363:201–216, 1995.
- [100] S. Aid et al. Transverse energy and forward jet production in the low x regime at HERA. *Phys. Lett. B*, 356:118–128, 1995.
- [101] C. Adloff et al. Diffractive dijet production at HERA. *Eur. Phys. J. C*, 6:421–436, 1999.
- [102] C. Adloff et al. Measurement of internal jet structure in dijet production in deep inelastic scattering at HERA. *Nucl. Phys. B*, 545:3–20, 1999.
- [103] C. Adloff et al. Forward jet and particle production at HERA. *Nucl. Phys. B*, 538:3–22, 1999.
- [104] C. Adloff et al. Measurement and QCD analysis of jet cross-sections in deep inelastic positron - proton collisions at $\sqrt{s}$ of 300 GeV. *Eur. Phys. J. C*, 19:289–311, 2001.
- [105] C. Adloff et al. Measurement of inclusive jet cross-sections in deep inelastic ep scattering at HERA. *Phys. Lett. B*, 542:193–206, 2002.
- [106] S. Chekanov et al. Inclusive jet cross-sections in the Breit frame in neutral current deep inelastic scattering at HERA and determination of α_s . *Phys. Lett. B*, 547:164–180, 2002.
- [107] S. Chekanov et al. An NLO QCD analysis of inclusive cross-section and jet-production data from the zeus experiment. *Eur. Phys. J. C*, 42:1–16, 2005.
- [108] S. Chekanov et al. Inclusive-jet and dijet cross-sections in deep inelastic scattering at HERA. *Nucl. Phys. B*, 765:1–30, 2007.
- [109] A. Aktas et al. Measurement of inclusive jet production in deep-inelastic scattering at high Q^2 and determination of the strong coupling. *Phys. Lett. B*, 653:134–144, 2007.

-
- [110] F. D. Aaron et al. Jet Production in ep Collisions at High Q^2 and Determination of α_s . *Eur. Phys. J. C*, 65:363–383, 2010.
- [111] F. D. Aaron et al. Jet Production in ep Collisions at Low Q^2 and Determination of α_s . *Eur. Phys. J. C*, 67:1–24, 2010.
- [112] V. Andreev et al. Measurement of multijet production in ep collisions at high Q^2 and determination of the strong coupling α_s . *Eur. Phys. J. C*, 75(2):65, 2015.
- [113] Frédéric A. Dreyer, Gavin P. Salam, and Grégory Soyez. The Lund Jet Plane. *JHEP*, 12:064, 2018.
- [114] Matteo Cacciari, Gavin P. Salam, and Gregory Soyez. FastJet User Manual. *Eur. Phys. J. C*, 72:1896, 2012.
- [115] B. R. Webber. Factorization and jet clustering algorithms for deep inelastic scattering. *J. Phys. G*, 19:1567–1575, 1993.
- [116] G. Breit. The Effect of Retardation on the Interaction of Two Electrons. *Phys. Rev.*, 34:553–573, 1929.
- [117] F. J. Ernst, R. G. Sachs, and K. C. Wali. Electromagnetic form factors of the nucleon. *Phys. Rev.*, 119:1105–1114, 1960.
- [118] R. G. Sachs. High-Energy Behavior of Nucleon Electromagnetic Form Factors. *Phys. Rev.*, 126:2256–2260, 1962.
- [119] R. Devenish and A. Cooper-Sarkar. *Deep inelastic scattering*. 2004.
- [120] Baptiste Cabouat and Torbjörn Sjöstrand. Some Dipole Shower Studies. *Eur. Phys. J. C*, 78(3):226, 2018.
- [121] Christian Bierlich et al. Robust Independent Validation of Experiment and Theory: Rivet version 3. *SciPost Phys.*, 8:026, 2020.
- [122] Peter Skands, Stefano Carrazza, and Juan Rojo. Tuning PYTHIA 8.1: the Monash 2013 Tune. *Eur. Phys. J. C*, 74(8):3024, 2014.
- [123] Matteo Cacciari, Gavin P. Salam, and Gregory Soyez. The Catchment Area of Jets. *JHEP*, 04:005, 2008.

-
- [124] P. J. Mulders and R. D. Tangerman. The Complete tree level result up to order $1/Q$ for polarized deep inelastic leptonproduction. *Nucl. Phys. B*, 461:197–237, 1996. [Erratum: *Nucl.Phys.B* 484, 538–540 (1997)].
- [125] John C. Collins, Davison E. Soper, and George F. Sterman. Transverse Momentum Distribution in Drell-Yan Pair and W and Z Boson Production. *Nucl. Phys. B*, 250:199–224, 1985.
- [126] S. Catani, L. Trentadue, G. Turnock, and B. R. Webber. Resummation of large logarithms in e^+e^- event shape distributions. *Nucl. Phys. B*, 407:3–42, 1993.
- [127] Roberto Bonciani, Stefano Catani, Michelangelo L. Mangano, and Paolo Nason. Sudakov resummation of multiparton QCD cross-sections. *Phys. Lett. B*, 575:268–278, 2003.
- [128] A. Banfi, G. P. Salam, and G. Zanderighi. Semi-numerical resummation of event shapes. *JHEP*, 01:018, 2002.
- [129] Edward Farhi. A QCD Test for Jets. *Phys. Rev. Lett.*, 39:1587–1588, 1977.
- [130] S. Brandt, C. Peyrou, R. Sosnowski, and A. Wroblewski. The Principal axis of jets. An Attempt to analyze high-energy collisions as two-body processes. *Phys. Lett.*, 12:57–61, 1964.
- [131] L. Clavelli. Jet Invariant Mass in Quantum Chromodynamics. *Phys. Lett. B*, 85:111–114, 1979.
- [132] Paul E. L. Rakow and B. R. Webber. Transverse Momentum Moments of Hadron Distributions in QCD Jets. *Nucl. Phys. B*, 191:63–74, 1981.
- [133] S. Catani, G. Turnock, and B. R. Webber. Jet broadening measures in e^+e^- annihilation. *Phys. Lett. B*, 295:269–276, 1992.
- [134] T. Chandramohan and L. Clavelli. Consequences of Second Order QCD for Jet Structure in e^+e^- Annihilation. *Nucl. Phys. B*, 184:365–380, 1981.
- [135] L. Clavelli and D. Wyler. Kinematical Bounds on Jet Variables and the Heavy Jet Mass Distribution. *Phys. Lett. B*, 103:383–387, 1981.
- [136] Andrea Banfi, Gavin P. Salam, and Giulia Zanderighi. Generalized resummation of QCD final state observables. *Phys. Lett. B*, 584:298–305, 2004.

-
- [137] S. Catani, B. R. Webber, Yuri L. Dokshitzer, and F. Fiorani. Average multiplicities in two and three jet $e^+ e^-$ annihilation events. *Nucl. Phys. B*, 383:419–441, 1992.
 - [138] M. Dasgupta and G. P. Salam. Resummation of nonglobal QCD observables. *Phys. Lett. B*, 512:323–330, 2001.
 - [139] Mrinal Dasgupta and Gavin P. Salam. Resummed event shape variables in DIS. *JHEP*, 08:032, 2002.
 - [140] S. Catani, B. R. Webber, and G. Marchesini. QCD coherent branching and semiinclusive processes at large x . *Nucl. Phys. B*, 349:635–654, 1991.
 - [141] Jonathan Gaunt, Maximilian Stahlhofen, Frank J. Tackmann, and Jonathan R. Walsh. N-jettiness Subtractions for NNLO QCD Calculations. *JHEP*, 09:058, 2015.
 - [142] Iain W. Stewart, Frank J. Tackmann, and Wouter J. Waalewijn. N-Jettiness: An Inclusive Event Shape to Veto Jets. *Phys. Rev. Lett.*, 105:092002, 2010.
 - [143] Stefano Catani. The Singular behavior of QCD amplitudes at two loop order. *Phys. Lett. B*, 427:161–171, 1998.
 - [144] Zhong-Bo Kang, Xiaohui Liu, Sonny Mantry, and Jian-Wei Qiu. The 1-Jettiness event shape for DIS with NNLL resummation. *Int. J. Mod. Phys. Conf. Ser.*, 25:1460041, 2014.
 - [145] Andrea Banfi, Jeffrey R. Forshaw, and Jack Holguin. The one-jettiness distribution contains super-super-leading logarithms. 11 2025.
 - [146] Thomas Becher, Patrick Hager, Matthias Neubert, and Dominik Schwienbacher. Factorization Beyond Coherence. 3 2026.
 - [147] Teppo T. Jouttenus, Iain W. Stewart, Frank J. Tackmann, and Wouter J. Waalewijn. The Soft Function for Exclusive N-Jet Production at Hadron Colliders. *Phys. Rev. D*, 83:114030, 2011.
 - [148] V. Ravindran, J. Smith, and W. L. Van Neerven. Next-to-leading order QCD corrections to differential distributions of Higgs boson production in hadron hadron collisions. *Nucl. Phys. B*, 634:247–290, 2002.
 - [149] Darcy Peake. Mathematica notebooks for the validation of ARES NNLL coefficients h_{10} , h_{21} and h_{22} . <https://github.com/darcypeake/1jettiness-coefficient-comparison>, 2026.

- [150] Jonathan R. Gaunt, Maximilian Stahlhofen, and Frank J. Tackmann. The Quark Beam Function at Two Loops. *JHEP*, 04:113, 2014.
- [151] Jonathan Gaunt, Maximilian Stahlhofen, and Frank J. Tackmann. The Gluon Beam Function at Two Loops. *JHEP*, 08:020, 2014.